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THEORETICAL TOOLS IN DB RESEARCH

Theoretical Tools in DB research

Lecture notes distilled from the SIGMOD 2026 full-corpus survey

Version 1.0 — First complete edition.

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Introduction: What DB Researchers Prove, and How

0.0 What these notes are

These lecture notes are a *toolbox tour* of the theoretical machinery that database researchers actually use, extracted mechanically from the complete proceedings of **SIGMOD 2026**: 385 research papers (PACMMOD Vol. 4, rounds 3–4, plus Vol. 3 issues 4–6, rounds 1–2). Every paper was fetched (arXiv $\text{\LaTeX}$ sources when available, ACM camera-ready PDFs otherwise), machine-extracted into formal statements, proofs, and bound sentences (2,405 statements, 1,130 proofs), and 228 theory-relevant papers were tagged with a controlled vocabulary of properties, tools, domains, and roles. The aggregates behind every figure in this chapter come from that survey; every worked example in the later chapters is a real paper from the corpus, cited in that chapter’s reference list.

Two honesty markers used throughout:

- $\text{\texttt{PODS}}$ marks exemplars from the **PODS’26 issue** (PACMMOD Vol. 4 No. 2, DOI band 10.1145/3801890–3801919): 29 of the 228 tagged papers (12.7%). They are part of the proceedings bundle; they often provide particularly clean exhibits of a tool, but they carry the theory-first culture of a sibling conference.
- Formal statements are transcribed from machine extraction of the papers; where the extraction was noisy we paraphrase conservatively rather than decorate.

0.1 The survey questions

1. **Modeling** — how do papers formalize a systems problem (cost model, adversary, noise model, distance metric)?

2. **Properties** — what do they prove about the artifact: correctness, complexity bounds, approximation/competitive ratios, statistical guarantees?
3. **Tools** — which theoretical machinery delivers the proof?
4. **Role** — does theory motivate the work, drive the design, serve as the evaluation yardstick, or *is* it the contribution?

0.2 *Properties: first correct, then fast*

Figure 1 (left) ranks what the 228 tagged papers prove. The headline surprise of the full-corpus count: **correctness is #1** (98 papers, 43%) — ahead of upper bounds (75) and probabilistic guarantees (42). The typical SIGMOD theory section is not a big theorem; it is a small invariant certifying that a component preserves a property (monotonicity of timestamps, order-independence of index construction, merge-safety of compressed blocks). Hardness (36) plays a different role: the standard *license for heuristics* in graph search, repair, and mining papers. The tail is where culture lives: **tightness** (11 — matching upper and lower bounds, a $5.5\times$ jump over the arXiv-only view of the corpus) and freshly activated slots such as competitive ratio and regret bounds (2 papers each, entering through systems doors: when to adapt, when to stop).

0.3 *Tools: a trinity plus a riser*

Figure 1 (right) ranks the machinery. A *trinity* dominates: **reductions** (52 papers), **concentration inequalities** (39), and **induction** (37) — respectively the tools of the hardness culture, the statistical-certification culture, and the component-correctness culture. The big riser once the ACM-only mass arrived is **exchange-greedy** (4→17 papers): prove a selection problem is a submodular/metric instance, inherit the approximation factor. Chapter 0.7 (the index table) maps each tool to its chapter.

0.4 *Domains and dialects*

Figure 2 (left) shows where theory lives. Graph systems (56 papers) are now the dominant dialect — hardness → branch-and-bound and parameterized algorithms → index-correctness lemmas. Privacy (26) is a near-monoculture of composition accounting (69% of its papers), increasingly in Rényi/f-DP currency. Streaming and sketching splits into a pure-theory wing (matching lower bounds) and estimator systems;

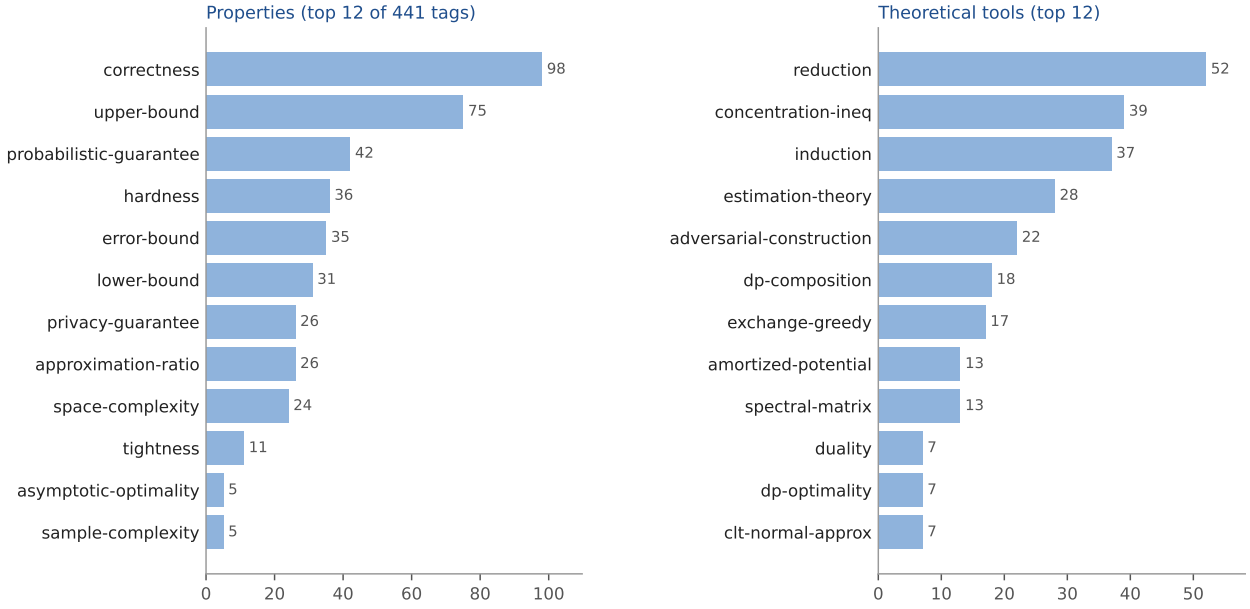


Figure 1: Left: most-studied properties (top 12 of 441 property tags on the 228 tagged papers). Right: most-used theoretical tools (top 12) — each later chapter is one tool.

vector search (25) splits between CLT-style certification and structural correctness of graphs and routers.

0.5 The playbook: *property* × *tool* pairings

Figure 2 (right) is the survey’s central cross-tab: which tool delivers which property. The two workhorses are **hardness** × **reduction** (32 papers) and **correctness** × **induction** (29). The pattern to internalize: DB theory compresses into about eight such pairings, and each chapter of these notes ends with the pairing(s) its tool owns.

0.6 Where theory sits

66% of tagged papers use theory as a *design driver*; in 25% the formal result *is* the paper (concentrated in the PODS issue and the pure streaming enclave); motivation-only and evaluation-yardstick roles cover the rest. The role split matters when reading exemplars: a design-driver paper proves what it builds; a theory-as-contribution paper builds nothing and proves everything.

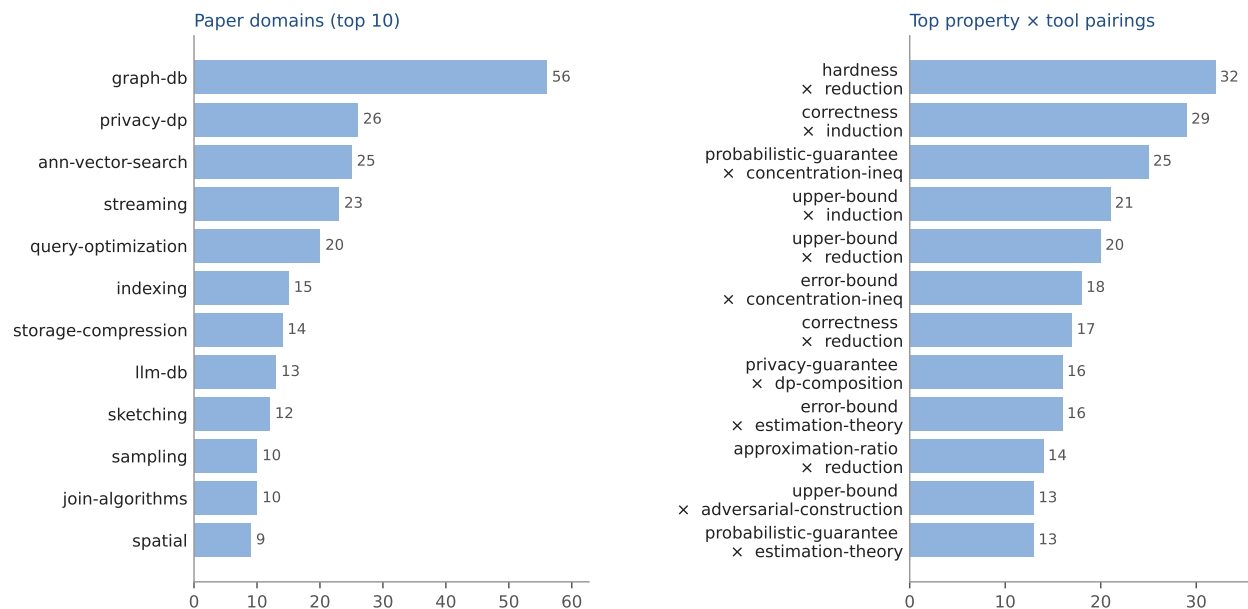


Figure 2: Left: paper domains (top 10). Right: top property $\times$ tool pairings (top 10) — the “playbook”.

0.7 Chapter map

Each of the following chapters is one tool: the formal core, then real SIGMOD 2026 exemplars with minimal context, then a “when to reach for it” box. Exemplars are transcribed from the papers’ own statements, with PODS-issue exemplars capped at one per chapter (two for communication complexity, whose corpus presence is small); a paper may recur in a second chapter when it is a clean exhibit of two tools — each chapter treats it under its own lens, and the ^{PODS} marker identifies PODS-issue exemplars in the text.

Tools not large enough for their own chapter in this corpus but worth knowing by name: adversarial-adjacent duality/LP arguments (7 papers), dynamic programming optimality (7), CLT/normal approximations (7), dyadic peeling (6), sketch mergeability theory (6), balls-bins-hashing (5) — several appear as guest stars inside the thirteen chapters.

0.8 How these notes were made

The survey was scripted end to end: the 385-paper list came from OpenAlex, sources were fetched (arXiv $\text{\LaTeX}$ when available, ACM camera-ready PDFs otherwise), statements, proofs, and bound sentences were machine-extracted (2,405 statements, 1,130 proofs), and the 228 theory-relevant papers were tagged against the controlled vocabulary above;

Table 1: Chapter map, ordered by how often the corpus uses each tool. “Papers” = tagged corpus papers using the tool; “Ex.” = exemplars featured in the chapter.

Ch.	Tool	Papers	Ex.
1	Reductions	52	6
2	Concentration inequalities	39	5
3	Induction	37	6
4	Estimation theory	28	6
5	Adversarial constructions	22	6
6	DP composition	18	6
7	Exchange-greedy arguments	17	6
8	Spectral / matrix tools	13	6
9	Amortized-potential analysis	13	6
10	Coresets & randomized NLA	8	6
11	Information-theoretic bounds	5	5
12	Communication complexity	4	4
13	Online decisions (competitive & regret)	4	4

every figure and count in this chapter is a direct aggregate of those tags.

These lecture notes are typeset with the `tufte-book` class of the Tufte- $\text{\LaTeX}$ project (Apache 2.0), as bundled in Boaz Barak’s panbook (MIT); the layout conventions — margin-column learning objectives, themed remark and recap blocks, end-of-proof markers, per-chapter references — follow Barak’s *Introduction to Theoretical Computer Science* notes (text licensed CC BY-NC-ND 4.0; we borrow the conventions only, no content). License notices for the distributed class files ship with the sources.

1

Reductions

Most “impossible” statements in database research are not proved from scratch: they are *imported*. A reduction exhibits a computable translation from a problem already believed hard — CLIQUE, Set Cover, inner-product estimation, hyperclique detection — into the problem at hand, so that any algorithm, approximation, private mechanism, or sampler for the target would also solve the source. The translation is a two-way conduit: instances travel forward through the gadget, and whatever the paper cares about — running time, approximation ratio, output count, memory, accuracy — travels backward, inheriting the source’s hardness. Run in the constructive direction, the same machinery turns new problems into old ones with known algorithms. Reductions are the most used theoretical tool in our SIGMOD’26 corpus: 52 of 228 tagged papers rely on them, 32 of those pair the reduction with an explicit hardness claim, and the heaviest concentration is graph analytics (14 papers).

This chapter builds the ladder of reduction types first — each rung named by the *currency* it preserves — and walks four corpus exemplars up it; Section 1.5 adds the advanced rungs (fine-grained hypotheses, reduction chains, algebraic transfer lemmas) as extension reading.

1.0 The tool: reductions

The problem class is uniform across applications. Somewhere there is a source problem H that a community has already failed to solve — CLIQUE, Set Cover, Disjointness, Hamiltonian path — together with evidence of its hardness: a proof (NP-completeness), a widely believed conjecture (the Unique Games Conjecture), or a quantitative hypothesis (SETH, combinatorial Boolean matrix multiplication, communication lower bounds). The paper’s target problem P is the one the reader actually cares about. What a reduction supplies is a pair of polynomial-time maps: a *gadget* f that builds an instance of P out of every instance of H , and a *decoder* g that reads a solution of quality q for $f(x)$ back into

LEARNING OBJECTIVES

- Recognize claims that need *imported* hardness, and pick the source problem.
- Match the rung to the currency: decision, ratio, count, time, space.
- Audit a gadget: instances forward, quality backward, parameters in every regime.
- Run reductions backwards to certify tractability — then glue both sides into a dichotomy.

a solution of quality $g(q)$ for x . Composing the two, any solver for P of quality q is a solver for H of quality $g(q)$; if H 's hardness statement rules that out, it rules out the solver for P . Figure 1.1 schematizes the two halves.

Hypotheses and currencies

Throughout, the *source* H carries a hardness statement at some resolution: qualitative (NP-hardness, #P-completeness), ratio-quantitative (no r -approximation under the Unique Games Conjecture), or fully quantitative (no $O(n^{2-\epsilon})$ algorithm under SETH; no $o(m)$ -communication protocol for Disjointness). The *currency* of a reduction is the quantity that travels backward through the decoder: yes/no membership, objective value (hence approximation ratio), output count, running time, memory, or estimation error. The target inherits exactly the source's resolution — no more: an NP-hardness source cannot yield an inapproximability ratio, and a conditional source yields a conditional bound.

1.0.1 The ladder

The classical rung preserves decision membership [6, 8, 10].

Definition 1.1 (Polynomial-time many-one reduction). Let L_1, L_2 be decision problems. We write $L_1 \leq_m^p L_2$ if there is a polynomial-time computable function f with $x \in L_1 \iff f(x) \in L_2$ for all x . ■

Proposition 1.2 — Hardness transfer. If L_1 is NP-hard and $L_1 \leq_m^p L_2$, then L_2 is NP-hard. ■

Proof. A polynomial-time algorithm for L_2 , composed with f , decides L_1 ; a polynomial composition of polynomials is polynomial. ■

This rung already reaches into databases, at the very origin of query-evaluation complexity [4]:

Example 1.3 (Conjunctive-query evaluation is NP-complete). Given a graph G and k , build in polynomial time the database D_G holding the edge relation of G , and let Q_k be the conjunctive query

$$Q_k = \exists v_1 \dots \exists v_k. \bigwedge_{1 \leq i < j \leq k} E(v_i, v_j). \quad \blacksquare$$

Then G has a k -clique iff $Q_k(D_G) \neq \emptyset$. Since CLIQUE is NP-complete, CQ evaluation is NP-hard (and it lies in NP).

Database problems, however, usually carry objectives — cost, error, latency — so the reduction must preserve *quality*, not just membership.

The standard rung is the approximation-preserving, or L -, reduction [6, 13]:

Definition 1.4 (L -reduction). An optimization problem A L -reduces to B if there are polynomial-time maps f (instances) and g (solutions) and constants $\alpha, \beta > 0$ such that, for every instance x of A , writing c_A, c_B for the objective values: (i) $\text{OPT}_B(f(x)) \leq \alpha \cdot \text{OPT}_A(x)$; and (ii) every solution y of $f(x)$ maps to a solution $g(y)$ of x with $|c_A(g(y)) - \text{OPT}_A(x)| \leq \beta \cdot |c_B(y) - \text{OPT}_B(f(x))|$. ■

Proposition 1.5 — Ratio transfer. If A L -reduces to B with constants α, β and B admits a polynomial-time r -approximation, then A admits an $\alpha\beta r$ -approximation. Hence inapproximability of A — e.g. Set Cover’s $\ln n$ threshold — transfers to B . ■

Proof. Apply the B -solver to $f(x)$ and decode by g : chaining (i) and (ii), the excess of $c_A(g(y))$ over $\text{OPT}_A(x)$ is at most $\alpha\beta$ times the excess of $c_B(y)$ over $\text{OPT}_B(f(x))$, so the achieved cost is within a $1 + \alpha\beta(r - 1)$, hence an $\alpha\beta r$, factor of the optimum. ■

1

When polynomial versus exponential time is too coarse, the source carries a quantitative hypothesis, and the reduction must *audit parameters* [14]:

Proposition 1.6 — Fine-grained transfer. Suppose a polynomial-time gadget f translates instances of H with parameters π into instances of P , and a solver for P can be leveraged (with polynomial postprocessing) using $N(\pi)$ calls on produced instances of size $s(\pi)$. Then any algorithm for P running in time T on size- $s(\pi)$ inputs yields an algorithm for H of cost $\text{poly}(\pi) + N(\pi) \cdot T(s(\pi))$. In particular, if this beats the time lower bound assumed for H in every parameter regime, the translated bound holds for P under the same hypothesis. ■

Proof idea. Count the cost of the emulation; the caveat is the whole point — a gadget whose parameter blow-up is benign in one regime and wild in another proves nothing there (cf. Section 1.1 for the audit done right). ■

Two further rungs complete the ladder. For counting problems the currency is the output count, and the natural notion is oracle access:

Definition 1.7 (Counting (Turing) reduction). A counting problem $\#A$ reduces to $\#B$, written $\#A \leq_{\#}^T \#B$, if $\#A(x)$ is computable in polynomial time given oracle access to $\#B$ on instances constructible in polynomial time. ■

¹ A *gap* reduction is the same idea with the ratio built into the gadget: f maps yes-instances to solutions of cost $\leq c$ and no-instances to instances where every solution costs $> \gamma c$, so any $(\gamma - \varepsilon)$ -approximation for B would decide H . Gap form and L -form are interchangeable currency for ratios; the corpus exemplar of Section 1.2 uses the L -form against Set Cover.

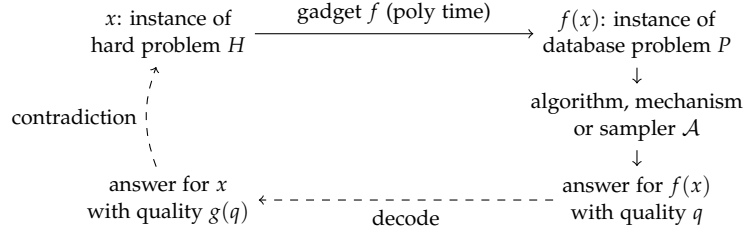


Figure 1.1: Every reduction has two halves: instances travel forward through the gadget; the currency — membership, cost ratio, count, time, space, or accuracy — travels backward, so quality achieved for P becomes quality for H , which the source’s hardness forbids.

Proposition 1.8 — #P-hardness transfer. If $\#A$ is #P-complete and $\#A \leq_{\#}^T \#B$, then $\#B$ is #P-hard. ■

Proof. Substitute the oracle: a polynomial-time algorithm computing $\#B$ turns the Turing reduction into a polynomial-time algorithm for the #P-complete $\#A$. ■

Finally, for streaming algorithms the classical bridge converts space into communication:

Proposition 1.9 — One pass, one message. If a one-pass streaming algorithm with s bits of memory computes a stream function F , then the two-party problem “Alice holds a stream prefix x , Bob holds the suffix y , compute $F(x \cdot y)$ ” has a deterministic protocol of communication s . A t -pass algorithm gives communication $O(s \cdot t)$. Consequently, any $\Omega(c)$ communication lower bound for the two-party problem forces $\Omega(c)$ (resp. $\Omega(c/t)$) space for one-pass (resp. t -pass) streams. ■

Proof. Alice simulates the algorithm on her prefix and sends Bob the final memory state — s bits, from which Bob resumes and finishes the pass. Multiple passes cross the channel once per pass per direction. ■

1.0.2 How to choose the rung

The selection procedure is mechanical once two questions are answered. *First, what must the claim rule out?* An algorithm (any quality) — rung of Definition 1.1; an approximation at some ratio — Definition 1.4 or its gap form; a fast algorithm at a refined running time — Proposition 1.6 under a quantitative hypothesis; a counter — Definition 1.7; a memory-frugal streaming algorithm — Proposition 1.9 against a communication bound. *Second, what does the source’s hardness actually say?* The target inherits the source’s resolution exactly: from Set Cover one may import a $\ln n$ inapproximability (under UGC) but not NP-hardness of approximation beyond it; from Disjointness, an $\Omega(m)$ space bound but

nothing about time; from SETH, quantitative time bounds conditioned on the hypothesis. Picking the source closest in *shape* to the target is what keeps the gadget small: covering structure suggests Set Cover, subgraph patterns suggest CLIQUE or hypercliques, private estimation suggests InnerProduct, enumeration suggests Hamiltonian path.

When to Reach for Reductions

Reach whenever a claim of the form “no system can do X ” or “doing X requires resource Y ” must be justified — or, constructively, whenever a new problem should inherit an existing algorithm. Then run the recipe:

1. Pick the canonical source problem closest in shape: Set Cover for covering and outlier budgets, CLIQUE/biclique for subgraph patterns, InnerProduct for private estimation, #Set Cover or the permanent for counting, Disjointness for stream space, Hamiltonian path for exact combinatorial targets.
2. Build the gadget f (instances forward) and the decoder g (quality backward); fix the currency and the matching rung of the ladder.
3. State explicitly which hypothesis the reader inherits: a proof (NP-, #P-), a conjecture (UGC), or a quantitative hypothesis (SETH, combinatorial BMM, a communication bound). Conditional claims must wear their conditions.
4. Audit the parameter blow-up of the gadget in *every* regime, and the ratio algebra ($\alpha\beta r$, Proposition 1.5) — a sloppy parameter map makes the bound vacuous somewhere.
5. Run the machinery backwards for the upper-bound side (reduce to LP feasibility, submodular maximization, dynamic programming); a dichotomy is two reductions glued at a common frontier.

1.1 Gadget reductions: privacy under continual release

The exemplar is a PODS’26 paper on differential privacy for streaming graphs [2]^{PODS}. When sensitive data arrives as a stream — friendship edges, transactions, sensor readings — a private system cannot wait for the stream to end: it must keep answering queries while the data grows, and every published answer leaks a little. The *continual release* model asks how accurate such a stream of answers can be while satisfying differential privacy throughout. Prior work had established

lower bounds only in restricted settings; this paper improves them to polynomial bounds for three graph statistics systems actually maintain — the maximum matching size, the degree histogram, and the k -core number — proving that any private incremental mechanism must incur additive error polynomial in the stream length T or the graph size n . It also gives the first lower bounds for Top- K norm estimation, extends the technique to monotone symmetric norms, and matches the lower bounds with binary-tree mechanisms whose error is polynomial with better-behaved constants — so the frontier it draws is essentially tight.

Problem. Work in the continual-release streaming model: an n -vertex graph evolves over T timesteps through a stream of edge insertions (*incremental streams*), and at every timestep the mechanism must output an estimate of a graph statistic g (*maintaining it*). Privacy is event-level differential privacy: any two streams differing in a single update event must yield indistinguishable output distributions. The mechanism has *additive error* α if, with probability at least $2/3$, at every timestep its estimate is within α of the true value $g(G)$ of the current graph.

2

Guarantee.

Theorem 1.10 — Continual-release lower bound, Theorem 1 of the paper. Let $\varepsilon \in [0, 1]$ and $\delta \in [0, 1/3]$. Any (ε, δ) -DP incremental mechanism in the exact setting for maintaining the size of the maximum matching, the degree histogram, or the core number of a vertex must have additive error

$$\alpha = \Omega(\min\{T^a, n^b\}),$$

for problem-specific constants $a, b \in [1/8, 1/2]$. ■

How the reduction is applied. The source problem is INNERPRODUCT: a secret dataset $x \in \{0, 1\}^d$ and queries $q^1, \dots, q^m \in \{0, 1\}^d$, with the mechanism answering every $\langle x, q^i \rangle$; a packing argument shows any (ε, δ) -DP mechanism for it has error $\tilde{\Omega}(\sqrt{d})$. The gadget is an *extendable bitwise-AND gadget*: a family of graphs H_x^i and Q_x^i , built from $v(d)$ vertices and $t(d)$ edges, in which the statistic's value jumps by exactly $\langle x, q^i \rangle$ between two queryable stream prefixes, and flipping one bit of x changes one stream event — event-level neighboring streams, so privacy of the graph mechanism implies privacy of the inner-product answers. The transfer is the paper's workhorse lemma:

Lemma 1.11 — Gadget transfer, Lemma 1 of the paper, notation cleaned. Let g be a graph problem that admits an extendable bitwise-AND gadget with $v(d)$ vertices and $t(d)$ edges per inner-product dimension. Any

² The “exact setting” in the paper’s statements means the mechanism maintains the exact statistic up to this additive error — no multiplicative slack is allowed. The exponents a, b are problem-specific: $1/4$ and $1/4$ for matching, $1/4$ and $1/2$ for the degree histogram, $1/8$ and $1/4$ for k -core.

(ε, δ) -DP event-level mechanism that maintains g in the exact setting, on n -node graphs over T timesteps, with additive error α , satisfies

$$\alpha = \tilde{\Omega}(\min\{t^-(T), v^-(n)\}), \quad f^-(x) = \sup\{d \mid f(d) \leq x\}. \quad \blacksquare$$

Proof sketch. Query the mechanism just before and just after inserting the edges that encode q^j ; the two estimates differ by $\langle x, q^j \rangle$ up to 2α , so the graph mechanism is an (ε, δ) -DP mechanism for INNERPRODUCT with error 2α ; the packing bound $\tilde{\Omega}(\sqrt{d})$ forces $\alpha \geq \tilde{\Omega}(\sqrt{d})$ with d set to the largest dimension the budget T, n admits. (Full proof in Section 1.6.) $\blacksquare$

The δ -tuning analogue is the *parameter audit*: d is chosen exactly as $d = \min\{t^-(T), v^-(n)\}$, the largest dimension whose gadget fits both the timestep budget T and the vertex budget n . Lemma 2 of the paper supplies the gadgets — $v(d) = t(d) = O(d^2)$ for matching, $v(d) = O(d)$ and $t(d) = O(d^2)$ for the degree histogram, $v(d) = O(d^2)$ and $t(d) = O(d^4)$ for k -core — and the audit turns them into exponents: matching gives $\alpha = \tilde{\Omega}(\min\{T^{1/4}, n^{1/4}\})$, the histogram $\alpha = \tilde{\Omega}(\min\{T^{1/4}, n^{1/2}\})$, and k -core $\alpha = \tilde{\Omega}(\min\{T^{1/8}, n^{1/4}\})$ — the constants $a, b \in [1/8, 1/2]$ of Theorem 1.10, with the polylogarithmic factors absorbed.

What it buys. The system designer learns that continually maintaining these statistics privately is polynomially expensive, *and where the escape hatches are*: the paper’s companion upper bounds (Theorem 2 of the paper) achieve error $O_{\varepsilon, \delta}(\min\{T^c, n^d\})$ with $c, d \in [1/6, 1/2]$ via binary-tree mechanisms and DP composition, while relaxing to the static setting — or allowing a $(1 + \zeta)$ multiplicative approximation — admits mechanisms with polylogarithmic error. The reduction did not just import hardness; it located the exact regime where hardness lives.

1.2 Gap reductions: clustering with set outliers

The exemplar is a SIGMOD’26 paper on robust clustering [12]. Clustering with outliers is a workhorse of robust analytics: alongside the k centers, the system may discard some data as unrepresentative. Prior models discard individual points; this paper discards whole *sets* from a candidate family — hyper-rectangles covering a spatial region (a maintenance window, a blackout area), or sets of tuples induced by a relational join, so that clustering can run directly over the join without materializing it. The paper develops near-linear-time $(2 + \varepsilon, 2, O(1))$ -approximation algorithms, coresets that shrink the instance to $O(\min\{n, k^2z, km\})$ points, and high-probability algorithms for the relational variants — and justifies the shape of these guarantees with a hardness result: under the Unique Games Conjecture, no polynomial-time algorithm can

keep all three quality dials at their optimum, and the outlier budget in particular cannot escape a logarithmic blow-up.

Problem. In *clustering with set outliers*, the input is a set P of n points, a family $\mathcal{H}$ of m subsets of P whose union covers P , and integer budgets $k, z > 0$. A solution picks at most k centers and at most z outlier sets from $\mathcal{H}$; all points covered by the chosen sets are discarded, and the goal is to minimize the clustering cost (here, the k -center objective) of the remaining points. An (α, β, γ) -approximation inflates the cost by at most α and the two budgets by at most β and γ . The relational variant takes P to be the answer set of an acyclic join query accessed through counting and sampling oracles.

3

³The Unique Games Conjecture (UGC) asserts that a certain two-prover game is hard to distinguish with high probability; taken with it, Set Cover admits no polynomial-time $(1 - \varepsilon) \ln n$ -approximation for any $\varepsilon > 0$. It is the standard conditional source for covering-type inapproximability.

Guarantee.

Theorem 1.12 — Inapproximability under UGC, Theorem of the paper. If the Unique Games Conjecture is true, there exists no $(1, f - \varepsilon, \gamma)$ -approximation algorithm that runs in polynomial time for the problem, where $f = o(\log n)$, $\gamma = o(|\mathcal{H}|)$ is any sublinear value, and ε any positive real number. ■

How the reduction is applied. The source is Set Cover, on the ratio rung: the outlier-set budget is the coordinate that carries the Set-Cover ratio, because inflating the budget by a factor f is exactly allowing $f \cdot z^*$ sets to cover the elements. The transfer is clean enough to state in full:

Lemma 1.13 — Set-Cover transfer, Lemma of the paper. If Alg is a polynomial-time $(1, f - \varepsilon, \gamma)$ -approximation algorithm for the set-outlier clustering problem, with $\gamma = o(|\mathcal{H}|)$ sublinear, then there exists a polynomial-time $(f - \varepsilon)$ -approximation algorithm for Set Cover. ■

Proof sketch. The gadget is a line-point encoding: element x_i becomes the point p_i at coordinate i of the real line; each set y becomes the outlier set $\{p_i : x_i \in y\}$; padding points, each with its own singleton outlier set, complete the family. With no centers, the cost is zero exactly when the chosen outlier sets cover every element-point, so a clustering solution within the inflated budgets is a set cover of the corresponding size. (Full proof in Section 1.6.) ■

The UGC-conditional $(1 - \varepsilon) \ln n$ threshold for Set Cover then crosses over: no $f = o(\log n)$ outlier-budget factor, even with sublinear center-budget slack. The same paper runs the machinery constructively — the run-backwards half of the recipe: an LP feasibility relaxation with greedy rounding and a weight-charging argument delivers $(2 + \varepsilon, 2f, 2 +$

ϵ)-approximations, coresets shrink the instance, and the relational variant reuses the LP through join sampling and counting oracles, with high-probability guarantees.

What it buys. The lower bound tells the system *which budget must pay*: the paper’s algorithms inflate the outlier budget by a logarithmic factor, and Theorem 1.12 certifies that (conditioned on UGC) essentially nothing better is possible — so the practitioner can treat $\Theta(\log n)$ outlier slack as the price of admission rather than an artifact of the analysis, and the join-based variants inherit the same frontier.

1.3 Both directions at once: volume-based answer diversity

The exemplar is a SIGMOD’26 paper on diversifying query answers [1]. When a query returns millions of tuples, a system shows the user a few *diverse* answers rather than the top ones; choosing them means maximizing a diversity function over answer sets. The familiar diversity measures are distance-based — pairs of answers compared under Hamming or Euclidean distance — but these functions fail to be submodular, so the greedy heuristics systems actually run come without guarantees, and the best provable ratios depend on the metric. The paper proposes *volume-based* diversity: each answer maps to a measurable set (its “volume”), and the diversity of an answer set is the measure of its union — a direct generalization of the classic multi-attribute notion. The payoff is a single theorem pair: diversity is always monotone and submodular, so one greedy routine delivers the optimal $(1 - 1/e)$ approximation for *every* polynomially computable volume assignment — and a matching Max-Coverage reduction shows no better ratio is possible.

Problem. Fix a query Q over a database D with answer set $Q(D)$. A *volume assignment* attaches to every answer s a measurable set $\beta(s)$ in a measure space $(\mathcal{U}, \mu)$:

Definition 1.14 (Volume-based diversity, Definition of the paper). Given a volume assignment $\Omega = (\mathcal{U}, \mu, \beta)$, the diversity of a finite answer set S is

$$\delta_\Omega(S) = \mu\left(\bigcup_{s \in S} \beta(s)\right). \quad \blacksquare$$

DiverseSet asks for $S \subseteq Q(D)$ with $|S| = k$ maximizing $\delta_\Omega(S)$; *DiverseTopk* asks for ranked enumeration.

Guarantee.

Proposition 1.15 — Diversity is submodular, Proposition of the paper. Let Ω be any volume assignment over a universe $\mathcal{U}$ of possible solutions. Then δ_Ω is monotone and submodular: for $S_1 \subseteq S_2$ and any answer a ,

$$\delta_\Omega(S_1 \cup \{a\}) - \delta_\Omega(S_1) \geq \delta_\Omega(S_2 \cup \{a\}) - \delta_\Omega(S_2). \quad \blacksquare$$

Corollary 1.16 — One ratio for all volumes, results of the paper. For every volume assignment Ω computable in polynomial time and every $k \geq 1$, the greedy algorithm computes S with $\delta_\Omega(S) \geq (1 - \frac{1}{e}) \cdot \max_{|S'|=k} \delta_\Omega(S')$ in polynomial time. No polynomial-time algorithm achieves ratio $(1 - \frac{1}{e} + \epsilon)$ for all volume assignments unless $P = NP$. $\blacksquare$

How the reduction is applied. This exemplar runs the tool in *both directions*. Backwards, it reduces DiverseSet to monotone submodular maximization: Proposition 1.15 certifies the structure, and the classical greedy analysis of Nemhauser, Wolsey, and Fisher supplies the algorithm and the $(1 - 1/e)$ ratio — one inherited theorem for the whole family of measures. Forwards, it imports hardness from Max Coverage on the ratio rung: the reduction encodes a coverage instance into a volume assignment over the atom algebra of the sets, so that each tuple’s volume is exactly what it covers and $\delta_\Omega(S)$ is the covered mass — and since Max Coverage cannot be approximated beyond $(1 - 1/e)$ unless $P = NP$, neither can volume-based diversity. A second, purely combinatorial gadget handles the exact variant: for the acyclic path query $Q(x_1, \dots, x_n) : -E(x_1, x_2), \dots, E(x_{n-1}, x_n)$, answers are walks in the input graph and the diversity of a single answer counts the distinct vertices it visits, so a walk of diversity n exists exactly when the graph has a Hamiltonian path — NP-hardness already for $k = 1$. The workhorse is the measure-algebra proof of Proposition 1.15: the marginal gain of adding a is $\mu(\beta(a) \setminus \bigcup_{s \in S} \beta(s))$, which shrinks as S grows (full proof in Section 1.6).

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⁴ For acyclic CQs the greedy step itself reduces further: finding the answer that maximizes marginal diversity is cast as a sum-product query over the tropical semiring, solvable in $O(|Q| \cdot |D|)$ time — so k greedy steps cost $O(k \cdot |Q| \cdot |D|)$, with ranked-enumeration preprocessing for DiverseTopk.

What it buys. The system keeps *one* diversification operator — greedy marginal-diversity — with a single certified ratio $(1 - 1/e)$ for every volume measure a user may define, instead of per-metric heuristics with per-metric guarantees; the matched lower bound certifies the ratio is optimal, and the tropical-semiring evaluation makes each step run in linear time over acyclic joins.

1.4 Counting reductions: Shapley values for aggregate queries

The exemplar is a SIGMOD’26 paper on explaining query answers [11]. When an aggregate answer surprises a user — why is this average so high? — the natural question is *which input facts drove it*. The Shapley

value, imported from cooperative game theory, is the canonical fair answer: it averages a fact's marginal contribution over all orders in which the endogenous facts could have arrived. Prior work pinned the complexity of Shapley values for Boolean conjunctive queries; this paper extends the analysis to *aggregate* queries — sum, max, min, and quantiles, with value functions applied at head positions — and draws the full frontier: computation is polynomial for hierarchical queries and #P-complete just past the frontier, with #P-completeness already for concrete small queries. Along the way it derives closed-form Shapley formulas for averages and quantiles, and uses the axioms of the Shapley value itself as an algorithmic tool on the tractable side.

Problem. A database D splits into an exogenous part D_{ex} (fixed) and an endogenous part D_{endo} (the facts that may be added or removed). For an aggregate query A the attribution question is:

Definition 1.17 (Shapley value of a fact, Definition of the paper). For $f \in D_{\text{endo}}$, let $\sigma_{<f}$ be the endogenous facts preceding f in a uniformly random permutation σ . The Shapley value of f is

$$\varphi(f, A) = \frac{1}{|D_{\text{endo}}|!} \sum_{\sigma \in \Sigma(D_{\text{endo}})} \left[A(D_{\text{ex}} \cup \sigma_{<f} \cup \{f\}) - A(D_{\text{ex}} \cup \sigma_{<f}) \right],$$

the expected amount by which inserting f changes the aggregate. ■

The aggregate queries are $A_\alpha = \alpha \circ Q$ for a self-join-free conjunctive query Q and $\alpha \in \{\text{sum}, \text{max}, \text{min}\}$ or a fixed quantile, composed with a *localized* value function applied to the values at head positions of Q .

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Guarantee.

Theorem 1.18 — Tractability frontier, Theorems of the paper, notation cleaned. Let Q be a CQ without self-joins, $A_\alpha = \alpha \circ Q$ with $\alpha \in \{\text{sum}, \text{max}, \text{min}\}$ or a fixed quantile. If Q is hierarchical, then $\varphi(f, A_\alpha)$ is computable in polynomial time, given D and a fact f , for every localized value function. Otherwise, there is a localized value function such that computing $\varphi(f, A_\alpha)$, given D and a fact f , is #P-complete. ■

How the reduction is applied. The currency is a *count*, so the rung is Definition 1.7 — the number of set covers, a #P-complete quantity, must be recoverable from Shapley values. The source is #Set-Cover:

Lemma 1.19 — #Set-Cover transfer, Lemma of the paper, notation cleaned. There is a polynomial-time algorithm that, given a #Set-Cover instance $(X, \mathcal{Y})$ with $|X| = n$, $|\mathcal{Y}| = m$, constructs databases $D_{q,r}$, $q \in [n]$, $r \in [m]$, each

⁵ A query is *hierarchical* if whenever two variables share an atom, the variables of one are contained in the other's atom set — equivalently, the connected-variable sets form a laminar family. It is the classical frontier between easy and hard conjunctive-query evaluation, and this paper shows the same line governs explanation.

with a distinguished fact f , such that the numbers

$$Z_{i,j} = \#\{ \mathcal{Z} \subseteq \mathcal{Y} : |\mathcal{Z}| = j, \bigcup_{Z \in \mathcal{Z}} Z \text{ covers exactly } i \text{ elements of } X \}$$

— in particular the number of set covers, $\sum_j Z_{n,j}$ — are computable in polynomial time from the Shapley values $\varphi(f, A)$ over the $D_{q,r}$. Hence computing Shapley values is #P-hard. ■

Proof sketch. Each database is built so that a coalition’s aggregate encodes how much of X the corresponding subfamily covers: chosen sets become endogenous facts, sentinel facts weight each covered element, and the fact f flips the answer exactly when the subfamily covers everything — so $\varphi(f, A)$ is a known linear combination of the $Z_{i,j}$, and the family $D_{q,r}$ provides enough independent combinations to solve for all of them. Theorem 1.18’s hardness side follows, together with a sibling reduction from the permanent (counting exact covers) for average-type aggregates. ■

On the tractable side the reduction runs backwards, through the axioms of the Shapley value itself: linearity decomposes an aggregate into indicator queries, symmetry equalizes the values of interchangeable facts, and null players — facts that never flip the answer — contribute zero; for hierarchical Q these identities collapse the sum of Definition 1.17 into a polynomial-time dynamic program over the query’s variable tree.

What it buys. An explanation engine learns exactly when fact-ranking is cheap: within the hierarchical fragment, responsibility scores come at evaluation cost, and one step outside it they are #P-complete even for two-atom queries — so a system should either restrict explanations to the fragment or budget for approximation, and the classical query-tractability map doubles as the explanation-tractability map.

1.5 *Advanced usage and further reading*

Three refinements round out the toolbox: quantitative hypotheses when the target claim is a refined running time, chained reductions when one hop is not enough, and transfer lemmas when the reduction’s purpose is to build an algorithm. This section states the pattern, sketches two harder exemplars, and points to the literature.

1.5.1 *Fine-grained and conditional reductions*

When the claim is not “no polynomial algorithm” but “no $O(N^{2-\varepsilon})$ / quasilinear / constant-delay algorithm”, the source must be quantitative: SETH, the hyperclique hypothesis, combinatorial Boolean matrix

multiplication, or listing lower bounds. The machinery is Proposition 1.6 with the parameter audit as the proof of correctness. The corpus uses it for query-evaluation dichotomies: direct access to conjunctive query answers in aggregate-driven orders admits quasilinear preprocessing exactly when the query has no chordless path between order variables — the hard side encodes hyperclique detection under the Hyperclique hypothesis and SETH [3]^{PODS}. A second shape runs the reduction against a *randomized* solver: for uniform sampling from join-project queries, a sampler restarted $O(\text{OUT} \log \text{OUT})$ times would list all answers — the coupon-collector argument turned backwards — so the assumed listing lower bound rules out fast samplers, and combinatorial BMM instantiates the hypothesis for matrix queries [7]^{PODS}.

1.5.2 Two advanced exemplars, sketched

Both exemplars of this section push past a single many-one hop; each is sketched in one paragraph, and the reduction chain of the first is transcribed in the Appendix (Section 1.6).

Reduction chains: pseudodeterministic streaming [5]. A streaming algorithm is *pseudodeterministic* if on every input it outputs a *canonical* value with probability at least $2/3$ — the reproducibility a debugger or an audit needs — and *representation oblivious* if its output distribution depends only on the multiset of stream elements, not their order. The paper proves that any one-pass representation-oblivious pseudodeterministic (ROPD) algorithm (ϵ, δ) -approximating F_0 (the number of distinct elements) requires $\Omega(n)$ space — against the $O(\text{polylog } n)$ classics — and $\Omega(n/t(n))$ for $t(n)$ passes, matching a deterministic algorithm. The argument is a *chain* on the communication rung (Proposition 1.9): an ROPD F_0 estimator becomes an ROPD protocol for Promise-Union (do Alice’s and Bob’s sets union to more than $3n/4$ of the universe?), which becomes a pseudodeterministic protocol for Disjointness, contradicting the classical $\Omega(m)$ communication bound; obliviousness is what carries equivalence classes of streams through each hop, and pseudodeterminism is what makes the final contradiction deterministic. The constructive side checks that the classical sketches (BJKST, AMS, linear sketches) *are* representation oblivious — the restriction bites exactly where canonical outputs are impossible.

Transfer lemmas: one algorithm for many query problems [9]. Counting satisfying subsets, evaluating tuple-independent probabilistic databases, and maximizing a repaired query’s multiplicity under bag-set semantics look like three problems, but over hierarchical queries they are

one: the paper annotates facts in a *two-monoid* algebra of decomposable provenance trees and runs a single variable-elimination algorithm (eliminate private variables, merge duplicate atoms). Correctness is itself a reduction statement — a transfer lemma: any map φ from provenance trees to another two-monoid that commutes with the two operations on decomposable trees carries the algorithm’s output to the correct value for the target problem, so each application (probability, counting, repair) is a homomorphism instantiation rather than a new algorithm, and the whole runs with a linear number of monoid operations. The frontier is closed with a many-one reduction in the other direction: for every non-hierarchical query, the repair problem is NP-complete via a biclique gadget — a full dichotomy in the reachbox’s sense.

1.5.3 Further reading

- Garey and Johnson [6] — the NP-completeness handbook; the gadget-construction craft of Sections 1.1 and 1.2 is their subject in the small.
- Karp’s twenty-one problems [8] — the original demonstration that one reduction technique spans seemingly unrelated problems.
- Papadimitriou [10] — computational complexity including the counting classes behind Section 1.4.
- Vazirani [13] — L -reductions and the approximation-preserving technology of Section 1.2.
- Vassilevska Williams and Williams [14] — the fine-grained equivalences program underlying Section 1.5.1.
- Chandra and Merlin [4] — conjunctive queries and homomorphisms, where database reduction arguments begin (Example 1.3).

Chapter Recap

- **What the tool is.** A reduction is a two-way conduit: a polynomial gadget carries instances of a hard source problem into the target, and a decoder carries quality backward — so any solver for the target solves the source, importing the source’s hardness at exactly its resolution.
- **Which rung when.** Ruling out algorithms: many-one (Definition 1.1). Ruling out approximations: L -gap reductions against a covering source (Definition 1.4). Ruling out counters: #P Turing reductions (Definition 1.7). Refined time bounds: fine-grained transfer under SETH/BMM/hyperclique hypotheses (Proposition 1.6). Stream space: one-pass-to-communication

conversion (Proposition 1.9).

- **Why the audit matters.** The gadget’s parameter blow-up must be checked in every regime — the privacy gadget of Section 1.1 chose its dimension $d = \min\{t^-(T), v^-(n)\}$ exactly so that both budgets bind, turning gadget sizes directly into the exponents of Theorem 1.10.
- **What the corpus showed.** Continual-release privacy (Section 1.1): InnerProduct gadgets import error bounds into DP stream maintenance. Set-outlier clustering (Section 1.2): a line-point encoding transfers Set Cover’s UGC threshold to the outlier budget. Answer diversity (Section 1.3): one problem reduced to submodular maximization (algorithm) and from Max Coverage (hardness) — both directions at once. Shapley attribution (Section 1.4): #Set-Cover counting reductions draw a dichotomy at the hierarchical frontier.
- **Advanced usage.** Fine-grained dichotomies under SETH/hyperclique/BMM hypotheses and coupon-collector reversals against randomized solvers (Section 1.5.1); reduction chains for pseudodeterministic streaming and algebraic transfer lemmas that unify algorithms (Section 1.5.2). Deferred proofs live in the Appendix (Section 1.6).

1.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Lemma 1.11 (gadget transfer, Section 1.1)

Statement (recalled). Any (ϵ, δ) -DP event-level mechanism maintaining exactly a graph problem g that admits an extendable bitwise-AND gadget with $v(d)$ vertices and $t(d)$ edges, on n -node graphs over T timesteps, has additive error $\alpha = \tilde{\Omega}(\min\{t^-(T), v^-(n)\})$, where $f^-(x) = \sup\{d \mid f(d) \leq x\}$. ■

Proof. Let $\mathcal{A}$ be such a mechanism with additive error α , and let $(x, q^1, \dots, q^m)$ be an INNERPRODUCT instance of dimension d , with d chosen as $d = \min\{t^-(T), v^-(n)\}$, so the gadget’s $v(d)$ vertices and $t(d)$ edges fit the budgets n and T . Emulate an (ϵ, δ) -DP mechanism for INNERPRODUCT: initialize the stream with the gadget’s H_{init} ; insert the edge e_i iff the data bit $x_i = 1$, giving the graph H_x^1 ; then, for each query q^j , (i) query $\mathcal{A}$, obtaining an estimate $\hat{e}^j$ of $g(H_x^j)$; (ii) insert the query edges S^j (those whose index corresponds to a query bit set to

one), giving Q_x^j , query $\mathcal{A}$ again, obtaining $\hat{q}^j$; (iii) output $\hat{q}^j - \hat{e}^j$ as the estimate of $\langle x, q^j \rangle$; (iv) insert the neutral edges T^j , giving H_x^{j+1} , ready for the next query. By the gadget's guarantee, the true values satisfy $\text{dec}_j(g(Q_x^j)) - \text{dec}_j(g(H_x^j)) = \langle x, q^j \rangle$, and the accuracy guarantee gives $|\hat{q}^j - g(Q_x^j)| \leq \alpha$ and $|\hat{e}^j - g(H_x^j)| \leq \alpha$ simultaneously with probability $2/3$, so every inner product is answered within 2α . Changing one bit of x changes one event of the stream (edge e_i inserted or not), so event-level DP of $\mathcal{A}$ makes the emulation (ε, δ) -DP — contradicting the packing lower bound $\tilde{\Omega}(\sqrt{d})$ for privately answering INNERPRODUCT unless $\alpha \geq \tilde{\Omega}(\sqrt{d}) = \tilde{\Omega}(\min\{t^-(T), v^-(n)\}^{1/2})$. ■

Proof of Lemma 1.13 (Set-Cover transfer, Section 1.2)

Statement (recalled). A polynomial-time $(1, f - \varepsilon, \gamma)$ -approximation for set-outlier clustering with sublinear γ yields a polynomial-time $(f - \varepsilon)$ -approximation for Set Cover. ■

Proof. Given a Set Cover instance $(X, \mathcal{Y})$, $X = \{x_1, \dots, x_{n'}\}$, build the clustering instance: for each element x_i , a point p_i at coordinate i of the real line; ℓ additional padding points $q_1, \dots, q_\ell$ at coordinates $2n' + 1, \dots, 2n' + \ell$; for each set $y \in \mathcal{Y}$, the outlier set $H_y = \{p_i : x_i \in y\}$; and singleton outlier sets $\{q_j\}$ for the padding points. Set the center budget to zero, so the objective is to cover all points by outlier sets, and the cost of any uncovered point exceeds any cover-based optimum (with the padding singletons forcing ℓ of the budget to be spent canonically, ℓ chosen polynomially bounded). *Yes to yes:* if $\mathcal{C} \subseteq \mathcal{Y}$ covers X with $|\mathcal{C}| \leq z$, then choosing the outlier sets $\{H_y\}_{y \in \mathcal{C}}$ plus the ℓ padding singletons covers every point within budget $z + \ell$, at cost zero. *Solution to cover:* conversely, any solution of cost zero (the cost factor 1 forces this) covers every element-point using some outlier sets; discarding the padding singletons, the corresponding sets y form a cover of X whose size is within the outlier-budget factor of the optimum cover size. Running the assumed $(1, f - \varepsilon, \gamma)$ -approximation on this instance (the sublinear center inflation γ is spent only on the padding construction) therefore returns a cover of size at most $(f - \varepsilon)$ times the optimum — a polynomial-time $(f - \varepsilon)$ -approximation to Set Cover, which the UGC-conditional $(1 - \varepsilon) \ln n$ threshold rules out for $f = o(\log n)$. ■

Proof of Proposition 1.15 (submodularity, Section 1.3)

Statement (recalled). For every volume assignment $\Omega = (\mathcal{U}, \mu, \beta)$, the diversity $\delta_\Omega(S) = \mu(\bigcup_{s \in S} \beta(s))$ is monotone and submodular. ■

Proof. *Monotonicity:* $S_1 \subseteq S_2$ implies $\bigcup_{s \in S_1} \beta(s) \subseteq \bigcup_{s \in S_2} \beta(s)$, and μ is monotone. *Submodularity:* let $S_1 \subseteq S_2$ and $a \notin S_2$ (if $a \in S_2$ both sides

are zero). For any S ,

$$\delta_\Omega(S \cup \{a\}) - \delta_\Omega(S) = \mu\left(\beta(a) \setminus \bigcup_{s \in S} \beta(s)\right),$$

by disjoint decomposition of the measure ($\mu(B \cup C) = \mu(B) + \mu(C \setminus B)$). Since $\bigcup_{s \in S_1} \beta(s) \subseteq \bigcup_{s \in S_2} \beta(s)$, we have $\beta(a) \setminus \bigcup_{s \in S_2} \beta(s) \subseteq \beta(a) \setminus \bigcup_{s \in S_1} \beta(s)$, and monotonicity of μ gives $\mu(\beta(a) \setminus \bigcup_{s \in S_2} \beta(s)) \leq \mu(\beta(a) \setminus \bigcup_{s \in S_1} \beta(s))$, which is the stated inequality. (This is the paper's proof; the Corollary 1.16 guarantee then follows from the classical greedy analysis of monotone submodular maximization.) ■

From estimator to protocol (Section 1.5.2)

Statement (recalled). If a one-pass ROPD streaming algorithm (ϵ, δ) -approximates F_0 over universe $[n]$ using s bits, then there is a pseudo-deterministic protocol of communication $O(s)$ for a Promise-Union $_n$ problem whose communication complexity is $\Omega(n)$; hence $s = \Omega(n)$ (Theorem 3.1 of the paper), and t -pass algorithms similarly need $\Omega(n/t)$ space. ■

Proof. Estimator to protocol. Alice holds $A \subseteq [n]$, Bob holds $B \subseteq [n]$. Alice converts A to its canonical stream $D(A)$ (the elements in lexicographic order), runs the estimator on it, and sends Bob the final memory state — s bits, exactly Proposition 1.9. Bob continues the execution with $D(B)$, so the algorithm processes the stream of $A \cup B$, and outputs 1 iff the returned estimate is at least $n/2$. On the promise (unions of size $\geq 3n/4$ versus $\leq n/4$), any (ϵ, δ) -approximation of $F_0 = |A \cup B|$ with $\epsilon, \delta \leq 1/3$ decides correctly with probability $\geq 2/3$. *Obliviousness carries the class:* $D(A) \cdot D(B)$ is a canonical representative of the R_0 -equivalence class of streams over $A \cup B$, so representation obliviousness makes the output distribution depend only on the union; pseudodeterminism supplies a canonical value g , and Bob's output equals the fixed predicate $h(A, B) = \mathbf{1}[g(D(A) \cdot D(B)) \geq n/2]$ with probability $2/3$ — the protocol itself is pseudodeterministic. *The final hop:* disjointness embeds into the promise gap by padding both players' inputs so that disjoint inputs land above the high threshold and intersecting ones below it, yielding a pseudodeterministic protocol for Disj $_m$ with communication $o(m)$ — contradicting the classical $\Omega(m)$ bound. The multi-pass bound ships the state back and forth once per pass: communication $O(s \cdot t)$, so $s = \Omega(n/t)$. ■

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2

Concentration Inequalities

Concentration inequalities bound the probability that a random quantity strays far from its expectation. For a systems paper they upgrade a randomized component that is correct *in expectation* — a sampler, sketch, random walk, or noise mechanism — into one correct *with high probability (whp)*: a quantified contract such as “ $(1 \pm \varepsilon)$ -accurate with probability $1 - 1/\text{poly}(n)$ ” or “running time $O(T)$ with probability $1 - 1/N$.” The recipe is constant: name the bad event, bound it by a tail inequality, union-bound over all copies of the component. In the SIGMOD’26 corpus this is among the most-used theoretical tools (39 tagged papers), directly supplying the probabilistic-guarantee property in 25 of them.

This chapter presents the basic ladder and the standard exemplars first; Section 2.4 adds the advanced rungs (martingales, two harder exemplars) as extension reading.

2.0 The tool: concentration inequalities

The problem class is uniform across applications: we can compute (or bound) the expectation of some random quantity X — the number of sampled points falling in a region, the error of a private estimate, the load of a hash bucket — but an algorithm must succeed on the *actual* draw, not the average. What concentration delivers is a bound of the form

$$\Pr[|X - \mathbb{E}X| \geq t] \leq \text{some decreasing function of } t,$$

plus the craft of making that function decrease fast enough. How fast is “fast enough” is the organizing question of this chapter: a bound of δ per bad event survives a union bound over $1/\delta$ events, so a component replicated $\text{poly}(n)$ times needs per-event failure $1/\text{poly}(n)$, which in turn needs tails that decay *exponentially* in the deviation.

LEARNING OBJECTIVES

- Recognize when a component needs a *whp* contract, and formalize the bad event.
- Prove the ladder Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff/Hoeffding, and match each rung to the dependence structure.
- Tune per-event failure to $\delta/(\text{events})$; union-bound over all copies.
- Dissect a paper’s argument: random object, independence, rung, δ budget.

Notation

“whp” (with high probability) always means probability at least $1 - 1/\text{poly}(n)$, or $1 - 1/N$ for a system of N components; the polynomial in the failure probability is set by the union bound over the whole system.

2.0.1 The basic ladder

Every bound in the family is Markov’s inequality in disguise [2, 4, 13].

Theorem 2.1 — Markov. Let $X \geq 0$ be a random variable with $\mathbb{E}X < \infty$. Then for every $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbb{E}X}{a}. \quad \blacksquare$$

Proof. $\mathbb{E}X \geq \mathbb{E}[X \cdot \mathbf{1}\{X \geq a\}] \geq a \Pr[X \geq a]$; divide by a . $\blacksquare$

One moment, one side, polynomial tail. Applying it to $(X - \mu)^2$ instead of X gives the two-sided, second-moment rung:

Corollary 2.2 — Chebyshev. Let $\mu = \mathbb{E}X$ and $\text{Var}(X) < \infty$. Then for every $t > 0$,

$$\Pr[|X - \mu| \geq t] \leq \frac{\text{Var}(X)}{t^2}. \quad \blacksquare$$

Proof. Markov (Theorem 2.1) applied to the nonnegative variable $(X - \mu)^2$ with $a = t^2$. $\blacksquare$

Polynomial tails suffice for constant success probability — boosted to high probability by repetition and medians — but they cannot survive a union bound over $\text{poly}(n)$ events. Exponential tails come from applying Markov to $e^{\lambda X}$:

Lemma 2.3 — Chernoff method. For any random variable X , any $a \in \mathbb{R}$ and any $\lambda > 0$,

$$\Pr[X \geq a] = \Pr[e^{\lambda X} \geq e^{\lambda a}] \leq e^{-\lambda a} \mathbb{E}[e^{\lambda X}],$$

and one optimizes over $\lambda > 0$ (the *moment generating function* controls the tail). $\blacksquare$

Proof. Markov (Theorem 2.1) applied to the nonnegative variable $e^{\lambda X}$ at level $a' = e^{\lambda a}$; the inequality is monotone in λ ’s sign, so only $\lambda > 0$ is useful. $\blacksquare$

Evaluating the moment generating function of a sum of independent bounded variables yields the two workhorses of randomized systems. The multiplicative form is the one to memorize for sampling and sketching [2, 3, 13]:

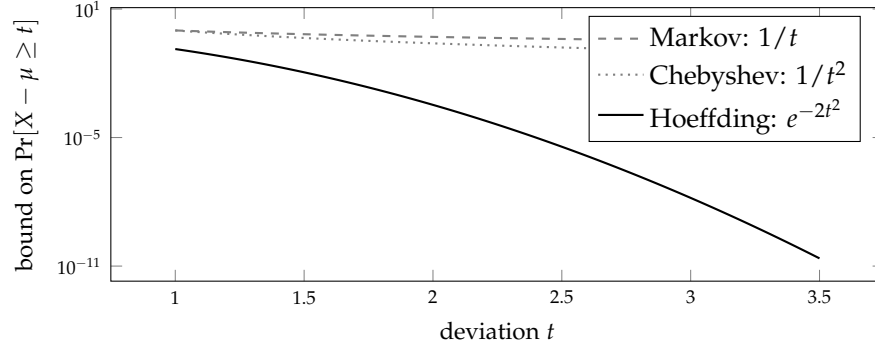


Figure 2.1: Tail decay for a unit-scale random quantity: moment bounds fall polynomially, independence buys the exponential curve — the gap that makes $\text{poly}(n)$ -way union bounds affordable.

Theorem 2.4 — Chernoff, multiplicative form. Let $X_1, \dots, X_m$ be independent random variables with $X_i \in [0, 1]$, $X = \sum_{i=1}^m X_i$, $\mu = \mathbb{E}X$. Then for every $\varepsilon \in (0, 1)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/3}, \quad \Pr[X \leq (1 - \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/2}. \quad \blacksquare$$

Proof idea. Chernoff method (Lemma 2.3) on X , then the elementary bound $(1 + \varepsilon)\ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ (and its lower-tail twin) on the optimized exponent; the full derivation is in the Appendix (Section 2.5). $\blacksquare$

The additive form is interchangeable whenever the summands are bounded but not naturally counts [6, 11, 13]:

Theorem 2.5 — Hoeffding, additive form. Let $X_1, \dots, X_m$ be independent with $X_i \in [a_i, b_i]$ a.s. and $X = \sum_i X_i$. Then for every $t > 0$,

$$\begin{aligned} \Pr[X - \mathbb{E}X \geq t] &\leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right), \\ \Pr[X - \mathbb{E}X \leq -t] &\leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right). \end{aligned} \quad \blacksquare$$

Proof idea. Chernoff method plus *Hoeffding's lemma*, $\mathbb{E}[e^{\lambda(X_i - \mathbb{E}X_i)}] \leq e^{\lambda^2(b_i - a_i)^2/8}$; optimize λ (proof in Section 2.5). $\blacksquare$

1

When independence fails but each random choice shifts the running statistic by a bounded amount, the ladder has two more rungs — Azuma–Hoeffding and McDiarmid — built on martingales; they are stated, with usage guidance, in Section 2.4.

2.0.2 How to choose a rung

The selection procedure is mechanical once two questions are answered. *First, what depends on what?* If the random object is a sum of independent

¹ The entry fee for both forms is *independence plus bounded summands*. The tunable currency is the exponent $-\varepsilon^2\mu$ or $2t^2 / \sum_i (b_i - a_i)^2$ — which must be pushed past $\log(\#\text{events}/\delta)$ for the failure probability to amortize. When the summands are unbounded but light-tailed (Gaussian, Laplace, sub-exponential, sub-gamma), the same Chernoff-method machinery applies with a moment-generating function that exists near 0; corpus papers use sub-gamma tails for approximate counters (Morris-type) and Laplace tails for privacy noise, and we will see the latter in Section 2.4.2.

bounded indicators — samples retained, points in a region, hashed keys — use multiplicative Chernoff when the bound should be relative to the expectation μ (sampling rates, subsampling), and additive Hoeffding when it should be relative to the range parameters $b_i - a_i$ (weighted sums, real-valued measurements). *Second, what moments are available?* If only the mean (or an exponent-optimized mean of a transform) is computable, Markov is the floor — and, applied to $e^{\lambda X}$, it is secretly the ceiling too.

When to Reach for Concentration

Reach whenever a randomized component must be certified — sampling/sketching estimators, random walks, sampled data structures, hashing, privacy noise — and the deliverable is a probability-qualified contract: accuracy $(1 \pm \varepsilon)$ whp, or time/space bounds holding with probability $1 - 1/N$. Then run the recipe:

1. Name the random object X and the bad event $\{|X - \mathbb{E}X| \geq t\}$.
2. Check the dependence structure: independent bounded summands $\rightarrow$ Chernoff/Hoeffding; adaptive exposure or capped per-variable influence $\rightarrow$ the martingale rungs of Section 2.4; only moments available $\rightarrow$ Markov/Chebyshev — constant probability, boostable by repetition with medians (and Markov on $e^{\lambda X}$ is the engine of every exponential-tail argument anyway).
3. Tune parameters so each bad event fails with probability at most $\delta/(\text{events})$ — i.e. push the exponent past $\log(\text{events}/\delta)$; only exponential tails amortize over $\text{poly}(n)$ copies, polynomial ones do not.
4. Union-bound over all copies of the component.

2.1 Trimmed statistics in a turnstile stream

The exemplar is a PODS'26 paper on robust streaming statistics [8]^{PODS}. Frequency moments, $F_p = \sum_i |x_i|^p$, are the workhorses of database analytics: query optimization, data mining, and network monitoring all rely on them, and the classical streaming results estimate them over the *whole* frequency vector. Many analytics tasks, however, want order-restricted quantities instead: statistics built from only the heaviest (or the central) part of the data, with the extremes treated as outliers — the streaming analogue of the trimmed mean from robust statistics. The paper studies such *trimmed statistics* — notably the F_p moment of the top- k coordinates, a Ky-Fan-type norm behind impact measures like the h -index — in the turnstile streaming model, and shows that for every

$p \in [0, 2]$ they can be estimated in a single pass using $\text{poly}(\log n/\varepsilon)$ bits of space.

Problem. Work in the *strict turnstile* streaming model: an n -dimensional frequency vector $\mathbf{x} \in \mathbb{Z}^n$ starts at zero and receives a long stream of updates of the form “add Δ to coordinate i ” (with Δ of either sign); the algorithm sees updates one at a time and must answer queries about the final $\mathbf{x}$ using space far less than n . Let $a_1, \dots, a_k$ be the k coordinates of $\mathbf{x}$ of largest absolute value. The *trimmed statistic* of order (p, k) is $\sum_{i=1}^k |a_i|^p$ for $0 \leq p \leq 2$: the F_p mass contributed by the top k coordinates — the part of the data that survives trimming away the tail. Such statistics are the streaming analogue of trimmed estimators in robust statistics.

Guarantee.

Theorem 2.6 — Top- k F_p estimation, Theorem 4 of the paper. Let $\mathbf{x} \in \mathbb{Z}^n$, $p \in [0, 2]$, $k \geq 0$, $\varepsilon \in (0, 1)$. Suppose $\|\mathbf{x}_{-k}\|_2^2 \geq (\varepsilon/\log n)^c \cdot k$ for a constant c , where $\mathbf{x}_{-k}$ excludes the top- k coordinates. Then there exists a *linear sketch* using $\text{poly}(\log n/\varepsilon)$ bits of space that estimates $\sum_{i=1}^k |a_i|^p$ to a $(1 \pm \varepsilon)$ factor with high constant probability. ■

2

How concentration is applied. The algorithm partitions coordinates by magnitude into *geometric level sets* whose boundaries are shifted by a random offset ζ uniform in $[1/2, 1]$, then *dyadically subsamples* each level: each coordinate of level S_j is kept independently with probability $r = \min\{1, C \log n/(\varepsilon^2 s_j)\}$ where $s_j = |S_j|$ and C is a constant. Survivors are found with a CountSketch³ fed with $\Theta(\log n)$ buckets per subsampling rate. The concentration step is the following lemma; its random objects are the independent retention indicators X_i of the coordinates of S_j , each $\text{Bernoulli}(r)$, so the survivor count has mean $s_j r$.

Lemma 2.7 — Level-size estimation, Lemma 4.4 of the paper, notation cleaned. Let S_j be a level set with $s_j = |S_j|$, and subsample each coordinate independently with probability $r = \min\{1, C \log n/(\varepsilon^2 s_j)\}$. If z coordinates of S_j survive and $\tilde{s}_j = z/r$, then with probability $1 - 1/\text{poly}(n)$,

$$(1 - \varepsilon) s_j \leq \tilde{s}_j \leq (1 + \varepsilon) s_j. \quad \blacksquare$$

Proof sketch. Write $u_1, \dots, u_{s_j}$ for the coordinates of S_j , and let X_i be the indicator that u_i is sampled, so $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff gives $\Pr[|\sum_i X_i - s_j r| \geq \varepsilon s_j r] \leq 2 \exp(-\varepsilon^2 s_j r/3) \leq 2 \exp(-C \log n/3) \leq 1/\text{poly}(n)$; divide by r . (Full proof in Section 2.5.) ■

² A *linear sketch* stores $A\mathbf{x}$ for a random matrix A ; it is the standard route to mergeable, one-pass, deletion-tolerant stream summaries. The condition on $\|\mathbf{x}_{-k}\|_2^2$ holds automatically once $k \geq \text{polylog } n$, so the theorem covers the practically interesting regime.

³ CountSketch (Charikar–Chen–Farach-Colton) recovers, from hashed counters, all coordinates whose value exceeds a θ -fraction of the residual ℓ_2 norm (a *heavy hitter* structure); the paper needs only this black box.

The δ tuning is explicit: r is chosen exactly so that the Chernoff exponent $\epsilon^2 s_j r$ equals $C \log n$, i.e. per-level failure $1/\text{poly}(n)$; there are only $O(\log_{1+\epsilon} m)$ levels (magnitude scales), so a union bound over levels preserves the polynomial failure probability. A companion lemma (Lemma 4.5 of the paper) uses the same Chernoff bound on the binomial survivor count to show that fewer than $O(\log n/\epsilon^2)$ coordinates survive per level whp — which is what makes the heavy-hitter structure, and hence the space bound, $\text{poly}(\log n/\epsilon)$.

What it buys. The system gets a one-pass, sublinear-space summary of trimmed statistics — robust aggregate mass that ignores the tail — with a certificate strong enough to union-bound into larger algorithms; the randomness of the offset ζ (rather than an adversarial grid) is what makes the level-set decomposition itself immune to worst-case input placement.

2.2 Maximum range sum by randomized grid shifting

The exemplar is a SIGMOD’26 paper on spatial query processing [5]. The *maximum range sum* (MaxRS) problem sits behind every “find the best location” query: given weighted points — customers, incidents, sensors — and a fixed-size geometric region Q , place Q anywhere in space so that it captures as much total weight as possible; that placement is where to site a store, a base station, or an emergency facility. The problem is well studied by both the database and the computational-geometry communities, and the paper revisits it in three practically motivated variations: *dynamic* MaxRS, where points are inserted and deleted and a good placement must be maintained under updates; *batched* MaxRS, where many query regions must be answered at once; and *colored* variants that group the points into classes. Its headline results are a randomized $(\frac{1}{2} - \epsilon)$ -approximation for dynamic MaxRS with polylogarithmic update time — the first algorithm of this kind — together with conditional lower bounds showing that the batched problem cannot be solved substantially faster than the trivial approach.

Problem. In the *maximum range sum* (MaxRS) problem, the input is a set P of n points in $\mathbb{R}^d$, each carrying a weight, and a query shape Q — here a d -ball of given radius. The goal is to place Q (translate it anywhere) so as to maximize the total weight of the points of P covered.

Exact MaxRS is expensive: for Q a 2-ball the colored version carries a conditional $\Omega(n^2)$ lower bound, and assuming that $(\min, +)$ -convolution on length- n sequences requires $\Omega(n^2)$ time, the batched variant on n points and m query radii requires $\Omega(mn)$ time — almost matching the trivial $O(mn \log n)$ upper bound.

Guarantee.

Theorem 2.8 — Static MaxRS, one representative case of the paper. For static MaxRS with Q a d -ball and any $0 < \varepsilon < 1/2$, there is a randomized $(\frac{1}{2} - \varepsilon)$ -approximation algorithm running in time $O(\varepsilon^{-2d-2} n \log n)$; the approximation factor holds with high probability, i.e. $1 - n^{-\Omega(1)}$. The dynamic version achieves the same factor with amortized update time $O(\varepsilon^{-2d-2} \log n)$. ■

How concentration is applied. The algorithm randomly shifts a grid over the point set and, per cell, takes a random sample of the cell's points; the candidate balls it examines are drawn from a finite set of $k \leq n$ possibilities determined by the shift. Two facts combine. First, a *geometric covering lemma*: if C is a d -ball of radius ρ and B a unit d -ball whose center lies at distance ρ^2 from the center of C , then the $(d-1)$ -dimensional measure of the cap $B \cap \partial C$ is at least a $(\frac{1}{2} - \Theta(\varepsilon))$ fraction of the measure of ∂C — i.e., a random point of C 's boundary lands in any fixed candidate ball with probability at least $\frac{1}{2} - \Theta(\varepsilon)$. Second, the concentration step:

Lemma 2.9 — Sampling concentration, Lemma of the paper, notation cleaned. Let $S = \{p_1, \dots, p_t\}$ be such that each point of S lies inside the candidate ball B independently with probability at least $\frac{1}{2} - \Theta(\varepsilon)$, and let $Y_B = \sum_{i=1}^t \mathbf{1}\{p_i \in B\}$. Then with high probability, simultaneously for all $k \leq n$ candidate balls,

$$Y_B \geq (1 - \varepsilon) \left(\frac{1}{2} - \Theta(\varepsilon) \right) t. \quad \blacksquare$$

Proof sketch. The $X_i = \mathbf{1}\{p_i \in B\}$ are independent indicators with $\mathbb{E}[Y_B] \geq (\frac{1}{2} - \Theta(\varepsilon))t$; multiplicative Chernoff gives $\Pr[Y_B < (1 - \varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 t/2}$, and a union bound over the $k \leq n$ balls keeps this whp provided $t = \Theta(\varepsilon^{-2} \log n)$. (Full proof in Section 2.5.) ■

The random objects are the per-point sampling/shift indicators; the structure is full independence across points (the random shift acts independently per cell); the rung is multiplicative Chernoff; and the δ tuning sets $t = \Theta(\varepsilon^{-2} \log n)$ so each of the n candidate balls fails with probability $n^{-\Omega(1)}$ — the same “push the exponent past $\log(\text{events})$ ” accounting as in Section 2.1, with the exponent coming from sample size rather than subsample rate.

What it buys. The spatial engine answers best-placement queries in near-linear time with a certified $(1/2 - \varepsilon)$ fraction of the optimum whp — and supports updates in polylog amortized time — where

⁴ *Conditional lower bound:* a reduction from $(\min, +)$ -convolution — widely conjectured to need quadratic time — transfers that hardness to MaxRS. This is the paper's evidence that *approximation* is necessary, and approximation is exactly where concentration enters.

exact computation faces (conditional) quadratic barriers. The same Chernoff-plus-union-bound skeleton, re-instantiated with trapezoidal maps, gives the paper’s $(1 - \epsilon)$ -approximation for the colored variant.

2.3 *User-level private query processing with sampled tuples*

The exemplar is a SIGMOD’26 paper on private query processing [12]. Answering select–join–aggregate (SJA) queries under differential privacy is a basic building block of privacy-preserving analytics: the system must return an accurate aggregate while guaranteeing that no adversary can confidently infer whether any one *user’s* data was present or absent. The state of the art achieves strong, instance-optimal accuracy at this *user level*, but pays for it by solving expensive optimization programs over the whole database — prohibitive at scale. Sampling is the classic escape hatch, buying a tunable trade-off between accuracy and computational cost, yet combining it with private processing is subtle: it is unclear *what* to sample for the best accuracy under a given compute budget, and the existing mathematical tool chains were not designed with sampling in mind. The paper builds a scalable user-level-DP SJA engine that privately truncates each heavy user’s contribution and Poisson-samples tuples to accelerate the truncation search, with an error analysis that cleanly separates sampling error from privacy noise.

Problem. Consider a select–project–join–aggregate workload over a relational instance whose rows belong to *users*, each user contributing many tuples. The privacy contract is differential privacy at the granularity of users:

Definition 2.10 (ϵ -differential privacy, user level). A randomized mechanism M satisfies ϵ -differential privacy if for all *neighboring* inputs D, D' and all measurable sets S ,

$$\Pr[M(D) \in S] \leq e^\epsilon \Pr[M(D') \in S].$$

For *user-level* DP, “neighboring” means *all* tuples of one user added or removed — much stricter than tuple-level DP.

Unbounded per-user contributions then blow up sensitivity, so the system [12] truncates each user’s contribution to a threshold τ chosen privately by a search (sparse vector technique), and accelerates the search by Poisson-sampling tuples at rate q before privatizing.

5

⁵ Poisson sampling also *amplifies* privacy: a mechanism that is ϵ -DP on the sample is (ϵ') -DP on the full data with $\epsilon' = \ln(1 + q(e^\epsilon - 1)) < \epsilon$; the paper’s Sample-and-Explore comparison baselines use this. The proof is a binomial tail computation — again concentration.

Guarantee.

Theorem 2.11 — Error of the sampled truncation, Algorithm 2 of the paper, compressed. The mechanism satisfies $(\varepsilon, 0)$ -DP. Let n be the number of records, m the number of users, and q the sample rate. Write $\tau^*(T) = \max_u \sum_{t \in T} \mathbf{1}[u \prec t]$ for the true maximum per-user contribution, and $\varepsilon_i^* \geq \varepsilon/L$ for the noise-calibration budget over the L search iterations. With probability $1 - \beta$, the released answer $\hat{f}(T)$ obeys

$$|\hat{f}(T) - f(T)| = \tilde{O}\left(\sqrt{\frac{n}{q}} + \frac{1}{q} + \frac{1}{\varepsilon_i^*} \left(\tau^*(T) + \sqrt{\frac{\tau^*(T)}{q}}\right)\right),$$

where $\tilde{O}(\cdot)$ hides constants and polylogarithmic factors in m and $1/\beta$. ■

How concentration is applied. The random objects are the independent sampling indicators $\mathbf{1}\{t \text{ sampled}\} \sim \text{Bernoulli}(q)$ over the n tuples. The sampled statistic $f(S) = \sum_{t \in T} w_t \mathbf{1}[t \text{ sampled}]$ satisfies $\mathbb{E}[f(S)] = q f(T)$ and $\text{Var}[f(S)] = q(1-q) \sum_t w_t^2 \leq qn$: a sum of independent bounded variables with an explicit variance proxy — the territory of the Bernstein/Chernoff rung, whose bound gives $|f(S) - qf(T)| \lesssim \sqrt{qn \log(3L/\beta)}$ with probability $1 - \beta/(3L)$; dividing by q yields the $\sqrt{n/q}$ term. The δ -budgeting is again structural: the truncation-threshold search runs L iterations, each is allocated failure probability $\beta/(3L)$, and a union bound over iterations upgrades the per-iteration bound to the global $1 - \beta$ contract — with the same per-event log factor $\log(3L/\beta)$ appearing inside the noise scale. The $\tau^*(T)/\varepsilon_i^*$ terms come from the Laplace-style calibration of the final release against the (privately learned) per-user cap.

What it buys. The query engine can offer user-level — not merely tuple-level — privacy at a modest accuracy price: the sampling error $\sqrt{n/q}$ decouples from the privacy noise, so tuning q trades compute for accuracy without touching the privacy guarantee, and the explicit $\tau^*(T)$ dependence quantifies exactly how much heavy user-contributors cost the answer.

2.4 Advanced usage and further reading

Everything so far needs independence. Two extensions belong in every practitioner’s backpack: the martingale rungs that replace independence with bounded per-step influence, and the light-tailed regime where summands are unbounded but have a moment generating function. This section states the rungs, sketches two harder exemplars, and points to the literature.

⁶ Recall: a martingale difference sequence (Y_i) with respect to a filtration $(\mathcal{F}_i)$ has $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0$; the partial sums $M_j = \sum_{i \leq j} Y_i$ then satisfy $\mathbb{E}[M_j \mid \mathcal{F}_{j-1}] = M_{j-1}$.

2.4.1 Martingale rungs: Azuma–Hoeffding and McDiarmid

Expose the randomness sequentially and book the shifts as a *martingale*⁶:

Theorem 2.12 — Azuma–Hoeffding. Let $(Y_i)_{i=1}^m$ be a martingale difference sequence with $|Y_i| \leq c_i$ a.s. Then for every $t > 0$,

$$\Pr \left[\sum_{i=1}^m Y_i \geq t \right] \leq \exp \left(- \frac{t^2}{2 \sum_{i=1}^m c_i^2} \right). \quad \blacksquare$$

Proof idea. Chernoff method on $\sum_i Y_i$, Hoeffding’s lemma applied to each Y_i *conditionally on the past* (boundedness is all that is needed; independence is replaced by iterated conditioning), then optimize λ (Section 2.5). $\blacksquare$

The function-of-independent-variables form is the rung most used for adaptive algorithms and data structures, where the analyzed quantity is not a sum but a function whose per-variable influence is capped [2, 4, 10]:

Theorem 2.13 — McDiarmid, bounded differences. Let $X_1, \dots, X_m$ be independent random variables and $f: \prod_i \mathcal{X}_i \rightarrow \mathbb{R}$ a measurable function satisfying the bounded differences condition: for each i there is $c_i \geq 0$ with

$$|f(x_1, \dots, x_i, \dots, x_m) - f(x_1, \dots, x'_i, \dots, x_m)| \leq c_i$$

for all inputs and all x'_i . Then for every $t > 0$,

$$\Pr [f(X_1, \dots, X_m) - \mathbb{E}f \geq t] \leq \exp \left(- \frac{2t^2}{\sum_{i=1}^m c_i^2} \right). \quad \blacksquare$$

Proof idea. Apply Azuma–Hoeffding (Theorem 2.12) to the *Doob martingale* $Y_i = \mathbb{E}[f \mid X_1, \dots, X_i] - \mathbb{E}[f \mid X_1, \dots, X_{i-1}]$, whose increments are bounded by c_i exactly because changing X_i moves f by at most c_i (Section 2.5). $\blacksquare$

When these rungs earn their keep. If the algorithm is *adaptive* — later random choices conditioned on earlier estimates, sampling without replacement, stopping times — the summands are dependent but each step’s effect is bounded, and Azuma–Hoeffding applies. If the quantity is a *function* of many independent inputs (a data structure’s quality, a clustering cost, a query plan’s runtime) with per-input influence at most c_i , McDiarmid converts that directly. Hashing with limited independence is the classic gap: the indicators are not fully independent, but exposure by hash key gives bounded increments.

2.4.2 Two advanced exemplars, sketched

Both exemplars of this section are privacy papers that push past the sum-of-independent-bounded-summands regime; each is sketched in

one paragraph, and the round-by-round argument of the first is transcribed in the Appendix (Section 2.5).

Private graph pattern counting under LDP [7]. Counting subgraph patterns — triangles, paths, stars — is a cornerstone of network analysis (community detection, fraud detection), and releasing such counts leaks information about individual edges. Under *local* differential privacy there is no trusted curator: each node of the graph randomizes its own messages before anything is aggregated, and existing LDP mechanisms were ad hoc, handling only specific patterns such as triangles and stars. The paper gives the first general mechanism for counting arbitrary *acyclic* patterns under LDP. It runs k rounds of message passing, adding fresh Laplace noise — *sub-exponential* tails, so the Chernoff method still applies — at every node in every round. The δ -accounting is the recipe of Section 2.0.2 in its purest form: failure budget β'/N per node per round, union bound over the kN node-rounds, then an *induction on rounds* that grows the noise scale geometrically but controllably. The resulting estimate is unbiased with error $\tilde{O}(\sqrt{N d(G)^{k-1}})$ — the square root of the maximal possible count. For simple paths, the tail is only polynomial in $1/\beta$ (a variance/Markov argument), illustrating the price of dropping exponential tails.

Product noise as a concentration bound [9]. Adding calibrated noise to a released statistic is the most fundamental route to differential privacy: the system outputs $f(x) + \mathbf{n}$, and the guarantee rests on the tail of the *privacy-loss random variable* $L_M = \ln \frac{dP}{dQ}$ — (ϵ, δ) -DP holds exactly when $\Pr[L_M \geq \epsilon] \leq \delta$, a tail bound in disguise. For Gaussian noise L_M is itself a univariate Gaussian, and this is where standard analyses stop. The paper instead exploits the spherical symmetry of typical noise distributions to generate $\mathbf{n}$ from a *product* of two independent components — a radius controlling the noise magnitude and a direction controlling its angle relative to the sensitivity vector — which re-expresses L_M as a product measure and lets the tail be optimized directly. The paper never writes the word “Chernoff,” but its argument — factorize L_M into independent terms, bound $\mathbb{E}[U^q]$ for a transform U , and *minimize the resulting δ over the exponent q* — is exactly the Chernoff method (Lemma 2.3) on a likelihood ratio. The payoff: expected noise smaller by a $\Theta(\log(1/\delta))$ factor than the Gaussian mechanism at equal privacy.

2.4.3 Further reading

- Boucheron, Lugosi, and Massart [2] — the comprehensive non-asymptotic treatment; the martingale and supremum versions live here.

- Dubhashi and Panconesi [4] — the randomized-algorithms viewpoint, with the exposure/Doob-martingale technique foregrounded.
- Mitzenmacher and Upfal [11] — textbook treatment with CS applications and all the Chernoff variants used in systems papers.
- The originals: Chernoff (1952) [3], Hoeffding (1963) [6], Azuma (1967) [1], and McDiarmid’s bounded-differences survey [10].
- For a two-page refresher with proofs of the additive forms, the Stanford CS229 notes [13].

Chapter Recap

- **What the tool is.** Concentration inequalities convert expectations into with-high-probability contracts: $\Pr[|X - \mathbb{E}X| \geq t]$ bounded by a tail function, tuned so that per-event failure $\delta/(\text{events})$ survives a union bound over all copies of the randomized component.
- **Which rung when.** Independent bounded indicators/sums: multiplicative Chernoff (relative error) or additive Hoeffding (range-based). Adaptive or dependent exposure with bounded per-step effect: Azuma–Hoeffding. Function of many independent inputs with capped per-input influence: McDiarmid. Only moments available: Markov and Chebyshev — constant-probability results, boostable by medians. Light-tailed unbounded noise (Laplace, Gaussian, sub-gamma): the Chernoff method still applies via the moment generating function.
- **Why exponential tails.** Only $e^{-\Omega(\text{exponent})}$ tails make $\text{poly}(n)$ -way union bounds affordable; polynomial tails ($1/t$, $1/t^2$) cannot amortize — visible in the corpus as the gap between $\tilde{O}(\cdot)$ guarantees (Laplace/Chernoff paths) and $1/\beta$ -type errors (variance-only paths).
- **What the corpus showed.** Trimmed-statistic sketching (Section 2.1): Chernoff on subsampling survivors, union over $O(\log_{1+\epsilon} m)$ levels. Spatial MaxRS (Section 2.2): Chernoff on sampled point-in-ball indicators, union over $\leq n$ candidate balls. User-level private queries (Section 2.3): Bernstein/Chernoff on Bernoulli-sampled sums with $\beta/(3L)$ budgets over search iterations.
- **Advanced usage.** The martingale rungs (Section 2.4.1) replace independence with bounded per-step influence; the privacy exemplars of Section 2.4.2 run the Chernoff method on Laplace

noise and on a likelihood ratio. Long proofs of everything deferred live in the Appendix (Section 2.5).

2.5 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 2.4 (Chernoff, multiplicative form)

Statement (recalled). Let $X_1, \dots, X_m$ be independent with $X_i \in [0, 1]$, $X = \sum_i X_i$, $\mu = \mathbb{E}X$. Then $\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/3}$ and $\Pr[X \leq (1 - \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/2}$. ■

Proof. By the Chernoff method (Lemma 2.3), for $\lambda > 0$, $\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\lambda(1+\varepsilon)\mu} \prod_i \mathbb{E}[e^{\lambda X_i}]$. For $X_i \in [0, 1]$ independence gives $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, so the product is at most $e^{\mu(e^\lambda - 1)}$ and, choosing $\lambda = \ln(1 + \varepsilon)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq \left(\frac{e^\varepsilon}{(1 + \varepsilon)^{1+\varepsilon}} \right)^\mu \leq e^{-\varepsilon^2\mu/3},$$

using $(1 + \varepsilon)\ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ for $\varepsilon \in (0, 1)$. The lower tail is symmetric, with exponent $\varepsilon^2/2$. ■

Proof of Theorem 2.5 (Hoeffding, additive form)

Statement (recalled). Let $X_1, \dots, X_m$ be independent with $X_i \in [a_i, b_i]$ a.s. and $X = \sum_i X_i$. Then for every $t > 0$, $\Pr[X - \mathbb{E}X \geq t] \leq \exp(-2t^2 / \sum_i (b_i - a_i)^2)$, and the same for the lower tail. ■

Proof. Hoeffding's lemma: if $Y \in [a, b]$ a.s. and $\mathbb{E}Y = 0$, then $\mathbb{E}[e^{\lambda Y}] \leq e^{\lambda^2(b-a)^2/8}$. (Proof: $e^{\lambda y}$ is convex, so $e^{\lambda Y} \leq \frac{b-Y}{b-a}e^{\lambda a} + \frac{Y-a}{b-a}e^{\lambda b}$ pointwise; take expectations and minimize over a shifted λ .) Applying the lemma to each centered summand and multiplying by independence, $\mathbb{E}[e^{\lambda(X - \mathbb{E}X)}] \leq e^{\lambda^2 \sum_i (b_i - a_i)^2/8}$; the Chernoff method then gives $\Pr[X - \mathbb{E}X \geq t] \leq e^{-\lambda t + \lambda^2 \sum_i (b_i - a_i)^2/8}$, and optimizing at $\lambda = 4t / \sum_i (b_i - a_i)^2$ yields the stated $e^{-2t^2 / \sum_i (b_i - a_i)^2}$. ■

Proof of Theorems 2.12 and 2.13 (martingale rungs)

Statement (recalled). (i) If $(Y_i)_{i=1}^m$ is a martingale difference sequence with $|Y_i| \leq c_i$ a.s., then $\Pr[\sum_i Y_i \geq t] \leq \exp(-t^2/2 \sum_i c_i^2)$. (ii) If f has bounded differences (c_i) over independent $X_1, \dots, X_m$, then $\Pr[f - \mathbb{E}f \geq t] \leq \exp(-2t^2 / \sum_i c_i^2)$. ■

Proof. Let $M_j = \sum_{i \leq j} Y_i$ with $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0$ and $|Y_i| \leq c_i$. Iterated conditioning gives $\mathbb{E}[e^{\lambda M_m}] = \mathbb{E}[e^{\lambda M_{m-1}} \mathbb{E}[e^{\lambda Y_m} \mid \mathcal{F}_{m-1}]]$, and *conditionally on $\mathcal{F}_{m-1}$* the variable Y_m is centered and bounded in an interval of length $2c_m$, so Hoeffding's lemma applies with $(b-a) = 2c_m$. Unrolling, $\mathbb{E}[e^{\lambda M_m}] \leq e^{\lambda^2 \sum c_i^2 / 2}$, and optimizing λ gives the stated tail. For McDiarmid, set $Y_i = \mathbb{E}[f \mid X_1, \dots, X_i] - \mathbb{E}[f \mid X_1, \dots, X_{i-1}]$: this is a martingale difference sequence with $|Y_i| \leq c_i$ precisely because resampling only X_i moves f by at most c_i (take conditional expectations over the two values of X_i), and $\sum_i Y_i = f - \mathbb{E}f$. Azuma–Hoeffding applied to (Y_i) is McDiarmid's inequality. ■

Proof of Lemma 2.7 (level-size estimation, Section 2.1)

Statement (recalled). Subsample each coordinate of a level set S_j , $|S_j| = s_j$, independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$. If z coordinates survive, then $\tilde{s}_j = z/r$ satisfies $(1 - \varepsilon)s_j \leq \tilde{s}_j \leq (1 + \varepsilon)s_j$ with probability $1 - 1/\text{poly}(n)$. ■

Proof. Denote the coordinates of S_j by u_i , $i \in [s_j]$, and let X_i be the indicator that u_i is sampled into substream P , so the X_i are independent with $\mathbb{E}[X_i] = r$ and $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff (Theorem 2.4) gives

$$\begin{aligned} \Pr\left[\left|\sum_i X_i - s_j r\right| \geq \varepsilon s_j r\right] &\leq 2 \exp(-\varepsilon^2 s_j r / 3) \leq 2 \exp(-C \log n / 3) \\ &\leq 1/\text{poly}(n), \end{aligned}$$

where the second inequality uses $s_j r = C \log n / \varepsilon^2$ (when the minimum in the definition of r is attained at this branch; otherwise $r = 1$ and there is nothing to estimate). Dividing by r converts the event into $(1 - \varepsilon)s_j \leq \tilde{s}_j \leq (1 + \varepsilon)s_j$. ■

Proof of Lemma 2.9 (sampling concentration, Section 2.2)

Statement (recalled). If each point of $S = \{p_1, \dots, p_t\}$ lies in the candidate ball B independently with probability at least $\frac{1}{2} - \Theta(\varepsilon)$, then simultaneously for all $k \leq n$ candidate balls, $Y_B = \sum_i \mathbf{1}\{p_i \in B\} \geq (1 - \varepsilon)(\frac{1}{2} - \Theta(\varepsilon))t$ whp. ■

Proof. Fix a candidate ball B and let $X_i = \mathbf{1}\{p_i \in B\}$; the X_i are independent, and by the covering property $\mathbb{E}[Y_B] = \sum_i \mathbb{E}X_i \geq (\frac{1}{2} - \Theta(\varepsilon))t$. Multiplicative Chernoff gives

$$\Pr[Y_B < (1 - \varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 \mathbb{E}Y_B / 2} \leq e^{-\Theta(\varepsilon^2 t)}.$$

With $t = \Theta(\varepsilon^{-2} \log n)$ this is $n^{-\Omega(1)}$ per ball, and a union bound over the $k \leq n$ candidate balls gives the simultaneous statement $Y_B \geq (1 - \varepsilon)(\frac{1}{2} - \Theta(\varepsilon))t$ for all of them, with high probability. ■

Round-by-round noise control (Section 2.4.2)

Statement (recalled). The k -round walk mechanism of [7] is unbiased for the walk count W_k and, with probability $1 - \beta$, errs by at most $k\gamma\sqrt{N}(d(G) + \gamma)^{k-1}$, where $\gamma = 2k\sqrt{8\log(2kN/\beta)}/\varepsilon$. ■

Proof. Write $X_i^{(\ell)}$ for node i 's round- ℓ value. The mechanism gives, for each node i and round $\ell \in \{1, \dots, k\}$,

$$X_i^{(\ell)} = \sum_{j \in N(i)} X_j^{(\ell-1)} + \text{Lap}(2k\|X^{(\ell-1)}\|_\infty/\varepsilon),$$

so $\sum_{j \in N(i)} X_j^{(\ell-1)} \leq d(G)\|X^{(\ell-1)}\|_\infty$ always. Allocate failure β'/N per node per round: the Laplace tail $\Pr[\text{Lap}(b) > u] \leq \frac{1}{2}e^{-u/b}$ gives, with probability $1 - \beta'/N$ per node-round, $2k\text{Lap}(\|X^{(\ell-1)}\|_\infty/\varepsilon) \leq \gamma\|X^{(\ell-1)}\|_\infty$ for the stated γ (with $\beta' \approx \beta/k$). Union-bounding over the $\leq kN$ node-rounds, all these events hold simultaneously with probability $\geq 1 - (k-1)\beta'$, and on that event an induction gives

$$\|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)\|X^{(\ell-1)}\|_\infty \implies \|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)^\ell$$

for all $\ell \leq k-1$. The released count aggregates the N per-node values, whose centered Laplace contributions are independent sub-exponential variables each of scale at most $O(\gamma(d(G) + \gamma)^{k-1})$; their sum concentrates around zero at scale $\sqrt{N} \cdot \tilde{O}(\gamma(d(G) + \gamma)^{k-1})$ by the same Chernoff-method argument applied to light-tailed variables, which yields the stated $k\gamma\sqrt{N}(d(G) + \gamma)^{k-1}$ error radius with probability $1 - \beta$. ■

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3

Induction

Induction turns a claim about an unbounded process — millions of cache fills, node eliminations, rewrite steps, recursive calls — into two finite checks: the claim holds at the start, and one more step cannot break it. For a systems paper this is the instrument of *component* correctness. No grand named theorem is involved; the corpus pattern is thousands of small, unnamed invariants — half-page lemmas asserting that one more elimination step, recursive call, or rewrite keeps a data structure consistent — with each cache, rule, or decomposition step verified locally and the global guarantee assembled from the local ones. Induction is also the standard scaffolding for complexity proofs: the same step-by-step argument that certifies what a component computes carries the per-step cost accounting that becomes a running-time bound.

In the SIGMOD'26 corpus, 37 tagged papers lean on induction — the third most-used tool, behind reductions and concentration inequalities — spanning graph analytics, join processing, entity resolution, and query optimization. The pairing *correctness* $\times$ *induction*, with 29 papers, is the single most frequent property-tool co-occurrence of the survey, and induction also powers 21 papers tagged *upper-bound*: the invariant proves the algorithm right, the accompanying decreasing measure proves it fast. This chapter develops the three substrates and the craft first; Section 3.5 adds strengthened and nested invariants and two harder exemplars.

3.0 The tool: induction and invariants

The problem class is uniform across applications. A component evolves by discrete local steps — an algorithm executes operations, an elimination pipeline removes one node, a rewrite system fires one rule, a recursive evaluator makes one call — and the claim we need is global: *every* answer produced is correct, *every* intermediate structure stays within its budget, *no* reachable configuration drifts. The number of configurations is astronomically larger than the code, so the claim cannot

LEARNING OBJECTIVES

- Recognize the three substrates: execution steps, recursive structure, measures.
- State an invariant, close base and step cases, audit every operation.
- Strengthen an invariant that will not close; pair it with a measure for costs.
- Dissect a paper's proof: induction variable, invariant, step case, payoff.

Figure 3.1: An invariant travels down every run: I holds at s_0 , each operation preserves I , so I holds at every reachable state — and I implies the specification.

$$\begin{array}{ccccccc} s_0 & \xrightarrow{\text{op}_1} & s_1 & \xrightarrow{\text{op}_2} & \cdots & \xrightarrow{\text{op}_t} & s_t \\ I(s_0) & & I(s_1) & & \cdots & & I(s_t) \end{array}$$

be checked state by state. What induction delivers is a predicate I on single configurations — the *invariant* — with two provable properties: it holds initially, and each single step preserves it. Those two finite checks certify the unbounded behavior. Every inductive argument in the corpus is an instance of this shape, differing only in *what the steps are*: states of a run, pieces of a recursively built object, or values of a decreasing counter.

Notation

I always denotes the predicate being propagated (the invariant); Γ the set of operations; $m: \mathcal{C} \rightarrow \mathbb{N}$ a measure. *Leaf-peeling* is the default move on tree-shaped objects: remove a leaf, invoke the hypothesis on the smaller object, extend by one step.

3.0.1 The three substrates

Substrate 1: execution steps. The component is imperative: it induces a sequence of states $s_0, s_1, \dots, s_t$, one per operation. This is the inductive-assertion method of Floyd [4] and Hoare [5], the loop-invariant discipline of algorithms texts [2], and Manna–Pnueli *safety* reasoning — nothing bad ever becomes true [9].

Theorem 3.1 — Invariant preservation. Let s_0 be an initial state and Γ a set of operations on states. Suppose the predicate I satisfies (i) $I(s_0)$, and (ii) $I(s) \Rightarrow I(\text{op}(s))$ for every $\text{op} \in \Gamma$. Then $I(s)$ holds for every state s reachable from s_0 by a finite sequence of operations; consequently, any specification P implied by I ($I \Rightarrow P$) holds at every reachable state. ■

Proof. Induction on the length t of the run $s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_t$. The base case $t = 0$ is (i). For the step, $s_t = \text{op}(s_{t-1})$ for some operation, and (ii) propagates $I(s_{t-1})$ to $I(s_t)$. ■

Substrate 2: structural recursion. The object is built recursively — a tree decomposition, an elimination forest, a recursive pseudo-tree, a relational expression — and the property is proved on the whole from the property on the parts. The database workhorse is leaf-peeling: remove a leaf bag or leaf vertex, apply the hypothesis to the smaller object, then extend by one step.

Proposition 3.2 — Structural induction. Let $\mathcal{C}$ be a class of objects generated from atoms by formation rules. If a property P holds for every atom,

and each formation rule maps ingredients satisfying P to a result satisfying P , then every object of $\mathcal{C}$ satisfies P . In particular, on trees it suffices to prove P for leaves and to prove that a node whose children's subtrees satisfy P satisfies P itself.

Proof. Induction on the number of formation steps used to build the object. Atoms use zero steps. For a composite object, each immediate ingredient is built with fewer steps, so the hypothesis gives P on the ingredients, and the preservation assumption lifts P to the composite. The leaf-peeling form is the same argument read from the root downwards. ■

Substrate 3: a decreasing measure. Many arguments are phrased not over runs or recursions but over a quantity that strictly decreases — nodes left to eliminate, bags left to peel, rewrite steps remaining. Structural induction is routinely converted into this form by choosing a measure [10], and the classical least-counterexample principle closes both:

Lemma 3.3 — Strong induction. Let $m: \mathcal{C} \rightarrow \mathbb{N}$ be a measure. Suppose that for every object X , if $P(X')$ holds for all X' with $m(X') < m(X)$, then $P(X)$. Then $P(X)$ holds for every object X . ■

Proof. If not, the set of counterexamples has an m -minimal element X^* (the natural numbers are well-ordered). Every X' with $m(X') < m(X^*)$ satisfies P , so the hypothesis yields $P(X^*)$ — a contradiction. ■

1

3.0.2 How to choose a substrate

The selection is mechanical once one question is answered: *what does one step look like?* If the component is an explicit state machine — a sweep with a stack, an incrementally maintained index — take execution steps and a state invariant (Substrate 1). If the object of study is recursively defined and the claim is about the finished object — decompositions, pseudo-trees, expressions — take structural induction, leaf-peeling by default (Substrate 2). If the argument is really about a process with no convenient state — an elimination order, a rewrite sequence, a peeling procedure — phrase it as strong induction on the natural measure (Substrate 3). All three are the same lemma in different clothes; pick the clothing that makes the step case a purely local check.

¹ The measure earns a second salary: since each step decreases m by at least one, the component terminates in at most $m(s_0)$ steps, and attaching a per-step cost to the induction turns the correctness proof into a running-time proof. This is how the same argument underwrites both the *correctness* pairing (29 papers) and the *upper-bound* pairing (21 papers) of this chapter's corpus.

When to Reach for Induction

Reach whenever a component has state evolving by local steps and the claim is “nothing drifts” — caches, elimination pipelines, rewrite rules, recursive evaluators — or whenever a guarantee must hold for *all* inputs of unbounded size. Then run the recipe:

1. Name the substrate and fix the induction variable precisely: the step index i of a run, the subtree rooted at a node A , the number of bags/nodes/rules left.
2. Write the invariant I as a property of *one* state or *one* subtree, strong enough to imply the specification at final states (or at the whole object) and concrete enough to check in one step. Audit the step case over *every* operation — and the trivial base cases (empty input, single bag, zero eliminations): one missed case silently breaks the entire argument.
3. If the step case will not close, *strengthen* I : carry more information (the exact block shape, the exact cache contents, both directions of an equivalence). Strengthening is the normal iteration of the method, not a failure — see Proposition 3.13.
4. Pair I with a strictly decreasing measure to obtain termination and per-step cost accounting — the same induction then yields the time/space bound, not just correctness.

3.1 Resistance-distance labels on small-treewidth graphs

The exemplar is a SIGMOD’26 paper on exact graph analytics over small-treewidth graphs [8]. Resistance distance — the effective resistance between two nodes when the graph is viewed as an electrical network — is a workhorse similarity measure in graph databases: link prediction, recommendation, and graph-feature extraction all use it because it averages over *all* paths between two nodes, not just the shortest one. Computing it exactly requires the pseudo-inverse of the graph Laplacian, a matrix inversion that does not scale to large graphs, which is why practitioners settle for approximations. The paper observes that on graphs of small treewidth — road networks, hierarchies, many social and biological networks — Gaussian elimination ordered along a tree decomposition factors the computation *once*, so that afterwards every pair query is answered from two small labels alone. Its headline results: an index built in $O(n \cdot d_{\max} \cdot h_G^2)$ time, labels of total size $O(n \cdot h_G)$, and *exact* single-pair queries in $O(h_G)$, where h_G is the height of the tree decomposition and $d_{\max}$ the maximum degree. The correctness of the

elimination pipeline is where induction enters.

Problem. Let $G = (V, E)$ be a graph with n nodes, Laplacian $L = D - A$, and $r(s, t) = (\mathbf{e}_s - \mathbf{e}_t)^\top L^+ (\mathbf{e}_s - \mathbf{e}_t)$ the resistance distance, where L^+ is the Moore–Penrose pseudo-inverse. The input includes a tree decomposition of G of height h_G (the maximum number of bags on a root-to-leaf path). The task is to preprocess G into one label $\ell(v)$ per node so that any query $r(s, t)$ is answered exactly from $\ell(s)$ and $\ell(t)$ alone, with preprocessing, label size, and query time parameterized by n , $d_{\max}$, and h_G .

Guarantee.

Theorem 3.4 — Labelling scheme, complexity lemmas of the paper. Given G and a tree decomposition of height h_G , the index construction computes labels of total size $O(n \cdot h_G)$ in time $O(n \cdot d_{\max} \cdot h_G^2)$, and any single-pair query $r(s, t)$ is answered *exactly* in $O(h_G)$ time from the two labels; a single-source query over all t costs $O(n \cdot h_G)$. ■

2

How induction is applied. The algorithm eliminates the nodes one at a time, in an order given by a traversal of the tree decomposition, maintaining a shrinking *state matrix* S initialized to the inverse of the Laplacian restricted to the current node set; eliminating node v applies the rank-one update $S \leftarrow S[:, v] S[:, v]^\top / S[v, v]$. The induction variable is the number of eliminated nodes, and the induction hypothesis is stated as an explicit invariant of the state:

Lemma 3.5 — Elimination-state invariant, Lemma of the paper, notation cleaned. Let $U_1 \subseteq U_2 \subseteq V$. After the elimination of the nodes of $U_2 \setminus U_1$, the state matrix satisfies

$$S_{|U_2 \setminus U_1|} = \begin{pmatrix} (L_{|U_1|})^{-1} & 0 \\ 0 & 0 \end{pmatrix},$$

i.e. all rows and columns of eliminated nodes are zero, and the block on the surviving nodes is the inverse of the Laplacian restricted to U_1 . ■

Proof sketch. Induction on the number k of eliminated nodes. Base case $k = 0$: the state is $(L_{|U_2|})^{-1}$ by initialization. Step: partition the state around the next eliminated node v ; the rank-one update zeroes row and column v and replaces the surviving block by a Schur complement, which the block-inverse formula identifies with the inverse of the Laplacian restricted to the remaining nodes. (Full proof in Section 3.6.) ■

² Why exactness is plausible here: on a tree decomposition, eliminating a node interacts only with its bag, so the inverse Laplacian factors into per-path pieces — the random-walk interpretation (resistance is a commute time) then reconstructs $r(s, t)$ from the pieces on the two root paths, via cut properties: a vertex cut of s, t zeroes the cross terms.

The labels store the elimination history: for each node, the columns of the state at the moments its neighbors were eliminated (a subtree-contiguity property of the traversal keeps each column segment addressable), and the query walks the two root paths recombining those pieces through the resistance formula. Note the craft: the invariant is an *exact block shape*, not “approximately correct” — strong enough to pin every entry of the state, which is what makes the reconstruction exact. The running-time bound rides the same induction: each of the n elimination steps touches at most $d_{\max}$ already-processed neighbors on segments of length $O(h_G)$, giving the $O(n \cdot d_{\max} \cdot h_G^2)$ total.

What it buys. The engine answers exact resistance-distance queries in time proportional to the decomposition height rather than the graph size, after near-linear preprocessing on low-treewidth data — no approximation error, no sampling. The pattern — eliminate along a decomposition, store the elimination history as labels, reconstruct from labels by an invariant — transfers to any computation that factors through Gaussian elimination on small-treewidth structure.

3.2 *Space–time plans for sum–product queries*

The exemplar is a SIGMOD’26 paper on the space–time complexity of conjunctive-query evaluation [3]. A *sum–product query* is a conjunctive query evaluated over a semiring: ordinary joins (Boolean semiring), output counts (natural numbers), tuple probabilities, or optima (tropical semiring). Worst-case optimal join algorithms evaluate such queries in optimal time but materialize as they go, paying full space for intermediate results; systems instead want caching plans that trade time for space. The paper asks which points of the space–time plane a query can occupy: it defines a ladder of plan classes — pseudo-trees, pseudo-trees with caching, tree-decomposition plans, and recursive pseudo-trees (RPTs) — ordered by a domination preorder on their (s, t) exponent pairs, and identifies what each class achieves. Headline results: every tree-decomposition plan is dominated by an RPT plan (proved by the second induction exhibited below), the dominations are mostly strict, witnessed by explicit path queries, and a Triple- k -Clique conjecture yields conditional space–time lower bounds over the tropical semiring.

Problem. Fix a semiring $\mathbb{K}$ and a sum–product query $Q(\mathbf{X}) = \prod_i R_i(\mathbf{X}_i)$; the database annotates tuples with values of $\mathbb{K}$, an assignment multiplies its annotations, and Q sums over satisfying assignments. A *pseudo-tree* is a directed tree over the variables of Q in which every atom’s variables sit on a single root-to-leaf branch; a plan is a pseudo-tree, possibly with caches at some vertices and

applied recursively to subqueries, and its exponents s and t (defined recursively in the paper) guarantee evaluation in space $O(|D|^s)$ and time $O(|D|^t)$. Plan P_1 *dominates* P_2 if $t(P_1) \leq t(P_2)$ and $s(P_1) \leq s(P_2)$; a class dominates another if for every query and every plan of the latter there is a dominating plan of the former.

Guarantee.

Theorem 3.6 — RPT evaluation, Theorem of the paper, compressed. Every recursive pseudo-tree plan $(\mathcal{P}, \mathcal{C}, \rho)$ of Q with exponents $s = s(\mathcal{P}, \mathcal{C}, \rho)$ and $t = t(\mathcal{P}, \mathcal{C}, \rho)$ correctly evaluates Q over any semiring, in space $O(|D|^s)$ and time $O(|D|^t)$. ■

3

How induction is applied. Two inductions run side by side. The *correctness* half is execution-step induction over the run of the recursive algorithm: the induction variable is the sequence of function calls $\text{solve}(A, \mathbf{x})$, and the invariant says each call returns exactly the value of the subquery rooted at A , with the cache holding exactly that value:

Lemma 3.7 — Cache invariant, Lemma of the paper, paraphrased. During any run of the algorithm on an RPT $(\mathcal{P}, \mathcal{C}, \rho)$, every call $\text{solve}(A, \mathbf{x})$ returns exactly the value of the subquery of Q rooted at A , conditioned on the part of the partial assignment $\mathbf{x}$ over the variables of A ; after the subsequent fillCache call, the cache M_A stores exactly this value. Hence the top-level call $\text{solve}(r(\mathcal{P}), ())$ returns $Q(\mathbf{D})$. ■

Proof sketch. Structural induction on the RPT, i.e. on the number of variables in the subtree rooted at A . Base case (single variable): the call accumulates $\bigoplus_a a$ directly, which is the subquery value. Step: the call iterates over candidate values a , multiplies the — by hypothesis correct — returns of the children’s calls, and accumulates; the cache variant additionally uses that two partial assignments agreeing on A ’s interface define the same subquery value. (Full proof in Section 3.6.) ■

The *domination* half (every tree-decomposition plan is dominated by an RPT plan) is leaf-peeling structural induction on the number of bags of the tree decomposition T : with a single bag the plan is already a pseudo-tree plan (base case); otherwise let v be a leaf of T with parent p , eliminate the variables occurring in v ’s bag but not in p ’s into a single new atom over their joint neighbors, apply the hypothesis to the smaller decomposition, and attach the resulting RPT at the parent — each peeling step preserves plan value and exponents, so the constructed RPT dominates.

³ The benchmark being traded against: a worst-case optimal join evaluates a full query in time $O(|D|^{\rho^*})$ with ρ^* the fractional edge cover — but materializes intermediates, i.e. space exponent ρ^* as well. The paper’s sublinear-space plans give up time exponents to buy space back.

What it buys. The optimizer gets a certified menu of space–time trade-offs — sublinear-space evaluation of full queries by caching at the right granularity — rather than a single all-or-nothing algorithm. The cache invariant is the implementer’s contract: it says precisely what each cache entry means, so the recursion can be memoized (and its cost accounted) safely; and the strict separations tell designers when adding a plan class genuinely buys new exponent pairs.

3.3 Near-duplicate text alignment by skyline sweeps

The exemplar is a SIGMOD’26 paper on all-pair near-duplicate text alignment [11]. Finding all pairs of near-duplicate substrings in long texts is a bread-and-butter entity-resolution task: plagiarism detection, version tracking, deduplication of logs and records. Locality-sensitive hashing (LSH) is the standard sieve — hash positions so that similar stretches collide, then verify candidates — but the candidate generation itself is the bottleneck: the maximal hash-consistent stretches, which the paper calls *tight intervals*, live in a space of quadratically many intervals, and missing even one means a silently missed near-duplicate pair. The paper characterizes tight intervals through a *dominance skyline* of positions and gives a monotone-stack sweep that generates exactly the tight intervals of a text of length n in $O(n)$ time per hash function, with $O(\log n)$ expected stack — and an $O(mn)$ expected-time alignment pipeline with a completeness guarantee: no near-duplicate pair is missed.

Problem. Let T be a text of length n and h a hash function on positions (drawn from a random family); write $h(T[q])$ for the hash of the token at position q . Position q *dominates* p (for $p < q$) iff $h(T[q]) < h(T[p])$. An interval $[l, r]$ is *left-tight* iff $l = 1$ or $h(T[l - 1])$ is smaller than every hash on $[l, r]$, and *right-tight* iff $r = n$ or $h(T[r + 1])$ is smaller than every hash on $[l, r]$; for a composite $g = (h_1, \dots, h_m)$, $[l, r]$ is *tight* iff it is left-tight under some h_i and right-tight under some h_j . The task: enumerate, exactly once each, all tight intervals of T under h (and, for m functions, under g).

Guarantee.

Theorem 3.8 — Tight-interval generation, Theorems 1–2 of the paper. The monotone-stack sweep (Algorithm 1 of the paper) generates exactly the tight intervals of T under h — every generated interval is tight, and every tight interval is generated — in $O(n)$ time, producing $O(n)$ intervals, with the stack holding $O(\log n)$ positions in expectation. With m independent hash functions the full pipeline runs in $O(mn)$

expected time. ■

How induction is applied. The induction variable is the scan position r ; the state is the *ordered skyline* $S(r)$ — the positions $p \leq r$ not dominated by any later position, in increasing order, whose hash values increase along the skyline and whose last element is r . The invariant: at the end of each loop iteration, S is exactly the ordered skyline of the current position. The workhorse is the one-step update:

Lemma 3.9 — Skyline update, Lemma 6 of the paper. Let $S(r) = (p_1, p_2, \dots, p_s)$ be the ordered skyline at position r and $i = \min\{j \in [1, s] : h(T[p_j]) < h(T[r+1])\}$. Then the ordered skyline at position $r+1$ is $S(r+1) = (p_1, \dots, p_i, r+1)$. ■

Proof sketch. By the monotonicity of hashes along the skyline, every p_j with $j > i$ has $h(T[p_j]) > h(T[r+1])$, so the new position $r+1$ dominates exactly those — they leave the skyline. Positions are dominated only by later ones, so the prefix $(p_1, \dots, p_i)$ keeps its pairwise order, and $r+1$ (the latest position, with the largest hash on the new skyline) is appended last. ■

The base case is $S(0) = (0)$. The stack operations — pop while the top’s hash exceeds $h(T[r+1])$, then push $r+1$ — implement exactly the update of Lemma 3.9, so the invariant holds after every iteration. Exactness then follows by combining the invariant with two characterization lemmas of the paper: an interval $[l, r]$ is left-tight iff $l-1 \in S(r) \setminus \{r\}$, and right-tight iff $h(T[r+1]) < h(T[p])$ where p is the smallest skyline position $\geq l$ (whose hash is the minimum on $[l, r]$). The sweep emits an interval exactly at a pop, which is exactly when both characterizations fire — soundness and completeness at once. (Full proof in Section 3.6.)

What it buys. The alignment stage enumerates all LSH candidates in linear time per hash function, replacing quadratic scans, with $O(\log n)$ expected working memory — and the exactness theorem doubles as a *recall certificate* for the entity-resolution pipeline: no near-duplicate pair can be missed. The pattern — a dominance order on positions, the skyline as state, a monotone stack as its compressed representation — is the same skeleton as classical monotone-stack algorithms, here made inductive and analyzable.

3.4 Succinct representations of all width-1 join plans

The exemplar is a SIGMOD’26 paper on compressing the optimizer’s search space [6]. Join-order selection is NP-hard in general, so cost-based optimizers enumerate and cost candidate *join trees*; for star

queries the space is enormous — n^{n-1} join trees for n relations — and enumerating them one by one does not scale. The paper focuses on the plans that matter most, *width-1* plans whose intermediate results are bounded by $O(N)$ in the maximum base cardinality N , and asks for a succinct representation from which all of them can be enumerated without duplicates. Its headline results: width-1 plans exist exactly for acyclic queries; a *meta-decomposition* — a labeled tree of size $O(|E(H)|)$ computed in $O(|E(H)|^3)$ — represents all join trees of an acyclic hypergraph; and an enumeration procedure outputs exactly the join trees, so the optimizer searches a linear-size object instead of an exponential family.

Problem. Let Q be a conjunctive query with hypergraph H (edges = atoms). A *join tree* of H is a tree over the hyperedges such that the hyperedges containing any vertex form a connected subtree; H is *acyclic* iff it has a join tree. For a (physical) query plan P , its *width* $w(P)$ is the least exponent such that every intermediate result of P is $O(N^{w(P)})$. For a hyperedge e , its *outer part* is $o(e, H) = e \cap (\bigcup(E(H) \setminus \{e\}))$ — the portion of e shared with other edges — and an *ear* is a hyperedge the classical ear-removal characterization of acyclicity may peel off. The task: characterize the width-1 plans and represent all join trees succinctly.

Guarantee.

Theorem 3.10 — Width-1 plans and their enumeration, Theorems of the paper, compressed. (i) A conjunctive query Q has a width-1 query plan if and only if Q is acyclic. (ii) For acyclic H , the meta-decomposition M produced by the algorithm has $|V(M)|, |E(M)| = O(|E(H)|)$, is computed in $O(|E(H)|^3)$ time, and the enumeration procedure outputs exactly the join trees of H — every output is a join tree, and every join tree is output. ■

⁴ The n^{n-1} count for star queries is Cayley's formula — the paper pairs it with Prüfer sequences to count join trees of specific shapes, sharpening just how exponential the naive search space is.

4

How induction is applied. The elimination structure is the ear-removal order: the meta-decomposition tree records it (a node p is a descendant of q iff e_p is removed before e_q), and the correctness and completeness of the enumeration are proved by structural induction over that tree — the induction hypothesis at a node is that the subtree's physical nodes induce a connected subtree of every join tree, so local choices extend to full join trees and every join tree arises. The step case that lets the algorithm remove ears in *batches* is:

Proposition 3.11 — Batch ear removal, Proposition of the paper. Let H be an

acyclic hypergraph and $S = \{e_1, \dots, e_n\} \subseteq E(H)$, $n \geq 2$, a set of distinct ears with a common outer part $o(e_1, H) = \dots = o(e_n, H) =: o$. Then the hypergraph H' with $E(H') = (E(H) \setminus S) \cup \{o\}$ is again acyclic. ■

The enabling fact — which pins the shared structure of same-outer-part ears — is a two-line set-algebra lemma:

Lemma 3.12 — Ears with equal outer parts, Lemma of the paper. If two ears e_1, e_2 of an acyclic hypergraph H satisfy $o(e_1, H) = o(e_2, H)$, then $o(e_1, H) = o(e_2, H) = e_1 \cap e_2$. ■

Proof sketch. The common value lies in both ears (each outer part is contained in its own edge), and conversely $e_1 \cap e_2$ lies in the outer part of e_2 because e_1 is one of “the other edges” for e_2 — a double inclusion. (Full proof in Section 3.6.) ■

With the batch proposition, the peeling induction proceeds on the number of ears remaining: remove all same-outer-part ears at once, insert the single new ear o , witness a join tree of H' from one of H (the shared outer part glues the removed ears’ positions into one subtree), and recurse on the strictly smaller hypergraph — the measure $|E(H)|$ decreases at every step.

What it buys. The optimizer keeps a linear-size certificate that stands for an exponential family of plans: enumerating all width-1 — i.e. linear-cost — join plans becomes a traversal of the meta-decomposition, with the exactness of the enumeration (no invalid plan, no duplicates, nothing missed) inherited from the peeling induction. On top of it, the paper evaluates relation-dominated queries in $O(|db|)$ time by restricting plan search to the width-1 space the theorem certifies.

3.5 Advanced usage and further reading

The substrate lemmas of Section 3.0 are complete — nothing deeper is ever added. What distinguishes advanced usage is the *craft* on top: deliberate strengthening of the hypothesis, and nesting one induction inside another. This section states the strengthening principle, sketches two harder exemplars, and points to the literature.

3.5.1 The advanced rungs: strengthening and nesting

Proposition 3.13 — Strengthening. To certify a specification P it suffices to exhibit *any* inductive predicate I with $I \Rightarrow P$; the specification P itself need not be inductive. Formally: if $I(s_0)$, I is preserved by Γ , and $I(s) \Rightarrow P(s)$ for all states s , then P holds at every reachable state. ■

Proof. Theorem 3.1 applied to I gives I at every reachable state; $I \Rightarrow P$ transfers it to P . ■

The point is methodological. Target specifications are almost never inductive — “the output equals the subquery answer” says nothing about a half-finished run — so the prover iterates: add bookkeeping to the predicate until the step case closes. Every exemplar of this chapter did it: the exact block shape (Lemma 3.5), the exact cache contents (Lemma 3.7), the ordered skyline with its monotone hashes (Lemma 3.9), the ear-removal order with shared outer parts (Proposition 3.11). *Nesting* is the second rung: two induction variables, an outer structural induction with an inner execution-step induction per piece — the sum-product exemplar of Section 3.2 nests its cache-invariant run argument inside the leaf-peeling domination proof. Beyond the naturals, termination arguments move to well-founded measures on multisets; those live in the term-rewriting literature rather than in this corpus.

3.5.2 Two advanced exemplars, sketched

Both exemplars of this section push induction into proof machinery that is itself the contribution; each is sketched in one paragraph.

Fixed-parameter width checking [7]^{PODS}. This PODS’26 paper settles the parameterized complexity of two central width measures: it is fixed-parameter tractable to check $\text{ghw}(H) \leq k$ and $\text{fhw}(H) \leq k$ — generalised and fractional hypertree width, the decomposition measures behind beyond-acyclic query evaluation — parameterised by k , the rank of H , and its maximum predicate arity $\Delta(H)$ (Corollary 8 of the paper; the first FPT algorithms for these problems). The engine transplants the MSO-transduction treewidth algorithm of Bojańczyk and Pilipczuk from tree decompositions to *elimination forests*: a transduction rewrites a suboptimal forest into an optimal one, and its correctness is an induction over the forest whose step case is the paper’s Reduced-Hypergraph Dealternation Lemma — alternation depth (the *split* and *irregularity* parameters) stays bounded, with conflict-graph coloring and neutral-swap/pigeonhole arguments supplying the bound. The invariant carried by the induction: the output is always an elimination forest of the same hypergraph, of width no larger than the input’s.

Codd’s theorem over semirings, and its limit [1]. Codd’s theorem — relational algebra equals domain-independent relational calculus — is the foundation of declarative query languages, and provenance semirings (annotating tuples with probabilities, counts, confidences) are the standard way to give queries provenance semantics. The paper transports

Codd’s theorem to $\mathbb{K}$ -relations over semirings with *monus* (truncated subtraction) and *support*: RA-expressibility, domain-independent calculus definability, and calculus definability coincide exactly over zero-sum-free semirings (Theorem of the paper), and the equivalence fails without zero-sum-freeness. Both directions rest on structural induction on expressions and formulas: every algebra operator translates to a calculus connective and back, and the hypothesis “the translation defines the same $\mathbb{K}$ -relation” is preserved by each operation. The same technique then proves an *inexpressibility*: over the bag semiring $\mathbb{N}$, division is not definable in basic relational algebra. Witness databases make division produce a tuple of multiplicity 2^n , while an induction on the construction of any expression E bounds its support by $n^{l(E)}$ and its multiplicities by $p_E(n) \cdot 2^{l(E)}$ — growth that 2^n outruns. Induction here counts what expressions *can* do: the mirror image of its usual certifying use.

3.5.3 Further reading

- Floyd [4] and Hoare [5] — the origins of the invariant-assertion method: verification as base case plus preservation.
- Manna and Pnueli [9] — safety properties (“nothing bad ever happens”) as the invariant-based fragment of temporal logic.
- Cormen, Leiserson, Rivest, and Stein [2] — the loop-invariant discipline for algorithms, with worked analyses.
- Rosen [10] — a refresher on strong induction, structural induction, and the well-ordering principle they rest on.

Chapter Recap

- **What the tool is.** Induction converts claims about unbounded behavior into two finite checks — an invariant holds initially and survives one step — over whichever substrate the component offers: execution steps, recursive structure, or a decreasing measure (all three interderivable, Lemma 3.3).
- **Which substrate when.** Explicit state machine (sweep, incremental index): state invariant. Recursively built object (decomposition, pseudo-tree, expression): structural induction, leaf-peeling by default. Process without explicit state (elimination order, rewrite sequence): strong induction on the natural measure.
- **Why the craft matters.** Specifications are rarely inductive; strengthen the predicate — exact block shape, exact cache contents, monotone skyline, removal order — until the step case

closes (Proposition 3.13), audit every operation, and pair the invariant with a decreasing measure to get termination and cost bounds.

- **What the corpus showed.** Resistance-distance labels (Section 3.1): induction on eliminated nodes, block-shape invariant. Sum-product plans (Section 3.2): cache invariant over the run, leaf-peeling over decompositions. LSH alignment (Section 3.3): skyline invariant over the scan, monotone-stack update. Succinct plan representations (Section 3.4): ear-removal induction with batch peeling.
- **Advanced usage.** Strengthening and nesting (Section 3.5.1) are the rungs beyond the basics; the advanced exemplars of Section 3.5.2 run induction inside MSO transductions and use it to count what expressions can do. Deferred proofs are collected in the Appendix (Section 3.6).

3.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Lemma 3.5 (elimination invariant)

Statement (recalled). After eliminating the nodes of $U_2 \setminus U_1$ from the state initialized to $(L_{|U_2|})^{-1}$, the state matrix is $\begin{pmatrix} (L_{|U_1|})^{-1} & 0 \\ 0 & 0 \end{pmatrix}$ — zero rows and columns on eliminated nodes, the inverse restricted Laplacian on the survivors. ■

Proof. Induction on the number k of eliminated nodes. *Base case* $k = 0$: the state is $S_0 = (L_{|U_2|})^{-1}$ by initialization, and no rows or columns are zeroed. *Induction hypothesis*: after k eliminations, all rows and columns of the eliminated nodes are zero, and the block on the remaining node set U is the inverse of the Laplacian restricted to U . *Step*: eliminate $v = v_{k+1}$ and partition the state as $S_k = \begin{pmatrix} A & b \\ b^\top & c \end{pmatrix}$ with $c = S_k[v, v]$. The rank-one update gives

$$S_{k+1} = S_k - \frac{S_k[:, v] S_k[:, v]^\top}{c} = \begin{pmatrix} A - b b^\top / c & 0 \\ 0 & 0 \end{pmatrix},$$

so row and column v are zeroed. The surviving block $A - b b^\top / c$ is the Schur complement of c in S_k , and by the block-inverse identification — if $M = \begin{pmatrix} A' & B \\ C & D \end{pmatrix}$ has inverse $\begin{pmatrix} E & F \\ G & H \end{pmatrix}$, then $E^{-1} = A' - B D^{-1} C$ — the inverse of that block is the restriction of $(S_k)^{-1}$ to the remaining nodes, which by the induction hypothesis is the Laplacian restricted to $U \setminus \{v\}$. After all $|U_2 \setminus U_1|$ eliminations the block on U_1 is $(L_{|U_1|})^{-1}$. ■

Proof of Lemma 3.7 (RPT cache invariant)

Statement (recalled). Every call $\text{solve}(A, \mathbf{x})$ returns exactly the value of the subquery of Q rooted at A conditioned on the part of $\mathbf{x}$ over A 's variables, and after fillCache the cache M_A stores exactly this value; hence the top-level call returns Q . ■

Proof. Write, as in the paper, $Q[Y \mid \mathbf{y}]$ for the subquery of Q restricted to the variables Y and conditioned on the partial assignment $\mathbf{y}$, and let $\chi(A)$ be the variables of the subtree rooted at A . Induction on $|\chi(A)|$. *Base case* $|\chi(A)| = 1$: the call allocates an output relation and accumulates over the candidate values at A ; by the paper's subquery identity this accumulation is exactly $Q[\chi(A) \mid \mathbf{x}]$. *Step:* for $|\chi(A)| > 1$, the call iterates over candidates a and multiplies the returns of the children's calls,

$$\text{OUT} = \bigoplus_a \left(a \otimes \bigotimes_{B \in \text{child}(A)} \text{solve}(B, (\mathbf{x}, a)) \right),$$

which, by the induction hypothesis applied to the children (whose subtrees are strictly smaller) and the same subquery identity, equals $Q[\chi(A) \mid \mathbf{x}]$. For the cache clause, let $\mathbf{y}$ be the first argument reaching A whose restriction to A 's interface agrees with $\mathbf{x}$'s: the call stores $Q[\chi(A) \mid \mathbf{y}]$ in M_A , and every later call with the same interface returns it — correct because two assignments agreeing on the interface define equal subquery values, $Q[\chi(A) \mid \mathbf{y}] = Q[\chi(A) \mid \mathbf{x}]$. The top-level call has an empty assignment and the full tree, so it returns Q . ■

Proof of Theorem 3.8 (tight-interval exactness)

Statement (recalled). The sweep generates exactly the tight intervals of T : every generated interval is tight, every tight interval is generated; a text of length n is processed in $O(n)$ time with $O(n)$ intervals emitted and an $O(\log n)$ expected stack. ■

Proof. First, by induction on r : after the iteration processing r , the stack S is the ordered skyline $S(r)$. Base case $r = 0$: $S = (0)$. Step: the while-loop pops exactly the positions whose hashes exceed $h(T[r+1])$, and the push appends $r+1$, so by Lemma 3.9 the stack is $S(r+1)$. Second, soundness: the sweep emits $[l, r]$ when popping p , where $l-1$ is the new stack top, so $l-1 \in S(r) \setminus \{r\}$ and the paper's Lemma 1 gives left-tightness; the popped p is the smallest skyline position $\geq l$, its hash is the minimum on $[l, r]$, and the pop condition $h(T[r+1]) < h(T[p])$ gives right-tightness by the paper's Lemma 3. Third, completeness: fix a tight $[l, r]$. By Lemma 1, $l-1 \in S(r)$ at the start of iteration r ; right-tightness gives $h(T[r+1]) < h(T[p])$ for p the smallest skyline position $\geq l$, so the pop of p — which the paper's Lemma 4 shows

cannot be avoided — exposes $l - 1$ as the top, and $[l, r]$ is emitted. For the complexity: each position is pushed once and popped at most once, so the loop body executes $O(n)$ times and emits one interval per pop; the stack bound is the paper’s Lemma 7, $\mathbb{E} |S(r)| = O(\log r)$ over the random hash family. ■

Proof of Lemma 3.12 (equal outer parts)

Statement (recalled). If ears e_1, e_2 of an acyclic hypergraph H satisfy $o(e_1, H) = o(e_2, H)$, then the common value is $e_1 \cap e_2$. ■

Proof. Recall $o(e, H) = e \cap (\bigcup (E(H) \setminus \{e\}))$, so $o(e, H) \subseteq e$ for every edge e . Let $o = o(e_1, H) = o(e_2, H)$. Since $o = o(e_1, H) \subseteq e_1$ and $o = o(e_2, H) \subseteq e_2$, we get $o \subseteq e_1 \cap e_2$. Conversely, e_1 is one of the other edges for e_2 , so

$$e_1 \cap e_2 \subseteq e_2 \cap (\bigcup (E(H) \setminus \{e_2\})) = o(e_2, H) = o.$$

The two inclusions give $o = e_1 \cap e_2$. ■

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4

Estimation Theory

Database systems constantly answer questions with numbers they did not exactly compute: a sketch reports a frequency, a sample reports a cardinality, simulated diffusions report an influence score. Estimation theory says how far such a number can sit from the truth. It builds an *estimator* — a function of the randomness the system has chosen to expend — and asks the classical questions in order: is it *unbiased* (correct in expectation), how large is its *variance* or mean-squared error, and could *any* estimator do better [3, 8]? Where the previous chapter bounded how far a random quantity strays from its mean, this chapter supplies the upstream half: what the mean should estimate, and how many samples make the estimate trustworthy — together the standard systems promise, a $(1 \pm \epsilon)$ estimate holding with probability $1 - \delta$.

In the SIGMOD'26 corpus this is among the most-used theoretical tools: estimation theory is tagged in 28 of the 228 tagged papers, the most frequent companion of the *error-bound* (16 papers) and *probabilistic-guarantee* (13 papers) tags. The corpus workflow is constant: read an estimator off a sketch, sample, or simulation; prove it unbiased; bound its variance or range; convert to a confidence statement; union-bound over all keys, nodes, or rounds; and, in the strongest papers, match with a lower bound proving the error or space optimal. This chapter presents the ladder and four basic exemplars first; Section 4.5 adds sequential (anytime-valid) confidence and two harder exemplars as extension reading.

4.0 The tool: estimation theory

The problem class is uniform. Some parameter θ of the data — a frequency moment, a distance, an influence score — is inaccessible at the accuracy the system can afford, but the system can run a randomized experiment: draw a sample, hash into a sketch, simulate a diffusion, walk a graph. An estimator $\hat{\theta}$ is the function converting the outcome into a reported number. Estimation theory climbs a four-rung ladder,

LEARNING OBJECTIVES

- Recognize when a reported number is an *estimator*, and state its bias and variance.
- Build unbiased estimators by inverse-probability weighting; know why nonlinearity breaks them.
- Invert tail bounds to set sample counts analytically, not by tuning.
- Certify optimality with entropy-, communication-, or information-based lower bounds.

each rung a clause a paper must earn before printing a guarantee: *correctness in expectation* (unbiasedness), *how fast the error shrinks* (variance, MSE), *whether any estimator shrinks it faster* (Cramér–Rao-style efficiency), and *high-probability accuracy surviving union bounds* (concentration). The exemplars show papers on each rung.

4.0.1 The ladder

Definition 4.1 (Estimator, bias, MSE). Let the data be drawn from a distribution P_θ indexed by a parameter $\theta \in \Theta$. An *estimator* is a statistic $\hat{\theta} = \hat{\theta}(X)$ not depending on θ . Its *bias* is $\mathbb{E}_\theta[\hat{\theta}] - \theta$ and its *mean-squared error* is $\text{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$. The estimator is *unbiased* if $\mathbb{E}_\theta[\hat{\theta}] = \theta$ for all θ .

The two error components separate exactly, and every design decision in a sketch or sampler is a move between them:

Proposition 4.2 — Bias–variance decomposition. For any estimator $\hat{\theta}$ with finite variance,

$$\text{MSE}_\theta(\hat{\theta}) = \underbrace{(\mathbb{E}_\theta[\hat{\theta}] - \theta)^2}_{\text{squared bias}} + \underbrace{\text{Var}_\theta(\hat{\theta})}_{\text{variance}}. \quad \blacksquare$$

Proof. Expand $\mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{E}[(\hat{\theta} - \mathbb{E}\hat{\theta})^2] + (\mathbb{E}\hat{\theta} - \theta)^2 + 2(\mathbb{E}\hat{\theta} - \theta)\mathbb{E}[\hat{\theta} - \mathbb{E}\hat{\theta}]$; the cross term vanishes. $\blacksquare$

Unbiasedness is prized for a structural reason: it makes errors *add*. Independent unbiased pieces — per-key counters, per-round messages, per-node visits — combine into an unbiased average whose variance is the average of the variances:

Lemma 4.3 — Averaging independent copies. Let $\hat{\theta}_1, \dots, \hat{\theta}_t$ be independent, each with $\mathbb{E}[\hat{\theta}_i] = \theta$ and $\text{Var}(\hat{\theta}_i) \leq \sigma^2$. Then $\bar{\theta} = \frac{1}{t} \sum_{i=1}^t \hat{\theta}_i$ is unbiased and

$$\text{Var}(\bar{\theta}) = \frac{1}{t^2} \sum_{i=1}^t \text{Var}(\hat{\theta}_i) \leq \frac{\sigma^2}{t}. \quad \blacksquare$$

Proof. Unbiasedness is linearity of expectation; independence zeroes the cross terms in $\text{Var}(\sum_i \hat{\theta}_i) = \sum_i \text{Var}(\hat{\theta}_i) + \sum_{i \neq j} \text{Cov}(\hat{\theta}_i, \hat{\theta}_j)$. $\blacksquare$

¹ Unbiasedness is *fragile under nonlinearity*: $\mathbb{E}[g(X)] \neq g(\mathbb{E}X)$, so an unbiased estimate of x does not yield one of x^p by plugging in. Corpus papers either estimate the nonlinear functional directly (Section 4.3) or *truncate*, capping the estimator and booking the bias into the guarantee (Section 4.1).

¹

The second rung asks whether the variance is not just bounded but *good*. The classical benchmark is the Cramér–Rao inequality: no unbiased estimator can beat the inverse of the information the data carries about the parameter [3, 8].

Theorem 4.4 — Cramér–Rao lower bound. Let $\{p_\theta\}_{\theta \in \Theta}$ be a regular parametric family (support independent of θ ; differentiation and expectation exchange) and $\hat{\theta}$ an unbiased estimator with finite variance. Then

$$\text{Var}_\theta(\hat{\theta}) \geq \frac{1}{I(\theta)}, \quad I(\theta) = \mathbb{E}_\theta \left[(\partial_\theta \log p_\theta(X))^2 \right],$$

the *Fisher information*. An unbiased estimator attaining the bound is called *efficient*. ■

Proof idea. The score $S = \partial_\theta \log p_\theta(X)$ has mean zero and covariance 1 with any unbiased estimator; Cauchy–Schwarz gives $1 \leq \text{Var}(\hat{\theta}) \cdot \text{Var}(S) = \text{Var}(\hat{\theta}) I(\theta)$ (full proof in Section 4.6). ■

2

The third and fourth rungs convert moments into the contracts systems actually print. First the cheap, moment-only rung:

Proposition 4.5 — Variance to confidence: the Chebyshev rung. Let $\hat{\theta}$ be unbiased with $\text{Var}(\hat{\theta}) \leq \sigma^2$, and let $\bar{\theta}$ average t independent copies. Then for any accuracy $E > 0$ and failure probability $\delta > 0$, choosing $t \geq \sigma^2 / (\delta E^2)$ gives

$$\Pr[|\bar{\theta} - \theta| \geq E] \leq \delta. \quad \blacksquare$$

Proof. Chebyshev: $\Pr[|\bar{\theta} - \theta| \geq E] \leq \text{Var}(\bar{\theta}) / E^2 \leq \sigma^2 / (t E^2) \leq \delta$. ■

Chebyshev spends samples proportionally to $1/\delta$ — fine for one estimate, unaffordable over many. When the copies are bounded, the concentration rung spends only logarithmically:

Proposition 4.6 — Sample-count inversion: the Hoeffding rung. Let $\hat{\theta}_1, \dots, \hat{\theta}_t$ be i.i.d. copies of an unbiased estimator with $\hat{\theta}_i \in [a, a + r]$ a.s. (range width r), and $\bar{\theta}$ their average. Then

$$\Pr[|\bar{\theta} - \theta| \geq E] \leq 2 \exp\left(-\frac{2tE^2}{r^2}\right),$$

so $t = r^2 \ln(2M/\delta) / (2E^2)$ makes M such estimates simultaneously (E, δ) -accurate. ■

Proof idea. Additive Hoeffding on bounded i.i.d. summands [1, 7], then a union bound over the M estimates: each fails with probability at most δ/M . ■

This inversion is the most reusable formula of the chapter: samples = range squared $\times$ $\log(\text{events}/\delta)$ over twice the squared accuracy — derived, never tuned. Sections 4.3 and 4.4 run exactly this computation, with $r = d_v \Delta$ and $M = n$ nodes.

² In database papers the “information” is rarely Fisherian — the data model is combinatorial, not a smooth family. The Cramér–Rao *role* — no estimator can do better — is played by entropy-incompressibility (Section 4.2), communication reductions (Section 4.1), and min-max arguments: whatever the estimator must distinguish dictates the samples or bits it must pay.

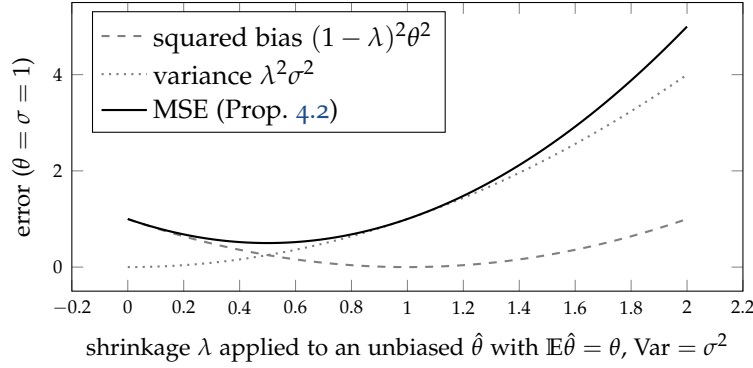


Figure 4.1: The bias–variance ledger: shrinking or capping an unbiased estimator ($\hat{\theta} \mapsto \lambda\hat{\theta}$, or the truncations of Sections 4.1 and 4.2) buys variance at the price of bias; the MSE-optimal $\lambda^* = \theta^2/(\theta^2 + \sigma^2)$ balances the two curves, and truncation estimators sit at an interior point chosen so the bias lands inside the stated factor.

Notation

The generic parameter of §4.0.1 is θ and its estimator $\hat{\theta}$. An (E, δ) -estimate means $|\hat{\theta} - \theta| \leq E$ with probability at least $1 - \delta$; a $(1 \pm \varepsilon)$ estimate is the multiplicative form. Sample counts appear as t (generic), ω_t (trees, Section 4.3), and θ again in $\theta_{\max}$ (sample-set threshold, Section 4.4) — always redefined locally. Space upper bounds are stated in machine *words*, lower bounds in *bits*.

4.0.2 How to choose an estimator

The selection procedure is mechanical once three questions are answered. *First, what is the parameter, and what is the randomness?* Name the quantity θ and the random experiment that touches it; the estimator is whatever function of the outcome you report. *Second, can the expectation be computed exactly?* If each sampled object carries a known inclusion probability, weight by its inverse (Horvitz–Thompson): the weighted sum is unbiased by linearity alone; a nonlinear functional is estimated directly per copy, not by plugging into a good estimate of something else. *Third, what do you control — the variance or the range?* Variance only: the Chebyshev rung (Proposition 4.5), $t \propto 1/(\delta E^2)$, boostable by medians. Bounded range: the Hoeffding rung (Proposition 4.6), $t = r^2 \ln(2M/\delta)/(2E^2)$ — derive the count, do not tune it. If neither is tame, cap the estimator and pay the bias into the guarantee, as Section 4.1 does.

When to Reach for Estimation Theory

Reach whenever a system reports numbers from a sample, sketch, simulation, or randomized mechanism — approximate query processing, cardinality and frequency estimation, graph distances, influence and pattern counts — and the deliverable is an error bar or an (ϵ, δ) contract; or when a lower bound must certify the design optimal. Recipe:

1. Name the parameter θ and the random experiment; write the reported number as a function $\hat{\theta}$ of the outcome.
2. Make $\hat{\theta}$ unbiased by inverse-probability weighting (indicator $\times 1/\Pr[\text{inclusion}]$); verify $\mathbb{E}[\hat{\theta}] = \theta$ by linearity.
3. If heavy tails blow up the variance, cap or shrink the estimator and book the induced bias into the approximation factor (the ledger of Proposition 4.2).
4. Bound what you control — Var (Chebyshev rung, $t \propto \sigma^2/(\delta E^2)$) or range width r (Hoeffding rung, $t = r^2 \ln(2M/\delta)/(2E^2)$) — and invert for the sample count analytically.
5. Union-bound over every key, node, or coordinate that must be simultaneously correct; the $\ln(2M/\delta)$ in the count is that union bound, pre-paid.
6. Certify optimality (entropy incompressibility, a communication reduction, or Theorem 4.4); matching within logarithmic factors is the gold standard.

4.1 Frequency moments from noisy, ambiguous streams

The exemplar is a PODS'26 paper on streaming statistics under noisy identity resolution [5]^{PODS}. Frequency moments, $F_p = \sum_j f_j^p$, are canonical streaming quantities behind cardinality, skew, and heavy-hitter analytics, and decades of sketching theory estimate them accurately when every item is identified exactly. Real pipelines resolve identities imperfectly: each item arrives *ambiguous*, compatible with several ground-truth identities, as after record matching or noisy deduplication. The paper introduces a per-instance noise measure, the F_p -*mismatch-ambiguity* Λ_p , and proves that F_p of the unknown ground truth is approximable to factor $(\epsilon + O(\Lambda_p))$ in one pass over m items with $O(\epsilon^{-2} m^{1-1/p})$ words of space, essentially optimally. The algorithm is estimation theory in one exhibit: an unbiased ordered-clique sampler whose variance is tamed by truncation, with Hölder supplying the

moment bound and Chebyshev the confidence.

Problem. A noisy dataset is a stream $\sigma = (\sigma_1, \dots, \sigma_m)$ of items over a universe U ; the ground truth is an assignment of each position to a true identity, $\mu = (\mu_1, \dots, \mu_m) \in U^m$. Writing $f_j = |\{i : \mu_i = u_j\}|$, the goal is $F_p(\mu) = \sum_{j=1}^n f_j^p$ for constant $p \geq 1$, but the algorithm sees only σ . Define two positions as *similar* ($\sigma_i \sim \sigma_{i'}$) when they are compatible with a common identity, and let $G^\sigma = ([m], E^\sigma)$ be the resulting *similarity graph*; write B_i^σ for the clique of position i in G^σ and B_i^μ for its ground-truth clique (the positions sharing i 's true identity). The paper's per-instance noise measure is:

Definition 4.7 (F_p -mismatch-ambiguity; Definition of the paper, notation cleaned). For $p \geq 1$,

$$\Lambda_p(\sigma, \mu) = \frac{1}{F_p(\mu)} \sum_{i \in [m]} \left(|B_i^\sigma \cup B_i^\mu|^{p-1} - |B_i^\sigma \cap B_i^\mu|^{p-1} \right).$$

When $\sigma = \mu$ the cliques coincide and $\Lambda_p = 0$; the quantity measures the relative excess clique mass the ambiguity injects. A $(c, 0.01)$ -approximation means an estimate within factor $(1 \pm c)$ of $F_p(\mu)$ with probability at least 0.99.

Guarantee.

Theorem 4.8 — One-pass estimation under mismatch ambiguity; Theorem of the paper, compressed. For any constant $p \geq 1$ and any noisy stream of length m with $\Lambda_p \leq 1/(3 \cdot p!)$, there is a one-pass algorithm computing an $(\varepsilon + O(\Lambda_p), 0.01)$ -approximation of $F_p(\mu)$ using $O(\varepsilon^{-2} m^{1-1/p})$ words of space. ■

Theorem 4.9 — Space lower bound; Corollary of the paper, compressed. For any constants $p > 1$, $C > 0$ and $\varepsilon \in (0, \frac{1}{3})$, any $O(1)$ -pass $(\varepsilon + C\Lambda_p, 0.48)$ -approximation algorithm for F_p over noisy streams of length at most m requires $\Omega(\varepsilon^{-1/p} m^{1-1/p})$ bits of space. ■

How estimation theory is applied. The random object is an *ordered p -clique* of the observable similarity graph — a tuple $(i_1, \dots, i_p)$ of positions that are pairwise similar. The key identity is that clique mass is the frequency moment: for noiseless data the ground-truth cliques partition $[m]$ into classes $V_1, \dots, V_{n_\mu}$ of sizes equal to the frequencies, so the ordered-clique count is $\sum_k |V_k|^p = F_p(\mu)$ (Fact of the paper); a divergence lemma bounds the noisy count, $|K_p^\sigma \setminus K_p^\mu| \leq \Lambda_p F_p$ and $|K_p^\mu \setminus K_p^\sigma| \leq \frac{p}{2} \Lambda_p F_p$ (Lemma of the paper), so the observable clique mass is a $(1 + O(\Lambda_p))$ -proxy for the target. The estimator is pure Horvitz–Thompson: sample I_1 uniformly from $[m]$ and, given the prefix, I_k uniformly from the *tail set* $\Phi(I_1, \dots, I_{k-1}) = \{i \geq I_{k-1} : i \sim I_{j'} \text{ for all } j' < k\}$.

Lemma 4.10 — Unbiased clique-count estimator; Lemma of the paper, notation cleaned. Let $R_1 = m$ and $R_k = |\Phi(I_1, \dots, I_{k-1})|$ for $k = 2, \dots, p$, and let $X = \prod_{k \in [p]} R_k$. Then

$$\mathbb{E}[X] = |K_p^\sigma|, \quad \blacksquare$$

the number of increasingly ordered p -cliques of G^σ .

Proof sketch. Each fixed increasingly ordered clique is sampled with probability exactly $\frac{1}{m} \prod_{k=2}^p \frac{1}{|\Phi(i_1, \dots, i_{k-1})|}$ and then contributes $X = \prod_k R_k$, the reciprocal of that probability; summing over cliques (linearity) counts each once. $\blacksquare$

The raw estimator is unbiased but heavy-tailed — a draw into a huge clique inflates X — so the paper truncates: $Y = \min\{X, m (d_I^\mu)^{p-1}\}$ with d_I^μ the ground-truth clique size of the sampled position I . The truncation level makes the bias land inside the Λ_p term of the guarantee while the second moment collapses to a Hölder consequence:

Lemma 4.11 — Truncation controls the variance; Claim of the paper, notation cleaned. With Y as above, $(1 - 2\Lambda_p)F_p(\mu) \leq \mathbb{E}[Y] \leq F_p(\mu)$ and

$$\text{Var}(Y) \leq \mathbb{E}[Y^2] \leq m^{1-1/p} F_p(\mu)^2. \quad \blacksquare$$

Proof sketch. The sandwich sums per-position clipping errors into $\Lambda_p F_p$; the moment bound chains $Y \leq m \max_k |V_k|^{p-1}$ with $\max_k |V_k|^{p-1} \leq F_p^{1-1/p}$ and Hölder's $m = \sum_k |V_k| \leq n_\mu^{1-1/p} F_p^{1/p}$ (full derivation in Section 4.6). $\blacksquare$

The confidence assembly runs on the ladder's rungs. *Lower tail:* Chebyshev (Proposition 4.5) on $t = \Theta(\varepsilon^{-2} m^{1-1/p})$ copies gives $\Pr[\bar{Y} < (1 - \varepsilon - 2\Lambda_p)F_p] \leq 10^{-5}$. *Upper tail:* a Rényi-style inequality due to Feige — for independent nonnegative Z_i with means $\mu_i \leq 1$, $\Pr[\sum_i Z_i < \sum_i \mu_i + \delta] > \min(\frac{\delta}{1+\delta}, \frac{1}{13})$ — shows the average is not too *large* with constant probability, boosted to 0.99 by repetition. The space bound is this t , and the lower bound (Theorem 4.9) is the estimation-theoretic mirror: DISJ reduces to F_p -under-ambiguity — each player's bits become items whose similarity pattern encodes set intersection — so beating the stated space would solve DISJ with too little communication.

What it buys. The streaming engine reports frequency-moment analytics over dirty, unresolved data with a certificate that degrades gracefully with the ambiguity: error $\varepsilon + O(\Lambda_p)$, one pass, sublinear space, and a lower bound proving nothing substantially better exists. The pattern — unbiased Horvitz–Thompson core, truncation paid into the approximation factor, Chebyshev below and a reverse-Markov bound above — transfers to any heavy-tailed estimator.

4.2 Epoch analysis and an entropy bound for expandable sketches

The exemplar is a SIGMOD’26 paper on sketches for unbounded streams [2]. Frequency-estimation (FE) sketches — Count-Min, CountSketch — are standard equipment, but they commit to a fixed width at initialization, while real streams have unknown, unbounded length and heavy skew: a fixed sketch either wastes memory up front or degrades silently as it overflows. The paper’s system, *Sublime*, is an expandable, CountSketch-style FE sketch that grows with the stream while keeping every per-key estimate unbiased, supported by a two-sided theory: a variance recursion over the expansion *epochs* shows the estimation error stays sublinear in the stream length, a matching memory analysis bounds the footprint of the skewed counters, and an entropy-incompressibility argument proves any expandable FE sketch must pay within a small logarithmic factor of *Sublime*’s space.

Problem. A stream of keys y (insertions and deletions) is processed by a sketch answering frequency queries $f(y)$ at any time; the total length N is unknown and unbounded, and the frequency distribution is skewed, so counters holding heavy keys must encode large values compactly. *Sublime* maintains d counter arrays whose width follows a *size function* $W(\cdot)$ — e.g. $W(N) = N/\epsilon$ (linear) or $W(N) = N^\alpha/\epsilon$ (sublinear, $0 \leq \alpha < 1$) — and expands over *epochs*: at epoch j the width is $W_j = W(N_j)$, N_j the keys seen by epoch j ’s start. Expansion rehashes and *decrements* counters rather than rebuilding, so old contributions stay visible; the price is that updates from epoch j survive into epoch i ’s counters only with probability $2^{i-j}/W_i$. The target is the per-key frequency $f(y)$, estimated CountSketch-style by signed counter reads.

Guarantee.

Lemma 4.12 — Variance recursion over epochs; Lemma of the paper, notation cleaned. The estimates are unbiased, like CountSketch’s. Moreover, letting $V(F)$ denote the estimation variance under size function $W(\cdot)$ and one array, on a stream with squared ℓ_2 -norm $F = \sum_y (f(y))^2$, at the end of epoch i :

$$V(F_{i+1}) \leq \frac{1}{W_i} \cdot \sum_{j=0}^i 2^{i-j} \cdot (F_{j+1} - F_j),$$

where F_j is the squared ℓ_2 -norm at the start of epoch j . ■

The recursion telescopes (geometric series in 2^{i-j} , evaluated per size function): for $W(N) = N^\alpha/\epsilon$ the error is $O(N/W(N))$ — sublinear —

while for $W(N) = N/\varepsilon$ it is $O(\varepsilon \log N)$. The complementary memory statement is:

Theorem 4.13 — Space lower bound for expandable FE sketches; Theorem of the paper. Consider an FE sketch with only overestimation errors that processes growing streams with an error bound of $E(N)$ and a confidence of $1 - \delta$. Such a sketch must use at least

$$\frac{N}{E(N)} \cdot \left[\Omega\left(\log \frac{1}{\delta} + \log \log \frac{N}{E(N)}\right) - \Theta(1) \right] \quad \blacksquare$$

bits of memory at some point in time while processing a stream.

How estimation theory is applied. The random objects are the hash placements and the ± 1 signs of the CountSketch read; unbiasedness is the classical sign-orthogonality computation ($\mathbb{E}[\sigma(x)\sigma(y)] = 0$ for $x \neq y$), re-run with the epoch structure: a key's contribution from epoch j reaches the current counter of a query only with probability $2^{i-j}/W_i$, so the variance integrates those geometric weights over the ℓ_2 -mass increments ($F_{j+1} - F_j$) — the variance term of Proposition 4.2, bookkept per epoch. Confidence is the Chebyshev rung (Proposition 4.5): a sketch with $|\mathbb{E}[\hat{f}(x)] - f(x)| \leq E(N)$ and $\text{Var}(\hat{f}(x)) \leq \delta E(N)^2$ guarantees error $2E(N)$ at confidence $1 - \delta$ by Chebyshev, and the theorem holds for any such sketch. The memory *upper* bound is where convexity earns its keep: counters are encoded as short *stubs* plus rare *extensions*, and since the logarithm is concave, Jensen bounds the total stub length by the logarithm of the *average* counter value — w counters with total absolute value k encode in $w \cdot [4.41 + 1.26 \log_2(k/w)]$ bits (Lemma of the paper). The lower bound is the entropy-incompressibility mirror in the Cramér–Rao role: a smaller sketch would compress sequences of $N/E(N)$ keys from a large universe below their entropy — using the law of total variance to find an epoch state with small second moment, then fixing the internal randomness so the state plus a few bits reconstruct the sequence. Information cannot be destroyed.

What it buys. The system gets a frequency-estimation sketch whose accuracy and memory are certified *while it grows*, with no prior knowledge of the stream's length or skew: near-sublinear error at confidence $1 - \delta$, a footprint of roughly $d \cdot W(N) \cdot [4.41 + 1.26 \log_2(N/W(N))]$ bits, and a lower bound within a small log factor of optimal.

4.3 Spanning-tree estimators for biharmonic distance

The exemplar is a SIGMOD'26 paper on graph distance estimation [4]. The *biharmonic distance* (BD) between two nodes is defined through the square of the Laplacian pseudoinverse; like resistance distance it

rewards *multiple* connecting paths, and it is more robust to noise and degree skew than shortest-path or resistance measures — attractive for similarity search and network analysis. But its definition seems to demand global matrix computations: exact BD needs the Moore–Penrose pseudoinverse ($O(n^3)$), and the fastest prior approximations need $\tilde{O}(m/\varepsilon^2)$ preprocessing over the whole graph — prohibitive when only a few pairs matter. The paper instead answers BD queries for a *queried set* of pairs locally: reformulate BD around a pivot via Sherman–Morrison, interpret each term as a statistic of absorbing random walks, and estimate those statistics by averaging sampled spanning trees — the tree count *derived* from the Hoeffding inversion of Proposition 4.6, pivot degree times diameter as the range.

Problem. Let $G = (V, E)$ be a connected undirected graph with Laplacian L and pseudoinverse $L^\dagger$. For distinct $s, t \in V$ and $\mathbf{b}_{st} = \chi_s - \chi_t$, the biharmonic distance is $\beta(s, t) := \mathbf{b}_{st}^\top (L^\dagger)^2 \mathbf{b}_{st}$ (Definition 2.1 of the paper, notation cleaned). Fix a *pivot* $v \in V$ and let L_v be L with row and column v deleted, and a the column of L linking v to the rest. The paper’s reformulation — Sherman–Morrison applied to the v -deleted Laplacian square $L_v^2 + aa^\top$ (Lemma 3.1 of the paper) — reduces BD to quantities of L_v^{-1} :

Theorem 4.14 — Pivoted reformulation of BD; Theorem 3.5 of the paper, notation cleaned. For any pivot v and $s, t \neq v$,

$$\beta(s, t) = \mathbf{x}_b^\top \mathbf{x}_b - \frac{(\mathbf{x}_b^\top \mathbf{x}_a)^2}{1 + \mathbf{x}_a^\top \mathbf{x}_a}, \quad \mathbf{x}_a = a^\top L_v^{-1}, \quad \mathbf{x}_b = \mathbf{b}_{st}^\top L_v^{-1},$$

i.e., BD is a rational function of the two vectors $\mathbf{x}_a, \mathbf{x}_b \in \mathbb{R}^{V \setminus \{v\}}$. ■

So the problem becomes: estimate the n coordinates of $\mathbf{x}_a$ and $\mathbf{x}_b$ well enough that the rational function is accurate. Two structural facts make the coordinates estimable without inverting anything. First, $(L_v^{-1})_{us} = \pi^v(u, s)/d_s$, where $\pi^v(u, s)$ is the expected number of visits to s of a random walk from u absorbed at v — the *absorbing-random-walk interpretation* of the inverse. Second, the paper’s *BackPush* computes the needed visit counts locally via the invariant

Lemma 4.15 — Pushback invariant; Lemma 4.1 of the paper. For each node $u \in V \setminus \{v\}$ and target x , throughout the algorithm,

$$\pi^v(u, x) = \hat{\pi}^v(u, x) + \sum_{w \in V} r(w, x) \pi^v(u, w), \quad \blacksquare$$

where $\hat{\pi}^v(u, x)$ is the (partial) estimate returned so far and $r(w, x)$ the pending residual at w .

Proof sketch. Induction over pushback operations: initially all mass is residual; pushing back at node y moves $r(y, x)/d_y$ to each neighbor’s

residual (and to $\hat{\pi}$ when $y = u$), and the walk interpretation of $\pi^v(\cdot, x)$ makes the two sides change by exactly the same amount. ■

The invariant certifies that when residuals are small the local estimate is exact (additive error $\sum_w \pi^v(u, w) r^{\max}$, Lemma 4.2 of the paper) — but for estimates, the paper’s *FastTree* replaces deterministic pushes by sampling.

Guarantee.

Theorem 4.16 — Tree-sampling accuracy; Theorem 6.3 of the paper. For error parameter $\varepsilon \in (0, 1)$ and failure probability $\delta \in (0, 1)$, let

$$\omega_t = \frac{d_v^2 \Delta^2 \log(2n/\delta)}{2\varepsilon^2},$$

where d_v is the pivot degree and Δ the graph diameter. Then sampling ω_t spanning trees yields estimators $\tilde{x}_a(u), \tilde{x}_b(u)$ satisfying, for every node u ,

$$|x_a(u) - \tilde{x}_a(u)| \leq \varepsilon \quad \text{and} \quad |x_b(u) - \tilde{x}_b(u)| \leq \varepsilon, \quad \blacksquare$$

and *FastTree* runs in $O(\omega_t \cdot (h_{s,v} + h_{t,v} + \sum_{w \in N(v)} h_{w,v}))$ time, where $h_{u,v}$ is the hitting time from u to v .

How estimation theory is applied. The random object is a uniform spanning tree.³ The bridge from trees to Laplacian entries is an electrical-network identity the paper proves with Kirchhoff’s laws:

Lemma 4.17 — Spanning-tree expression of the inverse; Lemma 6.1 of the paper, notation cleaned. Let N be the number of spanning trees of G , let $N_{sv}(a, b)$ be the number of spanning trees in which the unique path from s to v traverses the edge (a, b) in this order, and let P_{uv} be any path from u to v . Then

$$(L_v^{-1})_{su} = \frac{1}{N} \sum_{(a,b) \in P_{uv}} (N_{sv}(a, b) - N_{sv}(b, a)). \quad \blacksquare$$

Proof sketch. Treat G as a unit-resistor network with unit current injected at s and extracted at v : the current on edge (a, b) is exactly $(N_{sv}(a, b) - N_{sv}(b, a)) / N$ (a classical Kirchhoff fact), the voltage at u is the current summed along any path P_{uv} , and $(L_v^{-1})_{su}$ is that voltage. ■

By Lemma 4.17, each coordinate $x_a(u), x_b(u)$ of Theorem 4.14 is a sum of oriented edge indicators along paths of a random tree, so a single tree yields per-coordinate contributions whose expectations are the true coordinates, and averaging ω_t independent trees gives unbiased estimators $\tilde{x}_a(u), \tilde{x}_b(u)$ (Corollary 6.2 of the paper; its walk-based sibling averages the collision estimators of Lemma 5.1). The variance

³ Sampled by Wilson’s algorithm, which runs a loop-erased random walk to the root; its cost is governed by the same hitting times that appear in the running-time bound.

side is the Hoeffding rung in its purest form: each per-tree contribution is bounded, $|\tilde{x}_a^i(u)| \leq d_v \Delta$ and $|\tilde{x}_b^i(u)| \leq 2\Delta$ (range bounds of Corollary 6.2, at most d_v neighbor paths feeding $\mathbf{x}_a$). Additive Hoeffding then gives per-node failure δ/n at exactly the displayed ω_t , and the union bound over the n nodes — pre-paid as the $\log(2n/\delta)$ inside ω_t — delivers the simultaneous statement (proof in Section 4.6); the tuning is mechanical, $r = d_v \Delta$, $M = 2n$.

What it buys. The graph engine answers biharmonic-distance queries for targeted node pairs in time governed by hitting times between the query nodes and the pivot — no $O(n^3)$ pseudoinverse, no whole-graph preprocessing. Every coordinate carries its accuracy certificate, and the sample count was derived, not tuned — the chapter’s formula applied verbatim.

4.4 Concentration-sized sampling for influence blocking

The exemplar is a SIGMOD’26 paper on curbing misinformation spread [9]. When false information diffuses through a social network, harm depends on *time*: a node infected early contributes more damage, so interventions have a deadline. The paper studies *time-critical influence minimization* (TCIM): given a diffusion process with edge probabilities and delays, a seed set of misinformation, a deadline T , and a budget k , choose k nodes to block so that the total *activation-duration-weighted* spread is minimized. The objective is maximally hostile to exact optimization: evaluating the blocking benefit $D(B)$ is #P-hard, D is monotone but *non-submodular*, and no polynomial-time method can approximate TCIM within any constant factor unless $P = NP$. The algorithm, CBFM, therefore *sandwiches* the objective between submodular surrogates $D_L \leq D \leq D_U$, maximizes them greedily on randomly sampled *reverse-score sample sets*, and — the estimation-theoretic core — derives unbiased coverage estimators and concentration bounds that set the number of sample sets analytically and power anytime stopping, with the final guarantee holding with probability $1 - 3\delta$.

Problem. Diffusion follows the independent-cascade model with delays: each edge $\langle u, v \rangle$ is live with probability $p(u, v)$ and carries a random delay, inducing a distribution over *possible worlds* w ; node v ’s activation time t_v^w determines its weight $U(T - t_v^w)$ under a monotone weight function U , zero past the deadline T . The *AD influence spread* is $\sigma_G^T(S) = \sum_{v \in V} \mathbb{E}[U(T - t_v)]$, and the blocking benefit of $B \subseteq V \setminus S$ is $D(B) = \sigma_G^T(S) - \sigma_G^T[V \setminus B](S)$; the goal is a size- k B maximizing $D(B)$. Since D resists optimization, CBFM maximizes the submodular bounds $D_L \leq D \leq D_U$ (D_L from delayed activations, D_U from *time-aware reach-*

able trees; Lemmas 4.1–4.8 of the paper), where evaluating either bound is again #P-hard (Lemma 5.1 of the paper) — hence estimation. A *GRS/TRS sample set* R_i draws a source node with probability proportional to a computable weight and records, in a random possible world, which candidate blockers would *reverse-reach* the source before T , each with its potential score; $X_i(B)$ is the total score in R_i covered by B , and $\Lambda_R(B) = \sum_{i=1}^{\theta} X_i(B)$ the *coverage* of B over θ sample sets.

Guarantee.

Theorem 4.18 — Approximation guarantee of CBFM; Theorem 6.7 of the paper, notation cleaned. Given $0 \leq \varepsilon, \delta \leq 1$, CBFM returns B^* satisfying

$$\begin{aligned} \Pr \left[D(B^*) \geq \tau \left(1 - \frac{1}{e} - \varepsilon \right) \cdot D(B_o) \right] \\ \geq 1 - 3\delta, \end{aligned} \quad \blacksquare$$

where B_o is the optimal size- k blocker set and τ is the data-driven sandwich ratio

$$\tau = \max \left\{ \frac{D_L(B_o^L)}{D(B_o)}, \frac{D_U(B_o^U) (1 - \varepsilon)}{(1 + \varepsilon) D(B_o)} \right\}.$$

How estimation theory is applied. The random objects are the reverse-score sample sets; the estimated parameters are the surrogate objectives $\psi \in \{D_L, D_U\}$. The workhorse is an unbiasedness lemma of exactly the Horvitz–Thompson shape of the ladder:

Lemma 4.19 — Unbiased coverage estimation; Lemma 5.8 of the paper, notation cleaned. For any $B \subseteq V \setminus S$ and generalized sample set R_i ,

$$\psi(B) = I_{T^*}(S) \cdot \mathbb{E}[X_i(B)], \quad \blacksquare$$

the expectation over the randomness of R_i , where $I_{T^*}(S)$ is the computable total reverse-reachability weight — the normalizer of the weighted source sampling. Consequently, with θ sample sets, $I_{T^*}(S) \cdot \Lambda_R(B)/\theta$ is an unbiased estimator of $\psi(B)$.

Proof sketch. Each possible world enters the sample with probability proportional to its weight; the weighted source choice times the recorded score is the world’s contribution to $\psi(B)$ divided by its inclusion probability, so expectation sums the worlds back in (the RIS accounting of influence-maximization sampling, re-derived for time-aware scores). $\blacksquare$

The second ingredient converts the estimator into a *stopping rule*: concentration of the coverage around its mean, with the target value itself as the variance proxy.

Lemma 4.20 — Coverage concentration; Lemma 6.4 of the paper, compressed. For any $\lambda > 0$, the coverage deviation obeys a Bernstein-style two-tail bound:

$$\Pr \left[\left| \Lambda_R(B) / \theta - \psi(B) \right| \geq \lambda \cdot U(T-1) \right] \leq 2 \exp \left(- \Theta \left(\frac{\theta \psi(B) \lambda^2}{I_{T^*}(S) U(T-1)} \right) \right),$$

i.e. deviations decay exponentially in $\theta \cdot \psi(B)$, at range scale $I_{T^*}(S) U(T-1)$. ■

Inverting this bound — the same move as Proposition 4.6, variance proxy in place of squared range — yields the threshold analytically (Lemma 6.5 of the paper, compressed): $\theta_{\max} = \tilde{O}(n U(T-1) / (\varepsilon^2 \cdot \text{OPT}_L))$ with OPT_L a computable lower bound on the optimum ($U(1)$ times the k largest activation probabilities). The threshold certifies greedy on $\theta_{\max}$ sample sets $(1 - \frac{1}{e} - \varepsilon)$ -approximate for both surrogates with failure $\delta/3$ each; a third $\delta/3$ guards the anytime stopping rule, which estimates the sandwich ratio τ on the fly and stops once it cannot improve. A union bound over the three events assembles Theorem 4.18 — the 3δ , itemized.

What it buys. The system selects blockers on graphs with billions of edges with a *certified*, instance-adapted guarantee (the ratio τ measures how tightly the submodular sandwich hugs D), while the concentration-derived threshold and stopping rule mean it draws exactly as many sample sets as the accuracy target demands — the modern template for influence-style optimization under simulation access.

4.5 Advanced usage and further reading

Everything so far fixes the sample size in advance; the two papers sketched here instead let the *data* decide when to stop, or rest unbiasedness on a sampling invariant.

Betting confidence sequences as stopping rules [10]. When an expensive oracle — an LLM, a human — processes records behind a cheap proxy score, the system routes records above a cascade threshold θ to the oracle and must certify that recall meets a target T with few oracle calls. Checking a fixed-sample confidence interval after every record invalidates its δ : errors stack over repeated looks. The paper adopts *anytime-valid* confidence sequences derived from betting (empirical-Bernstein capital processes of Waudby-Smith and Ramdas, simplified): the capital $K(m, Y) = \prod_j (1 + \min(\lambda_j, \frac{3}{4m})(Y_j - m))$, bets tuned to the running variance, stays bounded in expectation unless the mean crosses

m , so a threshold on K certifies a mean bound at *every* sample count simultaneously. Estimation becomes a *sequential stopping rule*: sample, update the capital, stop when the certified recall lower bound clears the target; a without-replacement variant handles finite populations, and a companion lower bound prices any proxy-monotone sampler by its precision cost.

Reservoir unbiasedness for hypergraph triangle counting [6]. Counting triangles in a hypergraph stream — one-, two-, and three-hyperedge notions alike — is confounded by memory limits: the stream cannot be stored, so the count must be estimated from a reservoir sample, and the estimator is only as good as the reservoir’s *inclusion probabilities*. The paper’s foundation is a reservoir lemma: its partitioned sampler holds each hyperedge with equal probability $|G_s[i]|/m[i]$, so each hyperedge seen so far is in the reservoir with probability $|G_s^{(t)}|/m^{(t)}$ at *every* time t — including after the replacement cascades that evictions force. Unbiasedness is then immediate: each triangle’s indicator, weighted by the inverse joint inclusion probability of its hyperedges, has expectation equal to its presence, and linearity sums to $\mathbb{E}[\hat{c}] = c$ (Theorems of the paper); direct variance computations — e.g. $\text{Var}[\hat{c}_{\text{otr}}] \leq (2c_{\text{otr}}^2 - c_{\text{otr}}) \cdot \frac{m(m-1)(m-2)}{|G_s|(|G_s|-1)(|G_s|-2)}$ — complete the analysis. The lesson: *unbiasedness under sampling is a property of the sampler’s inclusion probabilities*, and maintaining them under memory pressure is itself a theorem.

4.5.1 Further reading

- Lehmann and Casella [3] — the classical treatment of point estimation: unbiasedness, sufficiency, and the decision-theoretic view of MSE.
- Tsybakov [8] — minimax lower bounds (Le Cam, Fano); the source for the role our corpus papers fill with entropy and communication bounds.
- Boucheron, Lugosi, and Massart [1] — the non-asymptotic concentration machinery behind this chapter’s rungs.
- Mitzenmacher and Upfal [7] — the randomized-algorithms view: sampling, Chernoff/Hoeffding, and the median trick, with CS applications.

Chapter Recap

- **What the tool is.** Estimation theory certifies numbers a system did not exactly compute: an estimator $\hat{\theta}$ of a parameter θ , its bias and variance ($\text{MSE} = \text{bias}^2 + \text{Var}$), and the (E, δ) -accuracy

contracts built from them by Chebyshev (samples $\propto 1/\delta$) or Hoeffding (samples $\propto \ln(\text{events}/\delta)$).

- **Which rung when.** Inclusion probabilities known: Horvitz–Thompson weighting, unbiased by linearity. Heavy tails: truncate, pay the bias into the approximation factor. Only a variance bound: Chebyshev, boost by medians. Bounded range: Hoeffding, $t = r^2 \ln(2M/\delta)/(2E^2)$ — derive the sample count, never tune it. Nonlinear functional: estimate it directly per copy; unbiasedness does not survive plug-ins.
- **Why unbiasedness first.** It makes errors add: independent unbiased pieces average to variance σ^2/t ; one computation then feeds every concentration argument.
- **What the corpus showed.** Noisy frequency moments (Section 4.1): unbiased ordered-clique sampling, truncation, Hölder + Chebyshev + a reverse-Markov upper tail, DISJ near-optimality. Expandable sketches (Section 4.2): epoch-indexed variance recursion, Jensen-bounded encodings, an entropy lower bound. Bi-harmonic distance (Section 4.3): Kirchhoff identities turn Laplacian entries into tree expectations; Hoeffding inversion outputs $\omega_t = d_v^2 \Delta^2 \log(2n/\delta)/(2\epsilon^2)$. Influence blocking (Section 4.4): unbiased coverage estimators plus Bernstein concentration setting the sample threshold and stopping rule; union bound $\Rightarrow 1 - 3\delta$.
- **Advanced usage.** Anytime-valid betting confidence sequences make estimation a sequential stopping rule; reservoir inclusion-probability lemmas make unbiasedness a property of the sampler (Section 4.5). Deferred proofs live in the Appendix (Section 4.6).

4.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 4.4 (Cramér–Rao bound)

Statement (recalled). For a regular family p_θ and any unbiased $\hat{\theta}$ with finite variance, $\text{Var}_\theta(\hat{\theta}) \geq 1/I(\theta)$ with $I(\theta) = \mathbb{E}_\theta[(\partial_\theta \log p_\theta(X))^2]$. ■

Proof. Write $S(X) = \partial_\theta \log p_\theta(X)$ for the score. Regularity (support independent of θ ; differentiation under the integral) gives $\mathbb{E}_\theta[S] =$

$\partial_\theta \int p_\theta = 0$, and, for unbiased $\hat{\theta}$,

$$\text{Cov}_\theta(\hat{\theta}, S) = \mathbb{E}_\theta[\hat{\theta} S] = \int \hat{\theta} \partial_\theta p_\theta = \partial_\theta \mathbb{E}_\theta[\hat{\theta}] = \partial_\theta \theta = 1.$$

Cauchy–Schwarz now gives

$$1 = \text{Cov}_\theta(\hat{\theta}, S)^2 \leq \text{Var}_\theta(\hat{\theta}) \cdot \text{Var}_\theta(S) = \text{Var}_\theta(\hat{\theta}) \cdot I(\theta),$$

since $\text{Var}(S) = \mathbb{E}[S^2] = I(\theta)$. Divide by $I(\theta)$. ■

Proof of Lemma 4.11 (truncated clique estimator)

Statement (recalled). With I uniform on $[m]$, $X = m(d_I)^{p-1}$ — a one-draw form of the clique-mass estimator of Lemma 4.10 — and $Y = \min\{X, m(d_I^\mu)^{p-1}\}$, we have $(1 - 2\Lambda_p)F_p(\mu) \leq \mathbb{E}[Y] \leq F_p(\mu)$ and $\text{Var}(Y) \leq \mathbb{E}[Y^2] \leq m^{1-1/p}F_p(\mu)^2$, where $d_i = |B_i^\sigma|$ and $d_i^\mu = |B_i^\mu|$. ■

Proof. Upper expectation. $Y \leq m(d_I^\mu)^{p-1}$ with I uniform, so $\mathbb{E}[Y] \leq \sum_i (d_i^\mu)^{p-1} = F_p(\mu)$, the last identity being the paper’s Fact (ground-truth cliques $V_1, \dots, V_{n_\mu}$ of sizes equal to the frequencies).

Lower expectation. $\mathbb{E}[X - Y] = \frac{1}{m} \sum_i \max\{(d_i)^{p-1} - (d_i^\mu)^{p-1}, 0\}$, and since $(d_i)^{p-1} \leq |B_i^\sigma \cup B_i^\mu|^{p-1}$ while $(d_i^\mu)^{p-1} \geq |B_i^\sigma \cap B_i^\mu|^{p-1}$, each summand is at most $|B_i^\sigma \cup B_i^\mu|^{p-1} - |B_i^\sigma \cap B_i^\mu|^{p-1}$; summing gives $\mathbb{E}[X - Y] \leq \Lambda_p F_p(\mu)$, and the same display applied to $|(d_i)^{p-1} - (d_i^\mu)^{p-1}|$ gives $\mathbb{E}[X] \geq (1 - \Lambda_p)F_p(\mu)$. Hence $\mathbb{E}[Y] = \mathbb{E}[X] - \mathbb{E}[X - Y] \geq (1 - 2\Lambda_p)F_p(\mu)$.

Second moment. $Y \leq m \max_i (d_i^\mu)^{p-1} = m \max_k |V_k|^{p-1}$, so $\mathbb{E}[Y^2] \leq m F_p^{1-1/p} \cdot \mathbb{E}[Y] \leq m F_p^{1-1/p} \cdot F_p$, using $\max_k |V_k|^{p-1} \leq F_p^{1-1/p}$ and $\mathbb{E}[Y] \leq F_p$; and Hölder gives $m = \sum_k |V_k| \leq n_\mu^{1-1/p} F_p^{1/p}$ with $n_\mu \leq m$, so $\mathbb{E}[Y^2] \leq m^{1-1/p} F_p(\mu)^2$ and $\text{Var}(Y) \leq \mathbb{E}[Y^2]$. (This already suffices for Theorem 4.8: Chebyshev on t copies gives failure $m^{1-1/p}/(t\epsilon^2) \leq 10^{-5}$ at $t = \Theta(\epsilon^{-2} m^{1-1/p})$.) ■

Proof of Theorem 4.16 (tree-sample size)

Statement (recalled). With $\omega_t = d_v^2 \Delta^2 \log(2n/\delta)/(2\epsilon^2)$, the averaged tree estimators satisfy $|\tilde{x}_a(u) - x_a(u)| \leq \epsilon$ and $|\tilde{x}_b(u) - x_b(u)| \leq \epsilon$ for every node u simultaneously with probability $1 - \delta$. ■

Proof. The per-tree contributions are bounded: $|\tilde{x}_b^i(u)| \leq 2\Delta$ (paths from s and from t) and $|\tilde{x}_a^i(u)| \leq d_v \Delta$ (range bounds stated with the estimators, Corollary 6.2 of the paper; at most d_v neighbor paths feed x_a), and they are i.i.d. with means $x_a(u)$, $x_b(u)$. Additive Hoeffding on ω_t i.i.d. copies with range widths $r_a = 2d_v \Delta$ and $r_b = 4\Delta$ gives

$$\Pr[|\tilde{x}_a(u) - x_a(u)| \geq \epsilon] \leq 2 \exp\left(-\frac{2\omega_t \epsilon^2}{4d_v^2 \Delta^2}\right) \leq \frac{\delta}{2n},$$

since $\omega_t = d_o^2 \Delta^2 \log(2n/\delta)/(2\varepsilon^2)$ makes the exponent exactly $\log(2n/\delta)$; similarly $\Pr[|\tilde{x}_b(u) - x_b(u)| \geq \varepsilon] \leq 2 \exp(-2\omega_t \varepsilon^2/(16\Delta^2)) \leq \delta/(2n)$, absorbing the constant into the logarithm. A union bound over the n nodes and the two coordinate families keeps every deviation below ε with probability $1 - \delta$. The running time is ω_t times one tree's cost: Wilson's algorithm runs in time proportional to the hitting times from the roots $\{s, t\} \cup N(v)$ to v . ■

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5

Adversarial Constructions

An upper bound is a promise; an impossibility result is the proof that no better promise exists. The second kind of theorem is not derived from the first — it is *built*. To show that no algorithm, language, or index in a model can do better, the author constructs the world in which every allowable procedure fails: a pair of instances that look identical, a distribution of inputs hiding a secret, a stream of updates engineered so that staying correct is provably expensive, a witness object so large that merely reporting it exceeds the budget. That engineered world is the *adversary*, and building it is a craft with a small number of recurring patterns. In the SIGMOD’26 corpus the tool is used by 22 tagged papers, and it owns the survey’s signature pairing: in 13 of them the same article proves an upper bound *and* the adversarial lower bound that matches it. The pairing is no accident — the upper bound tells you what the adversary must force; the adversary certifies that the upper bound is the end of the line.

This chapter presents the basic patterns and the standard exemplars first; Section 5.5 adds the advanced regimes (adversaries against computational indistinguishability and against structural properties) as extension reading.

5.0 The tool: adversarial constructions

The problem class is uniform across applications. There is a *model* $\mathcal{M}$ — an observation interface (queries into a graph, a stream seen once, a sketch, the syntax of an algebra), a class of permitted procedures, and a cost measure (queries, time, space, edges, update work) — and a task f with a required accuracy. The deliverable is a theorem of the form “no procedure in $\mathcal{M}$ achieves f within resource R ,” and its proof always exhibits an explicit counter-world: the adversary. What varies from paper to paper is only *what all procedures are forced to share* — an identical observation, an identical data-structure state, a bounded growth ceiling, or an output that is simply too large — and that invariant is the

LEARNING OBJECTIVES

- Recognize the four targets of an impossibility result: cost, expressivity, output size, update scope.
- Build the three plants: an instance pair, a hard distribution, a witness object.
- Match the pattern to the target; import adversaries via reductions when direct planting fails.
- Dissect a lower-bound section: model, family, indistinguishability, contradiction.

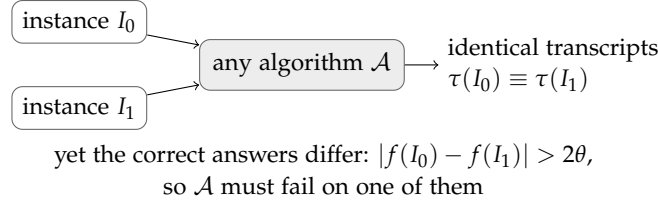


Figure 5.1: The two-instance pattern: the adversary plants instances that every procedure in the model must observe identically, while the demanded answers differ by more than the tolerance. Dynamic settings (Section 5.3) use the same picture with I_0, I_1 two futures of an update stream and τ the data structure’s state.

organizing question of this chapter.

Notation

“Adversary” always means a construction fixed *before* the algorithm’s coins: an instance pair, an input distribution, or an update sequence. A lower bound of $\Omega(g(n))$ is a statement about every procedure in the model; a *conditional* lower bound holds under a stated conjecture. Φ denotes an aspect ratio (largest over smallest nonzero distance), λ a doubling dimension, δ_G the degeneracy of a graph G .

5.0.1 The basic patterns

Every adversarial argument in the corpus is one of five patterns, stated here in their bare forms. The first is the atom from which the others are assembled.

Definition 5.1 (Observation model and transcripts). An *observation model* assigns to each instance I and each (possibly randomized) algorithm $\mathcal{A}$ the *transcript* $\tau_{\mathcal{A}}(I)$: the complete record of what $\mathcal{A}$ observes on I , including its own coins. Instances I_0, I_1 are *indistinguishable*, written $\tau(I_0) \equiv \tau(I_1)$, if for every procedure permitted by the model the distributions of $\tau_{\mathcal{A}}(I_0)$ and $\tau_{\mathcal{A}}(I_1)$ coincide.

Proposition 5.2 — Two-instance lower bound. Let f be a task solved with tolerance θ (the answer must fall within θ of $f(I)$). If there exist instances I_0, I_1 with $\tau(I_0) \equiv \tau(I_1)$ but $|f(I_0) - f(I_1)| > 2\theta$, then no algorithm in the model solves f . ■

Proof. On the shared observation, the algorithm’s output distribution is the same for both instances. Whatever value v it emits with positive probability, v cannot simultaneously lie within θ of $f(I_0)$ and within θ of $f(I_1)$; hence on one of the two instances the algorithm errs with positive probability. ■

1

Randomized algorithms do not escape the pattern — they route it through a distribution. *Yao’s minimax principle* [13] says that derandomizing the *input*, not the algorithm, loses nothing:

Theorem 5.3 — Yao’s minimax principle. For any cost function $\text{cost}(\mathcal{A}, x) \geq 0$,

$$\min_{\text{randomized } \mathcal{R}} \max_x \mathbb{E}[\text{cost}(\mathcal{R}, x)] \geq \max_{\text{distributions } \mathcal{D}} \min_{\text{deterministic } \mathcal{A}} \mathbb{E}_{x \sim \mathcal{D}}[\text{cost}(\mathcal{A}, x)]. \quad \blacksquare$$

Proof. Fix any $\mathcal{D}$ and any randomized $\mathcal{R}$. Averaging over the coins of $\mathcal{R}$ first,

$$\begin{aligned} \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}[\text{cost}(\mathcal{R}, x)] &= \mathbb{E}_{\text{coins}} \mathbb{E}_{x \sim \mathcal{D}}[\text{cost}(\mathcal{A}_{\text{coins}}, x)] \\ &\geq \min_{\mathcal{A}} \mathbb{E}_{x \sim \mathcal{D}}[\text{cost}(\mathcal{A}, x)]. \end{aligned}$$

The left side is at most $\max_x \mathbb{E}[\text{cost}(\mathcal{R}, x)]$; minimize over $\mathcal{R}$, then maximize over $\mathcal{D}$. $\blacksquare$

So against a randomized algorithm one exhibits a *hard distribution* and analyzes the best deterministic procedure against it — which is why so many lower bounds in the corpus are phrased as “draw the input from Σ ; any estimator with T queries fails” (the pattern of Section 5.4).

The next two patterns need no planting at all: the adversary *is* an object.

Proposition 5.4 — Witness counting. If some instance I of size n has an output set $\text{Out}(I)$ with $|\text{Out}(I)| \geq g(n)$, and a correct algorithm must report every element of $\text{Out}(I)$, then every correct algorithm takes $\Omega(g(n))$ time. $\blacksquare$

Proof. The algorithm must write $g(n)$ distinct answers; writing is a lower bound on running time. $\blacksquare$

The classical witness is Moon–Moser’s [10]: a graph on n vertices with $3^{n/3}$ maximal cliques, each of which an enumerator must output. The pattern reappears with a defect budget in Section 5.1, where the witness is generalized to k -defective cliques and certifies an enumeration algorithm as worst-case optimal.

When the object under attack is a *language* rather than a cost, the shared invariant is a growth ceiling: everything expressible in the model is small in some measure, and the target operation is not.

Proposition 5.5 — Growth ceiling. Let $\mu \geq 0$ be a measure on instances such that every expression of the model with description length ℓ , applied to inputs of size n , produces outputs with $\mu \leq n^{O(\ell)} \cdot c^\ell$ for a constant

¹ The tolerance in Proposition 5.2 is where approximation parameters enter: planting instances whose answers differ by just over 2θ is exactly how ε ’s in lower bounds are beaten — the gap must exceed the promised accuracy, not merely be nonzero.

c of the model. If the target operation T has, for every fixed ℓ , an input I with $\mu(T(I)) > n^{O(\ell)} \cdot c^\ell$ (for instance $\mu(T(I)) = 2^{|I|}$), then no expression of the model defines T . ■

Proof. Any candidate defining expression has some fixed length ℓ ; pick n large enough that the demanded μ -value exceeds its ceiling $n^{O(\ell)}c^\ell$, which exists because 2^n outgrows every polynomial in n times a constant. The expression's output is then too small in μ to equal $T(I)$. ■

² The corpus instance: on $\mathbb{N}$ -databases, every expression of basic relational algebra produces multiplicities at most polynomial in the input size times a factor exponential only in the expression's own length, while division demands multiplicities 2^n — so division is inexpressible [2]. Expressivity counterexamples of this kind go back to Codd [3]. No pair of instances is needed; growth rates alone separate.

2

Finally, when no direct planting suffices, the adversary is *imported*: a reduction converts instances of a believed-hard problem into instances of yours.

Proposition 5.6 — Reduction transfer. Suppose instances of a problem H , conjectured to require $T_H(n)$ time, map in $o(T_H(n))$ time to instances of P of size at most $s(n)$. Then any algorithm for P running in $T(n)$ time with $T(s(n)) = o(T_H(n))$ refutes the conjecture; equivalently, P requires $\Omega(T_H(s^{-1}(n)))$ time under the conjecture. ■

Proof. Compose the reduction with the algorithm: an instance of H of size n becomes an instance of P of size $\leq s(n)$, solved in $T(s(n)) + o(T_H(n)) = o(T_H(n))$ time, contradicting the conjecture. ■

The conjecture is usually a fine-grained one — combinatorial k -clique, OuMv [4], SETH — and the reduction is itself an adaptive adversary: it manufactures the update stream, and indistinguishability is inherited from the source problem rather than planted directly. When the source is a communication problem instead, the two-instance pattern arrives prefabricated: index, set-disjointness, and gap-Hamming transcripts are information-blind by construction [8], the route by which sketching lower bounds entered databases with the frequency-moment bound [1].

5.0.2 How to choose the pattern

The selection is mechanical once two questions are answered. *First, what is the resource under attack?* If it is running time or query complexity, ask whether the model restricts observation (a sketch, a query interface, one pass over a stream): it does $\rightarrow$ plant two instances or a hard distribution; it does not (arbitrary computation allowed) $\rightarrow$ import hardness via a reduction, since nothing else stops an unrestricted algorithm. If the resource is the *output*, use a witness object. If it is *expressivity*, use a growth ceiling or a symmetry. If it is *update work* in a dynamic setting, the shared observation is the data structure's state and the instances are futures of the update stream. *Second, is the algorithm randomized?* Then

Yao (Theorem 5.3) converts your task to exhibiting one hard distribution and beating the best deterministic procedure on it.

When to Build an Adversary

Reach whenever the deliverable is a no-better-is-possible statement — certifying an algorithm worst-case optimal, showing an index’s edge budget cannot shrink, separating a language fragment, proving an update operation cannot stay local — or when planning an upper bound and needing to know what the adversary will force. Then run the recipe:

1. Fix the model precisely: what procedures may observe, what they must output, the tolerance θ , the cost measure. Sloppy models make sloppy impossibilities.
2. Identify what *all* procedures must share: a transcript, a state, a growth ceiling, an output set. This shared object is where the adversary lives.
3. Plant the two instances (Proposition 5.2), a hard distribution (Theorem 5.3), a witness object (Proposition 5.4), or a ceiling-breaking instance (Proposition 5.5); if nothing plants, import via a reduction (Proposition 5.6).
4. Tune the parameters so the shared observation is exactly identical (not merely similar) while the demanded answers differ by more than 2θ ; verify no adaptive strategy can break the symmetry.
5. If every two-instance attempt fails, take the hint: the matching upper bound may simply exist. Try to prove it before fighting the lower bound further.

5.1 Defective clique enumeration: the output as adversary

The exemplar is a SIGMOD’26 paper on relaxed clique computation [5]. Exact cliques are famously brittle: one absent edge destroys one, so real communities — friendship groups, protein complexes, fraud rings — are better modeled by *defective* cliques, vertex sets that miss at most k edges for a small budget k . The paper studies enumerating all *maximal* k -defective cliques and searching for large ones, the way a graph engine would when k -hop tolerance matters. Its design question is classical branch-and-bound craft — how to pivot, how to prune — but its headline is a worst-case optimality guarantee: the search tree has at most $O(3^{n/3} \cdot n^k)$ nodes for constant k , and this is optimal because

no algorithm can beat the size of the output it must produce. The adversary here is not a pair of instances at all: it is a single witness graph, generalized from Moon–Moser’s, whose maximal defective cliques are so numerous that enumerating them dominates any analysis. The paper complements the witness with degeneracy-aware bounds that make the algorithm practical on sparse graphs.

Problem. Given a graph $G = (V, E)$ with n vertices and an integer $k \geq 0$, a vertex subset $S \subseteq V$ is a *k-defective clique* if $G[S]$ misses at most k edges, i.e. has at least $\binom{|S|}{2} - k$ edges. S is *maximal* if no superset is k -defective. The task is to enumerate all maximal k -defective cliques; the cost measure is the number of search-tree nodes (and total time) of any branching algorithm, judged against $\max_G |\text{Out}(G)|$, the worst-case output size.

3

³ Every k -defective clique of size q contains a clique of size $q - k$ — delete one endpoint of each missing edge. So the defective budget interpolates between clique enumeration ($k = 0$, Moon–Moser territory) and densest-subgraph-style relaxation.

Guarantee.

Theorem 5.7 — Worst-case optimal enumeration, combining the search-space and output-size theorems of the paper. For constant k , the worst-case output size of maximal k -defective clique enumeration is $\Omega(3^{n/3} \cdot n^k)$: there is a graph on n vertices with that many maximal k -defective cliques. The paper’s branching algorithm enumerates them all with a search tree of $O(3^{n/3} \cdot n^k)$ nodes in $O(m \cdot 3^{n/3} \cdot n^k)$ time, and its degeneracy-aware variant runs in $O(n \cdot 3^{\delta_G/3} \cdot \delta_G^{O(k)})$ time. ■

How the adversary is built. The witness is the Moon–Moser graph M_n : the complete $(n/3)$ -partite graph whose parts have size 3 (equivalently, the complement of $n/3$ disjoint triangles) [10]. The $k = 0$ case is the classical count — one vertex per part gives $3^{n/3}$ maximal cliques — and the defective budget k multiplies the harvest by a polynomial factor:

Lemma 5.8 — Witness count for defective cliques; output lemma of the paper, notation cleaned. For n a multiple of 3 with $n/3 \geq k$, the Moon–Moser graph M_n contains at least $3^{n/3} \cdot \binom{n/3}{k} = \Omega(3^{n/3} \cdot n^k)$ maximal k -defective cliques. ■

Proof sketch. Choose k of the $n/3$ parts; inside each chosen part keep *both* endpoints of one of its 3 vertex pairs (one missing edge), and in every unchosen part keep a single vertex. The resulting set misses exactly k edges — one per chosen part — and is maximal because the only addable vertices are the third vertices of chosen parts, each of which would create a second missing edge in its part. Counting the choices gives $\binom{n/3}{k} \cdot 3^k \cdot 3^{n/3-k}$ distinct maximal k -defective cliques. (Full proof in Section 5.6.) ■

Every maximal k -defective clique must appear in the enumeration, so Proposition 5.4 converts the witness count into the time lower bound of Theorem 5.7 — and the algorithm’s $O(3^{n/3}n^k)$ search tree matches it, with no gap. The upper-bound side then does the delicate work: a pivoting rule for defective cliques (every maximal solution through a partial set must touch the non-neighborhood of a pivot) and a branching-vector induction showing the tree size never exceeds the output bound, which is where the n^k factor is earned rather than assumed.

What it buys. The engine gets a certificate that its enumeration cost is a property of the problem, not of the algorithm: no competitor can be asymptotically faster at worst case. The degeneracy-aware bound turns the same analysis into a practical promise — on sparse real graphs the exponent collapses from $n/3$ to $\delta_G/3$ — and the witness graph itself becomes the natural stress test for any implementation claiming output sensitivity.

5.2 Proximity graphs: point sets that force edges

The exemplar is a SIGMOD’26 paper on proximity graphs for similarity search [9]. Vector-database indexes increasingly build a *proximity graph* over the point set and answer nearest-neighbor queries by greedy navigation: from any start point, repeatedly hop to the out-neighbor closest to the query, and stop when no neighbor is closer. The paper asks the quantitative question behind every such index: how many *edges* must a graph carry so that greedy navigation always lands on a $(1 + \varepsilon)$ -approximate nearest neighbor? It answers both directions. Positively, in d -dimensional Euclidean space a $(1 + \varepsilon)$ -proximity graph with $O((1/\varepsilon)^\lambda \cdot n)$ edges — linear in n — exists and can be built in near-linear time with high probability, where λ is the doubling dimension. Negatively, in general doubling metrics no such sparsity is possible: there are point sets on which *any* $(1 + \varepsilon)$ -proximity graph, no matter how much query time it is allowed, must carry nearly $n \log \Phi$ edges, where Φ is the aspect ratio of the point set. The lower bound is the paper’s separation theorem: it shows the Euclidean assumption is not cosmetic — drop it and the edge budget necessarily blows up by a factor $\log \Phi$.

Problem. Let P be n points in a metric space with aspect ratio Φ (largest over smallest nonzero pairwise distance) and doubling dimension λ . A directed graph G on P is a $(1 + \varepsilon)$ -proximity graph $((1 + \varepsilon)$ -PG) if, for every query point q and every start vertex, greedy navigation — hop to the out-neighbor nearest to q — terminates at a $(1 + \varepsilon)$ -approximate nearest neighbor of q in P . The task is to bound the minimum number

⁴ Doubling dimension λ : every ball of radius $2r$ is coverable by 2^λ balls of radius r ; it is the standard notion of intrinsic dimension for metric indexes. Navigability and $(1 + \varepsilon)$ -PG-hood are equivalent in the paper’s formulation, so the bound is about graphs, not about any particular search procedure.

of edges of a $(1 + \varepsilon)$ -PG over P ; a “2-PG” is the case $(1 + \varepsilon) = 2$.

4

Guarantee.

Theorem 5.9 — Size lower bound, Statement 1 of the paper’s lower-bound theorem, notation cleaned. For any powers of two $n \geq 2$ and Φ with $n^2 \leq \Phi \leq 2^n$, there is a set P of $\Theta(n)$ points with aspect ratio Φ , in a metric space of doubling dimension 1, such that any 2-PG for P must have $\Omega(n \log \Phi)$ edges, regardless of the query time allowed. ■

A companion statement extends the bound to a range of ε : with $n = s^d \cdot t$ points, doubling dimension at most $\log(1 + 2^d)$ and $\varepsilon = 1/(2s)$, any $(1 + \varepsilon)$ -PG needs $\Omega(s^d \cdot n)$ edges — again independent of query time.

How the adversary is built. The adversary is a planted point set in a tree-like metric — doubling dimension 1 is the regime where every ball splits into two half-radius balls, and the construction exploits exactly that binary hierarchy across the $\log \Phi$ distance scales spanned by the aspect ratio. The mechanism is *scale separation*: the two-instance pattern of Proposition 5.2 applied once per distance scale. The witness spans $\log \Phi$ scales, and a navigable graph must support greedy descent through every one of them: at each scale the construction places groups of points whose relevant distances sit at that scale, while an edge that could serve one scale cannot serve another because consecutive scales differ by more than the stretch factor. The necessity counting runs per scale and yields a linear edge count for each; summing over the $\log \Phi$ scales gives $\Omega(n \log \Phi)$ (the paper’s proof develops the counting in detail; we sketch the architecture). Two features make the bound robust. First, it is *non-geometric*: the witness metric is a hierarchy, not a point set in any $\mathbb{R}^d$, which is precisely why it does not contradict the Euclidean upper bound — the separation between general doubling metrics and Euclidean space is the theorem’s point. Second, the counted quantity is a property of the *object* (the edge set of any graph that is navigable at all), not of a procedure’s cost, which is what the “regardless of the query time allowed” clause buys: no amount of walk-length tolerance dodges an edge count. The generalization is honest work, as the paper notes: for ε -dependence (Statement 2), the hard input extends a classical query-time lower-bound witness for approximate nearest-neighbor data structures, but lifting a query-time witness to a *size* witness demands new ideas, since the adversary must now defeat graphs that may be arbitrarily dense in the wrong places.

What it buys. For index builders the theorem is a floor: on skewed-distance data (large aspect ratio), a proximity-graph index with near-linear edges simply does not exist in general metrics, so practitioners either exploit Euclidean structure or pay $\log \Phi$. The upper bound’s linear-in- n Euclidean graph with near-linear construction closes the design space from above, and the pair of results together tells system designers exactly which assumption — doubling dimension plus geometry versus doubling dimension alone — their edge budget depends on.

5.3 *Bi-core maintenance: an update pair that lies in wait*

The exemplar is a SIGMOD’26 paper on maintaining bipartite cores [12]. Bipartite graphs model user–item, author–paper, and customer–product relations, and their *bi-cores* are the standard cohesion notion: the (α, β) -core is the largest subgraph in which every vertex on the left has degree at least β and every vertex on the right at least α . When edges arrive and depart, a system wants to maintain all bi-core memberships without recomputing from scratch. The paper’s first contribution is an asymmetry theorem explaining why this is hard in one direction only: under the model of *locally persistent* algorithms — the model prior maintenance algorithms implicitly operate in, where an update is processed by examining local structure around the affected edge — handling *deletions* is bounded (work proportional to what actually changes), while handling *insertions* is unbounded. Its second contribution is a family of order-based algorithms that exploit the asymmetry: a maintained vertex order with “lack values” gives necessary conditions pruning which vertices can possibly change, taming insertion in practice.

Problem. Let $G = (V_1, V_2, E)$ be a bipartite graph; for a pair (α, β) , the (α, β) -core is the maximal subgraph where every $v \in V_1$ has degree $\geq \beta$ and every $v \in V_2$ has degree $\geq \alpha$; each vertex belongs to a set of *c-pairs*. Under single-edge insertions and deletions, maintain all *c-pairs*. An update algorithm is *locally persistent* (LP) if its processing of an update is confined to the persistent local information around the affected vertices; it is *bounded* if its cost per update is a function of the true change set ΔR alone — in particular $O(1)$ on updates that change no core membership.

5

Guarantee.

Theorem 5.10 — Insertion–deletion asymmetry, Theorems 3.1–3.3 of the paper.

⁵ The LP model abstracts what deployed maintenance algorithms actually do: identify candidate *c-pairs* near the edge, expand a candidate set, and update — never recomputing globally. “Unbounded” then means: there are updates where the algorithm must do $\omega(|\Delta R|)$ work, i.e. where local examination cannot certify that nothing changed.

(i) Given a bipartite graph G and all its bi-cores, updating them after inserting an edge (u, v) is *unbounded* under the model of locally persistent algorithms: some updates force $\omega(1)$ work while the true change set is empty. (ii) Updating the bi-cores after deleting an edge (u, v) is *bounded*, in time $O(|\text{CHANGED}| + |N_1(\text{CHANGED})|)$ for the natural deletion procedure. ■

How the adversary is built. The adversary is the two-instance pattern wearing dynamic clothes (Figure 5.1): the shared observation is the fully processed state of G , and the two instances are its two futures. The construction is distilled in the paper’s workhorse lemma.

Lemma 5.11 — The two-update trap, proof of Theorem 3.1 of the paper, notation cleaned. There is a bipartite graph G with n vertices and two single-edge insertions $\Delta G_1, \Delta G_2$ such that: (i) for each of ΔG_1 and ΔG_2 alone the true change set ΔR is empty; (ii) applied in sequence, $|\Delta R| = \Omega(n)$. Hence any LP algorithm that is bounded spends $O(1)$ on either update alone, yet any algorithm that maintains correctness must spend $\Omega(\ell)$ work across the two updates for a non-constant ℓ — a contradiction. ■

Proof sketch. Build G so that each insertion is individually harmless — neither moves any vertex across a core threshold by itself — but the two jointly push a cascade of $\Omega(n)$ vertices over their thresholds. A bounded LP algorithm must return in $O(1)$ time after the first update (since $\Delta R = \emptyset$ certifies there is nothing to do), leaving its local state unprepared; when the second update arrives, correctness forces the full cascade to be processed, which cannot be charged to either update’s change set. (The paper’s complete construction appears in its extended version.) ■

The trap is exactly Proposition 5.2 with τ = the algorithm’s internal state after processing G : the two futures ΔG_1 and ΔG_2 are locally indistinguishable (each looks like a no-op), so the algorithm must behave identically on both — but the demanded maintenance work differs, and one of the two futures punishes it.

What it buys. The theorem tells core-maintenance systems that insertion cannot be both local and worst-case bounded — explaining the observed bottleneck (long BFS expansions) in prior algorithms, which the paper measures at dominating proportions of update time. The fix it motivates is the order-based design: since boundedness is impossible in general, prune with necessary conditions (an update can cascade only along monotone paths in a maintained order, and only through vertices whose “lack values” hit zero), reducing practical insertion cost to $O(|N_1(\text{AFF})| \cdot |N_2(\text{AFF})|)$ over the actually affected sets.

5.4 Personalized PageRank: hard distributions with hidden splits

The exemplar is a PODS'26 paper on single-source personalized PageRank [6]^{PODS}. Personalized PageRank (PPR) asks where a random surfer who teleports back to a source s with probability α at every step will land: the score $\pi(s, t)$ powers recommendation, ranking, and graph mining over large graphs, and systems from web search to graph databases estimate it millions of times a day. The single-source task estimates the whole vector $\pi(s, \cdot)$ at once, with two standard accuracies — relative error for the coordinates above a threshold δ , or additive error ε — and cost measured in *queries*: the paper works in the arc-centric model, where the algorithm may ask for the k -th in- or out-edge of any node but cannot see the adjacency list wholesale. The paper is the corpus's pairing exhibit in its purest form: it gives an algorithm, DistWalks, answering relative-error queries in $O(\min\{\log n/\delta, m + n \log n \log(m\delta/n)\})$ expected queries — the first bound linear in m for dense graphs — and then proves matching lower bounds showing that, up to logarithmic factors, nothing better exists in this model. Both directions rest on the same adversary: a family of graphs whose useful information hides in a random classification that a sublinear number of queries cannot recover.

Problem. Fix a directed graph G (multigraphs allowed), source s , decay α . The PPR value $\pi(s, t)$ is the probability that an α -decay random walk from s (it stops at each step with probability α) ends at t . The SSPPR-R task: given error $c \in (0, \frac{1}{2})$, failure probability p_f , and threshold δ , output estimates $\hat{\pi}(s, \cdot)$ such that with probability at least $1 - p_f$, $|\hat{\pi}(s, t) - \pi(s, t)| \leq c \cdot \pi(s, t)$ for every t with $\pi(s, t) \geq \delta$. The algorithm accesses G through arc-centric queries (the k -th in- or out-edge of a named node); the cost is the expected number of queries. The lower bound may choose any node count n_0 , edge count $m_0 \in [n_0, O(n_0^2)]$, and threshold $\delta_0 \in (0, 1)$.

6

Guarantee.

Theorem 5.12 — Query lower bound for SSPPR-R, Theorem 1.6 of the paper, notation cleaned; upper bound Theorem 1.4 of the paper. For every n_0 , every $\delta_0 \in (0, 1)$ and every $m_0 \in \Omega(n_0) \cap O(n_0^2)$, any algorithm solving SSPPR-R with constant error c and constant failure probability p_f requires $\Omega(\min\{m_0, \log(1/\delta_0)/\delta_0\})$ expected queries on graphs with n_0 nodes and m_0 edges and threshold δ_0 . DistWalks answers in $O(\min\{\log n/\delta, m + n \log n \log(m\delta/n)\})$ expected queries — match-

⁶ DistWalks itself is a concentration showcase — Monte-Carlo discovery with Chernoff bounds and martingale error control — but its *optimality* is an adversarial construction; the two tool families meet in this paper, and Chapter 2 develops the former. Here we dissect the latter.

ing up to logarithmic factors. ■

Two sibling bounds complete the picture: additive-error estimation requires $\Omega(\min\{1/\varepsilon_0^2, m_0\})$ queries, and the single-target version $\Omega(\min\{m_0, n_0 \log n_0/\delta_0\})$ — the latter by reversing every edge of the hard family, since an arc-centric algorithm cannot tell a graph from its reverse.

How the adversary is applied. The construction runs Yao’s principle (Theorem 5.3) through a *hidden split*. The template graph $U(n, D, d)$ has a source s feeding a middle layer Y that points into a target layer X split into two equal classes X_1, X_2 : by the template, every node of X_1 receives D of its in-edges from the Y -layer, every node of X_2 only $D - d$. The PPR values separate exactly at the resolution the task demands — a calculation gives $\pi(s, x_1) = \frac{1-\alpha}{2} \cdot \frac{D}{(2D-d)n}$ and $\pi(s, x_2) = \frac{1-\alpha}{2} \cdot \frac{D-d}{(2D-d)n}$, a ratio $\frac{D-d}{D}$ — so an estimate within relative error c classifies every node of X : whoever can estimate PPR can recover the hidden split. Everything else about the graph is drawn uniformly at random: the distribution $\Sigma(n, D, d)$ puts a uniform prior on the split and uniform bijections on all half-edge connections, and an *r-padding* (extra edges into dummy node sets) lets the family be made dense without revealing edges, so the bound covers sparse and dense regimes alike. The argument then has two moves.

Move 1 (reduction): an algorithm using $o(\min\{m_0, \log(1/\delta_0)/\delta_0\})$ queries yields, after post-processing its PPR estimates, a split-recovery procedure using at most γnD arc-centric queries with average success at least $1 - 2p_f$ (Lemma 4.4 of the paper). The parameters satisfy $n(r + D) = \Theta(\min\{m_0, \log(1/\delta_0)/\delta_0\})$, so a sublinear query budget means $o(nD)$.

Move 2 (indistinguishability): the split cannot be recovered from $o(nD)$ queries — the workhorse lemma.

Lemma 5.13 — Switched-split indistinguishability; Lemma 4.6 of the paper, notation cleaned. For a sufficiently small constant γ , after any $T \leq \gamma nD$ arc-centric query answers about $U \sim \Sigma(n, D, d)$ there are $\Omega(n^2)$ *switched splits* — splits obtained from the true one by trading the classes of one node of X_1 and one node of X_2 — whose likelihood given the answers is within a factor $n^{1-\varepsilon}$ (of a constant $\varepsilon > 0$) of the true split’s likelihood. Consequently every split has posterior probability less than $1/n$: no procedure outputs the true split with useful probability. ■

Proof sketch. Each revealed edge adjacent to a switched node shifts the conditional density by at most $1 + O(d/D)$ (the X_1 versus X_2 in-degree discrepancy); all other edges contribute an exact factor 1 by the symmetry of the uniform bijections. Few nodes have many revealed edges

(“traversal is hard”), so the product over a switched pair’s revealed edges is at most $(1 + O(d/D))^{O(\gamma D)} \leq e^{O(\gamma d)} \leq n^{1-\varepsilon}$ using $d \leq \log n$. With $\Omega(n^2)$ switched splits each nearly as likely as the truth and a uniform prior, Bayes crushes the posterior of any single split below $1/n$. (Full proof in Section 5.6.) ■

Moves 1 and 2 contradict for large n ($1 - 2p_f$ versus $1/n$), so the assumed fast SSPPR-R algorithm cannot exist. The two “hard” intuitions the paper names — *traversal is hard* (sublinear queries leave most nodes’ in-edges almost entirely unseen) and *random walk is hard* (what is unseen is uniformly random, so it carries no class information) — are exactly the two-instance pattern sharpened into a likelihood ratio.

What it buys. The pairing certifies DistWalks as the end of the line for PPR query complexity in the arc-centric model, up to logarithmic factors — a systems-relevant answer, since it tells engines that the observed cost of PPR service is intrinsic, not an artifact of any implementation. The construction pattern transfers broadly: any system that must estimate many values from partially revealed graph structure (adjacency-list services, crawl-based analysis, API-rate-limited settings) inherits the same hidden-classification adversary.

5.5 *Advanced usage and further reading*

Everything so far plants instances that are *exactly* indistinguishable, counts objects, or imports conjectures. Two advanced regimes relax the pattern in different directions: the indistinguishability becomes *computational* (assumed, not exact), and the adversary attacks a *property* of a design rather than a cost. Each is sketched through one corpus exemplar.

5.5.1 *Computational adversaries: indistinguishability by assumption*

Proposition 5.2 needs transcript distributions that coincide exactly. The cryptographic weakening asks only that no *efficient* procedure tell the instances apart — and then the two-instance pattern survives with efficient adversaries.

Proposition 5.14 — Computational two-instance lower bound. If instances I_0, I_1 are indistinguishable to all polynomial-time procedures (no efficient test separates them with non-negligible advantage) but $f(I_0)$ and $f(I_1)$ differ beyond the tolerance, then no efficient algorithm computes f : an efficient correct algorithm would itself be a distinguisher. ■

Proof. Run the supposed algorithm for f on both instances; its answers, which must be accurate, separate I_0 from I_1 — contradicting indistinguishability. ■

Undetectable DP backdoors [11]. The exemplar is a SIGMOD’26 security paper asking whether a malicious vendor can backdoor a differentially private system without anyone noticing. The *Gaussian pancake mechanism* (GPM) replaces the Gaussian mechanism’s isotropic noise by noise that is Gaussian except for periodic density spikes — “pancakes” — concentrated along a secret direction w : noise drawn from the hCLWE distribution. To any polynomial-time auditor without the key w , GPM’s outputs are indistinguishable from an honest Gaussian mechanism, because distinguishing hCLWE from Gaussian is hard under lattice assumptions (SIVP or GapSVP versus quantum computation); that is Proposition 5.14 with I_0, I_1 the two noise distributions, and the reduction to hCLWE is the imported adversary. Yet with the key, the mechanism is grossly non-private: privacy loss $\epsilon \in \Theta(1/\beta^2)$ — exceeding 10^4 in practical regimes — because neighboring databases shift the pancake alignment along w , and the density ratio between aligned and misaligned peaks is enormous. The paper is the two-instance pattern *weaponized*: the adversary constructs the indistinguishable pair and profits from the gap only it can see. Its defensive value is symmetric — knowing the construction tells auditors what assumptions they must break or refuse to trust.

5.5.2 Adversaries against properties, not costs

A lower bound certifies optimality; a *counterexample wreck* certifies that a design principle is unachievable. The smallest adversaries in the corpus are single small graphs that kill a hoped-for structural property of an objective — monotonicity, submodularity, the absence of free riders — and thereby redirect algorithm design. The exemplar is the SIGMOD’26 community-search paper LMSC [7]: its objective, local sketch modularity $\text{LSM}(G, C, \tau) = l_C / (o_C \cdot |C|^\tau)$ (internal over external weight with a size penalty), is shown by explicit clique constructions to be *non-monotone* and *non-submodular* — so no greedy algorithm carries approximation guarantees, and the paper’s chain-based relaxation is the response. The same paper’s NP-hardness (a Set-Cover reduction) and its free-rider threshold — LSM provably resists query-independent free riders once $\tau > \log_{1+1/h}(h(h-1)/2)$ for size bound h — show the two regimes coexisting: an imported adversary for the complexity statement, a witness graph of a handful of vertices for the property statements. When your target is a property rather than a cost, the recipe of Section 5.0.2 shrinks to: write the property as an inequality between

two small instances, and search for the smallest graph that violates it.

5.5.3 Further reading

- Yao [13] — the minimax principle turning randomized lower bounds into hard distributions; one page, endlessly reused.
- Kushilevitz and Nisan [8] — the two-player adversary catalog (index, set-disjointness, gap-Hamming) from which communication-complexity lower bounds are imported.
- Alon, Matias, and Szegedy [1] — where sketching lower bounds began; the two-instance pattern in stream form.
- Moon and Moser [10] — the $3^{n/3}$ witness and its descendants in enumeration optimality.
- Henzinger, Krinninger, Nanongkai, and Saranurak [4] — the OuMv conjecture and the reduction-transfer pattern for dynamic problems.
- Codd [3] — the original expressivity question; counterexamples measuring relational languages ever since.

Chapter Recap

- **What the tool is.** Adversarial constructions prove that no better promise exists: a counter-world (instance pair, hard distribution, witness object, ceiling-breaking input, or imported reduction) in which every procedure permitted by the model fails or pays.
- **Which pattern when.** Observation-restricted model → plant two instances with identical transcripts (Proposition 5.2); randomized algorithms → harden into a distribution via Yao (Theorem 5.3); cost is the output itself → witness counting (Proposition 5.4); target is expressivity → growth ceiling (Proposition 5.5); unrestricted computation → import a conjecture (Proposition 5.6); dynamic setting → two futures of an update stream sharing a data-structure state.
- **Why it works.** Everything the algorithm does is a function of what it observes; two worlds with identical observations but different demands break every algorithm at once. The craft is making “identical” exact — or computational (Proposition 5.14) — while the demanded answers differ beyond tolerance.
- **What the corpus showed.** Defective-clique enumeration (Section 5.1): a Moon–Moser witness generalized by a defect budget, certifying worst-case optimality. Proximity graphs (Section 5.2): a tree-like point set forcing $\Omega(n \log \Phi)$ edges regardless of query

time, separating Euclidean from general doubling metrics. Bi-core maintenance (Section 5.3): a two-update trap making insertion unbounded for locally persistent algorithms while deletion stays bounded. Personalized PageRank (Section 5.4): a hidden split under a uniform distribution, with likelihood ratios over switched splits matching the upper bound up to logarithms — the corpus’s signature upper/lower pairing.

- **Advanced usage.** Computational indistinguishability turns the two-instance pattern into an assumption-backed weapon (the DP backdoor of Section 5.5.1); counterexample wrecks attack properties of designs rather than costs (Section 5.5.2). Deferred proofs live in the Appendix (Section 5.6).

5.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Lemma 5.8 (Moon–Moser witness count)

Statement (recalled). For n a multiple of 3 with $n/3 \geq k$, the Moon–Moser graph M_n (complete $(n/3)$ -partite graph, parts of size 3) contains at least $3^{n/3} \binom{n/3}{k} = \Omega(3^{n/3} n^k)$ maximal k -defective cliques. ■

Proof. Write $T_1, \dots, T_{n/3}$ for the parts of M_n : vertices in different parts are adjacent; the 3 pairs inside a part are non-edges. A set $S \subseteq V$ is k -defective exactly when at most k of the non-edges of M_n lie inside S . Construct S as follows: choose a set K of k parts ($\binom{n/3}{k}$ choices); in each part of K keep both endpoints of one of its 3 vertex pairs (3 choices each, the kept pair being one missing edge); in each of the remaining $n/3 - k$ parts keep one vertex (3 choices each). Then S contains exactly the k non-edges contributed by the parts of K and no others, so $G[S]$ misses exactly k edges: S is k -defective. It is maximal: the only vertices outside S are the third vertices of parts in K , and adding any such vertex v creates a second non-edge inside v ’s part (v is non-adjacent to both kept vertices of that part), so $S \cup \{v\}$ misses $k + 1$ edges and is not k -defective. Distinct choices yield distinct sets S (the parts of K are exactly those contributing two vertices), so the number of maximal k -defective cliques is at least

$$\binom{n/3}{k} \cdot 3^k \cdot 3^{n/3-k} = \binom{n/3}{k} \cdot 3^{n/3} = \Omega(3^{n/3} \cdot n^k),$$

where the last step uses $\binom{n/3}{k} = \Omega(n^k)$ for constant k . Since every maximal k -defective clique of M_n must be output, Proposition 5.4 converts the count into the enumeration time lower bound of Theorem 5.7. ■

Proof of Lemma 5.13 (switched-split bound)

Statement (recalled). For small enough constant γ , after any $T \leq \gamma n D$ arc-centric query answers about $U \sim \Sigma(n, D, d)$ there are $\Omega(n^2)$ switched splits with likelihood within a factor $n^{1-\varepsilon}$ of the true split's, and every split has posterior probability below $1/n$. ■

Proof. Work in the template family of Section 5.4: nodes of X_1 receive D in-edges from the Y -layer, nodes of X_2 receive $D - d$; the split and all half-edge bijections are uniform. Let $e_1, \dots, e_T$, $T \leq \gamma n D$, be the revealed edges and SP the true split. *Step 1 (traversal is hard).* Each revealed edge is incident to at most one node of X , so the T edges touch at most $\gamma n D$ in-edge slots of X ; by averaging, at least half the nodes of X are incident to at most $2\gamma D$ revealed edges, and because the split is uniform and independent of what queries reveal, both X_1 and X_2 contribute $\Omega(n)$ such low-incidence nodes: fix $X'_1 \subseteq X_1$, $X'_2 \subseteq X_2$ with $|X'_1|, |X'_2| \geq cn$ and every $x \in X'_1 \cup X'_2$ incident to at most $2\gamma D$ revealed edges. *Step 2 (likelihood ratio per switched pair).* Fix $x_{i_1} \in X'_1$, $x_{i_2} \in X'_2$ and let SP' be the switched split. By Bayes with a uniform prior and the chain rule,

$$\frac{\Pr[e_1, \dots, e_T \mid SP]}{\Pr[e_1, \dots, e_T \mid SP']} = \prod_{t=1}^T \frac{\Pr[e_t \mid SP, e_1, \dots, e_{t-1}]}{\Pr[e_t \mid SP', e_1, \dots, e_{t-1}]}.$$

For a revealed edge e_t not adjacent to x_{i_1} or x_{i_2} , the conditional laws under the two splits coincide — the uniform bijections are symmetric in the classes — and the factor is exactly 1. For an edge adjacent to a switched node, the discrepancy between the X_1 in-degree D and the X_2 in-degree $D - d$ changes the conditional density of seeing that particular edge by at most $\frac{D}{D-d} = 1 + O(d/D)$. At most $2 \cdot 2\gamma D$ factors differ from 1, so

$$\frac{\Pr[e_1, \dots, e_T \mid SP]}{\Pr[e_1, \dots, e_T \mid SP']} \leq \left(1 + O\left(\frac{d}{D}\right)\right)^{O(\gamma D)} \leq e^{O(\gamma d)} \leq n^{1-\varepsilon},$$

using $d \leq \log n$ and choosing γ a sufficiently small constant (with $\varepsilon \in (0, 1)$ a constant not depending on n). *Step 3 (posterior crash).* The $|X'_1||X'_2| = \Omega(n^2)$ switched splits are distinct and uniform under the prior, so Bayes gives

$$\begin{aligned} \Pr[SP \mid e_1, \dots, e_T] &= \frac{\Pr[e_1, \dots, e_T \mid SP]}{\sum_{SP''} \Pr[e_1, \dots, e_T \mid SP'']} \\ &\leq \frac{\Pr[e_1, \dots, e_T \mid SP]}{\Omega(n^2) \cdot \Pr[e_1, \dots, e_T \mid SP] / n^{1-\varepsilon}} < \frac{1}{n}. \end{aligned}$$

The same bound applies to every candidate split — each is the “true” split of some draw, or dominates an analogous family of switched alternatives — so any decoder’s success probability is below $1/n$. (Constants follow the paper’s appendix; the multigraph templates are lifted

to simple graphs by the paper’s 2-lift, which the arc-centric interface cannot distinguish.) ■

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6

Differential Privacy: Composition and its Calculus

A data system rarely releases one number. A private analytics engine answers many queries; an explanation tool selects candidates, then scores them, then publishes histograms; a join algorithm's *access pattern* is itself a leak. Differential privacy makes each single release safe, but the contract the auditor signs is end-to-end: everything the system ever emits must jointly satisfy one indistinguishability guarantee. The *composition calculus* is the small algebra that converts per-release budgets into that single contract: budgets of releases on the same data add (sequential composition); releases on disjoint slices cost only the largest budget (parallel composition); anything computed *after* a private release is free (post-processing immunity); and a guarantee against one-record changes degrades linearly when the adversary's inputs differ by a whole group (group privacy). Around this core sits the calibration rule that decides what a single release costs in the first place: sensitivity + calibrated noise, the Laplace and exponential mechanisms.

In the SIGMOD'26 corpus the tool is close to a monoculture: among the 26 papers tagged *privacy-dp*, 18 assemble their end-to-end privacy guarantee by composing mechanisms (69%), and the pairing *privacy-guarantee* $\times$ *dp-composition* is the tool's strongest, with 18 papers. This chapter builds the ladder and walks four full exemplars through it; Section 6.5 sketches two harder uses (oblivious joins, private shortest paths) and points to the literature.

6.0 The tool: DP composition

The problem class is uniform: a mechanism $\mathcal{M}$ must publish several interacting statistics of a dataset D — selections, counts, noisy parameters, even the sequence of memory cells it touches — and certify one contract over the *joint* output. The contract is indistinguishability of neighboring inputs:

LEARNING OBJECTIVES

- Recognize when a system's privacy proof is a *ledger*: several releases, each with its own budget, that must add up.
- Prove the ladder sequential $\rightarrow$ parallel $\rightarrow$ post-processing $\rightarrow$ group privacy, and match each rung to the data-access structure.
- Calibrate every release: compute the sensitivity, split the budget, then pick Laplace or exponential-mechanism noise.
- Dissect a paper's end-to-end proof: which rung carries each stage, and where budgets add rather than maximize.

Definition 6.1 (Differential privacy). A randomized mechanism $\mathcal{M}$ satisfies (ϵ, δ) -differential privacy if for all *neighboring* inputs $D \sim D'$ and all measurable sets S of outputs,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta.$$

With $\delta = 0$ the guarantee is *pure* ϵ -DP [4, 5].

Notation

Neighboring means “one unit apart” in whichever granularity the contract names: one tuple added or removed (the relational default; the *unbounded* setting, versus bounded when a size cap is public), one edge (edge-DP), or one node with all its incident edges (node-DP) — Section 6.3 lives entirely in the gap between the last two. Typical calibrations keep ϵ a constant and $\delta \leq 1/N$ for N data subjects. $\text{Lap}(b)$ is the Laplace distribution of scale b , and Lap noise is added independently per coordinate.

What a single release costs is governed by how much one unit of input change can move it:

Definition 6.2 (Global sensitivity). For a query Q on datasets, the *global sensitivity* is

$$GS_Q = \max_{D \sim D'} |Q(D) - Q(D')|,$$

the maximum over neighboring pairs, in the norm of the output space.

Counting queries have $GS_Q = 1$; a sum over values in $[0, U]$ has $GS_Q = U$; a histogram has sensitivity 1 in ℓ_1 . Two mechanisms convert sensitivities into releases:

Theorem 6.3 — Laplace mechanism. Let Q have global sensitivity GS_Q . Then $\mathcal{M}(D) = Q(D) + \text{Lap}(GS_Q/\epsilon)$ (i.i.d. noise per coordinate) satisfies ϵ -DP. ■

Proof sketch. For any output y , the ratio of densities at y under the two neighbors is at most $\exp(|Q(D) - Q(D')|/b) = e^{GS_Q/b} = e^\epsilon$ with $b = GS_Q/\epsilon$; integrating over $y \in S$ preserves the bound (full proof in Section 6.6). ■

For selecting a discrete object (an attribute, a plan, a candidate set), noise per coordinate is replaced by a biased lottery:

Theorem 6.4 — Exponential mechanism. Given candidates $\mathcal{R}$ and a quality function $q(D, r)$ of sensitivity Δ_q , the mechanism selecting $r \in \mathcal{R}$ with probability proportional to $\exp(\epsilon \cdot q(D, r) / (2\Delta_q))$ satisfies ϵ -DP. Moreover the selected r_E is nearly optimal:

$$\Pr[q(D, r_E) \leq \max_{r \in \mathcal{R}} q(D, r) - 2\Delta_q (\ln |\mathcal{R}| + t)] \leq e^{-t}. \quad \blacksquare$$

Proof sketch. Each candidate's selection-probability ratio between neighbors is at most e^ϵ because q moves by at most Δ_q and the exponent normalizes by $2\Delta_q$; the utility clause integrates the tail of the same lottery. (Statement as restated in [7], which cites the original.) ■

These are the coins; the calculus is how coins combine. The workhorse is sequential composition — note that it covers *adaptive* pipelines, where later mechanisms see earlier outputs:

Theorem 6.5 — Sequential composition. Let $\mathcal{M}_1: \mathcal{D} \rightarrow \mathcal{R}_1$ satisfy (ϵ_1, δ_1) -DP and let $\mathcal{M}_2: \mathcal{D} \times \mathcal{R}_1 \rightarrow \mathcal{R}_2$ satisfy (ϵ_2, δ_2) -DP in its first input for *every* fixed value of the second. Then $\mathcal{M}_3(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$ satisfies $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$ -DP; iterating, k adaptive releases satisfy $(\sum_i \epsilon_i, \sum_i \delta_i)$ -DP. ■

Proof sketch. Condition on $\mathcal{M}_1$'s output y : each conditional law obeys its own bound, and integrating over y multiplies the e^{ϵ_i} factors and adds the δ_i 's (full proof in Section 6.6). ■

When releases touch *disjoint* parts of the data, the arithmetic improves from sums to maxima:

Theorem 6.6 — Parallel composition. Let $\mathcal{M}_1, \mathcal{M}_2$ satisfy (ϵ_1, δ_1) - and (ϵ_2, δ_2) -DP, and let D_1, D_2 be disjoint subsets of the input domain. Then $\mathcal{M}'(D) = (\mathcal{M}_1(D_{D_1}), \mathcal{M}_2(D_{D_2}))$ — each mechanism seeing only the records of its slice — satisfies $(\max\{\epsilon_1, \epsilon_2\}, \max\{\delta_1, \delta_2\})$ -DP. ■

Proof. A unit input change lands in exactly one slice, say D_1 ; the other mechanism's output distribution is then identical under both neighbors, so only the ϵ_1 bound survives the product-measure factorization. ■

Theorem 6.5 charges for every release; the next rule charges for none:

Theorem 6.7 — Post-processing immunity. If $\mathcal{M}$ satisfies (ϵ, δ) -DP and g is any (deterministic or randomized) function that does not touch the raw data, then $g(\mathcal{M}(D))$ satisfies (ϵ, δ) -DP. ■

Proof. Disintegrate g against the output law: $\Pr[g(\mathcal{M}(D)) \in T] = \int \Pr[g(y) \in T] d\mathcal{M}(D)(y) \leq e^\epsilon \int \Pr[g(y) \in T] d\mathcal{M}(D')(y) + \delta$. ■

¹

When the two inputs differ by a whole group of λ units, the guarantee degrades linearly:

Theorem 6.8 — Group privacy. If $\mathcal{M}$ satisfies (ϵ, δ) -DP, then for any two inputs G, G' at neighborhood distance λ and any output set S ,

$$\Pr[\mathcal{M}(G) \in S] \leq e^{\lambda\epsilon} \Pr[\mathcal{M}(G') \in S] + \lambda\delta. \quad \blacksquare$$

¹ Post-processing immunity is what licenses *system artifacts* as private objects: clipped graphs (Section 6.3), pruned indices, and even access patterns of algorithms whose memory layout is a function of a private release (Section 6.5). No cleverness *after* the release can repair a broken release, either.

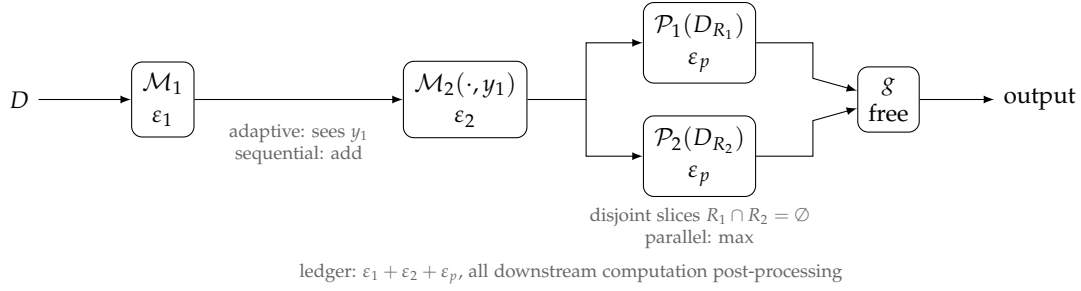


Figure 6.1: The composition ledger. Sequential releases (left) add budgets even when adaptive; releases on disjoint data slices (middle) cost only the maximum budget; everything after a private output (right) is post-processing and costs nothing.

Proof sketch. Insert λ intermediates $G = G_0 \sim G_1 \sim \dots \sim G_\lambda = G'$ and telescope the DP bound along the chain; the δ terms accumulate geometrically and are absorbed by $\lambda e^{(\lambda-1)\epsilon} \delta \geq \lambda \delta$ (Section 6.6). ■

Plain summation is not the end of the story: k adaptive releases can be accounted far better than $k\epsilon$, because the privacy-loss random variables concentrate (Chapter 2):

Theorem 6.9 — Advanced composition. For $\epsilon, \delta \in (0, 1)$ and $\delta' \in (0, 1)$, the k -fold adaptive composition of (ϵ, δ) -DP mechanisms satisfies $(\epsilon', k\delta + \delta')$ -DP for

$$\epsilon' = \sqrt{2k \ln(1/\delta')} \cdot (e^\epsilon - 1) \leq 2\epsilon \sqrt{2k \ln(1/\delta')}. \quad \blacksquare$$

Proof sketch. Write the log-likelihood ratio of outputs along the adaptive chain; each increment is bounded and (near-)centered, so Azuma-Hoeffding concentrates the total loss at $O(\epsilon\sqrt{k})$ with failure δ' [6]. Rényi DP composes the same loss in moments and converts back [3, 10]; subsampling buys a further discount [1]. ■

6.0.1 How to compose a privacy proof

Composing an end-to-end proof is mechanical once the system is drawn as the DAG of Figure 6.1. *First*, enumerate every channel that touches the raw data: released statistics, private hyperparameters that steer later computation, and — in oblivious settings — access patterns. *Second*, fix the neighborhood each channel is analyzed under and compute its query’s sensitivity; a wrong neighborhood (edge when the contract says node) silently breaks the ledger. *Third*, instantiate each channel with the mechanism matching its output type: Laplace for numeric aggregates, the exponential mechanism for selections, the sparse vector technique for threshold searches (Section 6.3). *Fourth*, read off the total: sums along adaptive chains, maxima across disjoint slices, linear inflation

across groups; post-processing edges carry zero. Budgets are *split*, not discovered: fix the total ϵ first and allocate shares so no single stage dominates the error.

When to Reach for DP Composition

Reach whenever a system releases *several* interacting statistics — candidate selections, noisy summaries, clipped releases, adaptive searches — and the deliverable is one end-to-end privacy contract. Then run the recipe:

1. Identify the releases: every mechanism that reads raw data, including private parameters that steer later stages and observable access patterns.
2. Split the budget: fix the target (ϵ, δ) , allocate (ϵ_i, δ_i) per release; spend less on stages whose error matters less.
3. Compute each query’s sensitivity under the neighborhood the contract names (tuple/edge/node; add/remove); calibrate Laplace or exponential-mechanism noise to it.
4. Compose: add along adaptive chains, take the maximum over disjoint data slices, multiply by group size when converting between neighborhoods; union-bound the failure probabilities over stages.
5. Post-process freely: clipping, rounding, index building, and all downstream analytics ride on the released values at zero cost — but never re-touch the raw data.

Skip the machinery when a single calibrated release suffices: composing there only overcounts. With many ($k \gtrsim 50$) releases, plain summation goes vacuous — switch the ledger to advanced or moment accounting (Theorem 6.9).

6.1 Private explanations for clusters

The exemplar is a SIGMOD’26 paper on explainable clustering under privacy [7]. Clustering is a workhorse of exploratory data analysis, but a bare cluster label tells a stakeholder nothing: analysts want an *explanation* — which attribute combination accounts for the partition, e.g. that customers grouped together because of age bracket and spending band. Computing such explanations requires scanning the tuples inside each cluster, and the explanation itself (“cluster 3 is explained by salary”) leaks information about those individuals, so it must be privatized like any other release. Prior private explanation tools could

explain one cluster at a time but not certify a *global* assignment — one attribute per cluster, jointly chosen — under a single privacy budget. The paper builds the first mechanism for such global histogram-based explanations (HBE), combining top- k candidate selection, an exponential mechanism over combinations, and noisy per-cluster histograms, and proves an end-to-end budget that is exactly the sum of three named dials. By direct sensitivity analysis it also shows that the standard divergence scores used in explanation systems are *high-sensitivity*, and replaces them with low-sensitivity surrogates so each dial can be calibrated tightly.

Problem. A dataset D and a clustering function f partitioning the tuples into clusters D_c , $c \in C$, are given; f is already public. A *single-cluster HBE candidate* is a tuple $(c, A, h_A(D \setminus D_c), h_A(D_c))$ pairing cluster c with an explanatory attribute A and two histograms; a *global HBE candidate* assigns one such candidate per cluster. Quality is scored by an *interestingness* (how skewed A 's distribution is inside c relative to the rest of D), a *sufficiency*, and a *diversity* score (how distinct the chosen attributes are across clusters), combined by convex weights λ (summing to one) into a global score $\psi_\lambda(D, f, \mathbf{e})$. The mechanism must choose a global candidate and publish supporting histograms under pure DP, neighbors $D \sim D'$ differing by one tuple.

Guarantee.

Theorem 6.10 — Privacy of the global explanation, Theorem of the paper. Given a clustering function f , a number of candidates k , and privacy parameters $\epsilon_{\text{CandSet}}, \epsilon_{\text{TopComb}}, \epsilon_{\text{Hist}}$, the algorithm GEN_GLOBAL_EXPLANATION is $(\epsilon_{\text{CandSet}} + \epsilon_{\text{TopComb}} + \epsilon_{\text{Hist}})$ -DP. ■

How DP composition is applied. The proof is the ledger of Figure 6.1 instantiated stage by stage. The top- k candidate selection is realized as $|C| \cdot k$ iterated exponential mechanisms, each at budget $\epsilon_{\text{CandSet}}/(|C|k)$; sequential composition (Theorem 6.5) totals $\epsilon_{\text{CandSet}}$. The combination selection is one exponential mechanism (Theorem 6.4) at unit sensitivity, costing $\epsilon_{\text{TopComb}}$. The histogram stage splits ϵ_{Hist} in half: per-attribute histograms all read the full dataset, so their budgets add sequentially to $\epsilon_{\text{Hist}}/2$; per-cluster histograms read *disjoint* clusters, so parallel composition (Theorem 6.6) caps their total at $\epsilon_{\text{Hist}}/2$ — maxima, not sums. The complementary histograms $\tilde{h}^{-c}$ are computed from already-released values, so post-processing immunity (Theorem 6.7) makes them free; summing the three stages gives the theorem. The unit-sensitivity input to the exponential mechanism is itself the paper's workhorse lemma:

Lemma 6.11 — Convex combinations of scores, Lemma of the paper. For $i = 1, \dots, m$ let $\alpha_i \geq 0$ be scalars and f_i functions of sensitivity Δ_i . Then $f = \sum_{i=1}^m \alpha_i f_i$ has sensitivity at most $\sum_{i=1}^m \alpha_i \Delta_i$. ■

Proof. For neighboring $D \sim D'$, the triangle inequality gives

$$\begin{aligned} |f(D) - f(D')| &= \left| \sum_i \alpha_i f_i(D) - \sum_i \alpha_i f_i(D') \right| \\ &\leq \sum_i \alpha_i |f_i(D) - f_i(D')| \leq \sum_i \alpha_i \Delta_i. \end{aligned}$$

■

Since ψ_λ is a convex combination (weights summing to one) of three sensitivity-1 scores, Lemma 6.11 bounds its sensitivity by 1, and the exponential mechanism runs at exactly one unit. The sensitivity-1 bounds are direct ℓ_1 histogram algebra: for the interestingness score $\iota(D, f, c, A) = \frac{1}{2} \|h_A(D_c) - \frac{|D_c|}{|D|} h_A(D)\|_1$, adding one tuple adds a unit basis vector h_a ($\|h_a\|_1 = 1$) to one histogram and rescales the other by $\|h_A(D)\|_1 = |D|$; a two-case analysis (tuple inside vs. outside the cluster) totals at most 1 in both cases. The replacement is justified by proving the *standard* scores cannot do better:

Proposition 6.12 — Jensen–Shannon sensitivity, Proposition of the paper, notation cleaned. The Jensen–Shannon distance d_{JS} , used as a diversity score on neighboring datasets, has sensitivity at least $\frac{1}{2}$: there are neighbors $D \sim D'$ with $|d_{JS}(D) - d_{JS}(D')| > \frac{1}{2}$. ■

Proof sketch. Take D where every tuple carries value a on attribute A and a singleton cluster $D_c = \{t\}$; both distributions are point masses, so the score is 0. Add one tuple t' with a fresh value a' : the dataset distribution becomes $(\frac{n}{n+1}, \frac{1}{n+1})$ and the cluster's becomes $(\frac{1}{2}, \frac{1}{2})$. The Jensen–Shannon divergence works out to $H_b(\frac{n}{2(n+1)} + \frac{1}{4}) - \frac{1}{2} H_b(\frac{1}{n+1}) - \frac{1}{2} H_b(\frac{1}{2}) \rightarrow H_b(\frac{1}{4}) - \frac{1}{2} \approx 0.311$ bits, so the distance exceeds $\sqrt{0.311} > \frac{1}{2}$ for large n (full computation in Section 6.6). ■

The same binary-entropy limit holds for the total-variation interestingness of prior work, so no divergence-based score can shrink the noise — the sensitivity-1 surrogates are within a factor of two of optimal *and* admit the clean algebra above.

What it buys. The deployed system gets a private explanation engine whose contract is three readable dials — candidates, combination, histograms — each independently tunable; parallel composition across disjoint clusters means adding clusters costs no histogram budget; and the exponential mechanism's utility theorem (Theorem 6.4) converts the unit sensitivity directly into near-optimal explanation quality.

6.2 Optimal pure-DP sparse histograms in deterministic linear time

The exemplar is a PODS’26 paper on private histogram release [9]^{PODS}. Releasing a histogram — how many users hold each value — is the most basic private analytics primitive, and it is delicate when the domain $[d]$ is enormous but any real dataset touches few values: the mechanism must publish a *sparse* histogram whose support tracks the data, not the domain. Two older constraints sat uneasily together: the samplers that produce pure ε -DP noise (no δ slack) have data-dependent running times, and the known linear-time mechanisms paid with randomized- or approximate-DP only — so timing side channels or approximate guarantees leaked in through the implementation. The paper closes both gaps at once: a pure-DP mechanism that releases a histogram over n users and domain $[d]$ in *deterministic* $O(n)$ time in the word-RAM model, with per-coordinate error asymptotically matching the known lower bound of Balcer and Vadhan. The privacy proof is a case-by-case dominance analysis over the supports of neighboring histograms, composed with a private upper-bound search.

Problem. The input is a dataset of n user-contributed elements from a domain $[d]$, viewed as its histogram $h \in \mathbb{Z}_{\geq 0}^d$; neighbors differ by one added or removed element (the unbounded setting, so the dataset size itself is sensitive). The mechanism must output a *sparse* $\hat{h}$ ($\|\hat{h}\|_0 \leq O(n)$) approximating h in every coordinate simultaneously under pure DP, in deterministic linear time — the running time may depend only on n , never on which values the users hold.

Guarantee.

Theorem 6.13 — Private sparse histogram, informal version of Theorem 6.1 of the paper. Let $n, d \in \mathbb{Z}_{>0}$ and $\varepsilon > 0$, with representations that fit in a constant number of machine words. Given a dataset of n elements from $[d]$, there is an ε -differentially private algorithm that runs deterministically in $O(n)$ time and outputs a histogram $\hat{h}$ with $\|\hat{h}\|_0 \leq O(n)$ and ℓ_∞ error asymptotically optimal — matching the lower bound $\Omega(\min\{\frac{1}{\varepsilon} \ln d, n\})$ of Balcer and Vadhan quoted in the paper:

$$\|\hat{h} - h\|_\infty = O\left(\frac{1}{\varepsilon} \ln d\right). \quad \blacksquare$$

How DP composition is applied. The noise source is the discrete cousin of the Laplace mechanism (Theorem 6.3): a *geometric* (discrete Laplace) variable with per-coordinate tail

$$\Pr[X \geq t] \leq \frac{2e^{-\varepsilon t}}{1 + e^{-\varepsilon}} \quad (t \geq 1)$$

(Fact of the paper, notation cleaned). A union bound over the d coordinates turns the per-coordinate tail into the simultaneous accuracy $\alpha = (c_\alpha/\epsilon) \ln(d/\beta)$ at confidence $1 - \beta$ — the (α, β) -simultaneous estimator of the formal theorem. The sampler itself is engineered to be *timing-oblivious*, with an indistinguishability contract of its own (the “purified” relaxed discrete Laplace sampler, informal Theorem of the paper): for each $t \in [n]$, $\Gamma(t-1)$ and $\Gamma(t)$ are $(\epsilon, 0)$ -indistinguishable, and for $\beta > 2\gamma$, $\Gamma(t)$ estimates t to within $\alpha = O(\frac{1}{\epsilon} \ln \frac{1}{\beta})$ with probability $1 - \beta$. Privacy of the histogram is *not* a one-line sensitivity argument, because neighboring datasets have different supports: writing i^* and j^* for the heaviest coordinates of h and h' and I_1, I'_1 for the two sampled supports, the output distributions μ, μ' are compared in *four cases* according to which of i^*, j^* lie in I_1, I'_1 ; in each case the mixture decomposes over the shared support and the per-coordinate dominance ratio $r_j \in (e^{-\epsilon/2}, e^{\epsilon/2})$ yields $\mu'(S) \leq e^\epsilon \mu(S)$. Finally, because the setting is unbounded, the algorithm first runs a private *upper-bound search* on the dataset size (an ϵ_1 -DP time-oblivious search quoting prior work, restated in the paper), truncates the input, and pads with the ϵ_2 -parameterized geometric mechanism; the paper quotes basic sequential composition (Theorem 6.5) to combine the two stages into the stated pure budget.

What it buys. Analytics stacks needing per-value counts — contingency tables, heavy-hitter dashboards, training features — get a black-box release whose running time is worst-case, so the implementation leaks nothing through timing; the optimal ℓ_∞ certificate means the noise is provably not improvable, and the sparse support keeps storage at $O(n)$ regardless of the domain.

6.3 Reducing node-DP to edge-DP

The exemplar is a SIGMOD’26 paper on private graph analytics [8]. Publishing graph statistics — triangle counts, degree moments, sub-graph counts — under differential privacy requires choosing the unit of protection: *edge-DP* hides one edge, while *node-DP* hides one vertex together with all its incident edges — the right contract when vertices are people. Node-DP is strictly harder: a single high-degree vertex can shift many statistics, so sensitivities explode with the maximum degree, and existing node-DP techniques mostly assume a predefined degree bound. Setting that bound conservatively destroys utility; the paper quotes the naive escape — dividing the budget by the bound via group privacy — as exactly this failure mode. The N2E framework instead *privately learns* a degree bound from the graph itself, clips the graph to it, and runs any off-the-shelf edge-DP mechanism on the clipped graph,

converting the whole edge toolbox to node-DP with error governed by the data's actual degree structure. The end-to-end proof is sequential composition over three steps, with group privacy and post-processing doing the conversion.

Problem. Work on graphs with two neighborhood relations: $G \sim_{\text{edge}} G'$ differ by one edge, $G \sim_{\text{node}} G'$ by one node and its incident edges; (ϵ, δ) -DP is Definition 6.1 under either. A *clipping* $\text{Clip}(G, \Delta)$ keeps, for each node, its Δ smallest-order edges. Given any graph task Q and any edge-DP mechanism $\mathcal{M}_Q^{\text{edge}}$ for it, N2E must output a node-DP estimate of $Q(G)$ whose error degrades with the true maximum degree $\deg(G)$, not a worst-case bound N .

Guarantee.

Theorem 6.14 — N2E accuracy and privacy, Theorems of the paper, notation cleaned. Given any ϵ, β, δ , for any graph G , task Q , and edge-DP mechanism $\mathcal{M}_Q^{\text{edge}}$: with probability at least $1 - \beta$, N2E outputs a node-DP result with error

$$|Q(G) - Q(\tilde{G})| + \text{Err}_Q^{\text{edge}}(\tilde{G}, \epsilon/(6\Delta^*), \delta/(4\Delta^*), \beta/3),$$

where $\tilde{G}$ is obtained by clipping the edges of $O(\log(\log(\deg(G))/\beta)/\epsilon)$ nodes of G , and the framework as a whole satisfies (ϵ, δ) -node-DP. ■

How DP composition is applied. The framework is a three-step ledger. *Step 1* privately approximates the maximum degree by a search over decreasing thresholds Δ , at budget $(2\epsilon/3, \delta/2)$; *step 2*, the clipping, consumes no budget — it is post-processing (Theorem 6.7) of Δ^* ; *step 3* runs the edge mechanism at $\epsilon/(6\Delta^*)$ and $\delta/(4\Delta^*)$. Group privacy (Theorem 6.8) is the rung doing the reduction: a node change becomes at most $\lambda \leq 2\Delta^*$ edge changes after clipping (with probability $1 - \delta/2$, by a clipping-distance lemma below), so an edge-DP mechanism at budget ϵ/λ is node-DP at $\epsilon/3$. Sequential composition (Theorem 6.5) totals the steps to (ϵ, δ) . The degree search is a sparse-vector-technique (SVT) instance whose queries are chosen to have unit sensitivity:

Lemma 6.15 — SVT over sensitivity-monotonic queries, Lemma of the paper, restating prior work. If the query sequence used in SVT is sensitivity-monotonic, then with threshold calibration $c = 1$ SVT preserves ϵ -DP. Moreover, given failure probability β , if some query satisfies $Q_k(G) \geq T + 4 \ln(2k/\beta)/\epsilon$, then with probability at least $1 - \beta$ SVT returns an index $\hat{i} \leq k$ with $Q_{\hat{i}}(G) \geq T - 4 \ln(2k/\beta)/\epsilon$. ■

The query family is $Q_{\text{Del-N}}(G, \Delta)$ — minus the number of nodes that must be deleted to cap the degree at Δ — which the paper shows is

monotone in Δ , has global sensitivity 1 under node-DP, and is *sensitivity-monotonic* (neighboring graphs order its values consistently): the three properties SVT needs. Because the search evaluates Δ on a *logarithmic* scale, only $\ln \log(4 \deg(G))$ iterations are ever consumed. For tasks where node deletion is combinatorial, the paper relaxes it to an LP over variables (x_v, y_e) — $x_v = \text{delete node } v$, $y_e = \text{drop edge } e$ — whose optimum $Q_{\text{LP-DeL-N}}(G, \Delta)$ keeps the same three properties, and rounds:

Lemma 6.16 — LP rounding, Lemma of the paper. For any graph G and any Δ , the LP optimum upper-bounds the deletion query in the sense that $|Q_{\text{DeL-N}}(G, 3\Delta)| \leq 3 |Q_{\text{LP-DeL-N}}(G, \Delta)|$. ■

Proof sketch. From the optimal LP solution, keep every edge with $x_v, x_{v'} \leq \frac{1}{3}$ (so $y_e \geq 1 - x_v - x_{v'} \geq \frac{1}{3}$) and delete every node with $x_v > \frac{1}{3}$ (paying $n_v \leq 3x_v$ per node). Each retained node's degree is at most $3 \sum_{e \ni v} y_e \leq 3\Delta$, and the number of deletions is at most $3 \sum_v x_v = 3 |Q_{\text{LP-DeL-N}}(G, \Delta)|$ — a feasible deletion scheme at threshold 3Δ . ■

The remaining ingredient — that clipping does not explode the edge distance — is the paper's clipping lemma; since a node change can saturate many nodes, the proof bounds the two directions of edge difference separately (deferred to Section 6.6):

Lemma 6.17 — Clipping distance, Lemma of the paper, notation cleaned. Let $G \subseteq G'$ be neighboring under node addition, and let k be the number of nodes of G with degree at least Δ . Then the edge distance between $\text{Clip}(G, \Delta)$ and $\text{Clip}(G', \Delta)$ is at most $\Delta + k$. ■

What it buys. Every edge-DP graph algorithm becomes node-DP — protecting individuals rather than single relationships — by plugging it into N2E, with error tracking the actual degree profile through the $O(\log(\log(\deg(G)))/\beta)/\epsilon$ -node clipping overhead instead of a worst-case bound; the LP-rounded variant keeps the framework polynomial-time even when the exact deletion query is NP-hard.

6.4 Per-record differential privacy

The exemplar is a SIGMOD'26 paper on heterogeneous privacy budgets [2]. Classical DP charges every record the same ϵ , but real deployments often want per-record budgets: a bank may owe wealthier clients stronger protection, and regulations may grade protection by record type. This *per-record DP* (PrDP) replaces the scalar ϵ with a budget function $\epsilon_{\mathcal{E}}: [U]^d \rightarrow [\epsilon_{\min}, \epsilon_{\max}]$ — now the privacy contract itself is a *function*, and so is the composition arithmetic. The paper gives a

general framework: privately discard the records whose budgets are too small to be worth noise, then run any existing DP mechanism on the survivors at the minimum surviving budget. For sums and counts it builds direct mechanisms that partition records into geometrically doubling budget slices, one Laplace mechanism per slice, and proves them near-optimal against down-neighborhood lower bounds — the same yardstick style as Section 6.3.

Problem. Records come from a universe $[U]^d$; a budget function $\varepsilon_{\mathcal{E}}$ assigns each record r a budget in $[\varepsilon_{\min}, \varepsilon_{\max}]$ (for the bank: a constant factor times wealth, capped). The *privacy-specified domain* $X_{[\varepsilon_a, \varepsilon_b]} = \{r : \varepsilon_{\mathcal{E}}(r) \in [\varepsilon_a, \varepsilon_b]\}$ collects records whose budgets fall in a band. The contract: $\mathcal{M}$ is $\varepsilon_{\mathcal{E}}$ -PrDP if for all neighbors $D \sim_r D'$ (differing in record r) and all outputs y , $\Pr[\mathcal{M}(D) = y] \leq e^{\varepsilon_{\mathcal{E}}(r)} \Pr[\mathcal{M}(D') = y]$. Queries of interest are sums and counts.

Guarantee.

Theorem 6.18 — General framework, Theorem of the paper. For any budget function $\varepsilon_{\mathcal{E}}(\cdot)$, $\beta \in (0, 1)$, and any DP mechanism $\mathcal{M}$ for query Q , the framework returns $\hat{Q}(D)$ such that, with probability at least $1 - \beta$,

$$|\hat{Q}(D) - Q(D)| \leq \underbrace{|Q(D) - Q(D')|}_{\text{(I)}} + \underbrace{\text{Err}_{\mathcal{M}}(D', \varepsilon_{\min}(D)/4, \beta/2)}_{\text{(II)}},$$

where D' is obtained by removing $O(\frac{1}{\varepsilon_{\min}(D)} \log \log(\cdot/\beta))$ records from D (hiding the argument of the iterated logarithm), and $\varepsilon_{\min}(D)$ is the smallest budget among records of D . ■

How DP composition is applied. Term (I) is a *bias* term — the query change caused by discarding the budget-poor records — and term (II) is the noise of the black-box mechanism run on the survivors: a sensitivity-vs-truncation trade, as in Section 6.3 but with budgets, not degrees, deciding the cut. To locate the cut, the count estimation searches over the doubling partition $X_i = X_{[2^{i-1}\varepsilon_a, 2^i\varepsilon_a]}$: it releases per-slice counts with slice-specific Laplace noise, thresholds them (Laplace tail bounds plus a union bound over the $\log(\varepsilon_{\max}/\varepsilon_{\min})$ slices), and certifies that the first surviving slice has budget $\hat{\varepsilon} \geq \frac{1}{2}\varepsilon_{\min}(D)$ while the discarded slices together hold only $O(\frac{1}{\varepsilon_{\min}(D)} \log \log(\cdot/\beta))$ records — the failure budget β split per slice and union-bounded, exactly the recipe of Section 6.0.1. The sum estimation composes per-slice mechanisms, with the composition arithmetic lifted from scalars to functions:

Lemma 6.19 — PrDP parallel composition, Lemma of the paper. Let $R_1, \dots, R_k$ be disjoint subdomains and $\mathcal{M}_1, \dots, \mathcal{M}_k$ $\varepsilon_{\mathcal{E}_i}$ -PrDP mechanisms. Their

parallel composition $\mathcal{M}(D) := (\mathcal{M}_1(D_{R_1}), \dots, \mathcal{M}_k(D_{R_k}))$ satisfies $\varepsilon_{\mathcal{E}'}$ -PrDP with $\varepsilon_{\mathcal{E}'}(r) = \max\{\varepsilon_{\mathcal{E}_i}(r)\}_{i=1}^k$. ■

Proof. A record r lives in exactly one slice, so changing r perturbs exactly one component, whose mechanism carries its own $\varepsilon_{\mathcal{E}_i}(r)$ bound; the other components are identical functions of unchanged data. ■

The sequential twin states $\varepsilon_{\mathcal{E}'}(r) = \sum_{i=1}^k \varepsilon_{\mathcal{E}_i}(r)$ (Lemma of the paper): sums along chains, maxima across slices — the same two rules as Theorems 6.5 and 6.6, now pointwise in the record. The sum mechanism calibrates each slice’s Laplace scale to the largest sensitivity that slice contributes; because slice budgets double, the per-slice noise forms a geometric series on the order of its largest term — error governed by the best-budgeted survivors, not by $\varepsilon_{\min}$. An LDP variant (Theorem of the paper) removes $O(\frac{\sqrt{n}}{\varepsilon_{\min}(D)} \log \log(\cdot/\beta))$ records instead, the $\sqrt{n}$ inflation being the standard price of aggregating n locally randomized responses. Near-optimality is certified against the *down-neighborhood lower bound* $L_\varepsilon(D) = \max_{M' \in \mathcal{M}} \min_{D' : d(D, D') \leq \varepsilon^{-1}} \{\tau : \Pr[|M'(D') - Q(D')| \leq \tau] \geq \frac{2}{3}\}$ (Definition of the paper, quoting prior work): no mechanism beats the error of the best mechanism on the best nearby dataset, and the paper’s mechanisms sit within a constant factor c of this bound $((\varepsilon, c)$ -down-neighborhood optimality).

What it buys. Regulated deployments can honor per-record protection levels — different ε per customer or record class — without giving up the machinery of standard DP: any existing mechanism drops into the framework, the error adapts to the budgets actually present in the data, and the optimality certificate says the residual error is the unavoidable price of the worst-protected records, not an artifact of the design.

6.5 Advanced usage and further reading

Everything so far composes *statistics*. Two harder corpus uses push the same calculus onto objects one would not at first call releases — memory access patterns and distance labels; each is sketched in one paragraph, and the chapter’s deferred proofs live in the Appendix (Section 6.6).

Differentially oblivious multi-way join [12]. Running a join in untrusted memory exposes not the output but the *access pattern* — which cells were read, and when. Fully oblivious (FO) algorithms erase that leak at the price of touching the whole memory; the paper defines *differentially oblivious* (DO) algorithms, whose access-pattern distribution satisfies DP over neighboring input instances, and asks how cheap a DO multi-way

join can be. The meta algorithm is the ledger of Figure 6.1 with an access pattern as the composed object: an $(\varepsilon_1, \delta_1)$ -DO subroutine releases an $(\varepsilon_2, \delta_2)$ -DP upper bound $\widehat{OUT}$ on the join size $|Q(I)|$, calibrated via smooth sensitivity — the paper proves smoothness lemmas for the join’s residual-set terms (RST/EST monotonicity under dropping attributes, plus a “relaxed smooth sensitivity” RRS damping local sensitivity by $e^{-\beta k}$ across k -step neighborhoods), and a hybrid construction picks the cheapest valid bound per subset of relations. The join itself then runs as an *advised FO* algorithm: an FO algorithm whose access pattern is a deterministic function of the input size $|I|$ and the advice $\widehat{OUT}$ only — so its leak is $|I|$ plus one DP release, and the construction is $(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2)$ -DO by sequential composition (Theorem 3.2 of the paper), with post-processing immunity licensing the pattern as a private object. The payoff: $O((N^w + |Q(I)| + \text{RRS}(I)) \cdot \log N)$ time under data complexity (w = fractional hypertree width) instead of full obliviousness.

Privacy-preserving shortest path via 2-hop labeling [11]. Two parties hold subgraphs of one weighted graph, connected by public cut edges, and must answer shortest-path distance queries between their vertices without revealing the private weights. Each party builds a 2-hop labeling of its subgraph — every vertex stores exact distances to higher-ranked *boundary* vertices — and the parties exchange secret-shared boundary graphs instead of raw data. The proof that this loses no information is an *induction on the optimal substructure of shortest paths* (Lemma 4.4 and Lemma 5.2 of the paper): any shortest path crossing partitions decomposes at boundary vertices, so the combined boundary graph preserves all boundary-to-boundary distances exactly. Privacy is the truncated Laplace mechanism: edge weights are perturbed with Laplace noise truncated to a bounded range — guaranteeing that the iterative label construction converges (no unbounded noise ever enters a distance) — and the released labeling satisfies (ε, δ) -DP (Theorem 5.4 of the paper), with weight-neighborhood $\|w - w'\|_1 \leq 1$. The composition lesson: the labeling is built by a pipeline of per-vertex noisy aggregations composed sequentially, and correctness is proved independently of privacy — post-processing immunity lets the exact-labeling argument ride on top of the noisy weights for free.

Further reading

- The monograph: Dwork and Roth [4] — the algorithmic foundations, with all composition proofs and the privacy-loss calculus done from first principles; the natural companion to this chapter.
- The originals: Dwork, McSherry, Nissim, and Smith [5] — sensi-

tivity and calibrated noise; Dwork, Rothblum, and Vadhan [6] — the advanced-composition theorem and the privacy-loss random variable.

- Beyond scalar budgets: Mironov’s Rényi DP [10] composes in moments and converts back to (ϵ, δ) ; Dong, Roth, and Su’s Gaussian/f-DP [3] makes the divergence curve itself the composed object — the direction in which budget accounting is moving.
- Subsampling: Balle, Barthe, and Gaboardi [1] — privacy amplification, the discount rule when mechanisms run on samples.
- For a refresher with the corpus exemplars in one place: the DP sections of the concentration chapter (Section 2.4.2 in Chapter 2) treat the noise tails; the present chapter treats the accounting.

Chapter Recap

- **What the tool is.** The DP composition calculus turns per-release guarantees into one end-to-end contract: budgets add along adaptive chains (sequential), collapse to maxima on disjoint data slices (parallel), survive any downstream computation unchanged (post-processing), and inflate linearly in group size (group privacy); single releases are calibrated by sensitivity — Laplace for numbers, the exponential mechanism for selections.
- **Which rung when.** Same data, several releases: sequential. Disjoint records/clusters/slices: parallel. Converting neighborhoods (node→edge, group→individual): group privacy. Anything after a private output: free. Many releases ($k \gtrsim 50$): advanced or moment accounting.
- **Why it works.** Indistinguishability multiplies: a product of per-release density ratios bounded by e^{ϵ_i} is bounded by $e^{\sum_i \epsilon_i}$; disjointness kills all but one factor; post-processing integrates against a fixed output law; distance- λ chains telescope.
- **What the corpus showed.** Cluster explanations (Section 6.1): a three-dial ledger with sequential + parallel + post-processing stages and sensitivity-1 scores by ℓ_1 algebra. Sparse histograms (Section 6.2): four-case dominance over neighboring supports composed with a private upper-bound search, optimal ℓ_∞ error. N2E (Section 6.3): SVT at logarithmic scale, group privacy at $\epsilon/(6\Delta^*)$, LP rounding of the deletion query. Per-record DP (Section 6.4): composition lifted from scalars to budget functions — pointwise sums and maxima — with down-neighborhood optimality.

- **Advanced usage.** The calculus also composes non-statistical objects: access patterns of differentially oblivious joins and noisy 2-hop distance labelings (Section 6.5); deferred proofs are collected in the Appendix (Section 6.6).

6.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 6.3 (Laplace mechanism)

Statement (recalled). If GS_Q is the global sensitivity of Q , then $Q(D) + \text{Lap}(GS_Q/\epsilon)$ satisfies ϵ -DP. ■

Proof. Let $b = GS_Q/\epsilon$ and write p_D for the density of $Q(D) + \text{Lap}(b)$ at a point y . For every coordinate j , $|(Q(D) - y_j) - (Q(D') - y_j)| \leq |Q(D) - Q(D')| \leq GS_Q$, so the density ratio is

$$\begin{aligned} \frac{p_D(y)}{p_{D'}(y)} &= \prod_j \exp\left(\frac{|Q(D')_j - y_j| - |Q(D)_j - y_j|}{b}\right) \\ &\leq \exp\left(\frac{\|Q(D) - Q(D')\|_1}{b}\right) \leq e^{GS_Q/b} = e^\epsilon. \end{aligned}$$

Integrating over any measurable S gives $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S]$. ■

Proof of Theorem 6.5 (sequential composition)

Statement (recalled). If $\mathcal{M}_1$ is (ϵ_1, δ_1) -DP and $\mathcal{M}_2(\cdot, y)$ is (ϵ_2, δ_2) -DP in its first input for every fixed y , then $\mathcal{M}_3(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$ is $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$ -DP. ■

Proof. For y in the range of $\mathcal{M}_1$ let $p_D(y) = \Pr[\mathcal{M}_1(D) = y]$ and let $q_D(S | y) = \Pr[\mathcal{M}_2(D, y) \in S]$. For any measurable S of the composed output space,

$$\Pr[\mathcal{M}_3(D) \in S] = \int q_D(S | y) dp_D(y).$$

Let $T = \{y : p_D(y) \leq e^{\epsilon_1} p_{D'}(y)\}$; by the DP bound on $\mathcal{M}_1$ (applied to the complement), $\Pr[\mathcal{M}_1(D) \notin T] \leq \delta_1$. Splitting the integral at T and using the two per-mechanism bounds on each part,

$$\begin{aligned} \Pr[\mathcal{M}_3(D) \in S] &\leq e^{\epsilon_2} \int_T q_{D'}(S | y) dp_D(y) + \delta_2 \\ &\leq e^{\epsilon_2} e^{\epsilon_1} \int q_{D'}(S | y) dp_{D'}(y) + e^{\epsilon_2} \delta_2 + \delta_1, \end{aligned}$$

which is at most $e^{\epsilon_1 + \epsilon_2} \Pr[\mathcal{M}_3(D') \in S] + (\delta_1 + \delta_2)$ for $\epsilon_2 \geq 0$. Iteration over k adaptive stages is this argument unrolled along the chain. ■

Proof of Theorem 6.8 (group privacy)

Statement (recalled). An (ϵ, δ) -DP mechanism satisfies $\Pr[\mathcal{M}(G) \in S] \leq e^{\lambda\epsilon} \Pr[\mathcal{M}(G') \in S] + \lambda\delta$ for inputs at neighborhood distance λ . ■

Proof. Induction on λ ; $\lambda = 1$ is DP itself. For the step, fix intermediates $G = G_0 \sim G_1 \sim \dots \sim G_\lambda = G'$ and apply DP twice:

$$\Pr[\mathcal{M}(G_0) \in S] \leq e^\epsilon \Pr[\mathcal{M}(G_1) \in S] + \delta \leq e^{2\epsilon} \Pr[\mathcal{M}(G_2) \in S] + e^\epsilon \delta + \delta.$$

After λ steps the error term is $\sum_{i=0}^{\lambda-1} e^{i\epsilon} \delta \leq \lambda e^{(\lambda-1)\epsilon} \delta$; since $\lambda e^{(\lambda-1)\epsilon} \geq \lambda$ this is absorbed by $\lambda\delta$ as claimed (the paper's statement). ■

Proof of Proposition 6.12 (JSD sensitivity bound)

Statement (recalled). The Jensen–Shannon distance score has sensitivity at least $\frac{1}{2}$: some neighbors $D \sim D'$ change it by more than $\frac{1}{2}$. ■

Proof. Let D have $n \geq 1$ tuples, all carrying value a on attribute A , and let $D_c = \{t\}$ be a singleton cluster. Both $P_A(D)$ and $P_A(D_c)$ are point masses on a , so $d_{JS}(P_A(D), P_A(D_c)) = 0$. Let $D' = D \cup \{t'\}$ and $D'_c = D_c \cup \{t'\}$ with $t'[A] = a' \neq a$. Then

$$P_A(D') = \left(\frac{n}{n+1}, \frac{1}{n+1}\right), \quad P_A(D'_c) = \left(\frac{1}{2}, \frac{1}{2}\right),$$

over $\{a, a'\}$, and the mixture puts mass $\frac{n}{2(n+1)} + \frac{1}{4}$ on a . Writing H_b for the binary entropy (bits),

$$JSD(P_A(D'), P_A(D'_c)) = H_b\left(\frac{n}{2(n+1)} + \frac{1}{4}\right) - \frac{1}{2}H_b\left(\frac{1}{n+1}\right) - \frac{1}{2}H_b\left(\frac{1}{2}\right).$$

As $n \rightarrow \infty$, the mixture mass tends to $\frac{3}{4}$, so the limit is $H_b(\frac{3}{4}) - 0 - \frac{1}{2} \cdot 1 = H_b(\frac{3}{4}) - \frac{1}{2} \approx 0.811 - 0.5 = 0.311$ bits, using $H_b(\frac{1}{n+1}) \rightarrow H_b(0) = 0$ and $H_b(\frac{1}{2}) = 1$. Hence for all large n the divergence exceeds 0.30 bits, so the Jensen–Shannon *distance* (its square root) exceeds $\sqrt{0.30} > \frac{1}{2}$ while it was 0 on D — a change of more than $\frac{1}{2}$ between neighbors, as computed in the paper. ■

Proof of Lemma 6.17 (clipping distance, N2E)

Statement (recalled). For node-neighboring $G \subseteq G'$ and $k =$ number of nodes of G of degree $\geq \Delta$, the clipped graphs $\text{Clip}(G, \Delta)$, $\text{Clip}(G', \Delta)$ differ in at most $\Delta + k$ edges. ■

Proof. Let u be the node of $G' \setminus G$. Count each direction of the difference. *Edges present after clipping G' but not after clipping G :* any such edge $e = (v_1, v_2)$ with both endpoints in V is among the Δ smallest-order edges of v_1, v_2 in both graphs, so it survives both clippings; hence every edge in this category touches u , and u keeps at most Δ edges — at

most Δ edges total. *Edges present after clipping G but not after clipping G'* : such an edge e is among the Δ smallest-order edges of its endpoints in G , yet is pushed out in G' only if one of its endpoints is *saturated* in G (degree $\geq \Delta$) and one of the new edges incident to u displaces it in that endpoint's smallest-order set. Each saturated node loses at most one edge per new edge of u , and the new edges number at most the number of saturated nodes they touch; since G has at most k saturated nodes, this category contributes at most k edges. Together, $\Delta + k$. ■

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7

Exchange Arguments and Greedy Algorithms

A greedy algorithm commits irrevocably: it keeps one plan out of thousands, merges two chunks, adds one edge to a forest, fixes the order in which an exact search tries values. The exchange argument is the certificate that makes such myopia safe: it shows that any solution avoiding the greedy's choice can be *rearranged* — one element, edge, or branch swapped for another — into a solution that honors it, at a loss that is either zero (the greedy is exactly optimal, or the pruning loses nothing) or bounded and countable (the greedy keeps a $1 - 1/e$ or $\ln$ fraction of the optimum). For a systems paper this converts “greedy worked well in our experiments” into a theorem, and the theory section often shrinks to checking a hypothesis: independence, diminishing returns, a covering constraint, metric distances, or symmetry. In the SIGMOD'26 corpus, 17 tagged papers use the tool; in the tag co-occurrences it pairs with hardness 10 times and with an approximation-ratio property 9 times — the signature of “NP-hard, but structured” — and with correctness 7 times, its exact-rung twin.

This chapter builds the ladder first: the exchange principle, greedy-stays-ahead, the matroid rung, and the three factor rungs (submodular, set-cover, metric swaps), each with its swap accounting. Sections 7.1–7.4 then dissect four corpus exemplars at full anatomy; Section 7.5 adds two sketched exemplars and further reading.

7.0 The tool: exchange arguments and greedy with certificates

The problem class is uniform across applications: a solution must be built or selected incrementally, under constraints, and each increment is irrevocable. The optimizer keeps a small set of plans, the sampler picks anchors one at a time under a budget, the temporal index decides edge by edge what a forest remembers, the exact search commits to a branching order. A greedy algorithm makes each choice by a

LEARNING OBJECTIVES

- Recognize when a selection or ordering problem admits an exchange certificate, and which structure buys it.
- Build the exchange: name the partner, count the swap's delta, pick the certificate.
- Match rung to structure: exact (matroid), $1 - 1/e$ (submodular), $\ln$ (cover), $3 + \epsilon$ (metric), complete (symmetry).
- Dissect a paper's greedy: what is committed, what is swapped, what the factor is against.

myopic rule — take the plan covering the most workload, the anchor with the best marginal fidelity, the heaviest edge that keeps a forest, the first value in a fixed order. What the exchange argument adds is the after-the-fact certificate: a proof that any solution avoiding the greedy's commitment can be rearranged into one that honors it. The organizing question of this chapter is which *structure* buys which *certificate*: independence (matroids) buys exactness, diminishing returns (submodularity) buys $1 - 1/e$, covering buys $\ln$, metrics buy $3 + \epsilon$, and symmetry buys prune-safety — completeness with no factor at all.

Notation

U is a ground set, $\mathcal{F} \subseteq 2^U$ a feasible family, w (linear objectives) or f (set functions) the objective to maximize unless said otherwise. S_i is greedy's set after i picks; $\Delta(u \mid S) = f(S \cup \{u\}) - f(S)$ is the *marginal gain*; y always names the *exchange partner* — the element of the optimal solution that a swap gives up; O denotes an optimal solution and OPT its value. $H(n) = \sum_{i=1}^n 1/i$ is the harmonic number. Maximization rungs quote factors ≤ 1 (as $1 - 1/e$); minimization rungs quote factors ≥ 1 (as $H(n)$, $3 + \epsilon$).

7.0.1 The ladder: from exact exchange to logarithmic coverage

The bottom rung is not a theorem about any one problem but the shape of every proof to come: an optimal solution can be rearranged, one swap at a time, to agree with the greedy's choices.

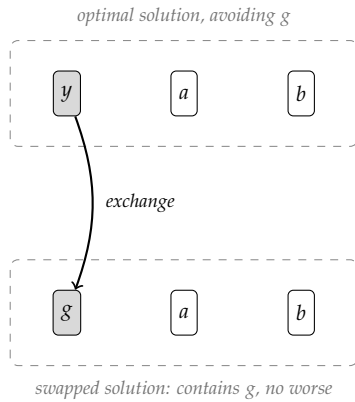


Figure 7.1: The exchange step. Greedy commits to g ; an optimal solution avoiding g is rearranged into a no-worse solution containing it, one swap at a time. Exact rungs pay nothing per swap; factor rungs lose a bounded, countable amount that compounds into the factor.

Proposition 7.1 — Exchange principle. Let $\mathcal{F} \subseteq 2^U$ be a feasible family and $w: \mathcal{F} \rightarrow \mathbb{R}$ an objective to maximize. Let g be greedy's first choice, and suppose that for every feasible S with $g \notin S$ there is an exchange partner $y \in S$ such that $S' = (S \setminus \{y\}) \cup \{g\}$ is feasible with $w(S') \geq w(S)$, and that the same condition holds recursively on the residual instance with ground set $U \setminus \{g\}$. Then greedy returns an optimal solution. ■

Proof. Induction on $|U|$. Let O be optimal. If $g \in O$, then $O \setminus \{g\}$ is optimal for the residual instance — any better residual solution would combine with g to beat O — and greedy on the residual returns it by hypothesis. If $g \notin O$, the swap produces an optimal $O' = (O \setminus \{y\}) \cup \{g\}$ containing g , which reduces to the first case. ■

The next rung makes the rearrangement concrete and shows the craft in miniature: order the choices well, and the partner swap loses nothing.

Theorem 7.2 — Greedy stays ahead. Among all sets of pairwise non-overlapping intervals, the greedy that repeatedly chooses the

compatible interval of *earliest finish time* returns one of maximum cardinality. ■

Proof. Write $g_1, g_2, \dots$ for greedy's picks, $o_1, o_2, \dots$ for any optimal solution's, and $f(\cdot)$ for finish times. By induction, $f(g_i) \leq f(o_i)$ for every i : it holds for $i = 1$ because greedy takes the earliest-finishing interval of all; and if it holds for i , then o_{i+1} starts at or after $f(o_i) \geq f(g_i)$, so o_{i+1} is compatible with $g_1, \dots, g_i$ and greedy's $(i+1)$ -st pick finishes no later than o_{i+1} . Greedy is never behind, so it cannot stop earlier: if the optimum had a $(k+1)$ -st interval, it would be compatible with $g_1, \dots, g_k$, and greedy would have had a choice at step $k+1$. ■

When the feasibility of each swap is governed by an *independence structure*, the principle industrializes into an exact algorithm for a whole problem class.

Definition 7.3 (Matroid). A pair $(U, \mathcal{I})$ with $\mathcal{I} \subseteq 2^U$ is a *matroid* if (i) $\mathcal{I}$ is hereditary — $S \in \mathcal{I}$ and $S' \subseteq S$ imply $S' \in \mathcal{I}$ — and (ii) it satisfies the *exchange axiom*: if $A, B \in \mathcal{I}$ with $|A| < |B|$, then $A \cup \{x\} \in \mathcal{I}$ for some $x \in B \setminus A$. Members of $\mathcal{I}$ are called *independent sets*.

Theorem 7.4 — Matroid greedy is exact, Edmonds. On a matroid, the greedy that scans elements in nonincreasing weight order and keeps each element that preserves independence returns a maximum-weight independent set — for every weight function. Conversely, if greedy is exact for every weight function, then the family of independent sets satisfies the exchange axiom, so the structure is a matroid [3]. ■

Proof idea. The exchange axiom is Proposition 7.1's one-for-one swap with zero loss: if the optimum “spends” y where greedy spent g with $w(g) \geq w(y)$, swap. The full induction is in Section 7.6. ■

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Outside matroids exactness dies, but diminishing returns keep a quantitative exchange alive: greedy's next pick matches the *average* marginal value of the optimum's remaining elements.

Definition 7.5 (Monotone submodular). $f: 2^U \rightarrow \mathbb{R}_{\geq 0}$ is *submodular* if $f(S \cup \{x\}) - f(S) \geq f(T \cup \{x\}) - f(T)$ whenever $S \subseteq T \subseteq U$ and $x \in U \setminus T$ (diminishing returns), and *monotone* if $S \subseteq T$ implies $f(S) \leq f(T)$.

Theorem 7.6 — Submodular greedy, Nemhauser–Wolsey–Fisher. Let f be monotone submodular with $f(\emptyset) = 0$, and let A_B be the set built by B greedy steps, each adding the $u \notin A$ that maximizes $\Delta(u \mid A)$. Then

$$f(A_B) \geq \left(1 - \frac{1}{e}\right) \max_{|A| \leq B} f(A), \quad \blacksquare$$

¹ The graphic matroid takes U to be a graph's edges and $\mathcal{I}$ its forests (acyclic edge sets); maximal independent sets are spanning forests, and the cycle rule — an edge that closes a cycle may replace the lightest edge of that cycle — is the exchange axiom in the wild. Kruskal's algorithm is the matroid greedy; the union-find structure that implements it reappears in Section 7.2.

and the factor $1 - 1/e$ is tight for maximum coverage unless $P = NP$ [6, 8].

Proof idea. Each greedy pick captures at least a $1/B$ fraction of the remaining optimality gap, so the gap contracts by $(1 - 1/B)$ per step; unrolling B steps gives $(1 - 1/B)^B \leq e^{-1}$ (Section 7.6). ■

When the objective is a covering constraint in disguise — cover the universe at minimum total weight — the same averaging against the optimum yields the harmonic factor:

Theorem 7.7 — Set-cover greedy, Chvátal. Given a universe of n elements and a family of sets with nonnegative weights, the greedy that repeatedly buys the set maximizing newly-covered elements per unit weight returns a cover of total weight at most $H(n) \cdot \text{OPT} \leq (1 + \ln n) \text{OPT}$, where $H(n) = \sum_{i=1}^n 1/i$; the bound sharpens to $H(d)$ when every set has at most d elements [2]. ■

Proof idea. While r elements remain uncovered, the sets of an optimal cover jointly cover them at total weight OPT , so one of them covers at rate at least r/OPT per unit weight — and greedy picks at least that well. Charging each element the price at which it was covered and summing the harmonic series gives $H(n)$ (Section 7.6). ■

The last rung replaces “take the best now” with “swap if it helps”: local search over exchanges, certified by the metric.

Theorem 7.8 — Swap local search for k -median, Arya et al.. In metric k -median, the local search that repeatedly swaps one center for a non-center whenever it lowers the total assignment cost terminates at a solution of cost at most $(3 + \varepsilon) \text{OPT}$; allowing multi-swaps of up to p centers improves the factor to $2 + \varepsilon$ [1]. ■

Proof idea. At a local optimum no single swap gains. Charge each optimal center to a nearby local center: the swap that trades one for the other reassigns points through two triangle inequalities, and the (non-improving) swap’s cost delta bounds OPT by three times the local cost. ■

The rungs share one mechanism — the swap — and differ only in the accounting: *rearrangement* (delta ≥ 0 , greedy optimal), *contraction* (delta $\geq \text{gap}/B$, compounding to e^{-1}), *averaging* (delta $\geq \text{residual}/\text{OPT}$, compounding to $H(n)$), *two triangle inequalities* ($3 + \varepsilon$). Two corpus exemplars below use rungs with no factor at all: the swap certifies that a construction is exact or that pruning is safe.

7.0.2 How to build the exchange

The construction procedure is mechanical once the greedy rule is written down. *First*, state what one step commits to — an element enters the solution, an edge enters a forest, a value order is fixed. *Second*, take any optimal solution O that avoids the commitment and *name the exchange partner y* : the element of O doing the job g would do — the optimal cover's set that greedy's set displaces, the optimal forest's edge on the cycle g closes, the optimal branch that tries the values in the other order. *Third*, count the swap's cost δ for $O' = (O \setminus \{y\}) \cup \{g\}$: feasibility of O' comes from the structure (the exchange axiom, heredity, the cycle property, automorphism invariance, plain coverage), and the δ comes from the objective's arithmetic — a weight comparison, a diminishing-returns inequality, a price-per-element average, two triangle inequalities, or identical neighborhoods. *Fourth*, read off the certificate type from the δ : zero loss with self-similar residuals gives exactness; a $1/B$ fraction of the gap gives $1 - 1/e$; a $1/\text{OPT}$ fraction of the residual gives $\ln$; a strictly earlier branch gives completeness. The audit effort goes into the *hypothesis*, not the algorithm: submodularity by pairwise marginal tests or a concave-gain decomposition, the exchange axiom by an independence oracle, metric structure by the triangle inequality, symmetry by automorphisms. Near-modular or noisy objectives void the certificate — and an honest paper then says so (Section 7.5).

When to Reach for Exchange/Greedy Arguments

Reach whenever a selection, covering, or clustering problem has been proved NP-hard (or an exact search explodes) but the objective carries *checkable* structure, and the deliverable is an algorithm whose quality is a theorem, not a benchmark: a factor-certified or exactness-certified polynomial-time procedure. Then run the recipe:

1. **Name the exchange partner.** Fix the greedy's next commitment g and an optimal solution O avoiding it; identify the element, edge, center, or branch y of O that plays g 's role.
2. **Identify the structure** that licenses the swap: matroid independence (exact), monotone submodularity ($1 - 1/e$), a covering constraint ($\ln$), metric distances ($3 + \varepsilon$ swaps), a symmetry group (pruning stays complete).
3. **Count the swap's cost δ .** Zero loss, a $1/B$ fraction of the gap, a $1/\text{OPT}$ fraction of the residual, two triangle inequalities, or identical neighborhoods — the count determines the certificate.

4. **State the certificate of near-optimality:** which factor, against which optimum, with which slack $(\ln \delta^{-1}, \epsilon)$ — inherited, not re-derived. If no structure checks out, say so and call the greedy a heuristic.

7.1 Risk coresets for robust query optimization

The exemplar is a PODS'26 paper on robust query optimization [9]^{PODS}. Query optimizers pick execution plans using cardinality estimates, and the estimates can be badly wrong; a plan that looks cheapest under the estimates may be terrible on the true cardinalities. Robust query optimization therefore scores each plan by *risk*: the probability, over the uncertainty about cardinalities, that the plan's cost exceeds a factor τ of the best plan for the realized instance. The paper asks for a small *coreset* of plans that covers a whole workload distribution — after seeing the coreset, a query-time inference step should pick, from a few precomputed plans, one nearly as robust as the best plan in the entire exponential space. Its main result is a greedy-cover algorithm that samples $O((\kappa^*/\delta^2) \ln N \ln(\delta^{-1}))$ queries from the workload and computes, with high probability, a coreset of size $O(\kappa^* \ln \delta^{-1})$ — within a logarithmic factor of the smallest coreset κ^* that exists — from which inference returns a plan of risk at most twice the optimum plus 4ϵ .

Problem. Work over a plan set Π with $|\Pi| = N$ and a workload distribution W over a query space $\mathcal{Q}$. Each query Q carries an *uncertainty profile* U_Q : a distribution over cardinality vectors s , against which plan costs $\text{Cost}(\pi, s)$ (linear in s) are judged. The *penalty* of π at s is its cost ratio against the best plan for s , $\text{Pnl}(\pi, s) = \text{Cost}(\pi, s) / \text{Cost}^*(s)$, and the τ -*risk* of π on Q is $\text{Risk}(\pi, Q; \tau) = \Pr_{s \sim U_Q}[\text{Pnl}(\pi, s) > \tau]$, with $\text{Risk}^*(Q; \tau)$ the minimum over all plans. A plan set $F \subseteq \Pi$ *covers* query Q when its best plan is nearly as robust as the overall best, $\min_{\pi \in F} \text{Risk}(\pi, Q; \tau) \leq \text{Risk}^*(Q; \tau) + \epsilon$; the failure probability $p(\epsilon, F)$ is the W -probability that F does not cover Q , and F is an $(\epsilon, \tau, \delta|W)$ -*coreset for risk* when $p(\epsilon, F) \leq \delta$. Let κ^* be the size of the smallest such coreset. The goal: compute a valid coreset of size $O(\kappa^* \cdot \text{polylog})$ from few workload samples.

Guarantee.

Theorem 7.9 — Risk coreset via greedy cover; Theorem 3.5 of the paper, notation cleaned. Let W be a workload distribution, Π a plan set with $|\Pi| = N$, parameters $\epsilon \in (0, 1]$, $\tau \geq 1$, $\delta \in (0, 1)$, and κ^* the size of the smallest $(\epsilon, \tau, \delta|W)$ -coreset for risk. For any constants $c_1, c_2 > 1$ there is an

algorithm that samples $O((\kappa^*/\delta^2) \ln N \ln(\delta^{-1}))$ queries from W and computes, in time $O((\kappa^*/(\varepsilon^2\delta^2)) \cdot N \ln^3 N \cdot \ln(\delta^{-1}))$, a set $C \subseteq \Pi$ of size $O(\kappa^* \ln(\delta^{-1}))$ such that, with high probability, C is a $(c_1\varepsilon, \tau, c_2\delta|W)$ -coreset for risk. ■

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How the greedy cover is applied. The algorithm runs in rounds indexed by a guess k_r that doubles every round; in round r it samples $n_r = (c_1 k_r / \delta^2) \ln N \ln(\delta^{-1})$ queries from W , estimates each plan's risk on each sampled query from $m_r = (c_2 / \varepsilon^2) \ln N \ln n_r$ profiles drawn from the query's uncertainty profile, and runs a maximum-coverage greedy with budget $b_r = c_3 k_r \ln(\delta^{-1})$ plans: repeatedly add the plan covering the most not-yet-covered sampled queries, terminating when at least a $(1 - 2\delta)$ -fraction is covered. Two estimation lemmas make the greedy's proxies faithful: Lemma 3.1 of the paper bounds the empirical failure probability by Hoeffding plus a union bound over all $\leq (b_r + 1)N^{b_r}$ plan subsets of size at most b_r , giving $|\hat{p}_r(\varepsilon', F) - p(\varepsilon', F)| \leq \delta/4$ simultaneously with probability $1 - N^{-O(1)}$; Lemma 3.2 of the paper does the same per (plan, query) pair, Hoeffding over the m_r risk indicators plus a union bound over the Nn_r pairs. So the greedy cover behaves almost as if it saw true risks. The exchange rung — averaging against the optimal coreset — is the workhorse:

Lemma 7.10 — Greedy termination; Lemma 3.4 of the paper. Assuming the conditions of Lemmas 3.1 and 3.2 of the paper hold in every round, the algorithm terminates with $k_r \leq 2\kappa^*$. ■

Proof sketch. By Lemma 3.1 the optimal coreset M , $|M| = \kappa^*$, covers at least a $(1 - 5\delta/4)$ -fraction n' of the sampled queries, so among M 's plans some covers at least a $1/\kappa^*$ fraction of whatever residual D_t remains; greedy picks at least that well, so $D_{t+1} \leq (1 - 1/\kappa^*)D_t \leq e^{-t/\kappa^*} n'$. After $t \geq c_3 \kappa^* \ln(\delta^{-1})$ rounds the coverage reaches $1 - 2\delta$, the termination condition — so the round with $k_r \geq \kappa^*$ terminates, and doubling gives $k_r \leq 2\kappa^*$. (Full proof in Section 7.6.) ■

Deployment adds two transfer steps, both exchange-flavored. Lemma 4.2 of the paper converts *local* risks (against the coreset F) into *global* risks:

$$\text{Risk}(\pi, Q; \tau^2) \leq \text{L-Risk}_F(\pi, Q; \tau) + \min_{\pi' \in F} \text{Risk}(\pi', Q; \tau),$$

so minimizing over $\pi \in F$ bounds the global τ^2 -risk by twice the best τ -risk in F . Lemma 4.6 of the paper is a Markov step: if C is an $(\alpha, \delta_1 \delta_2 |W)$ -coreset for neighborhood cost, then the neighborhood error-rate $N_{\alpha, C}(Q)$ — the U_Q -probability that no plan of C is within

² A *coreset* is a small surrogate that preserves the quality of the full object; here the object is the plan space and the quality is workload-wide risk coverage. κ^* is instance-dependent — small when a few plans already cover the workload — which is why the guarantee is stated relative to κ^* and not to N .

factor α of optimal — satisfies $\Pr_{Q \sim W}[N_{\alpha, C}(Q) > \delta_1] \leq \delta_2$, because $\mathbb{E}[N_{\alpha, C}(Q)] \leq \delta_1 \delta_2$ and Markov caps the tail. Together these give the paper’s Theorem 4.1: the inference step, which scores each coreset plan by its empirical local risk on $\Theta((\ln |F| + \ln \delta^{-1})/\epsilon^2)$ sampled profiles, returns with probability $1 - 2\delta$ a plan $\hat{\pi}$ with $\text{Risk}(\hat{\pi}, Q; \tau^2) \leq 2 \text{Risk}^*(Q; \tau) + 4\epsilon$.

What it buys. The optimizer precomputes once a plan set of size $O(\kappa^* \ln \delta^{-1})$ — logarithmic slack over the best possible — and every arriving query is answered by argmin over that set, in $\Theta(|F|(\ln |F| + \ln \delta^{-1})/\epsilon^2)$ time, with a certified worst-case robustness contract. The greedy cover inherits the set-cover factor; the Hoeffding and Markov layers only make the cover’s input faithful.

7.2 Cohesive subgraphs in temporal graphs, exactly

The exemplar is a SIGMOD’26 paper on temporal graph queries [7]. Temporal graphs record when each edge was alive, and analysts routinely ask for *cohesive subgraphs* — connected components, k -cores, connected k -cores — restricted to an arbitrary time window: “the community active between t_s and t_e .” Answering from scratch per window means recomputing a decomposition for every query, which is wasteful when windows overlap heavily. The paper instead builds a single index that answers such *historical* cohesive-subgraph queries for any window and any of the three models: an *affiliated forest* — essentially a Kruskal-style decomposition of the temporal edges, maintained by union–find with inverse-Ackermann amortization — plus a compact search structure over the resulting 3-dimensional intervals (BTAS). The headline: $O(m \log m)$ construction, $O(m)$ space, and $O(n + \log^2 m)$ per query, exactly, with no approximation anywhere.

Problem. A temporal graph $G = (V, E)$ gives each edge a lifespan $[t_1, t_2]$; the snapshot $G[t_s, t_e]$ keeps exactly the edges whose entire lifespan lies inside the window. A *cohesive subgraph model* (CSM) P maps each graph to a set of subgraphs; P is *non-overlapping* when any two subgraphs it returns have disjoint vertex sets, and *monotonic* when passing to a supergraph only grows the returned subgraphs. Lemma 2.3 of the paper verifies both properties for the three models of interest — the connected component, the k -core, and the connected k -core (the k -core case itself by a mini exchange argument: two overlapping k -cores union to a larger one, contradicting maximality, so the k -core is unique). A *Historical-CSM* query gives a window $[t_s, t_e]$ and asks for $P(G[t_s, t_e])$. Each edge becomes a *spanning edge* (u, v, t_1, t_2) of an affiliated graph, and a lemma of the paper (its Lemma 3.4) reduces Historical-CSM

queries for any non-overlapping, monotonic CSM to a unified, CSM-free *Spanning-CC* query on the affiliated graph: find all vertex sets that induce connected components of the windowed affiliated graph.

Guarantee.

Theorem 7.11 — AF-BTAS index; Lemmas 3.6, 3.12, and 3.13 of the paper, compressed. Let $G = (V, E)$ be a temporal graph with n vertices and m edges, and P a non-overlapping, monotonic CSM. The construction BuildAF produces, for every window $[x, y]$, a spanning forest $(V, F_{x,y})$ of the affiliated graph such that a spanning edge $e = (u, v, t_1, t_2)$ belongs to $F_{x,y}$ if and only if $0 \leq x \leq t_1$ and $t_2 \leq y \leq \tau(e)$, where $\tau(e)$ is the eviction time recorded for e (Lemma 3.6 of the paper); the construction is amortized near-linear through union-find (Lemma 3.8 of the paper). The index AF-BTAS built on the triples $(t_1, t_2, \tau(e))$ takes $O(m \log m)$ time and $O(m)$ space and answers a Spanning-CC query for any window in $O(n + \log^2 m)$ time. ■

How the exchange argument is applied. The rung is the graphic matroid: spanning forests are the independent sets, and the cycle rule — an edge that closes a cycle may replace a lighter edge of that cycle — is the exchange axiom. The paper’s AFDecomp sweeps the spanning edges once, in an order ω_1 by interval start; a union-find structure maintains the current forest S . When the arriving edge $e_i = (u, v, t_1, t_2)$ connects two vertices already connected, it closes a cycle: the algorithm finds the ω_2 -minimal edge $e_{\min}$ on that cycle (the one whose usefulness ends earliest) and *exchanges* — if $\omega_2(e_i) > \omega_2(e_{\min})$ then $S \leftarrow (S \cup \{e_i\}) \setminus \{e_{\min}\}$ and the eviction time is recorded as $\tau(e_{\min}) = t_2 - 1$; otherwise e_i is discarded. Every swap is Kruskal’s rule applied to a different weight order, and Lemma 3.6 is precisely the invariant the swaps preserve: an edge survives in exactly those windows $[x, y]$ that contain its lifespan and end before its eviction. Edges never evicted get $\tau(e) = t_{\max}$. Correctness of the decomposition is proved by induction along the sweep (the paper’s proof of its Lemma 3.10 shows the running forest equals each $F_{0,y}$ in turn), and monotonicity of the CSM guarantees supersets of u – v windows stay u – v windows (Lemma 3.2 of the paper). The $\alpha(\cdot)$ -amortized union-find checks make the whole sweep near-linear, and the resulting triples $(t_1, t_2, \tau(e))$ are answered by BTAS — a binary tree whose nodes carry priority search trees for the 3-sided query $x \leq t_s \leq y \leq t_e \leq z$ — in $O(\log m)$ recursion rounds of $O(\log m)$ each plus output size.

³ The exchange here earns no approximation factor because nothing is approximated: the cycle swap maintains a *maximum* forest for every window simultaneously, and the certificate is completeness — the index answers every query exactly. This is the matroid rung of Section 7.0.1 deployed as a data structure.

What it buys. One $O(m \log m)$ preprocessing step turns any-window cohesive-subgraph queries into output-sensitive $O(n + \log^2 m)$ lookups, replacing per-query recomputation. The same index serves all three CSMs because the construction uses only non-overlap and monotonicity, and the union-find amortization keeps the certificate cheap to earn.

7.3 Representative sampling for large-scale vector data

The exemplar is a SIGMOD'26 paper on sampling for vector data [10]. Graph-based nearest-neighbor indexes navigate a dataset greedily — beam search hops from anchor to anchor toward the query — so everything depends on the anchors being well placed: globally spread, yet locally dense enough that navigation does not stall. The paper studies *representative sampling* under a budget: choose a subset of data points that balances a global coverage constraint against local fidelity, where fidelity is measured through the local intrinsic dimension (LID) of each point's neighborhood. It proves the exact selection problem NP-hard, then designs a fidelity objective F built from per-point gains and shows that (lazy-)greedy under the budget returns F at least $1 - 1/e$ of optimal. A companion analysis shows the same anchor densities that F rewards also make greedy and beam navigation progress with high probability, via $(1 - p)^m \leq e^{-mp}$ bounds.

Problem. Given a dataset of n vectors, a sampling ratio ρ (budget $B = \lfloor n\rho \rfloor$) and a coverage budget $C_{\max}$, Problem 1 of the paper (*Representative Sampling*) asks for a sample A of at most B anchors maximizing a local-fidelity objective $F(A) = \sum_s g_s(m_s(A))$ subject to a coverage constraint $\text{Cost}_{\text{cov}}(A) = \frac{1}{n} \sum_x \min_{a \in A} \text{dist}(x, a) \leq C_{\max}$, where $m_s(A)$ counts the selected anchors in point s 's neighborhood ball $B(s)$ and g_s is s 's per-anchor fidelity gain. The problem is intractable exactly: Theorem 2 of the paper proves Problem 1 NP-hard by a parsimonious reduction from metric k -median — set $\rho \leftarrow k/n$ so the budget forces $|A| = k$, and $C_{\max} \leftarrow T/n$ so the coverage constraint is the k -median objective — so a feasible sample of size k within budget is precisely a k -median answer.

Guarantee.

Theorem 7.12 — Lazy-greedy representative sampling; Theorem 3 of the paper. Let B be the budget and A_{gre} the size- B output of the (lazy-)greedy algorithm (Algorithm 2 of the paper). Then

$$F(A_{\text{gre}}) \geq \left(1 - \frac{1}{e}\right) \max_{|A| \leq B} F(A). \quad \blacksquare$$

How the submodular greedy is applied. The exchange structure is monotone submodularity, and it is earned, not assumed. The per-point gains come from an estimation analysis: Lemma 1 of the paper represents the LID estimator at s as $D \cdot (\sum_{j \leq m} Y_j)^{-1}$ with $Y_j \sim \text{Exp}(1)$ i.i.d. (so $Z = \sum_j Y_j \sim \Gamma(m, 1)$), and its gamma-sum moment identities give closed forms — mean $D m / (m - 1)$, variance $D^2 m^2 / [(m - 1)^2 (m - 2)]$ — for the estimator’s error at anchor count m . Lemma 4 of the paper shows the resulting mean-squared error $\text{MSE}_s(m)$ is strictly decreasing in $m \geq 3$ with decrements $\Delta g_s(m) = \text{MSE}_s(m) - \text{MSE}_s(m + 1)$ positive and decreasing, so the cumulative gain $g_s(m) = \sum_{r < m} \Delta g_s(r)$ is non-decreasing and concave. That concavity is exactly the submodularity hypothesis:

Lemma 7.13 — Submodularity of the fidelity objective; Lemma 5 of the paper, notation cleaned. Let $F(A) = \sum_s g_s(m_s(A))$ with each g_s nondecreasing and concave. Then $F: 2^U \rightarrow \mathbb{R}_{\geq 0}$ is nondecreasing and submodular. ■

Proof sketch. The marginal gain of adding u to A is

$$\Delta F(A; u) = \sum_{s: u \in B(s)} [g_s(m_s(A) + 1) - g_s(m_s(A))],$$

which is nonnegative; if $A \subseteq A'$ then $m_s(A) \leq m_s(A')$, so discrete concavity of g_s makes each bracket at A' no larger than at A — diminishing returns. ■

With Lemma 7.13 in hand, Theorem 7.12 is the Nemhauser–Wolsey–Fisher rung (Theorem 7.6) verbatim: the greedy pick captures at least $1/B$ of the remaining gap, $F(A^*) - F(A_i) \leq (1 - \frac{1}{B})(F(A^*) - F(A_{i-1}))$, and unrolling B steps gives e^{-1} (full proof in Section 7.6); the lazy implementation re-evaluates marginals only when stale, preserving the selected sequence and the factor. Finally, the paper ties the certificate to navigation: Lemma 3 of the paper bounds the probability that none of m i.i.d. neighbors falls in a spherical cap by $(1 - \mu)^m \leq e^{-m\mu}$, whence its Theorem 1 lower-bounds the one-layer progress probability by $1 - \exp(-\alpha m_s(s) p_{\text{geom}})$ — more anchors per region (larger m_s , which is what F rewards) means exponentially smaller stall probability for both greedy ($b = 1$) and beam ($b \geq 1$) expansion.

What it buys. A one-page certificate turns a selection problem the reduction just declared hopeless into a $(1 - 1/e)$ -certified sampler with lazy acceleration, while the coverage constraint keeps global spread. The moment analysis does double duty: it justifies the *objective* (why these gains), not just the algorithm, and the $(1 - p)^m \leq e^{-mp}$ step connects the certified sample to what actually happens at query time.

7.4 Symmetry breaking for maximum common subgraph

The exemplar is a SIGMOD'26 paper on exact graph matching [5]. The maximum common induced subgraph (MCIS) problem — find the largest vertex sets of two graphs whose induced subgraphs are isomorphic — is the fundamental model of structural similarity between graphs, and it is NP-hard. Exact solvers are branch-and-bound searches that incrementally build vertex correspondences and prune infeasible or suboptimal branches; on symmetric inputs (regular graphs, repeated motifs) the search tree explodes into branches that explore the same subproblem over and over. The paper accelerates the search by detecting *modular symmetry* — pairs of vertices with identical neighborhoods — in $O(n^2)$ time, and breaking variable and value symmetries during the search, with proofs that the breaking loses no solutions. The payoff is exact answers with orders-of-magnitude fewer branches.

Problem. MCIS (Definition 2.1 of the paper): given graphs G and H , find induced subgraphs $G[V']$ and $H[U']$ that are isomorphic with $|V'| = |U'|$ maximized. The solver builds a vertex mapping incrementally; Definition 3.1 of the paper demands two properties of any acceleration: *correctness* (only mappings between isomorphic induced subgraphs are deemed valid) and *completeness* (at least one mapping of maximum cardinality is found). Call vertices u, v *positively symmetric* when their open neighborhoods coincide, $N^+(u) = N^+(v)$ (adjacent twins), and *negatively symmetric* when $N^-(u) = N^-(v)$ (non-adjacent twins); Theorem 3.1 of the paper (mutual exclusiveness) shows no vertex can be both. Symmetry *breaking* prunes branches that try equivalent values in a different order.

Guarantee.

Lemma 7.14 — Modular symmetry is an automorphism; Lemma 1 of the paper. Two vertices $u, v \in V$ are modular symmetric (positively or negatively) if and only if the permutation that swaps u and v is an automorphism of G . ■

Theorem 7.15 — Symmetry breaking is safe; Theorems 3.2 and 3.3 of the paper, compressed. Variable symmetry breaking and value symmetry breaking do not violate the correctness or the completeness of the MCIS solution. ■

How the swap argument is applied. The greedy commitment is the *fixed branching order*: variables in one fixed order, values in another. The exchange shows every pruned branch has a symmetric branch that the search considers strictly earlier. Transcribed from the paper's proof:

suppose a branch b with partial mapping M is pruned because of a conflict between the correspondence pairs (v_1, u_2) and (v_2, u_1) . The swap

$$(v_1, u_2), (v_2, u_1) \mapsto (v_1, u_1), (v_2, u_2)$$

creates a symmetric mapping M' on a hypothetical branch b' — hypothetical because b' may itself have been pruned. But the search tree branches on all values of v at the same node in the same fixed order, so b' was considered *before* b ; repeated swapping produces earlier and earlier branches, and since the branch order is well founded, the descent must terminate at a branch the search actually explores. Every swap preserves cardinality and isomorphism because, by Lemma 7.14, it is the application of an automorphism — a rearrangement of the *same* solution. Hence some maximum mapping survives all pruning (completeness), and survivors are genuine isomorphisms (correctness). There is no cost delta to count; the certificate is exactness — the pruning is free. The $O(n^2)$ detection (Lemma 2 of the paper) hashes each vertex's neighborhood, so finding the exchange partners costs less than using them saves.

What it buys. Exact MCIS answers with a fraction of the explored branches, and — because Theorem 7.15 is a soundness theorem — the speedup is free: no approximation is paid for the pruning. The exchange partner is manufactured wholesale by the automorphism group, turning group symmetry into a completeness certificate for exact search.

7.5 Advanced usage and further reading

Two further corpus papers push the pattern in opposite directions: one runs the averaging step in weighted form inside an LLM-era retrieval pipeline, the other shows what an honest paper does when no structure checks out and the greedy carries no certificate at all.

Skyline retrieval meets set-cover chunk merging [11]. Long-context LLM question answering over external documents first retrieves, then merges, text *chunks* into a compact “RAG-sketch.” The paper decomposes queries into meta-queries, scores chunks on two dimensions (keyword presence and semantic relevance), takes a skyline per meta-query, and arrives at the merge step: choose a minimum-token-cost set of chunks covering all meta-queries. Theorem 3.8 of the paper proves this merge problem NP-hard by a polynomial-time, cost-preserving reduction from Weighted Set Cover — treat each chunk as the set of meta-queries it covers, with its token count as the weight; the forward and backward

maps preserve coverage and cost exactly. The algorithm then merges greedily by gain-per-cost $G_{t+1}(C) = |Q(C) \cap R_t|/T(C)$ against the residual meta-query set R_t , and Lemma 3.10 of the paper (efficiency) is the averaging step in weighted form: for every t , $G_{t+1}(C_{t+1}) \geq r_t/\text{OPT}$, because an optimal cover's sets jointly cover the r_t residual meta-queries at total cost OPT , so one of them covers at that rate — and greedy picks at least as well. Unrolling as in Theorem 7.7 gives the logarithmic guarantee, and an independence argument lower-bounds the expected newly answered queries by $p_{\min}$ per newly covered meta-query.

Stress-testing causal claims via cardinality repairs [4]. Given a causal claim — an average treatment effect (ATE) of T on O computed over a database — the paper asks how fragile it is: find a minimum set of tuples whose deletion moves the ATE into a target range $[\text{ATE}_d - \varepsilon, \text{ATE}_d + \varepsilon]$ (or a minimal pattern defining such a subpopulation). The optimization problems are NP-complete for AVG and ATE aggregates (Propositions of the paper), by a reduction from SUBSET-SUM: for each number x create tuples $t_x^1 = (T=1, O=x)$ and $t_x^2 = (T=0, O=0)$, add one anchor tuple $(1, -k)$, set the target to 0 — deleting exactly the subset of t_x^1 tuples summing to k zeroes the estimate. A lemma of the paper shows every tuple set is the extension of some pattern, tying the pattern variant to the tuple variant. The algorithms (SubCure-tuple, SubCure-pattern) are then influence-scored greedy heuristics: score each tuple by its influence $\text{influence}(t) = \text{ATE} - \text{ATE}_{\setminus\{t\}}$, delete the most influential, recompute periodically, sample to scale. No approximation factor is claimed — influence is not diminishing, submodularity fails, and the paper says so, verifying the final ATE exactly instead. The contrast with Sections 7.1 and 7.3 is the lesson: without checkable structure, hardness motivates the greedy but certifies nothing.

7.5.1 Further reading

- Edmonds [3] — matroids and the greedy algorithm; the exact rung and the exchange axiom, in four pages.
- Nemhauser, Wolsey, and Fisher [8] — the $1 - 1/e$ analysis that every submodular corpus paper silently re-runs.
- Chvátal [2] — the set-cover charging argument; the averaging step in its original home.
- Arya et al. [1] — swap-based local search for k -median and facility location; the canonical metric exchange.
- Krause and Golovin [6] — the survey of submodular maximization; lazy greedy, curvature, and non-monotone variants live here.

Chapter Recap

- **What the tool is.** Exchange arguments certify greedy choices by rearranging any better solution to agree with them, one swap at a time: name the exchange partner, count the swap's cost delta, inherit the certificate — exact, $1 - 1/e$, $\ln$, $3 + \varepsilon$, or complete.
- **Which rung when.** Matroid independence: exact greedy (Kruskal; the temporal-forest index of Section 7.2). Monotone submodular under a budget: $1 - 1/e$ (representative sampling, Section 7.3). Covering constraint: $\ln$ -factor greedy (risk coresets, Section 7.1; chunk merging in Section 7.5). Metric distances: swap local search at $3 + \varepsilon$. Symmetry group: pruning stays complete (Section 7.4).
- **Why it works.** Three swap accountings: rearrangement (delta ≥ 0 , greedy optimal), contraction (delta $\geq \text{gap}/B$, compounding to e^{-1}), averaging (delta $\geq \text{residual}/\text{OPT}$, compounding to $H(n)$); completeness certificates add a well-founded descent — each pruned branch swaps to a strictly earlier one.
- **What the corpus showed.** Risk coresets (Section 7.1): greedy coverage averaging against the optimal coreset, exponential residual decay, $k_r \leq 2\kappa^*$. Temporal cohesive subgraphs (Section 7.2): Kruskal cycle exchanges maintain every window's forest exactly, union-find amortized. Representative sampling (Section 7.3): concave-gain submodularity plus contraction-unrolling at $1 - 1/e$. MCIS symmetry breaking (Section 7.4): automorphism swaps map pruned branches to earlier branches — pruning certified complete.
- **Advanced usage.** Weighted set-cover merging with a per-step averaging bound (Section 7.5), and an honest certificate-free greedy when no structure checks out. Deferred proofs are collected in the Appendix (Section 7.6).

7.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 7.4 (matroid greedy)

Statement (recalled). On a matroid $(U, \mathcal{I})$, the greedy that scans elements in nonincreasing weight order and keeps each element preserving independence returns a maximum-weight independent set. ■

Proof. Let $G = (g_1, \dots, g_k)$ be greedy's output with weights nonincreasing, and $O = (o_1, \dots, o_m)$ a maximum-weight independent set, also sorted by nonincreasing weight. First, all maximal independent sets of a matroid have the same size, so $k = m$. Second, suppose for contradiction that $w(g_i) < w(o_i)$ for some i , and take the smallest such i . The prefix $A = \{g_1, \dots, g_{i-1}\}$ is independent, and so is $B = \{o_1, \dots, o_i\}$ (heredity), with $|B| = i > |A|$. By the exchange axiom some $x \in B \setminus A$ has $A \cup \{x\}$ independent, and $w(x) \geq w(o_i) > w(g_i)$. But then at step i the element x was still available, kept independence with the prefix, and outweighed g_i — so greedy would have taken it. Contradiction. Hence $w(g_i) \geq w(o_i)$ for all i , and $w(G) \geq w(O)$. (The converse — exactness for every weight function forces the exchange axiom — is Edmonds' characterization; see [3].) ■

Proof of Theorem 7.6 (submodular $1 - 1/e$)

Statement (recalled). If f is monotone submodular with $f(\emptyset) = 0$ and A_B is greedy's set after B steps, then $f(A_B) \geq (1 - 1/e) \max_{|A| \leq B} f(A)$. ■

Proof. Fix an optimal A^* with $|A^*| \leq B$ and write A_i for greedy's set after i steps. *Fundamental inequality:* adding A^* 's elements to A_i one at a time telescopes to $f(A^*) - f(A_i) \leq \sum_{u \in A^* \setminus A_i} \Delta(u \mid A_i)$, since each marginal along the telescoping is at most the marginal at the sparser set A_i (submodularity). *Greedy captures the average:* the sum has at most B terms, each at most the greedy pick's marginal, so

$$f(A_i) - f(A_{i-1}) \geq \frac{1}{B} \sum_{u \in A^* \setminus A_{i-1}} \Delta(u \mid A_{i-1}) \geq \frac{f(A^*) - f(A_{i-1})}{B},$$

which rearranges to the one-step contraction $f(A^*) - f(A_i) \leq (1 - \frac{1}{B})(f(A^*) - f(A_{i-1}))$. Unrolling from $A_0 = \emptyset$,

$$f(A^*) - f(A_B) \leq \left(1 - \frac{1}{B}\right)^B f(A^*) \leq e^{-1} f(A^*),$$

using $(1 - 1/B)^B \leq e^{-1}$. Rearranging gives the theorem. ■

Proof of Theorem 7.7 (set-cover greedy)

Statement (recalled). The greedy that repeatedly buys the set maximizing newly-covered elements per unit weight returns a cover of total weight at most $H(n) \cdot \text{OPT}$. ■

Proof. When greedy buys set C_t at weight $w(C_t)$ covering d_t new elements, charge each newly covered element e its price $\text{price}(e) = w(C_t)/d_t$; the total weight equals the sum of prices. Let R_t be the residual (uncovered) set before step t , $r_t = |R_t|$. The optimal cover

restricted to R_t covers every residual element at total weight at most OPT , so some optimal set C satisfies $|C \cap R_t|/w(C) \geq r_t / \text{OPT}$; greedy maximizes newly-covered-per-unit-weight, so its chosen set satisfies it too, and every element covered at step t pays at most OPT / r_t . Now order all elements $e_1, \dots, e_n$ by the step at which they were covered. When e_j is covered, at least $n - j + 1$ elements are still uncovered (those covered at or after e_j 's step, including e_j), so $\text{price}(e_j) \leq \text{OPT} / (n - j + 1)$. Therefore

$$\sum_{j=1}^n \text{price}(e_j) \leq \text{OPT} \sum_{j=1}^n \frac{1}{n - j + 1} = H(n) \cdot \text{OPT}. \quad \blacksquare$$

Proof of Lemma 7.10 (greedy cover termination)

Statement (recalled). Under the estimation conditions of Lemmas 3.1 and 3.2 of the paper, the greedy cover terminates with $k_r \leq 2\kappa^*$. $\blacksquare$

Proof. By definition of a coreset there is $M \subseteq \Pi$ with $|M| = \kappa^*$ and $p(\varepsilon, M) \leq \delta$. Lemma 3.1 of the paper (with $\varepsilon' = \varepsilon$) lifts this to the empirical estimate, $\hat{p}_r(\varepsilon, M) \leq \delta + \delta/4$, so on at least a $(1 - 5\delta/4)$ -fraction of the n_r sampled queries, $\min_{\pi \in M}$ of the proxy risk is within 2ε of Risk^* (Lemma 3.2 of the paper transfers the risk estimates): M covers at least $n' = (1 - 5\delta/4)n_r$ sampled queries under the algorithm's coverage test. Let $D_t = n' - |C_t|$ be the residual with respect to the coverable set after t greedy rounds. The κ^* plans of M jointly cover n' queries, so at round $t + 1$ one of them covers at least D_t / κ^* of the residual; greedy maximizes marginal coverage, hence

$$D_{t+1} \leq \left(1 - \frac{1}{\kappa^*}\right) D_t \leq \exp\left(-\frac{1}{\kappa^*}\right) n'.$$

For $t \geq c_3 \kappa^* \ln(\delta^{-1})$ with c_3 a suitable constant and $\delta \leq 1/2$, the covered fraction reaches $1 - 2\delta$, which is exactly the termination condition — so the first round with guess $k_r \geq \kappa^*$ terminates. Since the guess doubles each round, that round has $k_r \leq 2\kappa^*$. $\blacksquare$

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8

Spectral and Matrix-Analytic Arguments

Spectral and matrix-analytic arguments replace a sprawling object — a graph, a recursion, a point set — with the few numbers that govern its behavior under repetition: eigenvalues, singular values, spectral radii, eigengaps. Find the linear map inside the artifact — a random walk applies a transition matrix, a smoothness objective is a quadratic form — then read the answer off the spectrum: does iteration converge, how fast, and which subspace carries the object?

In the SIGMOD'26 corpus, 13 papers are tagged with this machinery, pairing it with graph analytics (spectral clustering of multi-relational graphs, exact SimRank-style similarity), high-dimensional vector search (subspace projections and collision counting), and fairness pre-processing through latent attributes — plus, a reminder that the tool respects no domain boundaries, the space utilization of B-tree leaves under batched insertions. The chapter builds the ladder (Section 8.0), dissects four exemplars (Sections 8.1–8.4), and closes with advanced sketches and further reading (Section 8.5).

8.0 The tool: spectral and matrix arguments

The problem class is uniform: a square matrix A sits inside the system as the transition applied by one update step, the quadratic form being minimized, or the structure behind a vectorized equation; the deliverable — limits, solvability, subspaces, distortion — is decided by where the eigenvalues of A sit (Figure 8.1).

Notation

$A \in \mathbb{R}^{d \times d}$; for symmetric A , eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$. Operator norm $\|A\|_2$, Frobenius norm $\|A\|_F$, spectral radius $\rho(A) = \max_i |\lambda_i(A)|$; $A \geq 0$ is entrywise. $G[A]$: an edge $i \rightarrow j$ iff $A_{ij} \neq 0$;

LEARNING OBJECTIVES

- Recognize when a system's state is a matrix in disguise: transitions, affinities, recursions, Gram matrices.
- Prove the ladder Rayleigh $\rightarrow$ Perron–Frobenius $\rightarrow$ Neumann $\rightarrow$ interlacing $\rightarrow$ low rank/projection.
- Turn spectral facts into guarantees: rates, invertibility, subspace optimality, distance preservation.
- Name the matrix, localize the spectrum, exploit the gap or the rank.

A is *irreducible* if $G[A]$ is strongly connected, *Metzler* if its off-diagonal entries are nonnegative. $A \otimes B$ is the Kronecker product, $A \oplus B = (A \otimes I) + (I \otimes B)$ the Kronecker sum, $\text{Vec}(\cdot)$ stacks columns.

8.0.1 The basic ladder

Eigenvalues are the extreme values of a scalar function of the direction.

Theorem 8.1 — Rayleigh quotient. Let $A = A^\top \in \mathbb{R}^{d \times d}$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$ and orthonormal eigenvectors $v_1, \dots, v_d$. For the Rayleigh quotient $R(x) = \frac{x^\top A x}{x^\top x}$, $x \neq 0$,

$$\lambda_1 = \max_{x \neq 0} R(x), \quad \lambda_d = \min_{x \neq 0} R(x). \quad \blacksquare$$

Proof. $R(x) = \sum_i \lambda_i c_i^2 / \sum_i c_i^2$ for $x = \sum_i c_i v_i$ is a convex combination of the eigenvalues — at most λ_1 , at least λ_d , with equality at $x = v_1, v_d$. $\blacksquare$

Most matrices in database papers are not symmetric but *nonnegative* — counts, probabilities, affinities — and there a structure theorem applies [3, 4]:

Theorem 8.2 — Perron–Frobenius. Let $A \geq 0$ be irreducible. Then $\rho(A) > 0$ is an eigenvalue of A ; it is simple; it admits entrywise positive left and right eigenvectors; and every other eigenvalue $\lambda \neq \rho(A)$ satisfies $\text{Re}(\lambda) < \rho(A)$. $\blacksquare$

Proof sketch. Irreducibility makes A eventually map the nonnegative cone into its interior, so $x \mapsto Ax / \|Ax\|_1$ has an interior simplex fixed point — a positive eigenvector for $\rho(A)$; a rank-one positive perturbation shows the eigenvalue is simple and dominates in real part [4]. $\blacksquare$

The B-tree exemplar ([1]; Section 8.1) has a negative diagonal instead — a shift by cI fixes that:

Corollary 8.3 — Metzler variant, Lemma of [1], notation cleaned. Let A be an irreducible Metzler matrix admitting an entrywise positive w with $w^\top A = r w^\top$, $r > 0$. Then r is a simple eigenvalue with a positive right eigenvector, and every other eigenvalue λ satisfies $\text{Re}(\lambda) < r$. $\blacksquare$

Proof. $B = A + cI \geq 0$ (with $c = \max_i |A_{ii}|$) is irreducible, and $w^\top B = (r + c) w^\top$ is a positive left eigenvector, which Perron–Frobenius (Theorem 8.2) assigns to $\rho(B) = r + c$: simple, positive right eigenvector, dominant in real part. Since $\sigma(B) = \sigma(A) + c$, the claims transfer. $\blacksquare$

The second rung turns spectral location into a rate: a spectrum strictly inside the unit disk makes the inverse of $I - A$ a geometric series [4].

Theorem 8.4 — Neumann series with truncation rate. Let $\rho(A) < 1$. Then $I - A$ is invertible with $(I - A)^{-1} = \sum_{k \geq 0} A^k$, and for any submultiplicative norm with $\|A\| \leq q < 1$ the partial sum $S_L = \sum_{k=0}^L A^k$ obeys

$$\|(I - A)^{-1} - S_L\| \leq \frac{q^{L+1}}{1 - q}. \quad \blacksquare$$

Proof. $(I - A)S_L = I - A^{L+1}$ and $\|A^{L+1}\| \leq q^{L+1} \rightarrow 0$; the tail is the geometric series $\sum_{k > L} q^k$. $\blacksquare$

The third rung governs *subspaces*: compressing a symmetric matrix to a smaller orthonormal basis sandwiches the eigenvalues between those of the original, making “which eigenvectors do I keep” a well-posed optimization [2, 4].

Theorem 8.5 — Courant–Fischer and Poincaré separation. (Courant–Fischer) For symmetric A with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$,

$$\lambda_k(A) = \min_{\substack{V \subseteq \mathbb{R}^d \\ \dim V = d - k + 1}} \max_{0 \neq x \in V} R(x) = \max_{\substack{V \subseteq \mathbb{R}^d \\ \dim V = k}} \min_{0 \neq x \in V} R(x).$$

(Poincaré separation) If $B \in \mathbb{R}^{d \times s}$ has orthonormal columns and $\mu_1 \geq \dots \geq \mu_s$ are the eigenvalues of $B^\top AB$, then $\lambda_i(A) \geq \mu_i \geq \lambda_{d-s+i}(A)$, $i = 1, \dots, s$. $\blacksquare$

Proof sketch. Courant–Fischer: the max–min is attained on the top k eigenvectors’ span, and no k -dimensional subspace guarantees more (exchange argument). Poincaré: $B^\top AB$ is R restricted to $\text{span}(B)$; Courant–Fischer inside that subfamily sandwiches each μ_i . Full proof in Section 8.6. $\blacksquare$

The fourth rung certifies throwing information away: the residual of a truncated decomposition is governed exactly by the first discarded singular values [4, 6, 11].

Theorem 8.6 — Eckart–Young–Mirsky. Let $A \in \mathbb{R}^{m \times n}$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ and singular vectors u_i, v_i . Then for every B of rank at most r ,

$$\|A - B\|_2 \geq \sigma_{r+1}, \quad \|A - B\|_F \geq \left(\sum_{i > r} \sigma_i^2 \right)^{1/2},$$

and the truncated SVD $A_r = \sum_{i \leq r} \sigma_i u_i v_i^\top$ attains both bounds. $\blacksquare$

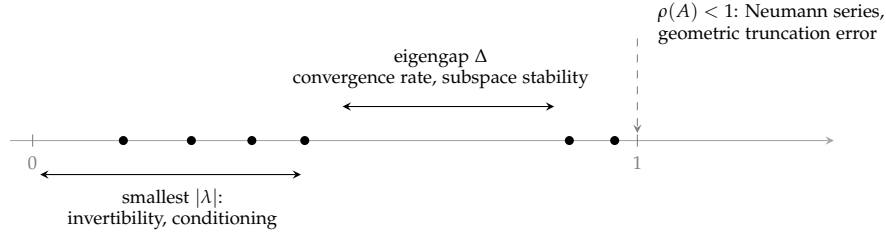


Figure 8.1: Reading a spectrum: distance from 0 certifies invertibility; a spectral radius below 1 licenses a Neumann expansion; a large eigengap Δ fixes rates and subspace stability.

Proof sketch. $\|A - A_r\| = \sigma_{r+1}$ by inspection of the tail; for the lower bound, the span of the top $r + 1$ right singular vectors meets the kernel of any rank- $\leq r$ B , and on a unit x there $\|(A - B)x\|_2 = \|Ax\|_2 \geq \sigma_{r+1}$. Full proof (both norms) in Section 8.6. ■

The last rung makes distances cheap: a random low-dimensional projection is an approximate isometry, with dimension set by accuracy and the number of events to survive [6, 11].

Theorem 8.7 — Distributional Johnson–Lindenstrauss. Let $A \in \mathbb{R}^{d \times D}$ consist of the first d rows of a uniformly random orthogonal $D \times D$ matrix. Then for every fixed $x \in \mathbb{R}^D$, $\mathbb{E}\|Ax\|_2^2 = \frac{d}{D}\|x\|_2^2$, and for $\varepsilon, \beta \in (0, 1)$ with $d \geq C\varepsilon^{-2} \ln(1/\beta)$,

$$\Pr \left[\left| \|Ax\|_2^2 - \frac{d}{D}\|x\|_2^2 \right| > \varepsilon \frac{d}{D}\|x\|_2^2 \right] \leq \beta. \quad \blacksquare$$

Proof sketch. A random orthogonal map takes $x/\|x\|$ to a uniformly random direction, so $\|Ax\|_2^2/\|x\|_2^2$ is distributed like the squared length of the first d coordinates of a uniform point on the sphere — approximately χ_d^2/d , whose concentration gives failure probability $2e^{-\Theta(\varepsilon^2 d)}$; full proof in Section 8.6. ■

8.0.2 How to choose the matrix lens

The selection is mechanical. *What are the entries?* Transition counts or probabilities point at Perron–Frobenius; an optimized quadratic form at Rayleigh/Courant–Fischer; a huge structured linear system at spectrum localization plus Neumann; fast-decaying singular values at low-rank truncation; expensive distances at random projection. *What property is needed?* Existence needs the spectrum clear of zero; a rate needs a radius or gap; a subspace recommendation needs interlacing; $(1 \pm \varepsilon)$ distances need the JL dimension. *Not symmetric?* Perron–Frobenius, Schur triangularization, or singular values.

When to Reach for Spectral Arguments

Reach whenever the object is governed by repetition — random walks, recursive similarity, iterative updates, insertion sequences — or a combinatorial objective (clustering, subspace selection) hides a quadratic form. Then run the recipe:

1. **Name the matrix:** write down the linear map that one step (one batch, one iteration, one hop) applies, and what its entries mean.
2. **Check the structure:** nonnegative and irreducible $\rightarrow$ Perron–Frobenius (Metzler variant if the diagonal is negative); symmetric $\rightarrow$ Rayleigh/interlacing; contraction ($\rho < 1$, or eigenvalues in $[0, 1]$) $\rightarrow$ Neumann; fast singular-value decay $\rightarrow$ truncated SVD; expensive distances $\rightarrow$ random projection.
3. **Localize the spectrum:** row sums, Gershgorin discs, stochasticity, or the Kronecker-sum identity $\lambda(A \oplus B) = \lambda_i(A) + \lambda_j(B)$ — place the spectrum relative to 0 and 1 and bound the dominant gap.
4. **Convert to the guarantee:** radius $\rightarrow$ geometric rate; gap $\rightarrow$ subspace stability and stopping rule; singular tail $\rightarrow$ rank- r error; projection dimension $\Theta(\varepsilon^{-2} \log(\text{events}/\delta))$, union-bounded over all points preserved.

8.1 B-tree fragmentation under batched insertions

The exemplar is a PODS’26 paper on the space utilization of B-trees [1]^{PODS}. When a B-tree leaf fills up and splits, the new leaves start out on average half empty — underfull leaves mean more cache misses and rebalancing downstream. Yao’s seminal result shows that even splitting against uniformly random single-key insertions achieves about 69% space utilization. Many applications, however, insert in *batches* — runs of consecutive keys at one position — a regime Yao’s fringe analysis does not cover. The paper generalizes fringe analysis to batched workloads: even splitting provably keeps working for many of them, and simple alternatives cover the remaining regimes.

Problem. A B-tree stores keys in leaves (blocks) of capacity B . The workload is random batched insertions: each batch is r consecutive keys inserted at a uniformly random position, and a *split strategy* decides how an overflowing block is partitioned. *Fringe analysis* tracks the histogram of block occupancies — after n keys, the vector $x^n = (x_k^n)$ counts blocks of each size k — and asks for the *expected final fullness*,

the limit of average block size normalized by B .

Guarantee.

Theorem 8.8 — Deferred even split fullness, Lemma of [1], notation cleaned. Let B and i be positive integers with batch size $r \in (B/(2i), B/(2i-1)]$. Under the deferred even split strategy (a block is split evenly only when it overflows), the expected final fullness is

$$\frac{2ir}{B} (H_{2i} - H_i), \quad H_k = 1 + \frac{1}{2} + \cdots + \frac{1}{k}. \quad \blacksquare$$

¹ Sanity checks: $r = B$ ($i = 1$) gives fullness 1; $r = 1$ ($i \approx B/2$) gives $\approx \ln 2 \approx 0.693$ — exactly Yao's classical 69%.

How spectral analysis is applied. The named object is the expectation vector $z_n = \mathbb{E}[x^n]$. Since one batch is linear on the histogram, z_n satisfies an exact *matrix recurrence*

$$z_{n+r} = \left(I + \frac{1}{n} A(B, r) \right) z_n,$$

where the k -th column of $A(B, r)$ records the net histogram change when a batch of r keys hits a block of size k . $A(B, r)$ is Metzler, and on the class S of reachable block sizes (a cycle modulo r in the pattern graph) the principal submatrix $A(B, r)_S$ is irreducible: $w_k = k$ satisfies $w^\top A(B, r)_S = r w^\top$ (a batch always adds r keys), so Corollary 8.3 makes r a simple dominant eigenvalue with positive eigenvector. Convergence of the recurrence is the workhorse.

Theorem 8.9 — Convergence of the matrix recurrence, Theorem of [1], notation cleaned. Let $v_{n+r} = (I + \frac{1}{n} A) v_n$ for $n \geq n_0$, where $r \in \sigma(A)$ is a simple positive eigenvalue dominating in real part. Then

$$\lim_{m \rightarrow \infty} \frac{v_{n_0+mr}}{n_0+mr} = P_r^A \left(\frac{v_{n_0}}{n_0} \right),$$

where P_r^A is the spectral projection onto the generalized eigenspace of r . ■

Proof sketch. Iterating, $v_{n_0+mr} = (\prod_{k=0}^{m-1} (I + \frac{1}{n_0+kr} A)) v_{n_0}$. In Jordan form the mode for λ with block size d evolves as scalars $(1 + \lambda/(n_0 + kr))$ times a nilpotent part; the product bound (Lemma 8.10) shows the dominant mode grows like m , every other like $o(m)$ or $(\ln m)^d$ — only the dominant spectral projection survives after division by m . ■

Lemma 8.10 — Scalar product bound, Lemma of [1], notation cleaned. For positive integers n_0, r and $\lambda \in \mathbb{C}$,

$$\left| \prod_{k=0}^{m-1} \left(1 + \frac{\lambda}{n_0 + kr} \right) \right| \leq m^{\operatorname{Re}(\lambda)/r} \cdot \alpha_{\lambda, n_0, r},$$

for a constant $\alpha_{\lambda, n_0, r}$ independent of m . ■

Proof sketch. Pointwise $|1 + t| \leq \exp(\operatorname{Re}(t) + \frac{1}{2}|t|^2)$; summing over k , $\sum_k \operatorname{Re}(\lambda/(n_0 + kr)) = \frac{\operatorname{Re}(\lambda)}{r} \ln m + O(1)$ (integral comparison) and $\sum_k |\lambda|^2/(n_0 + kr)^2 = O(1)$. Full proof in Section 8.6. ■

The last step is pure *eigenvector arithmetic*. By Theorem 8.9 the normalized histogram converges into the eigenspace of r , and a lemma of the paper pins the limit of the expected average block size to $\langle u, w \rangle_S / \langle u, 1 \rangle_S$ for *any* eigenvector u — a weighted average over any partition of S , hence at least the minimum class ratio. Three partitions give three bounds, each favorable in its regime ($r < 0.0058B$, $r < 0.21B$, $0.21B \leq r < 0.5B$), and the eigenvector recurrence $u_k = \frac{k-r}{k+r} u_{k-r}$ (Lemma of the paper) yields the $H_{2i} - H_i$ of Theorem 8.8; the knob r/B selects even, uneven, or the paper’s TargetSplit for long batches.

What it buys. The storage engine gets provable utilization numbers per batch-size regime — even splitting near-optimal for small r , uneven splits above 7/12 fullness, TargetSplit exact for long batches — all from one pattern: recurrence, dominant eigenvalue, product bound, eigenvector arithmetic.

8.2 Clustering multi-relational graphs by Dirichlet energy

The exemplar is a SIGMOD’26 paper on clustering large multi-relational graphs [5]. Multi-relational graphs (MRGs) model objects connected by many types of interaction at once; partitioning their N nodes into K clusters is a fundamental analytics task. Existing solutions either fuse the heterogeneous relations so crudely that result quality collapses, or lean on heavyweight deep models that cannot scale to billions of edges. The paper proposes DEMM and its scalable sibling DEMM+: first derive node feature vectors by minimizing a *multi-relational Dirichlet energy* (a smoothness functional over all relations), then cluster by minimizing Dirichlet energy over the resulting affinity graph — DEMM+ never materializes the dense $N \times N$ matrix and runs in linear time. The spectral content is threefold: a Neumann series computes the smoothed embedding, Courant–Fischer-type bounds certify the series is legal, and a Sinkhorn balancing makes the final step equivalent to plain k -means.

Problem. Let G be an MRG with N nodes and R relation types; relation r induces a symmetric normalized affinity $S^{(r)}$, and $S = \sum_{r=1}^R \omega_r S^{(r)}$ with $\sum_r \omega_r = 1$ is the fused affinity. The clustering objective is a normalized-cut-type quantity over node-cluster incidence matrices Y , equivalent to trace maximization $\max_Y \operatorname{trace}(Y^\top S Y)$ under combinatorial constraints — NP-hard, and it needs the dense matrix S . Stage one replaces hard clustering by continuous features Z : minimize

$\alpha \text{ trace}(Z^\top (I - S)Z) + \|Z - Z^0\|_F^2$ for smoothing parameter $\alpha > 0$ and input features Z^0 .

Guarantee.

Theorem 8.11 — Doubly stochastic affinity gives k -means, Theorem of [5], notation cleaned. If the affinity S_ω is doubly stochastic (all row and column sums equal 1) and $Z = Y^\top$ is the relaxed embedding, then optimizing the relaxed trace objective is equivalent to optimizing the k -means objective

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{v_i \in C_k} \|z_i - c^{(k)}\|_2^2, \quad c^{(k)} = \frac{1}{|C_k|} \sum_{v_j \in C_k} y_j,$$

where $c^{(k)}$ is the center of cluster C_k . ■

How spectral analysis is applied. Three matrices are named in turn. (i) *The fused affinity S .* Setting the derivative of stage one to zero gives the closed form $Z = \frac{1}{1+\alpha} (I - \frac{\alpha}{1+\alpha} S)^{-1} Z^0$ (Lemma of the paper) — an inverse that exists and expands as a Neumann series (Theorem 8.4) because the spectrum of S is confined to $[0, 1]$: a lemma of the paper proves $\lambda(S^{(r)}) \leq 1$ by Courant–Fischer (via $\sum_{ij} S_{ij} y_i y_j \leq \frac{1}{2} \sum_{ij} S_{ij} (y_i^2 + y_j^2)$), and the Lidskii-type inequality $\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B)$ lifts through the convex combination. (ii) *The truncated series.* Truncating at L terms gives the scalable solver:

Lemma 8.12 — Truncation error, Theorem of [5], notation cleaned. Let Z^L be the L -term truncation of the Neumann expansion of $\frac{1}{1+\alpha} (I - \frac{\alpha}{1+\alpha} S)^{-1} Z^0$. Then

$$\|Z - Z^L\|_F \leq \left(\frac{\alpha}{1+\alpha} \right)^{L+1} \|Z^0\|_F \cdot \max_{j \geq 1} \mu_{L,L+j},$$

$$\mu_{\ell_1, \ell_2} = \max_i |\lambda_i(S)^{\ell_1} - \lambda_i(S)^{\ell_2}|,$$

so the error decays geometrically at rate $\alpha/(1+\alpha)$, modulated by the eigengap quantity μ — itself decaying since $\lambda_i(S) \in [0, 1]$ ($\|S^{L+j} - S^L\|_2 = \mu_{L,L+j}$ exactly). ■

Proof sketch. The tail is $\frac{1}{1+\alpha} \sum_{j \geq 1} (\frac{\alpha}{1+\alpha})^{L+j} (S^{L+j} - S^L) Z^0$; bound the Frobenius norm by $\|S^{L+j} - S^L\|_2 = \mu_{L,L+j}$ (read off the eigendecomposition of S), then sum the geometric series in $\alpha/(1+\alpha) < 1$. ■

(iii) *The balanced affinity.* The hypothesis of Theorem 8.11 is manufactured, not assumed: Sinkhorn–Knopp proportional fitting on $SS^\top$ (alternating row and column normalizations) yields, by Sinkhorn’s theorem plus the symmetry of $SS^\top$, a *symmetric doubly stochastic* S_ω (Theorem of the paper). And the norms are never computed on $N \times N$

matrices: a CountSketch (the heavy-hitter hash of the concentration chapter) returns $\|PS^{(r)}\|_F^2$ to a $(1 \pm \varepsilon)$ factor with probability $1 - \delta$ using $m = O(r\varepsilon^{-4} \log(r/\varepsilon\delta) (r + \log(1/\varepsilon\delta)))$ columns (Corollary of the paper), with per-relation failure δ/R union-bounded over the R relations.

What it buys. The system clusters billion-edge multi-relational graphs in linear time without ever forming the dense affinity: a truncated Neumann embedding with certified geometric error, a balancing step with a uniqueness proof, and a theorem chain (spectrum in $[0, 1] \rightarrow$ Neumann legal $\rightarrow$ truncation rate $\rightarrow$ doubly stochastic $\rightarrow k$ -means).

8.3 Subspace collision for high-dimensional nearest neighbor search

The exemplar is a SIGMOD’26 paper on approximate nearest neighbor search [10]. ANNS in high-dimensional Euclidean space is a workhorse of vector databases, and *subspace collision* is a recently proposed framework for it: project the data onto many random low-dimensional subspaces and rank points by how often they “collide” with the query — fall within its projected neighborhood — across the subspaces. The framework performs well but is data-agnostic and query-oblivious, leading to imbalanced indexes and wasted probing. The paper’s system, TaCo, fixes both: a subspace-oriented data transformation equalizes the subspaces (balanced average entropy), making collision counting data-adaptive, with query-aware probing on top. Under the hood, index design is an eigenvalue allocation problem solved by interlacing, and query-side correctness is projection non-expansiveness plus Chernoff.

Problem. Given a dataset $\mathcal{D}$ of n points in $\mathbb{R}^d$, a query q , and integers k, N_s, s with $N_s \cdot s \leq d$: sample N_s subspaces of dimension s ; a point o *collides* with q in a subspace if the projected o^* is among the (αn) nearest neighbors of the projected q^* there; the SC-score of o counts the subspaces in which it collides, and the k highest SC-scores are the answer. Two things must be certified: the *index-side* choice of which dimensions to group (TaCo first applies an orthogonal transformation B), and the *query-side* fact that projected proximity reflects true proximity.

Guarantee.

Theorem 8.13 — Ordering preservation, Lemma and Theorem of [10], merged and notation cleaned. Let $B \in \mathbb{R}^{d \times (N_s \cdot s)}$, $N_s \cdot s \leq d$, have orthonormal columns, so that $BB^\top$ is an orthogonal projection. Suppose that for some $\varepsilon \in (0, 1)$ the local difference vectors satisfy $\|(I_d - BB^\top)(o_i -$

$o_j)\|^2 \leq \varepsilon \|o_i - o_j\|^2$ for all neighbors o_j of each $o_i \in \mathcal{D}$. Then

$$(1 - \varepsilon) \|o_i - o_j\|^2 \leq \|B^\top(o_i - o_j)\|^2 \leq \|o_i - o_j\|^2,$$

and consequently whenever $\|o_i - o_j\|^2 < (1 - \varepsilon)\|o_i - o_z\|^2$, the relative ordering survives the transformation: $\|B^\top(o_i - o_j)\| < \|B^\top(o_i - o_z)\|$. ■

Proof sketch. Pythagoras: $\|o_i - o_j\|^2 = \|B^\top(o_i - o_j)\|^2 + \|(I - BB^\top)(o_i - o_j)\|^2$; the hypothesis bounds the second term (lower bound), and projections are non-expansive (upper bound). Chain the bounds across o_j and o_z for the ordering claim. ■

How spectral analysis is applied. The named matrix is the sample covariance Σ of the transformed data: B 's columns are eigenvectors of Σ , which is why $BB^\top$ is an orthogonal projection and Theorem 8.13 applies. The index-side question — *which* eigenvectors to allocate to which subspace — is settled by the interlacing rung:

Theorem 8.14 — Eigensystem allocation is optimal, Theorem of [10], notation cleaned. Assume the sample covariance Σ has distinct eigenvalues, all ≥ 1 . Then the Eigensystem Allocation method (assigning eigenvectors of Σ to the N_s subspaces so that the subspace log-determinants $\log \det(B_j^\top \Sigma B_j)$ are balanced) solves the paper's balanced log-det optimization over all choices of the B_j . ■

Proof sketch. By Poincaré separation (Theorem 8.5), the eigenvalues of each compression $B_j^\top \Sigma B_j$ interlace those of Σ : $\lambda_i \geq \mu_i \geq \lambda_{d-s+i}$. Since log is monotone, maximizing the log-determinant means maximizing the product of compressed eigenvalues, and interlacing caps each factor at the corresponding top eigenvalue of Σ — a cap attained by taking B_j 's columns to be eigenvectors; a greedy pass solves the outer balanced partition. ■

On the query side, the SC-score is a binomial count: per-subspace collision probabilities p^* (true neighbor) versus $p < p^*$ (non-neighbor), and the paper's SC-score Separation lemma shows the optimal Type-I/Type-II errors of any decision rule decay at rate $\exp(-N_s \Delta(q)^2 / (8p(1-p)))$, $\Delta(q) = p^* - p$ — a Chernoff bound on the two binomials. The tunables: N_s sets the exponent, α moves p^* and p apart, and ε is the residual energy the eigensystem leaves outside B .

What it buys. The vector-search engine gets an index whose subspaces are provably balanced (equal average entropy, no wasted probing), a certificate that projected-space comparisons preserve neighbor orderings, and separation growing exponentially in the number of subspaces.

8.4 Exact graph similarity by Kronecker-sum inversion

The exemplar is a SIGMOD'26 paper on graph similarity [7]. Quantifying node-to-node similarity is central to graph analysis, and SimRank is its most-used recursive measure: two nodes are similar if they are referenced by similar nodes. The classical definition, however, only counts *equal-length* meeting paths, producing the “zero-similarity” problem — structurally related nodes score exactly zero when no symmetric in-neighbor paths exist. Existing fixes iterate path expansions and pay for expensive convergence checks. The paper proposes PJsims, an exact, closed-form measure summing meeting paths of all (possibly unequal) lengths, obtained by reformulating the recursion as a Sylvester matrix equation solved through structured decompositions; experiments report higher accuracy than iterative alternatives and speedups up to $3.22\times$ on synthetic and $2.15\times$ on real graphs. Every step is an eigenvalue-location argument.

Problem. Let Q be the backward transition matrix of a directed graph on n nodes and $c \in (0, 1)$ a decay factor. The PJsims similarity matrix S^{PJ} sums contributions over all pairs of meeting path lengths, which the paper derives to be the solution of the Sylvester equation

$$X S^{\text{PJ}} + S^{\text{PJ}} X^{\top} = 2(1 - c) I_n, \quad X = I_n - c Q.$$

The task: certify a unique solution, then compute it at scale — the vectorized system is $n^2 \times n^2$, and naive inversion costs $O(n^6)$.

Guarantee.

Theorem 8.15 — Vectorized solution, Theorem 3.1 and Corollary 3.1 of [7], notation cleaned. The PJsims Sylvester equation is equivalent, under the vectorization identity $\text{Vec}(ABC) = (C^{\top} \otimes A) \text{Vec}(B)$, to the Kronecker-sum linear system

$$(X \oplus X) \text{Vec}(S^{\text{PJ}}) = 2(1 - c) \text{Vec}(I_n), \quad X \oplus X = (X \otimes I_n) + (I_n \otimes X),$$

whose closed-form solution is $\text{Vec}(S^{\text{PJ}}) = 2(1 - c) (X \oplus X)^{-1} \text{Vec}(I_n)$; the inverse exists. ■

How spectral analysis is applied. The named matrix is the Kronecker sum $X \oplus X$ — the $n^2 \times n^2$ matrix hiding behind the two-sided recursion — and the rung is spectrum localization:

Lemma 8.16 — Spectrum of the Kronecker sum, Lemma of [7], notation cleaned. The eigenvalues of $X = I_n - cQ$ lie in $[1 - c, 1]$, and the eigenvalues of $X \oplus X$ are exactly the pairwise sums $\{\lambda_i(X) + \lambda_j(X) : i, j = 1, \dots, n\} \subseteq$

$[2(1 - c), 2]$. Since $c \in (0, 1)$, all eigenvalues of $X \oplus X$ are strictly positive, and $X \oplus X$ is invertible. ■

Proof sketch. The eigenvalues of the backward transition matrix Q lie in $[0, 1]$, so $\lambda_i(X) = 1 - c \lambda_i(Q) \in [1 - c, 1]$; the Kronecker-sum spectrum is the pairwise sum of the summands' spectra ($v_i \otimes v_j$ is an eigenvector of $A \oplus B$ for $\lambda_i(A) + \lambda_j(B)$), giving $[2(1 - c), 2] \subset (0, \infty)$. ■

Two consequences follow. (i) *Iterative approximation:* the spectrum of $I - (X \oplus X)$ lies within the unit disk (the paper expands $(X \oplus X)^{-1} = \sum_k [I_{n^2} - (X \oplus X)]^k$), so the Neumann rung (Theorem 8.4) yields a truncatable, geometrically convergent series — an anytime solver with no convergence checks. (ii) *Exact computation at scale:* apply the real Schur decomposition $X = VUV^\top$ (V orthogonal, U quasi-upper-triangular with 1×1 and 2×2 blocks, real even for complex eigenvalues). Substituting $Y = V^\top S^{\text{PJ}} V$ gives $UY + YU^\top = 2(1 - c)I_n$, solved by block back-substitution: the trailing block a smaller Sylvester equation, the off-diagonal blocks plain linear systems, each 2×2 block one 4×4 inversion — trivial against $O(n^6)$. The accuracy knob is the truncation length K ; the Schur route is exact.

What it buys. The graph-analytics engine gets a similarity measure with no zero-similarity blind spots, an existence proof costing three lines of eigenvalue arithmetic, and an exact solver — Schur plus back-substitution instead of $O(n^6)$ inversion — yielding measured 2–3 $\times$ speedups with higher accuracy than iterative predecessors.

8.5 Advanced usage and further reading

The four exemplars above use spectra of ordinary matrices. Two harder corpus papers push the same philosophy into tensor decompositions and random projections; each is sketched in one paragraph.

Fair pre-processing with latent attributes [12]. Fair pre-processing calibrates a dataset to a fairness policy so that sensitive attributes influence outcomes only through designated legitimate causal pathways — but real attribute spaces are imperfect: decision-relevant factors may be inadmissible or simply missing. LatentPre augments the policy with *latent* attributes and proves the augmentation recoverable: with one latent attribute and the observed attributes in four groups (conditional on (X_0, X_4) the first three are mutually independent), the local parameters of every X_i are *generically identifiable up to a global permutation of the latent states* (theorem of the paper). The core: conditioning on each value $X_4 = j$ yields a three-view latent-class model whose joint distributions form a 3-way tensor of low rank κ (the number of latent

states), and low-rank tensor decompositions are generically unique — the eigenvalues pin down the object. All slices share the same mixing distribution, so the per-slice permutations must agree, forcing one consistent global solution; estimation runs by a tailored EM with a Jensen-bound objective.

Skipping redundant distance computations [8]. In graph-based ANN search the distance comparison operation (DCO) — compute $\delta(x, q)$, compare against a dynamic threshold τ — is the bottleneck once SIMD makes full distance computations cheap. SkipComputing replaces the test by a projected-space screen: with $A \in \mathbb{R}^{d \times D}$ the first d rows of a random orthogonal matrix, Theorem 4.1 of the paper gives the two-probability contract $\Pr[\|A(p - q)\|_2^2 < (1 + \varepsilon) \frac{d}{D} \tau] > p_1$ when $\delta(p, q) < \tau$, and $< p_2$ when $\delta(p, q) > \tau$ — the Johnson–Lindenstrauss rung (Theorem 8.7) turning the threshold test into a cheap d -dimensional comparison at $d = \Theta(\varepsilon^{-2} \log(1/\beta))$. Survivors get lower-bound pruning (MS-distance: a monotone family of lower bounds refining to the exact distance), and the structure survives PCA pre-processing (Theorem 5.1) because the PCA rotation is orthogonal, hence norm-preserving.

8.5.1 Further reading

- Horn and Johnson [4] — the comprehensive reference: Perron–Frobenius, Kronecker calculus, Neumann, interlacing.
- Bhatia [2] — eigenvalue perturbation: Weyl, Lidskii, Davis–Kahan $\sin \Theta$.
- Chung [3] — spectral graph theory: random walks, Dirichlet energy, spectra of normalized affinities.
- Martinsson and Tropp [6]; Woodruff [11] — randomized NLA and sketching: truncated SVD, CountSketch, JL dimensions.
- Tropp [9] — matrix concentration for random matrices; the concentration chapter’s continuation.

Chapter Recap

- **What the tool is.** Name the linear map in a system and read the guarantee off its spectrum: convergence from radius and eigengaps, solvability from distance to zero, subspace choice from interlacing, distance fidelity from random projection.
- **Which rung when.** Nonnegative/irreducible transitions: Perron–Frobenius (Metzler variant for negative diagonals). Inverse of $I - A$ or smoothed embeddings: Neumann, legal once eigenval-

ues lie in $[0, 1]$ or $\rho < 1$. Choosing eigenvectors: Courant–Fischer and Poincaré separation. Discarding components: Eckart–Young–Mirsky. Cheap distances: distributional JL.

- **Why it works.** Repetition amplifies the dominant mode: Perron–Frobenius supplies a simple positive dominant eigenvalue, product bounds show every other mode grows strictly slower — limits, rates, and stability fall out of where the eigenvalues sit (Figure 8.1).
- **What the corpus showed.** B-trees (Section 8.1): histogram recurrence converged by the Metzler Perron–Frobenius rung, limit evaluated by eigenvector arithmetic. Multi-relational clustering (Section 8.2): spectrum in $[0, 1]$ licenses truncated Neumann; Sinkhorn balancing reduces to k -means. Subspace collision (Section 8.3): interlacing proves eigensystem allocation optimal. Graph similarity (Section 8.4): Kronecker-sum spectra certify invertibility; Schur replaces $O(n^6)$ inversion.
- **Advanced usage.** Tensor identifiability forces a consistent global solution in fair pre-processing; JL bounds turn distance thresholds into cheap projected-space tests (Section 8.5). Deferred proofs live in the Appendix (Section 8.6).

8.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 8.5 (Poincaré separation)

Statement (recalled). For symmetric A with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$ and $B \in \mathbb{R}^{d \times s}$ with orthonormal columns, the eigenvalues $\mu_1 \geq \dots \geq \mu_s$ of $B^\top AB$ satisfy $\lambda_i \geq \mu_i \geq \lambda_{d-s+i}$, $i = 1, \dots, s$. ■

Proof. Let $W = \text{span}(B)$; $y \mapsto By$ is an isometry onto W , and the Rayleigh quotient of the compression at y equals $R(By)$. *Upper bound:* by Courant–Fischer, $\mu_i = \max_{\dim U=i} \min_{0 \neq y \in U} R(By)$; each image BU is an i -dimensional subspace of $\mathbb{R}^d$, so the inner min is at most

$$\max_{\dim V=i} \min_{0 \neq x \in V} R(x) = \lambda_i(A).$$

Lower bound: $\mu_i = \min_{\dim U=s-i+1} \max_{0 \neq y \in U} R(By)$, where the images BU form a subfamily of the $(s-i+1)$ -dimensional subspaces (those inside W); a minimum over a subfamily is at least the minimum over all of them, which Courant–Fischer identifies as $\lambda_{d-s+i}(A)$. ■

Proof of Theorem 8.6 (Eckart–Young–Mirsky)

Statement (recalled). For every B of rank at most r , $\|A - B\|_2 \geq \sigma_{r+1}$ and $\|A - B\|_F \geq (\sum_{i>r} \sigma_i^2)^{1/2}$, with equality for the truncated SVD A_r . ■

Proof. Write $A = \sum_{i \geq 1} \sigma_i u_i v_i^\top$; the tail $A - A_r$ has spectral norm σ_{r+1} and Frobenius norm $(\sum_{i>r} \sigma_i^2)^{1/2}$, so A_r attains both. *Spectral-norm lower bound:* let $V_{r+1} = \text{span}(v_1, \dots, v_{r+1})$. Since $\text{rank}(B) \leq r$, $\dim \ker B \geq n - r$, so $\dim(V_{r+1} \cap \ker B) \geq 1$; pick a unit x in the intersection. Then $(A - B)x = Ax$ and $\|(A - B)x\|_2^2 = \sum_{i \leq r+1} \sigma_i^2 \langle v_i, x \rangle^2 = \sigma_{r+1}^2$. *Frobenius case:* let $P = BB^\top$ be the orthogonal projection onto the range of B . Projections do not increase Frobenius norm and $(I - P)B = 0$, so $\|A - B\|_F \geq \|(I - P)(A - B)\|_F = \|(I - P)A\|_F = (\|A\|_F^2 - \|PA\|_F^2)^{1/2}$. It remains to maximize $\|PA\|_F^2 = \text{trace}(PAA^\top)$ over rank- r orthogonal projections; summing Courant–Fischer over the top r eigenvalues (Ky Fan’s maximum principle) caps it at $\sum_{i \leq r} \sigma_i^2$, attained by $P = \sum_{i \leq r} u_i u_i^\top$. Hence $\|A - B\|_F^2 \geq \sum_{i>r} \sigma_i^2$. ■

Proof of Theorem 8.7 (random projection)

Statement (recalled). For A the first d rows of a uniformly random orthogonal matrix and fixed x : $\mathbb{E}\|Ax\|_2^2 = \frac{d}{D}\|x\|_2^2$, and the deviation probability is at most $2e^{-\varepsilon^2 d/12}$ for $\varepsilon \in (0, 1)$, so $d = \Theta(\varepsilon^{-2} \log(1/\beta))$ gives failure probability β . ■

Proof. By rotational invariance rotate x to $\|x\|_2 e_1$: $\|Ax\|_2^2 / \|x\|_2^2$ is the squared length $T = \sum_{i \leq d} u_i^2$ of the first d coordinates of a uniform unit vector u , i.e. $T \sim \text{Beta}(d/2, (D - d)/2)$ with mean d/D — the expectation claim. For the tail write $u = g / \|g\|_2$ with $g \sim \mathcal{N}(0, I_D)$, so $T = \chi_d^2 / (\chi_d^2 + \chi_{D-d}^2)$ with independent chi-squares. Condition on the total lying within a $(1 \pm \varepsilon/3)$ factor of its mean (failure $2e^{-\Theta(\varepsilon^2 D)}$, absorbed for $D \geq d$): then $|T - d/D| > \varepsilon d/D$ forces χ_d^2 outside $(1 \pm \varepsilon/2)d$. The chi-square Chernoff bound via $\mathbb{E}e^{\lambda(\chi_d^2 - d)} = e^{-\lambda d}(1 - 2\lambda)^{-d/2}$, optimized over λ , gives $\Pr[\chi_d^2 \geq (1 + \varepsilon)d] \leq e^{-d(\varepsilon^2/2 - \varepsilon^3/3)/2} \leq e^{-\varepsilon^2 d/12}$ for $\varepsilon \leq 1$, and the matching lower-tail bound holds likewise; union-bounding the two tails yields the claim. ■

Proof of Lemma 8.10 (recurrence product bound)

Statement (recalled). For positive integers n_0, r and $\lambda \in \mathbb{C}$, $|\prod_{k=0}^{m-1} (1 + \lambda/(n_0 + kr))| \leq m^{\text{Re}(\lambda)/r} \alpha_{\lambda, n_0, r}$ with α independent of m . ■

Proof. For complex t , $|1 + t|^2 = 1 + 2\text{Re}(t) + |t|^2 \leq e^{2\text{Re}(t) + |t|^2}$, hence $\ln|1 + t| \leq \text{Re}(t) + \frac{1}{2}|t|^2$. Apply with $t_k = \lambda/(n_0 + kr)$ and sum over

$k < m$:

$$\ln \left| \prod_{k=0}^{m-1} \left(1 + \frac{\lambda}{n_0 + kr} \right) \right| \leq \operatorname{Re}(\lambda) \sum_{k=0}^{m-1} \frac{1}{n_0 + kr} + \frac{|\lambda|^2}{2} \sum_{k=0}^{m-1} \frac{1}{(n_0 + kr)^2}.$$

Integral comparison gives $\sum_{k < m} \frac{1}{n_0 + kr} \leq \frac{1}{r} \ln m + \gamma_{n_0, r}$ with γ depending only on (n_0, r) , and the squared sum is $O(1)$; exponentiating yields the bound with $\alpha_{\lambda, n_0, r} = \exp(\operatorname{Re}(\lambda) \gamma_{n_0, r} + |\lambda|^2 \pi^2 / 12)$. (The paper's companion lemma covers nilpotent factors: order- j terms carry j -fold elementary symmetric sums of the $1/(n_0 + kr)$, each $O((\ln m)^j)$ — the $(\ln m)^d$ factor in Theorem 8.9.) ■

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Amortized Analysis and Potential Functions

Amortized analysis is the systems community’s favorite certificate that rebuilds, merges, and splits are *free*: one operation may detonate an $O(n)$ reorganization, but if many cheap operations must accumulate before the next detonation becomes possible, the cost per operation stays $O(1)$. The deliverable is a statement such as “amortized $O(1)$ update time” holding for *every* operation sequence, worst case included, certified by deterministic bookkeeping — a bank balance, a potential function, an epoch decomposition. In the SIGMOD’26 corpus thirteen tagged papers reach for the amortized/potential toolbox, owning one of the survey’s most lopsided tool–property pairings: *upper-bound* $\times$ *amortized-potential* occurs in ten of the eleven — the tool nearly always arrives to certify a per-operation bound that an honest worst-case statement could not quote.

This chapter builds the ladder from the aggregate method to epoch decompositions, with a dynamic-table worked example as the spine, then dissects four corpus exemplars — pattern-match enumeration, matrix sketches for streams, DeFi order routing, and quantile sketches — before Section 9.5 sketches two advanced usages.

9.0 The tool: amortized analysis

The problem class is uniform across applications. A data structure processes a long *sequence* of operations from an initial state D_0 ; almost every operation is cheap, but occasionally the structure must perform an expensive structural event — a bucket splits, an array doubles, a summary is re-decomposed, a segment is compacted. Worst-case analysis must quote the burst cost for *every* operation; amortized analysis proves instead that bursts are rare *because they require accumulation*, and charges each operation its share: we want f such that every sequence of m operations costs $O(mf)$ in total. Three framings and one structural

LEARNING OBJECTIVES

- Recognize when a per-operation contract needs an *amortized* argument, not a worst-case one.
- Prove the ladder aggregate $\rightarrow$ accounting $\rightarrow$ potential $\rightarrow$ epochs, and match rungs to burst structure.
- Design Φ : cheap deposits, withdrawals covering the burst, $\Phi \geq 0$ throughout.
- Dissect a paper’s argument: name the rare event and who pays for it.

refinement make up the ladder [1, 8].

9.0.1 The ladder

Theorem 9.1 — Aggregate method. Let $T(m)$ bound the total cost of any sequence of m operations starting from D_0 . Then the amortized cost per operation is $T(m)/m$. ■

Proof. Sum the sequence's costs and divide by m ; the hypothesis covers the worst sequence. ■

The aggregate method is a global sum with no accounting structure — it shines when the total has a clean combinatorial bound. The growable array is the canonical spine:

Example 9.2 (Dynamic table). A growable array holds num items in a buffer of capacity cap ; inserting into a full buffer allocates a double-capacity buffer and copies everything, cost $1 + \text{cap}$. After m insertions from empty the total copying cost is at most $\sum_{j \leq \lceil \log_2 m \rceil} 2^j < 2m$, so all insertions cost below $3m$: amortized $O(1)$ each by Theorem 9.1. The worst-case single insertion still costs $\Theta(m)$ — amortization smooths the *sequence*, not the operation.

The accounting method replaces the global sum by a local story about *who pays* for each future burst:

Theorem 9.3 — Accounting / banker's method. Assign each operation i a fictitious price p_i , and let the bank balance after i operations be $B_i = \sum_{j \leq i} p_j - \sum_{j \leq i} c_j$, where c_j is the true cost. If $B_i \geq 0$ for every prefix i , then the total true cost never exceeds the total price. ■

Proof. $B_m \geq 0$ is the claim at $i = m$; the prefix condition lets it be quoted at any horizon, since any prefix of a valid sequence is itself a valid sequence. ■

For the dynamic table charge 3 per insertion: 1 pays for the store, 1 for copying the new element at the next doubling, 1 for copying one earlier resident; at a doubling every buffered item holds a stored credit, so the burst is fully prepaid. The potential method instead makes the bank balance a function of the *state* rather than the history, which is what makes it composable and local:

Theorem 9.4 — Potential / physicist's method. Let Φ map data-structure states to $\mathbb{R}_{\geq 0}$ with $\Phi(D_0) = 0$, and define the *amortized cost* of operation i by

$$\hat{a}_i = c_i + \Phi(D_i) - \Phi(D_{i-1}),$$

i.e. amortized cost = true cost + potential change. Then for every sequence of m operations and every prefix of it,

$$\sum_{i \leq j} c_i \leq \sum_{i \leq j} \hat{a}_i \quad \text{for all } j \leq m. \quad \blacksquare$$

Proof. Telescoping: $\sum_{i \leq j} \hat{a}_i = \sum_{i \leq j} c_i + \Phi(D_j) - \Phi(D_0) = \sum_{i \leq j} c_i + \Phi(D_j) \geq \sum_{i \leq j} c_i$, since $\Phi(D_j) \geq 0$ and $\Phi(D_0) = 0$. $\blacksquare$

The design problem is explicit: find $\Phi \geq 0$ with $\Phi(D_0) = 0$ such that cheap operations raise Φ by at most their price (deposits) and every expensive event is paid by a large drop (withdrawals). For the dynamic table, $\Phi(D) = 2 \text{num}(D) - \text{cap}(D)$ makes each insertion's amortized cost at most 3 and each doubling's 0. The classic showcases are self-adjusting search trees and union-find with path compression — Φ summing imbalance or rank terms for amortized $O(\log n)$ per access and $O(\alpha(n))$ per operation — plus periodic global rebuilding as the degenerate case “rebuild cost over ℓ ” [6–8].

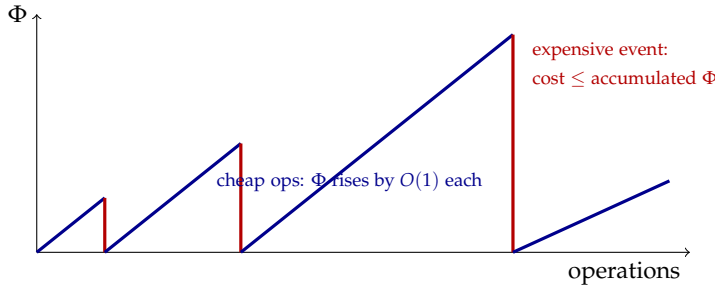


Figure 9.1: The sawtooth: cheap operations deposit into Φ ; an expensive event fires only when enough has accumulated, and its cost is paid entirely by the drop in Φ . Doubling the event cost doubles the inter-event gap.

The fourth rung packages the aggregate method for periodic structure — and it is the form most corpus papers actually use, because a batch schedule is easier to name than a potential:

Proposition 9.5 — Epoch decomposition. Partition every execution into consecutive *epochs*, epoch e having $\text{len}(e)$ operations and total cost $\text{cost}(e)$. Then the amortized cost per operation is $\sum_e \text{cost}(e) / \sum_e \text{len}(e)$. In particular, if every epoch costs $O(C)$ and contains at least ℓ operations, the amortized cost is $O(C/\ell)$; if an epoch's fixed setup S services T operations, its contribution is $O(S/T)$. $\blacksquare$

Proof. Sum epoch costs, sum epoch lengths, divide; the special cases substitute the uniform bounds. $\blacksquare$

¹ Sections 9.2 and 9.3 below are epoch arguments; Section 9.4 is a pure potential argument.

¹ Epochs are the aggregate method in periodic clothing, and the potential method is the accounting method with the balance read off the state instead of the history; the three proofs are one proof — sum a nonnegative quantity — and the choice is ergonomic: batch schedules state epochs, accumulation triggers state Φ .

Notation

Throughout, c_i is the true cost of operation i , $\hat{a}_i$ its amortized cost, and Φ a function from states to $\mathbb{R}_{\geq 0}$ with $\Phi(D_0) = 0$ on the initial state. “Amortized $O(f)$ ” always means total cost $O(mf)$ for every m and every sequence of m operations from D_0 — a deterministic statement about worst sequences, with nothing averaged and nothing probabilistic (per-operation *expected* cost is a different, randomized contract).

The price of every rung is the same: the bound is a promise about *sequences*, not single operations. A deployment needing every request answered within a deadline requires *deamortized* variants that spread each burst over neighboring operations [6].

9.0.2 How to choose the method

The selection is mechanical once two questions are answered. *First, can you sum a whole worst-case sequence?* If the total cost of m operations has a clean combinatorial bound, the aggregate method or its epoch packaging (Proposition 9.5) is the shortest honest proof. *Second, can you name who pays?* If the expensive event is triggered by an accumulation you can point at — rows since the last re-decomposition, excess of a counter over its budget, rounding slack of a cached value — define Φ on exactly that quantity and verify the three properties of Theorem 9.4. Use accounting when the payer is naturally an operation (“this insert pays for its own future copy”), potential when it is naturally a state (“every byte above the watermark”), and epochs whenever the algorithm itself has a batch schedule.

When to Reach for Amortization

Reach whenever a component must be certified with a *per-operation* time or space bound, the algorithm has an occasional expensive structural event — split, merge, rebuild, compaction, re-decomposition, full scan — and that event only fires after many cheap operations accumulate a trigger:

1. Identify the expensive rare operation: name the burst, its cost C , and the measurable quantity that must accumulate before it can fire again.
2. Define the potential on the state: Φ is that accumulated quantity (or the epoch schedule that bounds it), with $\Phi(D_0) = 0$.
3. Charge deposits and withdrawals: every cheap operation raises Φ by at most $O(1)$; every burst drops Φ by at least its cost C —

so the inter-burst gap is $\Omega(C)$ and the amortized cost stays $O(1)$ per deposit.

4. Verify $\Phi \geq 0$ initially and throughout: this is what makes the telescoping safe for every prefix, and it is the only place amortized proofs fail — one reachable state with $\Phi < 0$ voids the bound.

Caveats before shipping the bound: amortized $\neq$ worst case (budget for deamortization if tail latency matters), and insertion-only accumulation can shatter under deletions — check the fully dynamic lower bounds first.

9.1 Enumerating regular pattern matches with amortized constant delay

The exemplar is a PODS'26 paper on pattern matching [3]^{PODS}. Extracting all matches of a *pattern with variables* — fixed constants separated by variables standing for arbitrary substrings — is the engine behind regex-capture queries, log extraction, and transformation languages; when the fixed parts are plain strings the pattern is called *regular*. A match assigns each variable a substring so that the concatenation equals the text T , recorded by the k -tuple of positions where the constants sit. Enumeration algorithms are judged on preprocessing time, *delay* between outputs, and space; the paper pushes all three to optimal for this class, including an *amortized* constant-delay variant and an indexed version answering each later pattern in time near-linear in the pattern alone — reported as the first optimal combined-complexity algorithms for a non-trivial subcase of enumerating regular-expression matches with capture variables.

Problem. Let $T[1..n]$ be a text and $\alpha = w_0 \diamond x_1 w_1 \diamond \dots \diamond x_k w_{k+1}$ a regular pattern with $\sum_{i=1}^k |w_i| \leq n$. A *valid embedding* is a k -tuple $(j_1, \dots, j_k)$ such that w_i occurs at position j_i for each i , the occurrences are consecutive and non-overlapping, w_0 is a prefix of T , and w_{k+1} is the suffix starting after $j_k + |w_k| - 1$. The task is to enumerate the set S of all valid embeddings. The difficulty is the output: $|S|$ can reach n^k , so materializing occurrence lists up front risks blowing past $O(n)$, while the delay must stay constant however the outputs are distributed.

Guarantee.

Theorem 9.6 — Optimal enumeration, Theorems 1.1 and 1.2 of the paper, notation cleaned. (i) The set S of all valid embeddings of α in T can be enumer-

ated with constant delay after preprocessing in $O(n + |S|)$ time, using $O(n + \min\{|S|, n^k\})$ space. (ii) Alternatively, S can be enumerated with amortized constant delay after preprocessing in $O(n)$ time and the same space: for every $s \leq |S|$, the first s embeddings are reported in $O(n + s)$ total time. ■

2

² A third theorem of the paper upgrades to *worst-case* constant delay with $O(n)$ preprocessing at the price of $O(n \min\{|S|, n^k\})$ space — the deamortization trade of Section 9.0.

How amortization is applied. The variant (ii) is an aggregate-method argument over the enumeration phase, and its engine is a counting bound on where embeddings can live. The paper greedily computes a *left-canonical* embedding $(\ell_1, \dots, \ell_k)$ (each ℓ_i the leftmost feasible occurrence of w_i) and a symmetric *right-canonical* embedding $(r_1, \dots, r_k)$; every valid embedding (j_i) satisfies $\ell_i \leq j_i \leq r_i$. Write $O_i = \{p \in [\ell_i : r_i] \mid w_i \text{ occurs at } p\}$; the workhorse is the following bound.

Lemma 9.7 — Embedding counting bound, Lemma 3.4 of the paper, notation cleaned. With O_i and S as above, $\sum_{i=1}^k (|O_i| - 1) \leq \min\{kn, |S|\}$. ■

Proof sketch. $|O_i| \leq n$ gives the kn side. For the $|S|$ side, each $j_i \in O_i \setminus \{\ell_i\}$ substituted into the canonical hybrid yields a valid embedding, one distinct embedding per pair (i, j_i) . (Full proof in Section 9.6.) ■

The algorithm runs in two phases. In preprocessing, the intervals $[\ell_i : r_i + |w_i|)$ are partitioned into *epochs* (Definition 3.7 of the paper): maximal runs of intervals overlapping transitively, whose left and right parts are separately pairwise disjoint, so the total epoch length is $O(n)$. Each epoch is preprocessed *locally*, in time linear in its length: per-epoch suffix-array and suffix-tree ranges for the w_i (Lemma 2.2 of the paper), plus a range-maximum structure enumerating with worst-case constant delay all suffix-array entries above a threshold (Lemma 2.4). The amortization lives in the enumeration phase: rather than materializing all lists O_i during preprocessing (their total length can exceed n), the algorithm outputs the *simple* embeddings — those differing from the canonical skeletons in exactly one coordinate — *while* building the lists, one output per list element. Lemma 9.7 bounds the simple embeddings by $\min\{kn, |S|\}$, so every unit of list-materialization work is paid for by an emitted embedding, up to $O(n + k)$ overhang: the first s embeddings cost $O(n + s)$ total. Once the lists exist, a radix sort plus constant-delay enumeration (Lemma 3.6) finishes.

What it buys. Preprocessing linear in n independent of the output size, amortized constant delay, and space $O(n + \min\{|S|, n^k\})$ — plus the indexed variant (each pattern answered in $O(|\alpha| + k \log n)$ plus constant delay), the shape a database system needs for repeated extraction queries.

9.2 Matrix sketches for streams: batching the re-decomposition

The exemplar is a SIGMOD'26 paper on streaming matrix sketches [10]. A matrix sketch tracks the covariance structure of a stream of d -dimensional row vectors — the substrate for PCA-style monitoring, regression, and anomaly detection — and the standard tool, Frequent Directions, periodically compresses its summary by an SVD. The pain: a faithful SVD costs time quadratic in the error parameter, too expensive to redo on every arrival. The paper proposes AeroSketch, keeping the same space and error contracts while driving the amortized per-update time down to what it calls near-optimal, across persistent streams, sliding windows (only the last N updates matter), and distributed sites coordinating through a hub. The guarantees hold with tunable high probability, inherited from iteration error bounds for randomized power iterations — and the time bounds are amortized, exactly the batch-schedule amortization of Proposition 9.5.

Problem. A stream presents row vectors $a_1, a_2, \dots \in \mathbb{R}^d$ one at a time; stacking the arrivals so far gives A . The sketch B must maintain the covariance-error guarantee $\|A^\top A - B^\top B\|_2 \leq \epsilon \|A\|_F^2$, preserving the top of the spectrum. In the *persistent* setting A is the entire history; in the *sliding-window* setting only $(T - N, T]$ counts; in the *distributed* setting m sites observe substreams while a coordinator keeps the global sketch with bounded communication. The cost model charges amortized time per update, space, and communication, with success probability $1 - \delta$ for tunable δ .

Guarantee.

Theorem 9.8 — Tracking matrix sketches, main theorems of the paper, notation cleaned, ϵ -dependence restored. AeroSketch solves the persistent matrix-sketch tracking problem with space $O(\frac{d}{\epsilon} \log \|A\|_F^2)$ and amortized time $O(\frac{d}{\epsilon} \log d)$ per update, with success probability at least $\frac{99}{100} (1 - \frac{2}{\sqrt{e}} \cdot \frac{1}{\sqrt{d \log_2 d}})$; with probability amplification it achieves amortized time $O(\frac{d}{\epsilon} \log d \log^2 \frac{1}{\delta})$ per update while guaranteeing success probability at least $1 - \delta$, for tunable $\delta \in (0, 1/100)$; the sliding-window variant pays an extra $\log R$ in space and amortized time. ■

How amortization is applied. The expensive rare operation is the re-decomposition: when the sketch matrix C has absorbed $2/\epsilon$ rows, an eigen-decomposition compresses it back to $1/\epsilon$ rows. The paper batches it exactly as in Proposition 9.5:

Lemma 9.9 — Batched re-decomposition, Proposition of the paper and its proof,

notation cleaned, ε restored. Between two consecutive re-decompositions at least $1/\varepsilon$ updates elapse (the row count grows by one per update, from $1/\varepsilon$ back up to $2/\varepsilon$). Each re-decomposition costs $O(d/\varepsilon^2)$, so its amortized contribution is $O(d/\varepsilon)$ per update; the per-update power iteration monitoring the top singular value uses $k = \log_2 d + 1$ steps and costs $O(d \log d)$. Together, the amortized per-update time is $O(\frac{d}{\varepsilon} \log d)$. ■

Proof sketch. Rows enter one per update; the decomposition fires exactly when the row count reaches $2/\varepsilon$ and resets it to $1/\varepsilon$, so every epoch has length at least $1/\varepsilon$ and one burst of $O(d/\varepsilon^2)$; divide (full proof in Section 9.6). ■

The same accounting controls space: a re-decomposition *dumps* directions Z_i (rank k_i) into a snapshot queue, and a dump fires only after the sketch's residual Frobenius mass above a threshold θ has accumulated by $k_i\theta/2$, so the dumped ranks sum to $O(\|A\|_F^2/\theta)$ — an implicit potential on the residual mass bounding the queue. The *whp* part: the re-decomposition only needs its top singular values approximately; power iteration with $\log_2 d + 1$ steps fails to halve the top squared singular value with probability at most $\frac{2}{\sqrt{e}} \cdot \frac{1}{\sqrt{d \log_2 d}}$ (a corollary in the paper), which is the success probability of Theorem 9.8; amplification repeats the iteration and costs the $\log^2(1/\delta)$ factor. The *composition* part:

Lemma 9.10 — Decomposability, Lemma of the paper, notation cleaned. Decompose $A = [A_1, A_2, \dots, A_m]$ into submatrices. If each B_i sketches A_i with covariance error $\|A_i^\top A_i - B_i^\top B_i\|_2 \leq \varepsilon_i \|A_i\|_F^2$, then $B = [B_1, B_2, \dots, B_m]$ sketches A with $\|A^\top A - B^\top B\|_2 \leq \sum_{i=1}^m \varepsilon_i \|A_i\|_F^2$. ■

Proof. $A^\top A - B^\top B = \sum_i (A_i^\top A_i - B_i^\top B_i)$; apply the triangle inequality to the spectral norm and substitute each block's guarantee. ■

Sliding windows and distributed sites both become substreams (snapshots are blocks whose per-block errors add, decomposability summing them), with communication bounded by the same epoch counting on rounds of a doubling threshold. The per-update error telescopes over the window: the accumulated spectral error is the sum of discarded singular values, which the dump-threshold accounting keeps within $O(\varepsilon)\|A\|_F^2$.

What it buys. Amortized $O(\frac{d}{\varepsilon} \log d)$ per update instead of an $O(d/\varepsilon^2)$ burst every $\frac{1}{\varepsilon}$ updates — one proof skeleton covering persistent, sliding-window, and distributed deployments, with tunable success probability.

9.3 Order routing in exchange graphs: epochs over path selections

The exemplar is a SIGMOD'26 paper on order routing in decentralized finance [5]. A decentralized exchange is a graph: token pairs are joined by pools quoting a price (a *weight*) with limited liquidity (a *capacity*). Routing a market order means repeatedly picking the cheapest path in this *exchange graph*, executing against it until a bottleneck capacity is exhausted, re-pricing, and picking again — and since every fill moves prices and depletes capacities, the ranking of paths goes stale after each selection, while re-scanning all candidates after every fill is quadratic in the path count. The paper builds ORDER, a path-indexing structure whose maintenance is *lazy* — cached weights and capacities refresh only when a correctness invariant demands it — certified by an amortized complexity theorem over *epochs* of selections: contradiction proofs on the global weight ordering prove the laziness safe, an epoch decomposition proves it cheap.

Problem. For each token pair (u, v) an *ordered edge group* $E_g(u, v)$ holds the parallel edges (feasible swaps) sorted by nondecreasing weight w_ℓ , each with capacity c_ℓ ; the *front* of a group is its minimum-weight edge with remaining capacity. A path π carries the ordering key $W(\pi) = \sum_{g \in \pi} w_g$ (sum of front weights) and the bottleneck capacity $C(\pi) = \min_{g \in \pi} c_g$. The iterative selection loop repeatedly extracts the minimum-key path, executes an amount, consumes capacities along the path (possibly advancing fronts), and re-ranks. ORDER maintains an *active bucket* — a heap over the top- K paths by key — plus *reserve buckets* holding the remaining candidates in coarser weight ranges, replenished by periodic rebuilds.

Guarantee.

Theorem 9.11 — Amortized complexity, Theorem 3.9 of the paper, notation cleaned.

Consider the iterative path-selection algorithm over $|P|$ indexed paths with active-bucket size K . One rebuild epoch with T selection iterations costs $O(|P|) + O(K \log K) + O(T \log K)$, so the amortized time per selection iteration is $O\left(\frac{|P|}{T}\right) + O\left(\frac{K \log K}{T}\right) + O(\log K)$. ■

How amortization is applied. The proof (transcribed in Section 9.6) is a textbook epoch decomposition in the sense of Proposition 9.5: an epoch is (1) one scan of all $|P|$ paths recomputing keys and admitting the top K into the active bucket via a KLL-style quantile sketch, (2) the build of that bucket, a heap over at most K paths, and (3) the T selections, each an extract-min plus a bounded number of localized heap updates

at $O(\log K)$ apiece. The amortization invariant is that the K admitted paths are all consumed *within that same epoch* — extracted or evicted — so the fixed costs (1)–(2) are spread over exactly the T selections they service. What keeps the lazy structure *correct* while cheap is a pair of contradiction arguments on the ordering of keys:

Lemma 9.12 — No-cross-bucket domination, Lemma 4.2 of the paper, notation cleaned. Given the prioritized buckets, for any two bucket indices $i < i'$, no path in $B_{i'}$ has a strictly smaller ordering key than a path in B_i : $\pi_p \in B_i, \pi_q \in B_{i'}, i < i' \Rightarrow W(\pi_p) \leq W(\pi_q)$. ■

Proof sketch. Buckets partition a global sort of paths by $W(\cdot)$ into index-ordered, non-overlapping ranges; a strict reversal would place π_q before π_p in that sort, forcing it into a bucket of index at most i , contradicting $\pi_q \in B_{i'}$ with $i' > i$. ■

Domination means selection never looks past the active bucket until it is exhausted; a companion rebuild-truncation lemma shows a rebuild can stop early while some reserve path is unaffected by recent updates, and an on-demand bottleneck lemma (Lemma 3.4 of the paper) shows the bottleneck read off the *current fronts* at selection time already equals the true one — so capacities too are maintained lazily, on demand.

What it buys. Each fill pays $O(\log K)$ -ish maintenance instead of a full re-ranking, while the routing output remains *exactly* optimal — laziness certified safe by the ordering lemmas, not traded against approximation — and the epoch schedule says when to expect the next heavier rebuild bill.

9.4 Quantile sketches: a potential over bucket counters

The exemplar is a SIGMOD'26 paper on quantile sketches [11]. Estimating quantiles over a stream — medians, percentiles, heavy thresholds — is a core analytics primitive, and the classical deterministic summaries (GK and its descendants, KLL) share a ceiling: within a bucket, rank is interpolated under a uniformity assumption, and that assumption is what their error bounds pay for. SplineSketch instead interpolates the rank function inside each bucket by a *spline* (monotone cubic by construction), which tracks skewed data far better; buckets carry counters, and the sketch stays at k buckets through two structural events — a *split* when a counter exceeds its budget and a *join* of adjacent buckets to reclaim space. The analysis questions are exactly amortization questions — how many splits can a stream force, and how much rank error does each deposit? — answered by an explicit potential over the bucket counters, the purest Φ -argument in this chapter's corpus.

Problem. A stream of n items arrives; the sketch keeps k buckets over the value range, bucket i being an interval $(\sigma_{i-1}, \sigma_i]$ between thresholds with a counter b_i of absorbed items. Arrivals are buffered and periodically merged into buckets; to keep at most k buckets, a bucket violating the *bucket bound* $b_i \leq C_b \cdot n/k$ must *split*, and adjacent buckets are *joined* to reclaim space — a pair is *joinable* if the merged counter satisfies $b_i + b_{i+1} \leq 0.75 \cdot C_b \cdot n/k$ and the threshold σ_i is not *protected* (freshly created by a split). Queries are answered by the counters plus spline interpolation inside each bucket. The task is to bound the number of splits — hence the protected thresholds, hence the ability to always find a joinable pair — and the resulting rank error.

Guarantee.

Theorem 9.13 — Rank error, from the paper’s error analysis, notation cleaned.

After processing n items (with a Misra–Gries sketch of size $O(k)$ filtering heavy hitters in parallel), for any threshold σ_i of the final sketch the rank error obeys

$$\varepsilon^n(\sigma_i) \leq O(\log \alpha \cdot n/k),$$

where α is the paper’s aspect-ratio parameter of the stream; the lower error is symmetric, and in the mergeable setting the bound is $O(\log(n/k) \cdot n/k)$. ■

How amortization is applied. Nothing here is random — this is deterministic bookkeeping. The paper defines the potential of a sketch state S at time t'' with $k(S)$ buckets as the total *excess of the counters over the join budget*:

$$\Phi(S) = \sum_{i=1}^{k(S)} \max\left\{0, b_i - 0.75 \cdot C_b \cdot t''/k\right\},$$

and verifies exactly the reachbox recipe: $\Phi(S) \in [0, t'']$ (nonnegativity); each of the $t - t'$ arrivals in an interval $(t', t]$ increments one counter by one, raising Φ by at most 1 (deposits, total increase $\leq t - t'$); joins never raise Φ (joinability keeps the merged counter at or below the join budget, where the max-terms vanish); and every forced split *decreases* Φ by at least $\beta \cdot C_b \cdot t/(2.5k)$ for a placement constant β (withdrawals). The workhorse is then pure division:

Lemma 9.14 — Split budget, from the paper’s analysis, notation cleaned. The number of splits during $(t', t]$ is at most $\frac{t-t'}{\beta C_b t/(2.5k)} = \frac{2.5k}{\beta C_b} \leq \frac{k}{18}$, with C_b chosen as a function of β . ■

Proof sketch. Divide the total possible deposit $t - t'$ by the minimum withdrawal per split; the withdrawal case analysis is in Section 9.6. ■

The split budget powers two conclusions. First, protected thresholds are bounded (each split creates $O(1)$), so a counting argument shows the sketch *always* contains a joinable pair — too many unjoinable adjacent pairs would need more protected thresholds than splits can create or more counter mass than t items supply. Second, the rank error is *charged along the split history*: a threshold’s error grows only at splits, and each ancestor in the merge tree contributes a geometrically shrinking share — the same arithmetic in log scale, giving the $O(\log \alpha \cdot n/k)$ bound. The spline itself contributes a feasibility lemma proved by a coefficient argument:

Lemma 9.15 — Spline midpoint bound, Lemma of the paper, notation cleaned. Let $P(x)$ be a cubic polynomial with $P(0) = 0$, $P(1) = 1$, non-decreasing on $[0, 1]$. Then $P(\frac{1}{2}) \in [\beta, 1 - \beta]$ with $\beta = \frac{1}{2} - \frac{\sqrt{3}}{4} \approx 0.067$. ■

Proof sketch. Write $P(\lambda) = a\lambda^3 + b\lambda^2 + c\lambda$ with $a + b + c = 1$; monotonicity forces $3a\lambda^2 + 2b\lambda + c \geq 0$ on $[0, 1]$, and evaluating that constraint at one well-chosen test point bounds $-(\frac{3a}{8} + \frac{b}{4})$, which is exactly $\frac{1}{2} - P(\frac{1}{2})$. (Full proof, with the test point, in Section 9.6.) ■

The lemma guarantees the interpolation inside any bucket is never arbitrarily skewed at its midpoint — the coefficient feasibility that makes the spline’s error arithmetic work with clean constants.

What it buys. $O(\log k)$ per operation via two heaps and a queue, deterministic and mergeable with the stated error guarantees, and — the experimental headline — noticeably more accurate quantiles than KLL and GK-style baselines at equal space, because the potential certifies a thrifty split schedule.

9.5 Advanced usage and further reading

The two usages sketched here compose the tool with other theory: an amortized bound purchased at the price of approximation, certified unavoidable by a lower-bound reduction; and an amortized update bound riding a DP space optimization.

Approximate query processing under updates [2]. Maintaining the answer to a conjunctive query over a semiring under updates is blocked, outside the tiny q-hierarchical class, by dynamic lower bounds: any non-q-hierarchical Boolean CQ needs amortized $\Omega(\sqrt{N})$ update time under fully dynamic sequences. The paper’s escape splits the free-connex join tree into an *exact connex part* (helper views kept exact, so enumeration stays constant-delay) and an *approximated part*, whose cached annotations are rounded to a factor b and refreshed lazily when the true

value outgrows the cache. The potential is the total rounding slack: an insertion deposits $O(1)$, a recomputation cascade fires only when some tuple's slack is exhausted, and each tuple can fire at most $\log_b M$ times, M the largest annotation ever cached. The guarantee (Theorem 4.7 of the paper): for any $b > 1$ and free-connex query, all auxiliary views update in amortized $O(\log_b M)$ time under insertion-only sequences (for the join size of a 3-way join, $O(\log_b N)$). The hypothesis is necessary: any multiplicative approximation on the semiring $(\mathbb{R}_{\geq 0}, +, \times)$ yields exact answers on the Boolean semiring, so the $\Omega(\sqrt{N})$ fully-dynamic lower bound transfers.

RadixGraph: amortized compaction over log segments [9]. RadixGraph is a dynamic graph storage structure: adjacency is compressed into radix-tree layers over bit prefixes of vertex ids, and updates append to log segments periodically *compacted* back into the layers. The layer structure is chosen by an occupancy model whose space objective is minimized by a DP over layer splits, computable in $O(nlx^2)$ time and $O(lx)$ space. On top sits the amortized bound (Theorem and proof sketch of the paper): edge insertion, update, and deletion cost amortized $O(1)$ — appending to a non-full log segment costs $O(1)$, and when a segment fills, the compaction costs $O(d)$ but is charged to the $\Omega(d)$ appends that must accumulate before it can fire, epoch counting in its barest form.

9.5.1 Further reading

- The amortized-analysis chapters of Cormen et al. [1] — aggregate, accounting, and potential methods with the dynamic-table example.
- Tarjan's survey [8], with the union-find and splay-tree classics [7].
- Overmars [6] — periodic rebuilding as the degenerate epoch schedule, and the deamortization caveats.
- Greenwald and Khanna [4] — quantile summaries where potential-bounded compaction first appeared in the DB literature; the ancestor of Section 9.4's analysis.

Chapter Recap

- **What the tool is.** Amortized analysis certifies per-operation time/space bounds over *sequences*: bursts are rare because they require accumulation, and the certificate is a bank balance (accounting), a state function $\Phi \geq 0$ (potential), or a batch schedule (epochs).
- **Which rung when.** Clean global sum of a worst-case sequence: aggregate. Explicit batch schedule (re-decompose every ℓ ar-

rivals, rebuild after T selections): epoch decomposition (Proposition 9.5). Measurable state trigger (counter excess, rounding slack, residual mass): potential (Theorem 9.4). Payer naturally an operation: accounting (Theorem 9.3).

- **Why it works.** Telescoping: $\sum_i c_i = \sum_i \hat{a}_i - \Phi(D_m) \leq \sum_i \hat{a}_i$ whenever $\Phi \geq 0$ and $\Phi(D_0) = 0$; the failure modes are a reachable state with $\Phi < 0$, or a burst that fires without its accumulation.
- **What the corpus showed.** Pattern enumeration (Section 9.1): a counting bound pays for list materialization with outputs — amortized $O(1)$ delay. Matrix sketches (Section 9.2): SVD bursts batched over $1/\varepsilon$ arrivals, composed across windows and sites. Order routing (Section 9.3): rebuild epochs over T selections, contradiction proofs certifying lazy maintenance. Quantile sketches (Section 9.4): a potential over bucket counters bounds splits by $k/18$; rank error is charged along the split history.
- **Advanced usage.** Amortization purchased with approximation ($O(\log_b M)$ updates at factor- b rounding, lower bound imported by a Boolean-semiring reduction), and amortized $O(1)$ graph updates riding a DP-optimal layout (Section 9.5). Deferred proofs are collected in the Appendix (Section 9.6).

9.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Lemma 9.7 (embedding counting)

Statement (recalled). For the valid occurrence sets $O_i \subseteq [\ell_i : r_i]$ of the constant strings w_i and the set S of all valid embeddings, $\sum_{i=1}^k (|O_i| - 1) \leq \min\{kn, |S|\}$. ■

Proof. The bound $\sum_i (|O_i| - 1) \leq kn$ is immediate from $|O_i| \leq n$. For the second bound, the paper’s canonical-embedding lemma gives that for $j_i \in O_i \setminus \{\ell_i\}$ the hybrid tuple $(\ell_1, \dots, \ell_{i-1}, j_i, r_{i+1}, \dots, r_k)$ is a valid embedding: anchoring the prefix at the leftmost and the suffix at the rightmost feasible occurrences leaves every intermediate valid occurrence of w_i feasible. So each i yields $|O_i| - 1$ valid embeddings, pairwise distinct — two for the same i differ in coordinate i , and two for $i_1 < i_2$ differ in coordinate i_1 . All lie in S , so $\sum_i (|O_i| - 1) \leq |S|$. ■

Proof of Lemma 9.9 (batched re-decomposition)

Statement (recalled). Re-decompositions fire at most once per $1/\varepsilon$ updates, each costs $O(d/\varepsilon^2)$, and the per-update power iteration costs $O(d \log d)$; the amortized per-update time is $O(\frac{d}{\varepsilon} \log d)$. ■

Proof. The sketch matrix C gains one row per update; the re-decomposition triggers when the row count reaches $2/\varepsilon$ and resets it to $1/\varepsilon$, so consecutive triggers are at least $\frac{1}{\varepsilon}$ updates apart — epochs of Proposition 9.5 with $\ell = 1/\varepsilon$. The trigger costs the eigen-decomposition of a $d \times \frac{2}{\varepsilon}$ matrix, $O(d/\varepsilon^2)$ in the paper's accounting, an amortized $O(d/\varepsilon)$ per update. Independently, every update runs a power iteration with $k = \log_2 d + 1$ matrix-vector products, $O(d \log d)$, against the dump threshold. Since $d \log d \leq \frac{d}{\varepsilon} \log d$ for $\varepsilon \leq 1$, the contributions sum to $O(\frac{d}{\varepsilon} \log d)$ amortized per update. ■

Proof of Theorem 9.11 (epoch accounting)

Statement (recalled). One rebuild epoch with T selections costs $O(|P|) + O(K \log K) + O(T \log K)$, so the amortized time per selection is $O(\frac{|P|}{T}) + O(\frac{K \log K}{T}) + O(\log K)$. ■

Proof. Decompose the epoch as in the paper's proof. (1) *Scan and admission*, $O(|P|)$: all $|P|$ paths are visited once to recompute weights and assign reserve-bucket ranges; the top K are selected through a KLL-style quantile sketch, which returns the admission threshold and emits the top K items. (2) *Active-bucket build*, $O(K \log K)$: a binary heap over at most K paths. (3) *Per-selection maintenance*, $O(T \log K)$: each selection is one extract-min plus $O(1)$ localized heap updates at $O(\log K)$ each. The amortization invariant — every path admitted in (1) is consumed within the same epoch — ensures the fixed costs (1)–(2) are covered by exactly the T selections they service; summing and dividing by T (Proposition 9.5) yields the per-iteration bound. ■

Proof of Lemma 9.14 (split budget)

Statement (recalled). With $\Phi(S) = \sum_{i=1}^{k(S)} \max\{0, b_i - 0.75 \cdot C_b \cdot t''/k\}$, the number of splits during $(t', t]$ is at most $\frac{2.5k}{\beta C_b} \leq \frac{k}{18}$. ■

Proof. Write $\tau_J = 0.75 \cdot C_b \cdot t''/k$ for the join budget at time t'' . *Deposits*: each of the $t - t'$ arrivals increments one counter by one, so Φ rises by at most 1 per arrival (total increase $\leq t - t'$), and $\Phi \geq 0$ throughout since every summand is nonnegative. *Joins*: joinability gives $b_i + b_{i+1} \leq \tau_J$, so the pre-join summands and the post-join one vanish; joins never increase Φ . *Withdrawals*: a split at time $t'' \in (t', t]$ fires on a bucket with $b_i > C_b t''/k$, whose summand contributes $b_i - \tau_J > 0.25 C_b t''/k$; the

split places the new threshold so each child receives at least a β -fraction of the parent's counter ($2.5 = 1/\beta$). If both children have counters $\geq \tau_j$, the decrease is exactly $\tau_j \geq 0.75 C_b t''/k$; otherwise the smaller child's summand vanishes and the decrease is at least that counter, $\geq \beta b_i > \beta C_b t''/k \geq \beta C_b t/(2.5k)$ (using $t'' \leq t$) — the paper's uniform withdrawal, met in both cases. *Division*: total decrease cannot exceed total increase, so splits number at most $\frac{t-t'}{\beta C_b t/(2.5k)} = \frac{2.5k}{\beta C_b} \leq k/18$, by the paper's calibration of C_b from β . ■

Proof of Lemma 9.15 (spline midpoint)

Statement (recalled). If P is cubic with $P(0) = 0$, $P(1) = 1$, non-decreasing on $[0, 1]$, then $P(\frac{1}{2}) \in [\frac{1}{2} - \frac{\sqrt{3}}{4}, \frac{1}{2} + \frac{\sqrt{3}}{4}]$. ■

Proof. By symmetry ($Q(x) = 1 - P(1 - x)$) is again cubic, non-decreasing, with $Q(\frac{1}{2}) = 1 - P(\frac{1}{2})$ it suffices to prove $P(\frac{1}{2}) \leq \frac{1}{2} + \frac{\sqrt{3}}{4}$. Write $P(\lambda) = a\lambda^3 + b\lambda^2 + c\lambda$: $P(1) = 1$ gives $a + b + c = 1$, and monotonicity gives $P'(\lambda) = 3a\lambda^2 + 2b\lambda + c \geq 0$ on $[0, 1]$. Since $P(\frac{1}{2}) = \frac{1}{2} - (\frac{3a}{8} + \frac{b}{4})$, it remains to bound $-(\frac{3a}{8} + \frac{b}{4})$. Substituting $c = 1 - a - b$ into the derivative constraint and evaluating at the test point $\lambda^* = \frac{3+\sqrt{3}}{6} \in [0, 1]$, where $3\lambda^{*2} - 1 = \frac{\sqrt{3}}{2}$ and $2\lambda^* - 1 = \frac{\sqrt{3}}{3}$, yields

$$0 \leq (3\lambda^{*2} - 1)a + (2\lambda^* - 1)b + 1 = \sqrt{3}\left(\frac{a}{2} + \frac{b}{3}\right) + 1,$$

so $\frac{a}{2} + \frac{b}{3} \geq -\frac{1}{\sqrt{3}}$; multiplying by $-\frac{3}{4}$ gives $-(\frac{3a}{8} + \frac{b}{4}) \leq \frac{3}{4\sqrt{3}} = \frac{\sqrt{3}}{4}$, hence the claim by symmetry. ■

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Coresets and Randomized Numerical Linear Algebra

Coresets and randomized numerical linear algebra (RNLA) answer the same question from two directions. The object is too big to optimize over — millions of rows, a dense affinity matrix, a plan space for every query shape, a join output too large to materialize — so replace it with a *small randomized proxy* on which every query of interest is answered within a $(1 \pm \epsilon)$ factor, with probability $1 - \delta$. The sampling culture (coresets) keeps a subset of the rows and reweights it, using *sensitivity* scores to decide who must be kept; the projection culture (sketches) mixes coordinates through a random matrix, shrinking the object while preserving norms and inner products. Both pay the same bill — per-query failure pushed below $\delta / (\text{queries})$ so one union bound certifies *all* queries at once — and both deliver *mergeable* summaries, which is what makes them the theory engine of distributed and streaming systems.

In the SIGMOD'26 corpus this book surveys, 8 tagged papers use this machinery — the survey's coreset-geometry tag (5 papers) merged here with randomized-nla (3) — and their pairing is uniform: every exemplar couples a proxy construction with a probabilistic-guarantee property, from set-outlier clustering over joins and robust query optimization to proximity graphs for similarity search. This chapter builds the two ladders first (Section 10.0), dissects four exemplars in full (Sections 10.1–10.4), and closes with two advanced sketches and further reading (Section 10.5).

10.0 The tool: coresets and randomized NLA

The problem class is uniform: a large object X — point set, matrix, workload, join output — and a family $\mathcal{F}$ of *queries*, each $f \in \mathcal{F}$ assigning a nonnegative cost $f(X)$, where computing on X directly is the

LEARNING OBJECTIVES

- Recognize when a workload admits a small proxy: a query family plus a cost to preserve for every query.
- Prove both ladders: sensitivity sampling $\rightarrow$ mergeable ϵ -coresets; random projections $\rightarrow$ sketch-and-solve $\rightarrow$ subspace embeddings.
- Size the proxy by total sensitivity or projection dimension, with per-query failure tuned to $\delta / (\text{queries})$ and union-bounded.
- Merge: distributed shards combine without re-reading the full data.

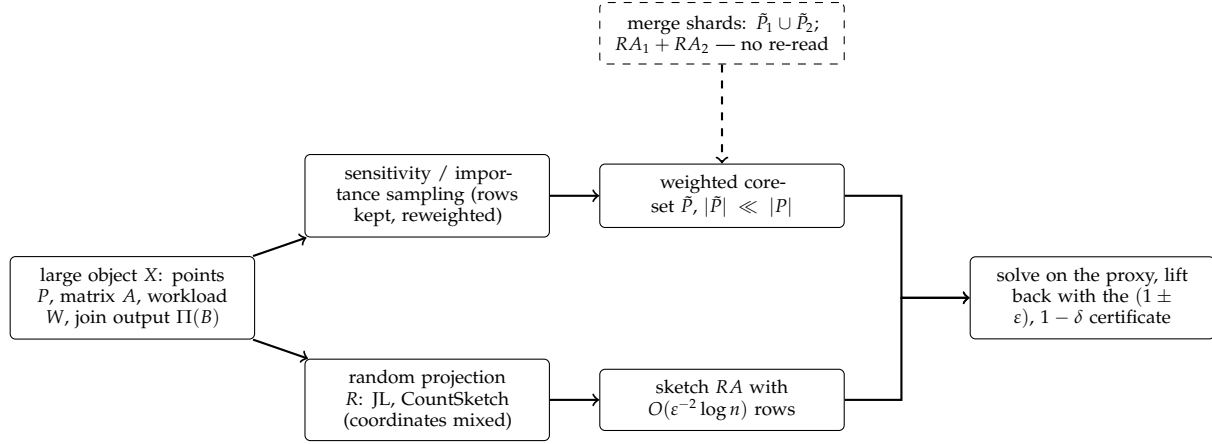


Figure 10.1: The two proxy tracks. Sampling keeps rows of the object and reweights them by sensitivity; projection mixes coordinates through a random matrix. Both end in the same place: solve on the small proxy, certify every query at once with one union bound, and merge shards of the proxy without revisiting the full object.

bottleneck. The deliverable is a small randomized $\tilde{X}$ with

$$|f(\tilde{X}) - f(X)| \leq \varepsilon f(X) \quad \text{for every } f \in \mathcal{F}, \quad \text{w.p. } 1 - \delta,$$

built in near-linear time. Everything here varies three choices: *what* to keep (rows vs. coordinate mixtures), *how* to bias the choice (sensitivities vs. projection distributions), and *how many* samples or dimensions the union bound demands (Figure 10.1).

Notation

$P = \{x_1, \dots, x_n\}$ is a weighted point set (w_i the weight of x_i); $\mathcal{F}$ a query family; $f(X)$ the answer of query f on X . “ $(1 \pm \varepsilon)$ -accurate” means relative error at most ε ; the failure probability is always δ , split across events by union bounds. On the RNLA side, R is the random sketch matrix, $\|\cdot\|_2$ the Euclidean/operator norm, $\|\cdot\|_F$ the Frobenius norm. For metric exemplars: the *doubling dimension* λ is the least λ such that every ball is covered by 2^λ balls of half the radius; the *aspect ratio* Φ is the diameter divided by the smallest inter-point distance.

10.0.1 The sampling ladder: from uniform to sensitivity

Uniform sampling is the first idea everyone has, and the first casualty of skew. Its failure is structural, not an edge case.

Proposition 10.1 — Uniform sampling fails on skew. For every n and every sample size $s \leq n/2$ there is a weighted set of $n + 1$ points in $\mathbb{R}$ such that, with probability at least $1/2$, a uniform sample of s points —

reweighted by $(n+1)/s$ — estimates the weighted cost of one of the two centers 0, 1 as 0 when its true cost is n^2 : no $(1 \pm \varepsilon)$ guarantee with constant ε is possible. ■

Proof. Take p_0 at the origin with weight n^2 , and n unit-weight points at coordinate 1. For the weighted sum-of-distances cost $f(c) = \sum_p w_p |p - c|$ we have $f(0) = n$ and $f(1) = n^2$ (the light points sit exactly at 1 and contribute 0 there). The sample misses p_0 with probability $(1 - \frac{1}{n+1})^s \geq 1 - \frac{s}{n+1} \geq \frac{1}{2}$, and on that event the estimated cost of center 1 is 0 against a true cost of n^2 . ■

Note what failed: not the sample budget but the sampling *distribution* — the heavy point above has sensitivity exactly 1, yet uniform sampling draws it with probability $\frac{1}{n+1}$. The repair is to sample where the cost concentrates [2, 7], with each point's worst-case share of a query's cost as the bookkeeping device:

Definition 10.2 (Sensitivity). For a weighted set P and a query family $\mathcal{F}$ of cost functions f , the *sensitivity* of point $x \in P$ is

$$s(x) = \max_{f \in \mathcal{F}} \frac{f(x)}{f(P)},$$

where $f(x)$ is x 's contribution to the answer of f . The *total sensitivity* is $S = \sum_{x \in P} s(x)$.

Sampling each x with probability proportional to $s(x)$ and reweighting is the workhorse theorem of the coreset culture. We state the finite-family version, whose proof is one Hoeffding step plus a union bound; sharper analyses replace $\ln |\mathcal{F}|$ by the VC-type complexity of $\mathcal{F}$ [7, 9].

Theorem 10.3 — Sensitivity sampling. Let $\mathcal{F}$ be a finite family of nonnegative cost functions on P with total sensitivity $S \geq 1$, and $\varepsilon, \delta \in (0, 1)$. Draw m i.i.d. points $x_1, \dots, x_m$, each equal to x with probability $s(x)/S$, and give x_i weight $w_i = \frac{S}{m} \cdot \frac{1}{s(x_i)}$. If $m = O(\frac{S^2}{\varepsilon^2} (\ln |\mathcal{F}| + \ln \frac{1}{\delta}))$, then with probability at least $1 - \delta$, simultaneously for every $f \in \mathcal{F}$,

$$\left| \sum_{i=1}^m w_i f(x_i) - f(P) \right| \leq \varepsilon f(P). \quad \blacksquare$$

Proof sketch. Fix f : the weighted sum is a sum of m i.i.d. mean- $f(P)$ variables, each in an interval of length $O(\frac{S}{m} f(P))$ because $f(x) \leq s(x)f(P)$ by Definition 10.2. Hoeffding gives per- f failure $\exp(-\Omega(m\varepsilon^2/S^2))$, and a union bound over $|\mathcal{F}|$ functions absorbs the $\ln |\mathcal{F}|$ term (full proof in Section 10.6). ■

1

What Theorem 10.3 produces has a name and a closure property that powers every distributed use.

¹ The sensitivity framework is the general theory of importance sampling for geometric objectives: after a bi-criteria approximation, the total sensitivity of k -median-type objectives is $O(k)$, so coreset size is $\text{poly}(k, 1/\varepsilon)$ — independent of n [7, 9]. Applied to the rows of a matrix, the same statistics gives *leverage scores* — sensitivity sampling with f ranging over residual directions [6].

Definition 10.4 (ε -coreset). A weighted multiset C is an ε -coreset of P for $\mathcal{F}$ if for every $f \in \mathcal{F}$, $|\sum_{x \in C} w_x f(x) - f(P)| \leq \varepsilon f(P)$.

Lemma 10.5 — Mergeability. If C_i is an ε -coreset of P_i for $\mathcal{F}$, $i = 1, 2$, then $C_1 \cup C_2$ with summed weights is an ε -coreset of $P_1 \cup P_2$. ■

Proof. For every f , the two absolute errors add over the disjoint union, and each is at most ε times its own part of $f(P_1 \cup P_2)$; the sum is at most $\varepsilon f(P_1 \cup P_2)$. ■

Mergeability is why coresets compose across shards of a distributed database: each shard builds its proxy locally, the proxies are shipped and unioned, and no shard is read twice. Linear sketches summaries of the form RX have the same closure by default — they add across shards — which is the design principle behind mergeable streaming summaries, from AMS sketches [3] through CountSketch [4] to Frequent Directions for matrices [12].

10.0.2 The projection ladder: from JL to subspace embeddings

The second culture shrinks the *dimension* rather than the row count, on the strength of the fact that Euclidean geometry is compressible: a random low-dimensional projection preserves all pairwise distances of a fixed point set [10].

Theorem 10.6 — Johnson–Lindenstrauss. For every $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^d$ and every $\varepsilon, \delta \in (0, 1)$ there is a random linear map $\Pi: \mathbb{R}^d \rightarrow \mathbb{R}^m$ with $m = O(\varepsilon^{-2}(\ln n + \ln \frac{1}{\delta}))$ such that, with probability at least $1 - \delta$, simultaneously for all pairs $i \neq j$,

$$(1 - \varepsilon) \|p_i - p_j\|_2^2 \leq \|\Pi p_i - \Pi p_j\|_2^2 \leq (1 + \varepsilon) \|p_i - p_j\|_2^2. \quad \blacksquare$$

Proof sketch. Take Π scaled Gaussian. For a *fixed* unit vector u , $\|\Pi u\|_2^2$ is chi-square with an exponential tail — a Chernoff-method fact (Chapter 2). Apply this to the $\binom{n}{2}$ difference directions $u_{ij} = (p_i - p_j) / \|p_i - p_j\|_2$ with per-pair failure $O(\delta/n^2)$ and union-bound (full proof in Section 10.6). ■

² The map need not be Gaussian: ± 1 signs suffice [1], and sparse sign matrices with one nonzero per column — CountSketch [4] — make the projection cost $O(\text{nnz})$ rather than $O(md)$; “database-friendly” projections are what make JL deployable inside a scan operator. The guarantee is also *oblivious*: the same distribution works for every P , so shards share one R and merge by summation.

²

JL preserves a *fixed finite set* of directions; numerical linear algebra needs an entire *subspace* — every direction in the column space of a data matrix — because the optimization variable ranges over a continuum of candidates [17, 20].

Theorem 10.7 — Subspace embedding by sparse sketches. Let $A \in \mathbb{R}^{n \times d}$ and $\varepsilon \in (0, 1)$. There is a distribution over CountSketch matrices $R \in \mathbb{R}^{m \times n}$

with $m = O(d^2/\epsilon^2)$ rows such that, with constant probability,

$$(1 - \epsilon) \|Ax\|_2 \leq \|RAx\|_2 \leq (1 + \epsilon) \|Ax\|_2 \quad \text{for all } x \in \mathbb{R}^d. \quad \blacksquare$$

Proof sketch. A net argument upgrades a fixed-direction JL guarantee to the whole d -dimensional column space — preserve a fine net of the sphere and push the distortion through products — and the heavy-hitter property of CountSketch tames the hash collisions; see [5, 20] for the complete argument. $\blacksquare$

Subspace embeddings convert optimization into *sketch-and-solve* [17]: compress, then optimize exactly on the small instance.

Corollary 10.8 — Sketch-and-solve for least squares. Let R be a $(1 \pm \epsilon)$ subspace embedding for the column space of $[A \ b]$, where $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^n$, and let $\hat{x} = \operatorname{argmin}_x \|RAx - Rb\|_2$. Then

$$\|A\hat{x} - b\|_2 \leq \frac{1 + \epsilon}{1 - \epsilon} \min_x \|Ax - b\|_2. \quad \blacksquare$$

Proof sketch. The embedding preserves *all* residual norms $\|Ax - b\|_2$ at once (the residuals live in a $(d+1)$ -dimensional space), so the minimizer of the preserved objective is near-minimal for the true one (full proof in Section 10.6). $\blacksquare$

The last rung, *iterative refinement*, is the practitioner’s caveat about Corollary 10.8: a $(1 \pm \epsilon)$ -embedded objective solved exactly on the sketch yields only constant-factor solutions. The RNLA workflow instead uses the sketch as a *preconditioner* — solve the sketched system once for a well-conditioned basis, then a few conjugate-gradient/LSQR iterations on the *original* residuals, converging at a rate set by the condition number of that basis [8, 15, 20]. Sketch once, refine cheaply.

10.0.3 How to choose the proxy

The selection is mechanical once two questions are answered. *What is the query family?* Geometric or combinatorial cost evaluations — clustering objectives, covering radii, range-space counts, plan-switching risks — point to the sampling track: bound the sensitivities, sample by them, reweight, size by Theorem 10.3. Linear-algebra quantities — norms, inner products, regression, low-rank factors, eigenvectors — point to the projection track: pick the sketch distribution (Gaussian / sign / CountSketch), set the dimension by JL or the embedding bound, and choose sketch-and-solve (Corollary 10.8) vs. precondition-and-refine by the accuracy the downstream consumer needs. *Where does the object live?* Rows on one machine: either track. Rows across shards or a stream: insist on mergeability (Lemma 10.5; linearity for sketches). Updates and

deletions: linear sketches survive them; row samples generally do not. Hybrids are the norm — leverage scores are sensitivities for residual directions and drive row-sampling regression [6], and the exemplar of Section 10.3 sketches an affinity matrix that a geometric routine balances first.

When to Reach for Coresets and Sketches

Reach whenever the cost of a computation is set by the size of an object that no consumer needs in full — materializing a join output, factorizing a dense affinity, enumerating a plan space, scanning every row per query — and the analysis tolerates $(1 \pm \varepsilon)$ error with a probability certificate. Then run the recipe:

1. **Identify the query family $\mathcal{F}$:** every quantity that must survive the compression, stated as a cost function on the object.
2. **Compute sensitivities or pick the projection:** bound each point's worst-case share $s(x)$ of a query (total sensitivity S), or fix the sketch distribution and dimension; if only leverage scores are available, that is the same decision for matrices [6].
3. **Verify the (ε, δ) guarantee:** per-query failure $\exp(-\Omega(\text{sample} \cdot \varepsilon^2 / S^2))$ or the JL tail; tune the size until it is below $\delta / (\text{queries})$, then union-bound over $\mathcal{F}$ (or its ε -net).
4. **Merge if distributed:** union the coresets / sum the sketches; by Lemma 10.5 and linearity the guarantee survives untouched — no shard is read twice.

10.1 Set-outlier clustering over relational data

The exemplar is a SIGMOD'26 paper on clustering with set outliers and its applications to relational clustering [18]. Clustering is made robust by allowing the algorithm to discard up to z outliers, but the standard model discards individual *points* — the wrong granularity for relational data, where one spurious tuple pollutes an entire set of join results and burning the outlier budget point by point leaves the damage in place. The paper introduces *clustering with set outliers*: exclude up to z sets from a given family, cluster what remains, and pay only for points no excluded set covers. For the abstract problem it gives LP-based approximation algorithms; for hyper-rectangle outlier sets in constant dimension, fast geometric algorithms; and for outliers induced by the tuples of an acyclic join, a coreset-based algorithm whose cost is near-linear in the database size rather than the quadratic-size join output. A Unique-Games-conditional hardness theorem shows the trade-offs are

nearly tight, and the coreset lens catches the two shrinkage steps inside: instance reduction, and the relational sample of the join output.

Problem. The abstract problem (Section 2 of [18], notation cleaned): given n points P in a metric space, a family $\mathcal{H}$ of m subsets covering P , and integers k, z , choose a center set $C \subseteq P$, $|C| \leq k$, and an outlier family $\mathcal{H}' \subseteq \mathcal{H}$, $|\mathcal{H}'| \leq z$, minimizing the k -center-style cost $\text{cost}(C, P \setminus \bigcup_{h \in \mathcal{H}'} h)$ — the maximum distance from any surviving point to its nearest center. An (α, β, γ) -approximation returns a solution of cost at most α times the optimum using at most βk centers and γz outlier sets. Two refinements matter: the *geometric* case, $P \subset \mathbb{R}^d$ with $\mathcal{H}$ a set of m hyper-rectangles covering P ; and the *relational* case, where P is the output $\Pi(B)$ of an acyclic join over a database B of size N and each outlier set is the set of join results that use a given tuple. Hardness: assuming the Unique Games conjecture, no polynomial-time $(1, f - \varepsilon, \gamma)$ -approximation exists for $f = o(\log n)$ and $\gamma = o(|\mathcal{H}|)$ (Theorem of the paper, via a Set-Cover reduction encoding set-element incidences as points on a line and sets as point sets).

Guarantee.

Theorem 10.9 — Set-outlier clustering with hyper-rectangles; Theorem of [18], notation cleaned. Given $P \subset \mathbb{R}^d$ (d constant), m hyper-rectangles covering P , and parameters k, z, ε , there is a $(2 + \varepsilon, 2, O(1))$ -approximation algorithm whose running time is near-linear in $n + m$ once the terms $\beta_1 = \min\{n, km\}$, $\beta_2 = \min\{n, kz\}$ and $\beta_3 = \min\{m, kz\}$ are isolated (the dependence on k and z lives in the β 's). ■

3

How coresets are applied. The engine room is a feasibility LP and its rounding; the coreset lens catches two shrinkage steps. (i) *LP feasibility with weight-charging rounding.* For a guessed radius r , an LP relaxation encodes “every point is r -covered by center mass or covered by outlier mass”; if $r \geq r_{k,z}^*$ (the optimum) the optimum itself witnesses feasibility, and greedy rounding converts feasibility into a solution with a charging certificate:

Lemma 10.10 — Rounding certificate; Lemma of [18], notation cleaned. If the feasibility LP is satisfiable for radius r , then greedy rounding returns C and $\mathcal{H}'$ with $\text{cost}(C, P \setminus \bigcup_{h \in \mathcal{H}'} h) \leq 2r$, $|\mathcal{H}'| \leq 2fz$ (for LP outlier inflation f), and $|C| \leq 2k$. ■

Proof sketch. Each greedy center is picked where it deactivates all active points within $2r$, giving the cost bound; the center count comes from charging to each selected center the LP mass of the r -ball around it —

³ The same LP-plus-rounding skeleton gives a $(2, 2, O(1))$ -approximation for the abstract problem, and — for bounded doubling dimension and disjoint outlier sets — a variant whose running time drops the n and m terms to the β 's (Theorem and Corollary of the paper): once the β 's are polylogarithmic the algorithm runs in *sublinear* time. That is the coreset dividend — the proxy, not the input, sets the cost.

the balls are disjoint, each carries mass at least $\frac{1}{2}$, and the total mass is at most k (full proof in Section 10.6). ■

(ii) *Instance shrinkage.* The β terms in Theorem 10.9 exist because the algorithm first computes a *sub-instance* — a coreset of the input itself. For a feasible radius $r \geq r_{k,z}^*(P, \mathcal{H})$ there are subsets $P' \subseteq P$ and $\mathcal{H}' \subseteq \mathcal{H}$ with $|P'| = O(\min\{n, k^2z, km\})$ and $|\mathcal{H}'| = O(\min\{m, kz\})$ such that the sub-instance’s optimum, for adjusted budgets, is at most $18r$ (Lemma of the paper): beyond the sets the optimum already excludes, only cluster boundaries matter, and $O(k^2z)$ points carry them. (iii) *The relational coreset.* For an acyclic join with $|\Pi(B)|$ results, the paper samples points of $\Pi(B)$ through counting/sampling oracles, builds a BBD-tree over the samples, and rounds:

Lemma 10.11 — Relational coreset; Lemma of [18], notation cleaned. Let Q be an acyclic join over a database B of size N , and $\delta = z/|\Pi(B)|$. One can compute $S \subseteq \Pi(B)$ with $|S| \leq k$ and $T \subseteq \Pi(B)$ with $|T| \leq (1 + \varepsilon)^2 \delta |\Pi(B)| = (1 + \varepsilon)^2 z$ such that $\text{cost}(S, \Pi(B) \setminus T) \leq 3(1 + \varepsilon)^2 r_{k,z}^*(\Pi(B))$, with probability at least $1 - 1/N$ (via counting/sampling oracles over the join). ■

The probability in Lemma 10.11 is where *randomized repetition* enters: a single sample catches all tuples generating the optimal centers with probability only about $2^{-g(k+z)}$ (each of the $\leq g(k+z)$ witness tuples is kept with probability $\frac{1}{2}$), so the algorithm repeats $\Theta(2^{g(k+z)} \log N)$ times and unions the per-run failure — repetition buying the exponent, exactly the “push per-event failure past $\log(\text{events})$ ” accounting used throughout this book. Chaining the coreset with the geometric algorithm yields the paper’s relational theorem: a $(1, 1 + \varepsilon, 3 + \varepsilon)$ -approximation with probability $1 - 1/N$.

What it buys. A query engine can answer “cluster this join, ignoring damaged regions” with cost near-linear in the database size N rather than the output size $|\Pi(B)|$: the coreset keeps k centers plus $(1 + \varepsilon)^2 z$ excluded results, and everything downstream — LP, rounding, binary search — runs on that proxy. The certified $(2 + \varepsilon, 2, O(1))$ trade-off, with UGC hardness showing the center/outlier budgets cannot all be kept at $1 + o(1)$, tells the system designer exactly which relaxations are free and which are impossible.

10.2 Coresets for robust query optimization

The exemplar is a PODS’26 paper on coresets for robust query optimization [16]^{PODS}. Query optimizers pick plans under cardinality estimates, and estimates are wrong: the plan that looks best can be catastrophic

on the true cardinalities. Robust query optimization hedges by maintaining a *portfolio* of plans such that whatever the workload brings, some plan in the portfolio is nearly as good as the best plan in the entire plan space. Portfolios cost memory and inference time, so the question is how small one can be. The paper’s answer is a coreset construction in an unusual direction — the object compressed is not the data but the *plan space* itself: a risk-aware portfolio $C \subseteq \Pi$ of size $O(\kappa^* \log \frac{1}{\delta})$ is computed from only $O(\frac{\kappa^*}{\delta^2} \ln N \ln \frac{1}{\delta})$ sampled queries, with high probability, where κ^* is the best possible portfolio size. The sizing argument is exactly the recipe of Section 10.0.3: Hoeffding per candidate portfolio, union bound over all of them. (The greedy selection procedure is dissected under the exchange-and-greedy lens in Section 7.1; this section keeps the coreset lens.)

Problem. Let Π be the set of N candidate plans for a query family, W a distribution over queries, and each query Q an *uncertainty profile*: a distribution U_Q over cardinality vectors s . The penalty of plan π on s is its cost ratio $\text{Pnl}(\pi, s) = \text{Cost}(\pi, s) / \text{Cost}^*(s)$ against the optimal plan for s , and the τ -risk of π on Q is $\text{Risk}(\pi, Q; \tau) = \Pr_{s \sim U_Q}[\text{Pnl}(\pi, s) > \tau]$ — the probability that π blows past tolerance τ . Writing $\text{Risk}^*(Q; \tau)$ for the best risk over all of Π , a set $F \subseteq \Pi$ is an $(\varepsilon, \tau, \delta|W)$ -coreset for risk (reconstructed from Lemmas 3.1 and 3.3 of the paper, notation cleaned) if

$$\Pr_{Q \sim W} \left[\min_{\pi \in F} \text{Risk}(\pi, Q; \tau) > \text{Risk}^*(Q; \tau) + \varepsilon \right] \leq \delta,$$

i.e., for all but a δ -fraction of the workload some portfolio plan is within ε of the best achievable risk; κ^* is the size of the smallest such set.

Guarantee.

Theorem 10.12 — Risk coresets; Theorem 3.5 of [16], notation cleaned. Let W be a workload distribution, Π a set of N plans, parameters $\varepsilon \in (0, 1]$, $\tau \geq 1$, $\delta \in (0, 1)$, and κ^* the size of the smallest $(\varepsilon, \tau, \delta|W)$ -coreset for risk. For any fixed constants $c_1, c_2 > 1$ there is an algorithm that samples $O(\frac{\kappa^*}{\delta^2} \ln N \ln \frac{1}{\delta})$ queries from W and computes, in time $O(\frac{\kappa^*}{\varepsilon^2 \delta^2} N \ln^3 N \ln \frac{1}{\delta})$, a set $C \subseteq \Pi$ of size $O(\kappa^* \ln \frac{1}{\delta})$ that is, with high probability, a $(c_1 \varepsilon, \tau, c_2 \delta|W)$ -coreset for risk. ■

How coresets are applied. The random object is the batch of sampled queries $Q_1, \dots, Q_{n_r} \sim W$ (plus, per query, an inner sample $s_1, \dots, s_{m_r} \sim U_Q$ used to estimate risks); the query family is, unusually, the family of *all subsets of Π up to size b_r* — every portfolio the algorithm might output is a query whose cost the sample must certify. Sizing is a two-layer Hoeffding argument:

Lemma 10.13 — Coreset sizing; Lemma 3.1 of [16], notation cleaned. Fix $\varepsilon' \in (0, 1)$ and round $r \geq 1$. Let $\hat{p}_r(\varepsilon', F)$ be the empirical fraction of the $n_r = \frac{c_1 k_r}{\delta^2} \ln N \ln \frac{1}{\delta}$ sampled queries on which $F \subseteq \Pi$ fails to be ε' -competitive, and $p(\varepsilon', F)$ its expectation. Then $\hat{p}$ approximates p within $\delta/4$ simultaneously for all $F \subseteq \Pi$ with $|F| \leq b_r = c_3 k_r \ln \frac{1}{\delta}$, with probability at least $1 - N^{-O(1)}$. ■

Proof sketch. $\hat{p}_r(\varepsilon', F)$ is a mean of i.i.d. $[0, 1]$ variables, so Hoeffding gives failure $2 \exp(-\delta^2 n_r / 8)$ per set; the number of candidate portfolios is at most $(b_r + 1)N^{b_r}$, and a union bound over them turns the exponent into $N^{-O(1)}$ once c_1 dominates c_3 (full proof in Section 10.6). ■

A companion lemma (Lemma 3.2 of the paper) runs the same Hoeffding-plus-union argument over the $\leq N n_r$ (plan, query) pairs for the inner risk estimates, with inner sample size $m_r = \frac{c_2}{\varepsilon^2} \ln N \ln n_r$; the δ -budgeting is explicit — accuracy $\delta/4$ for the outer fraction, $\varepsilon/10$ for the inner risks, constants tuned so both failures are polynomially small in every round. The substitution property — the reason the proxy can replace the full plan space at inference time — is a clean transfer between *global* risk (against all of Π) and *local* risk (against the portfolio):

Lemma 10.14 — Risk transfer; Lemma 4.2 of [16], notation cleaned. For any $F \subseteq \Pi$, $\pi \in F$ and query Q : $\text{Risk}(\pi, Q; \tau^2) \leq \text{L-Risk}_F(\pi, Q; \tau) + \min_{\pi' \in F} \text{Risk}(\pi', Q; \tau)$, where L-Risk_F measures the penalty of π relative to the best plan in F ; consequently $\min_{\pi \in F} \text{Risk}(\pi, Q; \tau^2) \leq 2 \min_{\pi \in F} \text{Risk}(\pi, Q; \tau)$. ■

Proof sketch. Factor the penalty as a product of the local ratio and the reference plan's global ratio; the event $\{\text{Pnl} > \tau^2\}$ is then contained in the union of the two factor events $\{\cdot > \tau\}$ — a union bound in disguise (full proof in Section 10.6). ■

Chaining Lemma 10.14 with an empirical version of the local risk (Hoeffding over the inner samples again) gives the paper's inference guarantee: the returned plan $\hat{\pi}$ obeys $\text{Risk}(\hat{\pi}, Q; \tau^2) \leq 2 \text{Risk}^*(Q; \tau) + 4\varepsilon$ with probability at least $1 - 2\delta$, in $\Theta(|F|(\ln |F| + \ln \frac{1}{\delta})/\varepsilon^2)$ time (Theorem 4.1 of the paper): solving on the proxy costs a factor 2 — the standard coreset tax — instead of a scan over all N plans.

What it buys. The optimizer carries a portfolio of $O(\kappa^* \log \frac{1}{\delta})$ plans instead of N , built from a logarithmic number of workload samples, with a certificate that for all but a δ -fraction of the workload some portfolio plan is within ε of the best achievable risk, and inference against the portfolio runs in time nearly linear in its size. The sizes

adapt to the workload (through κ^*) rather than to the plan space, so easy workloads get tiny portfolios for free.

10.3 Sketching the affinity before the eigenvalue work

The exemplar is a SIGMOD'26 paper on effective clustering for large multi-relational graphs [13]. A multi-relational graph links its nodes through many relation types at once, and partitioning the N nodes into K clusters is a standard analytics task that existing systems either solve inaccurately (crudely fused relations) or expensively (deep models that stall at millions of nodes). The paper's pipeline optimizes a normalized-cut-type objective through an affinity matrix and refuses to ever form the dense $N \times N$ object: the relaxed problem becomes a k -means instance after a balancing step, the smoothed node embedding behind it is a truncated series of sparse matrix products, and the Frobenius norms that calibrate everything are estimated from a CountSketch of the data. Chapter 8 dissects the same paper through the spectral lens (Neumann series, eigenvalue confinement $\lambda \leq 1$, Sinkhorn balancing); this section keeps the randomized-NLA lens: what is compressed, by which sketch, and with what probability certificate.

Problem. Let G be a multi-relational graph with N nodes and R relations; relation r induces a (row-normalized) affinity block $S^{(r)}$, fused into $S = \sum_{r=1}^R \omega_r S^{(r)}$ with $\sum_r \omega_r = 1$. The clustering objective is a normalized-cut quantity, equivalent to trace maximization $\max_Y \text{trace}(Y^\top S Y)$ over node-cluster incidence matrices — NP-hard, and the standard spectral relaxation needs a partial eigendecomposition of the dense $N \times N$ matrix S . The paper replaces this with a two-stage program. Stage one computes smoothed node features $Z = \frac{1}{1+\alpha} (I - \frac{\alpha}{1+\alpha} S)^{-1} Z^0$ (closed form; the inverse is a Neumann series because the spectrum of S lies in $[0, 1]$ — Lemma of the paper, proved by Courant–Fischer plus the Lidskii inequality $\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B)$). Stage two balances S to doubly stochastic by Sinkhorn–Knopp iterative proportional fitting (row and column scalings coincide by symmetry and Sinkhorn uniqueness, Theorem of the paper), which makes the trace objective *equivalent* to weighted k -means on the rows of Z . The computational obstacle: evaluating the objective terms and the truncation error requires Frobenius norms of dense $N \times N$ products.

Guarantee.

Corollary 10.15 — **Sketched objective; Corollary of [13], notation cleaned.** Let R

be a CountSketch matrix with

$$m = O\left(r \varepsilon^{-4} \log \frac{r}{\varepsilon \delta} \cdot \left(r + \log \frac{1}{\varepsilon \delta}\right)\right)$$

columns, where r is the target rank and $\varepsilon, \delta \in (0, 1)$. Then the sketched estimate of the squared Frobenius norm of the whitened product $\hat{M}^{(r)} = M(M - M^{(r)})^{-1/2}$ — the stage-one objective term, whose exact value is the trace quantity $\text{trace}(M(M - M^{(r)})^{-1})$ by the paper’s identity — is within a $(1 \pm \varepsilon)^2$ factor of its exact value, with probability at least $1 - \delta$. ■

How randomized NLA is applied. The random object is the CountSketch matrix — each row of the data is hashed to one of m buckets with a random sign — the oblivious projection of Theorems 10.6 and 10.7: the same distribution works for every M , so shards sketch independently and merge by summation. The structure exploited is that every quantity the pipeline needs is a squared Frobenius norm, and a squared Frobenius norm is a sum of inner products: the paper first converts its trace term into $\|\hat{M}^{(r)}\|_F^2$ (identity $\text{trace}(M(M - M^{(r)})^{-1}) = \|\hat{M}^{(r)}\|_F^2$, Lemma of the paper), then estimates it from the sketch, because CountSketch preserves pairwise inner products with high probability — for every pair of rows, the sketched inner product is within ε of the true one with probability $1 - \delta$ (the inner-product-preservation property the paper imports). The δ -budgeting is familiar: per-relation failure δ/R , union-bounded over the R relations, so the fused estimates inherit the certificate. Supporting the sketch, the eigenvalue side certifies that the sketched pipeline is legal: $\lambda(S^{(r)}) \leq 1$ (Courant–Fischer plus Lidskii) makes the Neumann series converge, and truncating after L terms costs at most $(\frac{\alpha}{1+\alpha})^{L+1} \|Z^0\|_F$ divided by the eigengap quantity $\mu_{L,L+j} = \max_i |\lambda_i(S)^{L+j} - \lambda_i(S)^L|$ of the fused spectrum (Theorem of the paper; the full spectral dissection, with proofs, is in Section 8.2).

What it buys. The engine clusters graphs with billions of edges in time linear in the input instead of quadratic in N : no dense $N \times N$ matrix is ever formed, the embedding is L sparse products with a certified geometric truncation error, and the calibration norms come from a mergeable sketch with an (ε, δ) certificate. Because the balancing step makes the objective exactly k -means, the final clustering solves the relaxed problem the sketch was built for — not a surrogate of it.

10.4 Proximity graphs from covering nets

The exemplar is a SIGMOD’26 paper on proximity graphs for similarity search [14]. Graph indexes are how vector databases answer nearest-neighbor queries at scale: build a graph over the data points

and answer a query by *greedy routing* — start anywhere, repeatedly hop to the neighbor closest to the query, stop when no neighbor is closer. The graph is a $(1 + \varepsilon)$ -proximity graph $((1 + \varepsilon)$ -PG) if the walk always ends at a $(1 + \varepsilon)$ -approximate nearest neighbor; the practical questions are how many edges it needs, how fast it builds, and how long the walk takes. The paper settles the trade-off for doubling metrics: $O(\frac{1}{\varepsilon^\lambda} n \log \Phi)$ edges are enough and constructible in near-linear time, the $\log \Phi$ is *necessary* in general (an $\Omega(n \log \Phi)$ lower bound) yet *vanishes* in constant-dimensional Euclidean space, and the query analysis is a distance-descent lemma on the greedy walk. The construction itself is a coreset in metric clothing: a hierarchy of covering nets keeping a bounded number of representatives per distance scale.

Problem. Let $(\mathcal{M}, D)$ be a metric space of doubling dimension λ , P an n -point subset with aspect ratio Φ , and $q \notin P$ a query. A directed graph G on P is a $(1 + \varepsilon)$ -PG if greedy routing — from any start, hop to the out-neighbor closest to q , halt when no out-neighbor is closer — terminates at a point p with $D(p, q) \leq (1 + \varepsilon) D(p^*, q)$ for an exact nearest neighbor p^* . The design goals: few edges (space), near-linear construction, and short greedy walks (query time), where the walk length counts distance evaluations.

Guarantee.

Theorem 10.16 — Proximity graphs for doubling and Euclidean metrics; Theorem of [14], notation cleaned. (i) For any metric space of constant doubling dimension λ and any n -point set P with aspect ratio Φ , there is a $(1 + \varepsilon)$ -PG with $O(\frac{1}{\varepsilon^\lambda} n \log \Phi)$ edges and greedy query time $O(\frac{1}{\varepsilon^\lambda} \log^2 \Phi)$, constructible in $O(\frac{1}{\varepsilon^\lambda} n \log n)$ time. (ii) For $P \subset \mathbb{R}^d$ under L_2 with $d = O(1)$, there is a $(1 + \varepsilon)$ -PG with only $O(\frac{1}{\varepsilon^\lambda} n)$ edges and query time $O(\frac{1}{\varepsilon^\lambda} \log^2 \Phi + \frac{1}{\varepsilon^{d-1}} \log n \log^2 \Phi)$; w.h.p. it is built in $O(\frac{1}{\varepsilon^\lambda} n \log n)$ time. ■

The complementary lower bound shows part (i) cannot lose its $\log \Phi$:

Theorem 10.17 — Size lower bound; Theorem of [14], notation cleaned. There is a set of $\Theta(n)$ points with aspect ratio Φ (a power of 2) in a metric space of doubling dimension 1 such that any 2-PG for it must have $\Omega(n \log \Phi)$ edges, *regardless of the query time allowed*. Also, for every $s \geq 2$, $t \geq 1$, $d \geq 1$ there is an $n = s^d \cdot t$ point set with doubling dimension at most $\log(1 + 2^d)$ such that any $(1 + \frac{1}{2s})$ -PG needs $\Omega(s^d \cdot n)$ edges. ■

How coresets are applied. The proxy is a *covering-net hierarchy*: nested subsets $Y_h \subseteq \dots \subseteq Y_1 \subseteq Y_0 = P$ where Y_i is a 2^i -net — every point of P is within 2^i of some point of Y_i — and $h = \log \Phi$ indexes the scales

from the minimum inter-point distance to the diameter. The graph keeps, for each point, edges to nearby net representatives at every scale; the decomposition is a coresets of the metric — a per-scale sample of the point set that preserves every point’s distance profile to within the scale resolution — and its sparsity is what sizes the graph.

Lemma 10.18 — Net out-neighbors; from the construction and its proof in [14], notation cleaned. In the covering-net construction, every point $p \in P$ keeps, at each scale i , an out-neighbor $y \in Y_i$ with $D(p, y) \leq 2^i$ (the covering property); the per-point out-degree is $O(\frac{1}{\epsilon^\lambda} \log \Phi)$, and the graph has $O(\frac{1}{\epsilon^\lambda} n \log \Phi)$ edges. ■

Proof sketch. Covering supplies the per-scale representative; the doubling property bounds how many net points matter in p ’s distance window at each scale, and there are $O(\log \Phi)$ scales (full proof in Section 10.6). ■

The query analysis is where the paper’s sharpest lemma lives — the greedy walk descends the net hierarchy, and each “jackpot” hop shrinks the *logarithm* of the distance to the true nearest neighbor:

Lemma 10.19 — Greedy descent and log-drop; Lemma of [14], notation cleaned. Fix a query q and $p \in P$ that is not a $(1 + \epsilon)$ -ANN of q ; let p^+ be the out-neighbor of p closest to q . Then (i) $D(p^+, q) < D(p, q)$; and (ii) for any $x \in P$ with $D(x, q) \leq D(p^+, q)$ that is not a $(1 + \epsilon)$ -ANN of q ,

$$\log D(x, p^*) < \log D(p, p^*)$$

for an exact nearest neighbor p^* of q . ■

Statement (ii) is the workhorse: between two greedy-descent configurations the (log) distance to the nearest neighbor strictly drops, so after $O(\log \Phi)$ drops the walk is within scale 1 of p^* , and each inter-drop segment is short because the walk is monotone in distance to q while the net hierarchy has only $O(\log \Phi)$ scales (the paper splits the walk into subsequences at jackpot hops and charges each subsequence $\Theta(\log n \log \Phi)$ hops). The Euclidean separation of Theorem 10.16(ii) removes one $\log \Phi$ from the *space*: in L_2 , an angle-separation lemma (points equidistant from p on nested query spheres are separated by angle $> \pi/8$) shows each scale needs only a bounded cone of representatives per point, while the adversarial metrics of Theorem 10.17 — recursively nested clusters — prove general doubling metrics enjoy no such savings.

What it buys. The vector-search index gets graphs that are simultaneously sparse (affordable memory), near-linearly constructible (affordable builds), and certifiably correct under greedy routing with

polylogarithmic query paths — and the matching lower bound tells the engineer that pushing space below $n \log \Phi$ in general doubling metrics is provably impossible, so optimization effort belongs in the Euclidean regime where the extra $\log \Phi$ genuinely vanishes.

10.5 Advanced usage and further reading

The four exemplars share one pattern: a coreset or sketch whose size is governed by complexity parameters ($k, z; \kappa^*$; rank r ; doubling dimension and Φ) rather than by the data size. Two further corpus papers use the same machinery in harder combinations; each is sketched in one paragraph.

Spanners as distance-preserving proxies on terrains [19]. The constrained shortest path problem CPTS asks for a shortest path on a terrain surface that visits a set of points of interest, and is NP-hard by reduction from Euclidean TSP (Theorem 3.3 of the paper). The proxy is a *spanner*: a sparse graph G over the terminal points with $d_g(u, v) \leq d_G(u, v) \leq (1 + \varepsilon) d_g(u, v)$ for all pairs, where d_g is the geodesic distance — exactly the coreset contract of Definition 10.4 with query family $\mathcal{F} = \{\text{pairwise distances}\}$, satisfied for *every query at once* by a deterministic construction. The spanner lemmas are proved by contradiction and induction on an iterative geometric procedure (with triangle-inequality charging of the per-iteration weights), and the final route is assembled on G by Kruskal’s MST plus a pre-order traversal, whose edge-weight accounting yields a factor-2 path over the spanner — solve on the proxy, lift back with a certificate, the signature move of this chapter in a deterministic register.

Sketches inside a privacy budget [11]. HeteroFedSyn synthesizes tabular data under differential privacy across heterogeneous participants. Its privacy accounting is zCDP composition (mechanisms with parameters ρ_i compose to $\rho = \sum_i \rho_i$, Definition of the paper) plus post-processing, which turns budget allocation across the $O(d^2)$ two-way marginals into an optimization problem; and its compression is this chapter’s: a random projection squeezes the two-way marginal vectors before the Gaussian noisy release, so the noise (calibrated to the projected sensitivity) covers fewer coordinates. An exact expectation calculation strips the projection and noise bias, yielding an unbiased estimator of the pairwise dependency measure InDif^2 (Theorem of the paper), and a greedy selection spends the budget where dependency error dominates noise error. The RNLA lesson: inside a privacy pipeline the sketch is not an optimization but a *variance* device — fewer released coordinates means less noise per coordinate at fixed ρ .

10.5.1 Further reading

- **Coresets:** the unified sensitivity framework of Feldman and Langberg [7]; the k -means/ k -median construction of Har-Peled and Mazumdar [9]; the geometric survey of Agarwal, Har-Peled, and Varadarajan [2].
- **Random projections:** the original Johnson–Lindenstrauss lemma [10] and the database-friendly ± 1 variant of Achlioptas [1].
- **Sketching for linear algebra:** Sarlós’s projection-based regression [17], input-sparsity subspace embeddings of Clarkson and Woodruff [5], leverage-score sampling [6], and Woodruff’s monograph [20].
- **The randomized-NLA survey of Mahoney [15]** and the randomized matrix-decomposition treatment of Halko, Martinsson, and Tropp [8] cover the iterative-refinement mode of Section 10.0.2.
- **Mergeable summaries:** AMS frequency-moment sketches [3], CountSketch [4], and Frequent Directions for matrix sketches [12].

Chapter Recap

- **What the tool is.** A small randomized proxy — a weighted row sample (coreset) or a random coordinate mixture (sketch) — that answers every query in a family within $(1 \pm \epsilon)$ with probability $1 - \delta$, built in near-linear time and sized by complexity parameters, not data size.
- **Which track when.** Geometric or combinatorial cost queries: sensitivity sampling (Theorem 10.3) — uniform sampling fails on skew (Proposition 10.1). Linear-algebra queries: JL for finitely many directions (Theorem 10.6), subspace embeddings for a continuum of candidates (Theorem 10.7); solve on the sketch for constant factors (Corollary 10.8), refine iteratively for high accuracy. Distributed or streaming: insist on mergeability (Lemma 10.5; linearity for sketches).
- **Why it works.** Per-query failure is pushed below $\delta/(\text{queries})$ — Hoeffding on the weighted sample, or the chi-square tail on the projection — and one union bound (over the query family, an ϵ -net, or all candidate portfolios) certifies every query simultaneously.
- **What the corpus showed.** Set-outlier clustering (Section 10.1): LP-guided instance shrinkage to $O(k^2z)$ points and a $k + (1 + \epsilon)^2z$ -result coreset of the join output, with randomized repetition over samples. Robust query optimization (Section 10.2):

Hoeffding-plus-union sizing of plan portfolios, with global-to-local risk transfer as the substitution property. Multi-relational clustering (Section 10.3): CountSketch estimates the objective norms so the dense affinity is never formed. Proximity graphs (Section 10.4): covering-net hierarchies (metric coresets) with a log-drop lemma bounding greedy query paths, plus $\Omega(n \log \Phi)$ optimality.

- **Advanced usage.** Deterministic distance proxies (spanners) satisfy the coreset contract without randomness; sketches inside privacy pipelines trade coordinates against noise at fixed budget (Section 10.5). Deferred proofs live in the Appendix (Section 10.6).

10.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 10.3 (sensitivity sampling)

Statement (recalled). Draw m i.i.d. points, each x with probability $s(x)/S$, weight $w_i = \frac{S}{m} \frac{1}{s(x_i)}$, with $m = \Theta(\frac{S^2}{\epsilon^2} (\ln |\mathcal{F}| + \ln \frac{1}{\delta}))$; then w.h.p. every $f \in \mathcal{F}$ has $|\sum_i w_i f(x_i) - f(P)| \leq \epsilon f(P)$. ■

Proof. Fix f and let $Z_i = w_i f(x_i)$. Each draw picks x with probability $p_x = s(x)/S$, so $\mathbb{E}[Z_i] = \sum_x p_x \frac{S}{ms(x)} f(x) = \frac{1}{m} \sum_x f(x) = \frac{f(P)}{m}$ — the estimator is unbiased. For the range, $f(x) \leq s(x)f(P)$ (Definition 10.2) gives $0 \leq Z_i \leq \frac{S}{m} f(P)$, so the Z_i are i.i.d. in an interval of length $b = \frac{S}{m} f(P)$. Hoeffding (Theorem 2.5) yields

$$\begin{aligned} \Pr\left[\left|\sum_i Z_i - f(P)\right| \geq \epsilon f(P)\right] &\leq 2 \exp\left(-\frac{2m^2 \epsilon^2 f(P)^2}{mb^2}\right) \\ &= 2 \exp(-2m \epsilon^2 / S^2). \end{aligned}$$

With $m \geq \frac{S^2}{2\epsilon^2} (\ln |\mathcal{F}| + \ln \frac{2}{\delta})$ this is at most $\delta/|\mathcal{F}|$ per query, and a union bound over all $f \in \mathcal{F}$ completes the proof. ■

Proof of Theorem 10.6 (Johnson–Lindenstrauss)

Statement (recalled). A scaled Gaussian map $\Pi \in \mathbb{R}^{m \times d}$ with $m = O(\epsilon^{-2} (\ln n + \ln \frac{1}{\delta}))$ preserves all $\binom{n}{2}$ pairwise squared distances of P to $(1 \pm \epsilon)$ with probability $\geq 1 - \delta$. ■

Proof. Let $\Pi = \frac{1}{\sqrt{m}} G$ with $G_{ij} \sim N(0, 1)$ i.i.d. For a fixed unit vector u , $Y_u = \|\Pi u\|_2^2 = \frac{1}{m} \sum_{j=1}^m g_j^2$ with $g_j \sim N(0, 1)$ i.i.d. The Chernoff method

(Lemma 2.3) applied to $\mathbb{E} e^{\lambda g^2} = (1 - 2\lambda)^{-1/2}$, optimized over λ , gives the chi-square tail: for $\varepsilon \in (0, 1)$,

$$\Pr[|Y_u - 1| > \varepsilon] \leq 2e^{-m\varepsilon^2/12}.$$

Apply this to each difference direction $u_{ij} = (p_i - p_j)/\|p_i - p_j\|_2$: since Π is linear, $\|\Pi p_i - \Pi p_j\|_2^2 = \|p_i - p_j\|_2^2 Y_{u_{ij}}$, so $(1 \pm \varepsilon)$ -preservation of $Y_{u_{ij}}$ is preservation of the pair. Union-bounding over the $\binom{n}{2} \leq n^2/2$ pairs, the total failure probability is at most $n^2 e^{-m\varepsilon^2/12} \leq \delta$ for the stated m . ■

Proof of Corollary 10.8 (sketch-and-solve)

Statement (recalled). If R is a $(1 \pm \varepsilon)$ subspace embedding for the column space of $[A \ b]$ and $\hat{x}$ minimizes $\|RAx - Rb\|_2$, then $\|A\hat{x} - b\|_2 \leq \frac{1+\varepsilon}{1-\varepsilon} \min_x \|Ax - b\|_2$. ■

Proof. Let x^* minimize $\|Ax - b\|_2$. The embedding hypothesis holds for every x : since $Ax - b$ lies in the column space of $[A \ b]$,

$$\begin{aligned} (1 - \varepsilon) \|A\hat{x} - b\|_2 &\leq \|RA\hat{x} - Rb\|_2 \leq \|RAx^* - Rb\|_2 \\ &\leq (1 + \varepsilon) \|Ax^* - b\|_2, \end{aligned}$$

where the middle inequality uses that $\hat{x}$ minimizes the sketched objective. Dividing by $1 - \varepsilon$ gives the claim. ■

Proof of Lemma 10.13 (coreset sizing)

Statement (recalled). The empirical failure frequency $\hat{p}_r(\varepsilon', F)$ over $\mathbf{n}_r = \frac{c_1 k_r}{\delta^2} \ln N \ln \frac{1}{\delta}$ i.i.d. queries approximates $p(\varepsilon', F)$ within $\delta/4$, simultaneously for all $|F| \leq b_r = c_3 k_r \ln \frac{1}{\delta}$, with probability $1 - N^{-O(1)}$. ■

Proof. For fixed F , $\hat{p}_r(\varepsilon', F)$ is a mean of $\mathbf{n}_r$ i.i.d. $[0, 1]$ indicators with expectation $p(\varepsilon', F)$, so Hoeffding gives

$$\begin{aligned} \Pr[|\hat{p}_r(\varepsilon', F) - p(\varepsilon', F)| > \delta/4] &\leq 2 \exp(-2\mathbf{n}_r(\delta/4)^2) \\ &= 2 \exp(-\delta^2 \mathbf{n}_r / 8). \end{aligned}$$

The number of subsets of Π of size at most b_r is at most $(b_r + 1)N^{b_r}$, so the probability that any F violates the accuracy is at most $2(b_r + 1)N^{b_r} \exp(-\delta^2 \mathbf{n}_r / 8)$. Substituting $\mathbf{n}_r = \frac{c_1 k_r}{\delta^2} \ln N \ln \frac{1}{\delta}$ and $b_r = c_3 k_r \ln \frac{1}{\delta}$, the exponent $\delta^2 \mathbf{n}_r / 8 - b_r \ln N$ is $\Theta(c_1 k_r \ln N \ln \frac{1}{\delta}) - \Theta(c_3 k_r \ln \frac{1}{\delta} \ln N)$, which is $\Theta(c_1 k_r \ln N \ln \frac{1}{\delta})$ once c_1 dominates c_3 — so the bound is $N^{-O(1)}$ as claimed (proof transcribed from Lemma 3.1 of [16]). ■

Proof of Lemma 10.14 (risk transfer)

Statement (recalled). For $F \subseteq \Pi$, $\pi \in F$, and query Q ,

$$\text{Risk}(\pi, Q; \tau^2) \leq \text{L-Risk}_F(\pi, Q; \tau) + \min_{\pi' \in F} \text{Risk}(\pi', Q; \tau),$$

hence $\min_{\pi \in F} \text{Risk}(\pi, Q; \tau^2) \leq 2 \min_{\pi \in F} \text{Risk}(\pi, Q; \tau)$. ■

Proof. Fix s and any $\pi' \in F$. Since the best cost in F is at least the best cost overall,

$$\text{Pnl}(\pi, s) = \frac{\text{Cost}(\pi, s)}{\text{Cost}(\pi', s)} \cdot \frac{\text{Cost}(\pi', s)}{\text{Cost}^*(s)} \leq \text{L-Pnl}_F(\pi, s) \cdot \text{Pnl}(\pi', s).$$

Define $E_1 = \{s : \text{Pnl}(\pi, s) > \tau^2\}$, $E_2 = \{s : \text{L-Pnl}_F(\pi, s) > \tau\}$, $E_3 = \{s : \text{Pnl}(\pi', s) > \tau\}$. If $s \in E_1$ then the product bound forces $s \in E_2 \cup E_3$ (both factors $\leq \tau$ would give $\text{Pnl} \leq \tau^2$). Hence, for $s \sim U_Q$,

$$\begin{aligned} \text{Risk}(\pi, Q; \tau^2) &= \Pr[E_1] \leq \Pr[E_2] + \Pr[E_3] \\ &= \text{L-Risk}_F(\pi, Q; \tau) + \text{Risk}(\pi', Q; \tau), \end{aligned}$$

and minimizing the last term over $\pi' \in F$ gives the first claim. Applying it to a minimizer π_F^* of the τ -risk over F , and using that the local penalty never exceeds the global one ($\text{L-Risk}_F(\pi_F^*, Q; \tau) \leq \text{Risk}(\pi_F^*, Q; \tau)$), yields the factor-2 form (proof transcribed from Lemma 4.2 of [16]). ■

Proof of Lemma 10.10 (weight charging)

Statement (recalled). If the feasibility LP for radius r is satisfiable, greedy rounding returns $C, \mathcal{H}'$ with cost $\leq 2r$, $|\mathcal{H}'| \leq 2fz$ and $|C| \leq 2k$. ■

Proof. *Cost:* the algorithm stops only when no active points remain, and each selected center deactivates every active point within $2r$; hence all surviving points are within $2r$ of C . *Outliers:* if more than $2fz$ sets were chosen, the corresponding LP variables would each exceed $\frac{1}{2f}$ and sum beyond z — contradicting the LP constraint. *Centers:* charge to each selected center p_i the LP mass $w_{p_i} = \sum_{p \in B(p_i, r)} x_p^*$ of the r -ball around it. For a point p_i not covered by chosen outlier sets, the LP forces $\sum_{p \in B(p_i, r)} x_p^* \geq \frac{1}{2}$ (its outlier variables sum to at most $\frac{1}{2}$, so the covering constraint must be met by center mass), so $w_{p_i} \geq \frac{1}{2}$. The balls of selected centers are pairwise disjoint (each selection deactivates everything within $2r$, so two centers are more than $2r$ apart), the charges do not overlap, and $\sum_i w_{p_i} \leq \sum_p x_p^* \leq k$; with each charge at least $\frac{1}{2}$, $|C| \leq 2k$. (Under the ε -approximate range queries the relational algorithm uses, the charge degrades to $\frac{1}{2} - \varepsilon$ and the bound to $2k/(1 - 2\varepsilon)$.) Proof transcribed from the rounding lemma of [18]. ■

Proof of Lemma 10.18 (net out-neighbors)

Statement (recalled). Every $p \in P$ keeps, at each scale i , an out-neighbor $y \in Y_i$ with $D(p, y) \leq 2^i$; per-point out-degree is $O(\frac{1}{\epsilon^\lambda} \log \Phi)$. ■

Proof. Fix p and scale i . By the covering property of the net Y_i there is $y \in Y_i$ with $D(p, y) \leq 2^i$, and the construction links p to such y — routing to y makes progress at scale i (the covering step the paper’s construction proof verifies). There are $h = O(\log \Phi)$ scales from the minimum inter-point distance to the diameter, and the doubling property bounds the number of relevant net points in p ’s distance window at each scale by $O(2^\lambda)$, taken over $(1 + \epsilon)$ -shells for the accuracy parameter, giving the stated per-point degree (argument follows the construction proof of [14]). ■

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11

Information-Theoretic Arguments

Information theory counts information. Its two workhorses — the *entropy* of a distribution (information content) and the *divergence* between two distributions (distinguishability) — set limits no algorithm, protocol, or noise mechanism can beat, in both directions. Downward: a quantity of H bits can be written down in about H bits — how sketches and compressible counters justify their footprint. Upward: a task implicitly identifying one of M objects forces $\log M$ bits — how space lower bounds and impossibility results are proved. Divergences do the same for pairs: differential privacy *is* a divergence bound between outputs on neighboring inputs; a synthesizer’s accuracy is a divergence to the truth.

In the SIGMOD’26 corpus the tool is tagged in a small cohort (5 papers) with a clean division of labor: error bounds in three of the five (frequency-estimation sketching, private query processing, private data-synthesis benchmarking), space lower bounds and privacy guarantees in two each, and a PODS’26 entry that turns Shannon inequalities directly into a join algorithm. Two exemplars (DP-S4S, Sublime) recur from Chapters 2 and 4, but their information-theoretic side is a different object from the side treated there. Section 11.5 adds entropy as a difficulty measure, and further reading.

11.0 The tool: information theory

The problem class is uniform. Somewhere in the system there is a random object — a stream prefix, a pair of neighboring databases, a hypothesis about the data — and the question is not how a computation behaves *on average* (Chapter 2) but how much *information* the object carries, or how *distinguishable* two of its instantiations are. An entropy of H bits is both a guarantee (an encoding of length $\approx H$ exists) and a barrier (nothing shorter suffices).

LEARNING OBJECTIVES

- Recognize information-theoretic claims: content bounds (bits) or distinguishability bounds (divergence).
- Prove the ladder: source coding $\rightarrow$ chain rule $\rightarrow$ KL/Rényi and Pinsker $\rightarrow$ incompressibility $\rightarrow$ Fano.
- Match measure to task: entropy for space, KL for error, Rényi for composed stages, Fano for impossibility.
- Dissect a paper’s argument: random object, measure, chain-rule telescope, final bound.

Notation

Logs are base 2 (*bits*) except for divergences D, D_α , in *nats* (the divergence/privacy convention). $H(X)$: entropy of X ; lowercase $h(X)$: the *entropy function* of a joint distribution viewed as a set function on attribute sets (the join-processing usage). $p \ll q$: $q(x) = 0 \Rightarrow p(x) = 0$.

11.0.1 The basic ladder

Definition 11.1 (Shannon entropy). For a discrete random variable X with distribution p , the *entropy* is

$$H(X) = - \sum_x p(x) \log_2 p(x) \quad (0 \log 0 := 0),$$

and the *conditional entropy* of X given Y is $H(X | Y) = \sum_y \Pr[Y = y] H(X | Y = y)$. Entropy measures the number of bits needed, on average, to pin down the value of X .

The first rung makes that sentence a theorem [2, 11]: no lossless code beats the entropy, and one within a bit exists (“code” = prefix-free binary description).

Theorem 11.2 — Source-coding bound. Let $\ell: \mathcal{X} \rightarrow \mathbb{N}$ be the length function of any prefix-free binary code for X . Then

$$\mathbb{E}[\ell(X)] \geq H(X),$$

with equality iff $\ell(x) = -\log_2 p(x)$. Conversely, the Shannon code $\ell(x) = \lceil -\log_2 p(x) \rceil$ is prefix-free with $\mathbb{E}[\ell(X)] < H(X) + 1$.

Proof. Codewords are disjoint leaves of a binary tree, so Kraft’s inequality gives $Z := \sum_x 2^{-\ell(x)} \leq 1$; set $q(x) = 2^{-\ell(x)} / Z$. Writing D_2 for the KL divergence of Definition 11.5 in bits,

$$D_2(p \| q) = \sum_x p \log_2 \frac{p}{q} = -H(X) + \mathbb{E}[\ell] + \log_2 Z \geq 0$$

by Gibbs (Lemma 11.6); $\log_2 Z \leq 0$ gives $\mathbb{E}[\ell] \geq H(X)$, equality iff $q = p$, $Z = 1$. Conversely $\ell(x) = \lceil -\log_2 p(x) \rceil$ satisfies $2^{-\ell(x)} \leq p(x)$, so Kraft holds, and $\ell < -\log_2 p + 1$. ■

The second rung lets entropy telescope over multi-part objects — the step that turns a pipeline into an equation.

Proposition 11.3 — Chain rule and conditioning. For any pair of random variables,

$$H(X, Y) = H(X) + H(Y | X), \quad \text{hence} \quad H(Y | X) \leq H(Y),$$

with equality in the second inequality iff X and Y are independent. ■

Proof. Write $\log_2 p(x, y) = \log_2 p(x) + \log_2 p(y \mid x)$ and take $-\mathbb{E}$; $H(Y|X) \leq H(Y)$ follows by Jensen (log concave) over the average in x . ■

The join-processing application needs the set-function form $h(X) = H(X)$, $X \subseteq V$, for a joint distribution over variables V .

Corollary 11.4 — Entropy is submodular; polymatroids. For all $X, Y, Z \subseteq V$:

$$h(XY) + h(XZ) \geq h(X) + h(XYZ), \quad h(XY) \geq h(X) \geq h(\emptyset) = 0.$$

That is, the entropy function of any joint distribution is a *polymatroid* (normalized, monotone, submodular); inequalities valid for all polymatroids are called *Shannon inequalities*. ■

Proof. $h(XY) + h(XZ) - h(XYZ) - h(X) = I(Y; Z \mid X) = H(Y|X) - H(Y|XZ) \geq 0$ by Proposition 11.3 applied twice; monotonicity is $H(Y|X) \geq 0$. ■

The third rung measures distinguishability of two distributions [2, 10].

Definition 11.5 (KL divergence). For distributions p, q with $p \ll q$, the *Kullback–Leibler divergence* is

$$D(p \parallel q) = \sum_x p(x) \ln \frac{p(x)}{q(x)} \quad (\text{nats}),$$

and $D(p \parallel q) = \infty$ if $p \not\ll q$.

Lemma 11.6 — Gibbs. $D(p \parallel q) \geq 0$ with equality iff $p = q$. ■

Proof. $-D(p \parallel q) = \sum_x p(x) \ln \frac{q(x)}{p(x)} \leq \ln \sum_x q(x) \leq 0$ by Jensen applied to the concave $\ln$ (the sum runs over $\text{supp } p$). ■

Two structural facts convert divergences along pipelines; both are one application of the log-sum inequality [2] and both recur in the exemplars.

Proposition 11.7 — Mixture convexity, chain rule, data processing.

1. (Convexity in the mixture.) If P_i, Q_i are distributions and $\beta_i \geq 0$, $\sum_i \beta_i = 1$, then $D(\sum_i \beta_i P_i \parallel \sum_i \beta_i Q_i) \leq \sum_i \beta_i D(P_i \parallel Q_i)$.
2. (Chain rule.) With the same mixing distribution on both sides, the joint divergence is the average of the conditional divergences:

$$D(p(x)P(z|x) \parallel p(x)Q(z|x)) = \sum_x p(x) D(P(\cdot|x) \parallel Q(\cdot|x)).$$

Consequently (data processing for a partition) $D(p \parallel q) \geq d(p(E) \parallel q(E))$ for every event E , with d the two-point divergence. ■

Proof sketch. Expand $\sum_{x,z} p(x)P(z|x) \ln \frac{P(z|x)}{Q(z|x)}$: the $p(x)$ terms cancel, giving (ii); (i) is the log-sum inequality applied per outcome. For the last claim, let z be the cell indicator of E and drop the second, nonnegative term of (ii). ■

Divergence controls total variation — the bridge from information to probability:

Theorem 11.8 — Pinsker. For all p, q ,

$$\text{TV}(p, q) := \frac{1}{2} \sum_x |p(x) - q(x)| \leq \sqrt{\frac{1}{2} D(p||q)}. \quad \blacksquare$$

Proof sketch. Data processing (Proposition 11.7) reduces to two points at the event E achieving total variation, $a = \Pr_p(E) > b = \Pr_q(E)$. The two-point divergence has $\partial_b d(a||b) = \frac{a-b}{b(1-b)}$, so $d(a||b) = \int_b^a \frac{a-t}{t(1-t)} dt \geq \int_b^a 4(a-t) dt = 2(a-b)^2$ since $t(1-t) \leq \frac{1}{4}$: $D(p||q) \geq 2 \text{TV}(p, q)^2$. ■

The fourth rung parametrizes the divergence by an order α — the calculus privacy accounting runs on [10, 12].

Definition 11.9 (Rényi divergence). For $\alpha \in (0, 1) \cup (1, \infty)$,

$$D_\alpha(p||q) = \frac{1}{\alpha - 1} \ln \sum_x p(x)^\alpha q(x)^{1-\alpha},$$

with $D_1 = D(p||q)$ (KL) and $D_\infty = \ln \max_x \frac{p(x)}{q(x)}$ (*max-divergence*).

Proposition 11.10 — Rényi calculus. $D_\alpha(p||q)$ is nondecreasing in α (Figure 11.1); D_∞ is what pure differential privacy bounds, and every $\alpha > 1$ composes additively [7, 12]. ■

The mixture fact that survives the passage from KL to Rényi — the engine of Section 11.2 — is quasi-convexity:

Lemma 11.11 — Quasi-convexity of Rényi divergence; van Erven–Harremoës. For any pairs (P, Q) , (P', Q') , any $0 < \beta < 1$, and $\alpha > 1$ (the case privacy accounting uses),

$$D_\alpha(\beta P + (1 - \beta)P' || \beta Q + (1 - \beta)Q') \leq \max\{D_\alpha(P||Q), D_\alpha(P'||Q')\}. \quad \blacksquare$$

Proof. Let $F(r, s) = r^\alpha s^{1-\alpha}$ on $\mathbb{R}_+^2$: its Hessian has determinant 0 and positive diagonal entries, so F is convex (linear along rays $s \propto r$). Jensen applies outcome by outcome: $\sum_x (\beta p + (1 - \beta)p')^\alpha (\beta q + (1 - \beta)q')^{1-\alpha} \leq \beta A + (1 - \beta)A' \leq \max\{A, A'\}$, with $A = \sum_x p^\alpha q^{1-\alpha} = e^{(\alpha-1)D_\alpha(P||Q)}$, similarly A' ; dividing exponents by $\alpha - 1 > 0$ preserves the inequality. ■

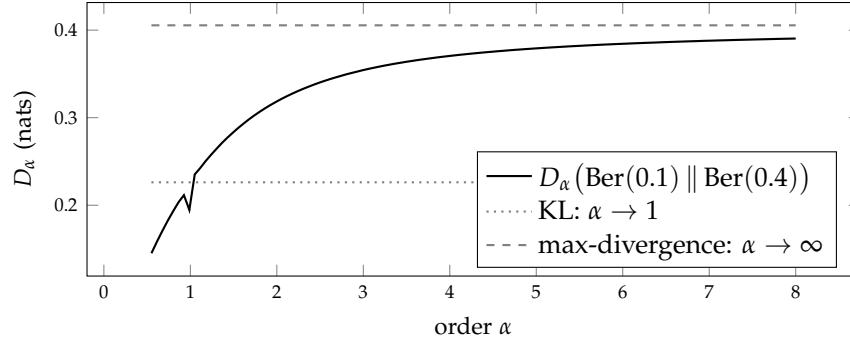


Figure 11.1: Rényi divergence vs. the order α for two Bernoulli distributions: nondecreasing, through KL at $\alpha = 1$, converging to the max-divergence pure DP bounds — Rényi-DP certifies one point of this curve.

The last two rungs convert information into impossibility. First, incompressibility — the counting reading of Theorem 11.2:

Corollary 11.12 — Incompressibility. If X is uniform over M outcomes then $H(X) = \log_2 M$, and any data structure with s bits that determines X must satisfy $s \geq \log_2 M$: s bits give fewer than 2^s states, and each state witnesses only one X . ■

Proof. Theorem 11.2 at the uniform distribution; the phrasing is pigeon-hole. ■

The method of *incompressibility lower bounds* runs this backwards: too small a sketch would be a code for objects of entropy beyond its bit count — Section 11.3 is a full exhibit. Second, Fano: an observation carrying little information about a hypothesis pins down no estimator.

Theorem 11.13 — Fano. Let X take $M \geq 2$ values, let Y be generated through any channel from X , and let $\hat{X}(Y)$ be any estimator. With $p_e = \Pr[\hat{X}(Y) \neq X]$,

$$p_e \geq \frac{H(X|Y) - 1}{\log_2 M}. \quad \blacksquare$$

Proof sketch. Condition on the error indicator $E = \mathbf{1}\{\hat{X} \neq X\}$: since (X, E) is determined by (X, Y) , $H(X|Y) \leq H(E|Y) + H(X|E, Y) \leq 1 + (1 - p_e) \log_2 M$ (E is binary; given $E = 0$, X is determined by Y). Rearrange. ■

Corollary 11.14 — Capacity-style impossibility. If X is uniform over M hypotheses and the observable Y satisfies $I(X; Y) = \log_2 M - H(X|Y) \leq c$, then every estimator errs with probability at least $1 - (c + 1)/\log_2 M$. ■

11.0.2 How to choose the information measure

The selection is mechanical once the deliverable is named. *Space, communication, or encoding bounds* ($\geq$ or $\approx$ bits): the source-coding pair (Theorem 11.2, Corollary 11.12) — upper bounds by exhibiting a code (Jensen bounds average code length), lower by incompressibility. *Distinguishability of two distributions* (privacy, testing, distributional error): KL for expected-loss statements, Rényi D_α when stages compose — privacy accounting lives here, α tuned at the final conversion. *Estimation impossibility*: Fano (Theorem 11.13) with the capacity reading of Corollary 11.14. *Structural size inequalities* (output sizes, join costs): the Shannon-inequality system of Corollary 11.4, optimized as a linear program.

When to Reach for Information Theory

Reach when the claim is about *information content* or *distinguishability* rather than average behavior: space lower bounds for sketches, “no mechanism can distinguish neighbors,” “no estimator recovers the hidden data,” “this encoding is within $O(1)$ of optimal.” Then run the recipe:

1. Name the random object: the stream prefix, the neighboring pair (D, D') , the hypothesis X — and what must be remembered, compared, or estimated about it.
2. Pick the measure: entropy for content (space/communication), KL for expected distributional error, Rényi D_α when stages compose (privacy accounting), Fano for estimation impossibility.
3. Write the chain-rule telescope along the structure: $H(X_1 \cdots X_k) = \sum_i H(X_i \mid \text{past})$, or the mixture decomposition over samples or stages.
4. Convert to the bound: $\mathbb{E}[\ell] \geq H$; $\text{TV} \leq \sqrt{D/2}$; $p_e \geq (H(X|Y) - 1) / \log M$ (Fano); $\epsilon' = \epsilon + \ln(1/\delta) / (\alpha - 1)$ (privacy conversion).

11.1 Shannon inequalities as a join algorithm

The exemplar is a PODS’26 paper on join query processing [5]^{PODS}. Evaluating conjunctive queries is the heart of database theory, and worst-case optimal algorithms alone ignore the *skew* real data exhibits: a degree constraint saying “each value of X has at most $N_{Y|X}$ matching Y s” can shrink the output by orders of magnitude. The PANDA framework exploits such constraints through the polymatroid inequalities of Corollary 11.4, but the original algorithm pays for its generality

with complex recursive partitioning. The paper introduces *PANDAExpress*: pick the inequality's weights by LP duality, and turn each Shannon-inequality proof step into a recursive hyperplane partition of the data. The result computes a query model meeting the polymatroid size bound B in time $O(N \log N + B \log N)$, recovering fractional hypertree-width and submodular-width runtimes as corollaries.

Problem. Queries are *disjunctive datalog rules* (DDRs)

$$Q(Z) :- \bigvee_{R(X) \in \Sigma_{\text{in}}} R(X)$$

An output *model* assigns tuples to the atoms $Q(Z) \in \Sigma_{\text{out}}$ so that every tuple t of the natural join $\bowtie_{\Sigma_{\text{in}}}$ is covered, i.e. $t_Z \in Q(Z)$ for some Z ; the goal is a *small* model. The instance satisfies *degree constraints* (Δ, N) : for each term $\delta = (Y|X) \in \Delta$, the max degree $\deg_R(Y|X=x)$ is at most N_δ in some input relation R . Writing h for the entropy function of any distribution consistent with the constraints, a *Shannon-flow inequality* is

$$\sum_{Z \in \Sigma_{\text{out}}} \lambda_Z h(Z) \leq \sum_{\delta \in \Delta} w_\delta h(\delta), \quad \|\lambda\|_1 = 1, \quad \lambda, w \geq 0 \text{ rational},$$

valid for all polymatroids, where $h(Y|X) = h(XY) - h(X)$; it certifies $|\Sigma_{\text{out}}| \leq B$ with $B = \prod_{\delta} N_\delta^{w_\delta}$ (take logs of $h(\delta) \leq \log N_\delta$).

Guarantee.

Theorem 11.15 — PANDAExpress, Theorem 6.4 of the paper. Given a DDR, an input instance over Σ_{in} of size N satisfying degree constraints (Δ, N) , and a Shannon-flow inequality with weights (λ, w) , PANDAExpress computes a model Σ_{out} for the DDR of size

$$\|\Sigma_{\text{out}}\| = O(B) \quad \text{in time} \quad O(N \log N + B \log N), \quad B = \prod_{\delta \in \Delta} N_\delta^{w_\delta}. \quad \blacksquare$$

How information theory is applied. Everything hangs on one move: the paper gives the Shannon inequality a *measure-theoretic* second life. A sub-probability measure is a nonnegative function with total mass at most 1; its marginals and conditionals are again such, and so is the geometric mean of k of them (AM–GM/Hölder on the masses). The weights come first: LP duality (Proposition 6.3 of the paper) minimizes $B = \prod_{\delta} N_\delta^{w_\delta}$ subject to the Shannon-flow inequality, and any optimal solution puts $w_\delta = 0$ on every constraint with $N_\delta > B$: the budget ignores loose constraints. The workhorse then converts the inequality into measures: instantiate each constraint $\delta = (Y|X)$ on the data as $p_\delta(y|x) = 1/N_\delta$ if (x, y) occurs in some input relation, 0 otherwise — a sub-probability conditional.

Lemma 11.16 — Measures from inequalities; Lemma 4.4 of the paper, notation cleaned. Let (λ, w) , $\|\lambda\|_1 = 1$, be nonnegative rational weights for which $\sum_Z \lambda_Z h(Z) \leq \sum_\delta w_\delta h(\delta)$ is a Shannon-flow inequality over V . Then for any collection of sub-probability measures p_δ , $\delta \in \Delta$, there exist sub-probability measures p_Z , $Z \in \Sigma_{\text{out}}$, such that

$$\prod_{Z \in \Sigma_{\text{out}}} p_Z(t)^{\lambda_Z} \geq \prod_{\delta \in \Delta} p_\delta(t)^{w_\delta} \quad \text{for all } t \in \text{Dom}_V. \quad \blacksquare$$

Proof sketch. Clear denominators to an *integral* Shannon-flow inequality over multisets; it has a *proof sequence* (Theorem 2.4 of the paper): single steps of submodularity, composition, decomposition, or monotonicity. Give each term of $\mathcal{D}$ its measure; at every step replace measures so the pointwise product only grows: monotonicity $h(XY) \rightarrow h(X)$ replaces p_{XY} by its marginal; submodularity $h(Y|X) \rightarrow h(Y|XZ)$ reconditions, $p_{Y|XZ}(y|xz) := p_{Y|X}(y|x)$; composition and decomposition split $p_{XY} = p_X \cdot p_{Y|X}$ or merge back. The final multiset contains $\mathcal{Z}$; group equal terms by geometric means (still sub-probability) and normalize exponents back to (λ, w) . Full case analysis in Section 11.6. $\blacksquare$

The bound follows by “taking logs”: for a join tuple $t \in \mathfrak{X}_{\Sigma_{\text{in}}}$, every $p_\delta(t) = 1/N_\delta$, so the right-hand product is $1/B$; since $\|\lambda\|_1 = 1$, some Z has $p_Z(t_Z) \geq 1/B$, and the model $Q(Z) = \{z : p_Z(z) \geq 1/B\}$ covers every join tuple and has at most B tuples because p_Z has mass ≤ 1 — Markov at the measure level.¹ The algorithm implements each proof step as a recursive partition along the step’s measure threshold (light tuples continue the proof sequence; the heavy side restarts via the paper’s Reset Lemma), and a potential $|\mathcal{D}| + |\mathcal{M}| + 3|\mathcal{S}|$ decreases at every restart, bounding the recursion depth by the proof length.

¹Why *hyperplanes*: no axis-parallel light/heavy split proves one two-atom cover, but the tilted hyperplane $h(C) = h(F)$ does; the geometric-mean construction finds such cuts automatically.

What it buys. Worst-case-optimal evaluation *with* skew at $O(N \log N + B \log N)$, where prior PANDA-style algorithms needed many bucket partitions per cut — and the theory delivers the algorithm, not just the bound: each proof step is executable.

11.2 Rényi-divergence accounting for sampled private queries

The exemplar is a SIGMOD’26 paper on private query processing [9]. Answering select-join-aggregate (SJA) queries under *user-level* differential privacy — no adversary may confidently infer the presence or absence of any one *user’s* entire tuple set — is demanding, because users contribute unboundedly many tuples and one heavy user can dominate the answer, and the state of the art attains strong accuracy only via expensive optimization over the whole database. The paper’s engine, DP-S4S, accelerates the search by Poisson-sampling tuples before privatizing, truncates each heavy user’s contribution at a threshold

found privately by a sparse-vector-technique (SVT) search, and releases through a smooth-sensitivity Gaussian mechanism. Its privacy analysis is pure information theory: privacy *is* a divergence bound between neighboring outputs, sampling turns each output into a *mixture* over samples, and the divergence of mixtures is controlled by Rényi quasi-convexity, with a binomial count deciding which components matter.

Problem. Privacy of a mechanism M is a bound on how distinguishable $M(D)$ and $M(D')$ are for *neighboring* inputs $D \sim D'$ (user-level: all tuples of one user added or removed). The information-theoretic formulation used here is Rényi differential privacy [7]:

Definition 11.17 (Rényi DP, user level). For $\alpha > 1$ and $\varepsilon > 0$, a randomized mechanism M is (α, ε) -RDP if for all neighboring D, D' ,

$$D_\alpha(M(D) \parallel M(D')) \leq \varepsilon,$$

with D_α as in Definition 11.9. An (∞, ε) bound is pure ε -DP; RDP converts back to $(\varepsilon + \frac{\ln(1/\delta)}{\alpha-1}, \delta)$ -DP for any $\delta > 0$.

The engine Poisson-samples each tuple independently with probability q (sample $S \subset T$), runs an SVT search M_1 over truncation thresholds on S , and releases the truncated aggregate through a Gaussian mechanism M_2 calibrated in Rényi terms. The pipeline is $L = \ell + 1$ stages (ℓ SVT iterations plus the release), each needing an amplified budget: sampling shrinks the effective divergence, and the analysis must say by how much.

Guarantee.

Corollary 11.18 — RDP-S4S for scalar SJA; Corollary of the paper, notation cleaned.

There is an (α, ε) -RDP version of the engine's main algorithm. Let n be the number of records, m the number of users, q the sample rate, $\tau^*(T) = \max_u \sum_{t \in T} \mathbf{1}[u \prec t]$ the maximum per-user contribution, and $\varepsilon_i^* \geq \varepsilon/(\ell + 1)$ the amplified noise-calibration budget. With probability $1 - \beta$,

$$|\hat{f}(T) - f(T)| = O\left(\sqrt{\frac{n}{q}} + \frac{1}{q} + \frac{\sqrt{\alpha}}{\varepsilon_i^*} \left(\tau^*(T) + \sqrt{\frac{\tau^*(T)}{q}}\right)\right),$$

where $O(\cdot)$ hides constants and polylogarithmic factors in m and $1/\beta$. ■

How information theory is applied. The random object is the pair of *output distributions on neighboring inputs* — and sampling makes each a mixture: $\Pr[M(T) = z] = \sum_S \Pr[\lambda(T)=S] \cdot \Pr[M(S) = z]$. Quasi-convexity

(Lemma 11.11) bounds the mixture's divergence by the worst component, $D_\alpha(\sum_S p_S M(S) \parallel \sum_S p_S M(S \cup S_u)) \leq \max_S D_\alpha(M(S) \parallel M(S \cup S_u))$; the reverse direction uses convexity in the second argument plus concavity of $\ln$ (the Proposition 11.7 logic, order by order). The per-component divergence is Gaussian, closed-form:

Lemma 11.19 — Gaussian Rényi divergence; Lemma of the paper, notation cleaned. For $\mu_1, \mu_2 \in \mathbb{R}^d$ and i.i.d. noise,

$$D_\alpha(\mathcal{N}(\mu_1, \sigma_1^2 I_d) \parallel \mathcal{N}(\mu_2, \sigma_2^2 I_d)) = \frac{\alpha \|\mu_1 - \mu_2\|_2^2}{2\sigma_{\text{mix}}^2} + \frac{d}{2} (\alpha - 1) \ln \frac{\sigma_1^{2(1-\alpha)} \sigma_2^{2\alpha}}{\sigma_{\text{mix}}^2}, \quad \blacksquare$$

where $\sigma_{\text{mix}}^2 = (1 - \alpha)\sigma_1^2 + \alpha\sigma_2^2$; the divergence is infinite if $\sigma_{\text{mix}}^2 \leq 0$. For equal scales σ , only the first term survives: $D_\alpha = \alpha \|\mu_1 - \mu_2\|_2^2 / 2\sigma^2$.

The count structure now enters: user u contributes k of its $\hat{\tau}$ sampled tuples to $S \cup S_u$ with probability $p_k = \Pr[\text{Bin}(\hat{\tau}, q) = k]$, and the truncated query moves by at most $\min\{\tau, k\}$ between components — the amplified budget is a binomial average of per-count Gaussian divergences, $\varepsilon'(\alpha) = \frac{1}{\alpha-1} \ln(\sum_{k \leq \hat{\tau}} p_k \exp((\alpha-1)c \min\{\tau, k\}^2) + \Pr[\text{Bin}(\hat{\tau}, q) > \hat{\tau}] e^{(\alpha-1)\varepsilon_s})$ with c set by the noise scale (“RDP Amplification” Lemma of the paper; some symbols dropped by the extractor, structure preserved). In the pure-DP limit this yields $\varepsilon' = \ln(1 + q'(e^{\varepsilon_1} - 1))$ with $q' = 1 - (1 - q)^{\hat{\tau}}$ — subsampling amplification with the collision probability q' in place of q . Finally the stages compose: RDP composition sums the L budgets (Proposition 11.10), and Definition 11.17's conversion returns to (ε, δ) -DP, α chosen once to optimize the final ε .

What it buys. User-level privacy whose accounting survives L -stage composition without the $\sqrt{L}$ loss of naive advanced composition, sampling amplification recognized as a divergence shrink, and — via the $\sqrt{\alpha}$ term in Corollary 11.18 — a dial trading Rényi order against noise scale.

11.3 Source coding and incompressibility for expandable sketches

The exemplar is a SIGMOD'26 paper on streaming frequency estimation [3], read here through the entropy lens rather than the estimator lens of Chapter 4. Frequency-estimation (FE) sketches such as Count-Min and CountSketch commit to a fixed width at construction, but real streams are unbounded and skewed: the length is unknown in advance,

heavy keys make some counters grow large while most stay tiny. The paper's system, *Sublime*, is an expandable CountSketch-style sketch that grows with the stream, and its two-sided theory is a textbook information argument. Upper bound, source coding: counters are stored as short stubs plus rare extensions, and Jensen (concavity of log) shows the array encodes in about $w \log_2(k/w) + O(w)$ bits — within a constant of the information content of w counters summing to k . Lower bound, incompressibility: any FE sketch achieving error $E(N)$ at confidence $1 - \delta$ on growing streams must at some point hold at least $\frac{N}{E(N)} [\Omega(\log \frac{1}{\delta} + \log \log \frac{N}{E(N)}) - \Theta(1)]$ bits, else its state would compress random key sequences below their entropy.

Problem. A stream of keys (insertions, deletions) of unknown ultimate length N is processed by an *expandable* FE sketch maintaining d counter arrays whose width follows a size function $W(\cdot)$ — e.g. $W(N) = N/\varepsilon$ or $W(N) = N^\alpha/\varepsilon$ with $0 \leq \alpha < 1$ — expanding over *epochs* at width $W_j = W(N_j)$. Skew makes the counter values far from uniform: a few large, many small, and the storage cost depends on how compactly that multiset is *written down*. The lower-bound adversary is any FE sketch with only overestimation errors guaranteeing error $E(N)$ at confidence $1 - \delta$ on a growing stream.

Guarantee.

Theorem 11.20 — Memory footprint; Theorem of the paper, notation cleaned. With a sublinear power size function, the sketch's memory footprint is

$$\approx d \cdot W(N) \cdot [4.41 + 1.26 \cdot \log_2 \frac{N}{W(N)}] \text{ bits;}$$

with a linear size function, $\approx d \cdot W(N) \cdot [4.41 + 1.26 \log_2 \frac{N \log_2 N}{W(N)}]$. The same bounds hold with $W(F)$ (the squared ℓ_2 norm) in place of $W(N)$, plus one bit per counter if counters are signed. ■

Theorem 11.21 — Space lower bound; Theorem of the paper. Any FE sketch with only overestimation errors that processes growing streams with error bound $E(N)$ and confidence $1 - \delta$ must use at least

$$\frac{N}{E(N)} \cdot \left[\Omega\left(\log \frac{1}{\delta} + \log \log \frac{N}{E(N)}\right) - \Theta(1) \right] \quad \blacksquare$$

bits of memory at some point in time while processing a stream.

How information theory is applied. The upper bound runs the source-coding direction of Theorem 11.2. A single counter value v is coded as an s -bit *stub* plus, when $v \geq 2^s$, a variable-length *extension* of $\lfloor v/2^s \rfloor$ in base 3 — at most $4 + 1.26 \log_2 v$ bits in total (Lemma of the paper; the

constant $1.26 = 2 \log_3 2$ is the base conversion). The array-level step is the entropy-flavored one:

Lemma 11.22 — Counter-array encoding; Lemma of the paper, notation cleaned. An array of w unsigned counters with total absolute value k encodes in

$$w \cdot \lceil 4.41 + 1.26 \cdot \log_2(k/w) \rceil \text{ bits,} \quad \blacksquare$$

and signed counters cost one extra bit each.

Proof sketch. Sum the per-counter bound and use the concavity of $\log_2$ twice: the stub length satisfies $s \leq 2 + \frac{1}{w} \sum_i \log_2 |v_i| \leq 2 + \log_2(k/w)$ (Jensen), and after replacing each per-counter term by the larger of the two scales, $\sum_i \log_2(\frac{k}{4w} + |v_i|) \leq w \log_2(\frac{5}{4} \cdot \frac{k}{w})$ (Jensen again); $1.26 \log_2 \frac{5}{4} \approx 0.41$ supplies the constant. Full proof in Section 11.6. $\blacksquare$

The lower bound runs the converse, Corollary 11.12, as a *reduction to compression*: a smaller sketch would compress sequences $\bar{x} = (x_1, \dots, x_n)$ of $n = N/(2E(N))$ keys drawn uniformly from a universe of size $u \cdot n$ — objects of entropy $n \log_2(un)$. Three steps. First, a *variance chain rule* (law of total variance over the sketch’s estimates across prefixes $\bar{x}_1, \dots, \bar{x}_\ell$ of geometrically growing lengths) exhibits one prefix state $\mathcal{S}_i^*$ whose estimates have second moment at most $2V / \log_2 n$. Second, *fixing the internal randomness*: averaging over the seed makes bias and variance small simultaneously, so one seed works on at least half of all sequences — a seed with $\mu(\bar{x}) \leq 64\delta / \log_2 n$ and $\nu(\bar{x}) \leq 64\delta / \log_2 n$ on at least half the sequences (Lemma of the paper). Third, the *code*: write $\bar{x}$ as (seed, state of $\mathcal{S}_i^*$); a decoder re-queries the sketch for every key and recovers $\bar{x}$ up to the error event, so the codewords are nearly injective over half of a set of entropy $n \log_2(un)$. The state must hold $n \log_2(un) - O(n \cdot E(N))$ bits minus the seed index, which unwinds to the stated $\frac{N}{E(N)} [\log \frac{1}{\delta} + \log \log \frac{N}{E(N)} - \Theta(1)]$: a sketch that answers queries is a code.

What it buys. A footprint certified *while it grows* — no prior knowledge of stream length or skew — within a small logarithmic factor of what any expandable FE sketch must pay; the same entropy accounting compresses its counters and fences off its competition.

11.4 Divergence budgets for private data synthesis

The exemplar is a SIGMOD’26 Experiments & Analysis paper benchmarking differentially private tabular data synthesis [1]. A DP synthesizer generates a fake table preserving useful statistics of a real one while protecting every individual’s row; synthesizers are pipelines — select marginals, discretize, infuse noise, compose the joint — and

competing systems make incomparable accuracy/privacy choices, so evaluations are hard to interpret. The paper benchmarks representative synthesizers under a unified treatment: every stage's privacy loss is an (α, ϵ) -RDP budget, stages compose additively, and accuracy is the *KL divergence* between synthesized and true joint. Its theory contribution is an ordering theorem: estimating a joint through noisy *conditionals* is never worse in KL error than a *product of noisy marginals* — the divergence version of “conditionals carry the dependence.” The extract records the statements without proofs; we transcribe conservatively and flag where the standard argument fills in.

Problem. The pipeline is a composition of randomized stages over a table with attributes $A_1, \dots, A_d$: per-attribute marginals are estimated noisily (Gaussian mechanism, exponential mechanism for selection, PrivTree partitions, rare-value merges), and the joint is then *estimated* from the noisy pieces — either *independently*, as the product $\hat{P}[A_i] \cdot \hat{P}[A_j]$ of noisy marginals, or *conditionally*, through noisy conditionals. Each stage j carries an (α, ϵ_j) -RDP guarantee (Definition 11.17), so the composed synthesizer is $(\alpha, \sum_j \epsilon_j)$ -RDP by composition and converts to (ϵ, δ) -DP at the end; accuracy is $D(\hat{P}[A_i, A_j] \parallel P[A_i, A_j])$, the KL error of the implied two-attribute joint against the true one.

Guarantee.

Theorem 11.23 — Estimation ordering; Theorem of the paper, notation cleaned. For any pair of attributes (A_i, A_j) , the KL-divergence error of conditional estimation is always no larger than that of independent estimation:

$$D(\hat{P}_c[A_i, A_j] \parallel P[A_i, A_j]) \leq D(\hat{P}_i[A_i, A_j] \parallel P[A_i, A_j]),$$

where $\hat{P}_c$ is the joint estimated through the noisy conditionals and $\hat{P}_i = \hat{P}[A_i] \cdot \hat{P}[A_j]$ is the product of the noisy marginals (the paper's display is partially mangled by extraction; the ordering claim and both comparands are as stated). ■

How information theory is applied. The workhorse is the mixture/conditional convexity of KL — Proposition 11.7 specialized, and transcribed by the paper as a standalone lemma:

Lemma 11.24 — KL convexity; Lemma of the paper, notation cleaned. If $P_1(z) = \sum_x p(x) P_1(z \mid x)$ and $P_2(z) = \sum_x p(x) P_2(z \mid x)$ share the same mixing weights, then

$$D(P_1(z) \parallel P_2(z)) \leq \sum_x p(x) D(P_1(z \mid x) \parallel P_2(z \mid x)). \quad \blacksquare$$

Proof sketch. Proposition 11.7(ii) with z marginalized away: the joint divergence over (x, z) equals the average of the conditional divergences (the mixing weights coincide), and marginalizing x is data processing, which cannot increase divergence. ■

Applied with x ranging over the values of A_i , the lemma converts the joint error of the conditional estimator into the *average of its per-condition errors*; comparing term by term against the error the product estimator pays for the same marginals yields the ordering of Theorem 11.23 — noise spent inside conditionals is never wasted, noise spent pretending independence is. The privacy side is divergence budgeting: each stage comes with a tabulated RDP guarantee — the Gaussian mechanism $(\alpha, \alpha\Delta_2^2/2\sigma^2)$ -RDP at noise scale σ , the exponential mechanism $(\alpha, \alpha\epsilon^2/8)$, rare-value merge $(\alpha, \alpha\Delta_2^2)$, PrivTree $(\alpha, \alpha\Delta_1)$ (statements of the paper, imported from prior work; Δ_p the ℓ_p sensitivity) — and the total budget $\sum_j \epsilon_j$ is what the benchmark allocates across stages before converting once to (ϵ, δ) -DP. The extract records no proofs for these; they are standard in the RDP literature [7].

What it buys. An apples-to-apples budget grammar — any synthesizer is a diversion of one $\sum_j \epsilon_j$ across stages — and a provable design rule: spend noise on conditional structure; Theorem 11.23 certifies it never loses KL accuracy against independence.

11.5 Advanced usage and further reading

The corpus’s fifth paper uses entropy as a difficulty measure, not a bound.

Entropy as difficulty: LM-approximated LIKE predicates [6]. SQL LIKE patterns (wildcard string matching) are costly to evaluate at scale, and language models suggest a shortcut: let the model approximate the predicate — when is that viable? The paper models a pattern’s difficulty by its *entropy*: Theorem 5.1 of the paper computes $H(P)$ for a LIKE pattern with p underscore and q percent wildcards over alphabet Σ — the underscores each contribute $\ln |\Sigma|$, the percents a combinatorial sum over the extra lengths they admit (the extract’s formula is partially mangled; the quantity is the logarithm of the number of strings the pattern generates, the pattern’s intrinsic uncertainty). The success model is a capacity ratio: a model with abstract capacity M internalizes task W with probability at least $c \cdot \min\{1, M/H(W)\}$ — the model must overwhelm the task’s entropy, the same M versus $\log_2$ (hypotheses) comparison Fano turns into impossibility, here read optimistically as a gate: LM approximation is worth trying exactly when the predicate’s

entropy fits the model's capacity. The framework is heuristic (capacity is assumed, not derived) — the corpus's cleanest entropy-as-predictor use.

11.5.1 Further reading

- Cover and Thomas [2] — the standard textbook: entropy, source coding, chain rules, KL, and the inequalities used here.
- Shannon [11] — the original: the source-coding theorem and the entropy calculus in their first form.
- Rényi [10] — the 1961 paper introducing the order- α divergences privacy accounting runs on.
- van Erven and Harremoës [12] — the modern treatment of Rényi divergence: monotonicity, quasi-convexity, conversions.
- Mironov [7] — Rényi differential privacy: the composition calculus both privacy exemplars run on.
- Fano [4] and Pinsker [8] — the originals behind Theorems 11.13 and 11.8.
- Yeung [13] — entropy functions and polymatroids, and the Shannon-type vs. entropic gap behind Section 11.1.

Chapter Recap

- **What the tool is.** Information-theoretic arguments bound information content (entropy, bits) and distinguishability (KL/Rényi divergence), in both directions: H bits guarantees an $\approx H$ -bit encoding and forbids anything smaller.
- **Which measure when.** Space/communication: entropy and source coding (upper by exhibiting codes, lower by incompressibility). Distributional error: KL. Composed stages and privacy: Rényi D_α — quasi-convex over mixtures, additive over stages, converted once at the end. Impossibility: Fano, capacity reading. Structural sizes: Shannon inequalities, optimized by LP duality.
- **Why it works.** Every rung telescopes: a chain rule (entropy, variance, or divergence over mixtures) decomposes the object along the pipeline, and Gibbs ($D \geq 0$) turns the telescope into a one-sided bound.
- **What the corpus showed.** PANDAExpress (Section 11.1): Shannon inequalities re-proved as sub-probability measures, each proof step an executable hyperplane partition. DP-S4S (Sec-

tion 11.2): privacy as Rényi divergence, sampling as mixtures tamed by quasi-convexity. Sublime (Section 11.3): counters encoded to $w[4.41 + 1.26 \log_2(k/w)]$ bits; any expandable sketch forced to $\frac{N}{E(N)}[\log \frac{1}{\delta} + \log \log \frac{N}{E(N)} - \Theta(1)]$. DP synthesis (Section 11.4): KL convexity orders conditional over independent estimation; RDP assigns per-stage budgets.

- **Advanced usage.** Entropy as a difficulty measure and capacity gate for LM-approximated predicates (Section 11.5); deferred proofs in the Appendix (Section 11.6).

11.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Lemma 11.16 (PANDA measures)

Statement (recalled). A Shannon-flow inequality with rational weights (λ, w) , $\|\lambda\|_1 = 1$, converts any sub-probability measures p_δ into p_Z with $\prod_Z p_Z^{\lambda_Z} \geq \prod_\delta p_\delta^{w_\delta}$. ■

Proof. Write $\lambda_Z = m_Z/d$, $w_\delta = m_\delta/d$ with positive integers m_Z, m_δ and common denominator d , and multiply through: an *integral* Shannon-flow inequality over multisets, Z appearing m_Z times and δ appearing m_δ times. By the proof-sequence construction (Theorem 2.4 of the paper) it is proved by steps turning $\sum_{\delta \in \mathcal{D}_i} h(\delta)$ into $\sum_{\delta \in \mathcal{D}_{i+1}} h(\delta)$; we carry measures along so that products only grow. Monotonicity $h(XY) \rightarrow h(X)$: replace p_{XY} by its marginal; then $p_{XY}(x, y) \leq p_X(x)$ pointwise. Submodularity $h(Y|X) \rightarrow h(Y|XZ)$: set $p_{Y|XZ}(y|xz) := p_{Y|X}(y|x)$; unchanged. Composition $h(X) + h(Y|X) \rightarrow h(XY)$: set $p_{XY}(x, y) := p_X(x) p_{Y|X}(y|x)$; unchanged. Decomposition is the reverse factorization via marginal and conditional; unchanged. Marginals and products of sub-probability measures are sub-probability, so every output is legal. The final multiset satisfies $\mathcal{D}_k \supseteq \mathcal{Z}$, so $\prod_{Z \in \mathcal{Z}} p_Z(t) \geq \prod_{\delta \in \mathcal{D}} p_\delta(t)^{m_\delta}$. Group the copies of each $Z \in \Sigma_{\text{out}}$ by the geometric mean $p_Z^{\text{gm}} := (\prod_{\text{copies}} p_Z)^{1/m_Z}$: it is sub-probability because, by Hölder and $\sum_x p_Z(x) \leq 1$ per copy, $\sum_x p_Z^{\text{gm}}(x) \leq \prod_{\text{copies}} (\sum_x p_Z(x))^{1/m_Z} \leq 1$. Dividing all exponents by d gives the claimed (λ, w) form. ■

Proof of Lemma 11.22 (counter-array encoding)

Statement (recalled). An array of w unsigned counters with total absolute value k encodes in $w[4.41 + 1.26 \log_2(k/w)]$ bits. ■

Proof. Per counter: values $v < 2^s$ occupy their s -bit stub; otherwise the stub keeps the s low-order bits and an extension codes $\lfloor v/2^s \rfloor$ in base 3 with $\log_3(v/2^s) + 2$ fragments of 2 bits, for at most $2(\log_3 2 \cdot \log_2(v/2^s) + 2) + s \leq 4 + 1.26 \log_2 v$ bits. Summing, $m \leq \sum_i \max(s, 4 + 1.26 \log_2 |v_i|)$. Tune the stub to the average counter: by Jensen ($\log_2$ concave), $\frac{1}{w} \sum_i \log_2 |v_i| \leq \log_2(k/w)$, so $s \leq 2 + \log_2(k/w) = 4 + \log_2(k/4w)$. Each per-counter maximum is at most $4 + 1.26 \log_2(k/(4w) + |v_i|)$ (the max of two scales is bounded by the log of their sum), and Jensen again gives

$$\frac{1}{w} \sum_i \log_2 \left(\frac{k}{4w} + |v_i| \right) \leq \log_2 \left(\frac{k}{4w} + \frac{k}{w} \right) = \log_2 \left(\frac{5}{4} \cdot \frac{k}{w} \right).$$

Hence $m \leq w[4 + 1.26 \log_2(\frac{5}{4} \frac{k}{w})] \leq w[4.41 + 1.26 \log_2(k/w)]$ since $1.26 \log_2 \frac{5}{4} \approx 0.41$. ■

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12

Communication Complexity

Communication complexity measures how many bits two or more parties must exchange about inputs they hold privately in order to compute a joint function of all of them. For a systems paper it is the canonical machine for impossibility proofs of one specific kind: proofs that *data must move* — any distributed protocol computing X exchanges $\Omega(\cdot)$ bits — or that *state must be large* — any streaming algorithm or sketch for X keeps $\Omega(\cdot)$ bits. The conversion is always a simulation argument: cut the input into pieces, hand one piece to each party, and let the algorithm’s memory contents and messages *be* the protocol’s transcript. The hard core is almost always one of a handful of canonical relations — disjointness, indexing, greater-than — whose communication complexity is known, so most of the craft lies in the embedding.

In the SIGMOD’26 corpus the tool is tagged in 4 papers — a small presence, but the method recurs inside the lower-bound halves of many streaming, distributed, and sampling papers where the headline tag went elsewhere. The three exemplars of this chapter show the three standard shapes: a multiparty disjointness reduction proving that noisy frequency moments need polynomial space and communication (Section 12.1); two- and multi-party disjointness reductions plus a coupon-collector transfer certifying output-optimal join sampling and counting (Section 12.2); and a three-step reduction chain that contradicts $\text{CC}(\text{DISJ}) = \Omega(m)$ to separate representation-oblivious pseudo-deterministic streaming from ordinary randomization (Section 12.3). Section 12.4 adds the advanced method — information cost, round elimination, asymmetric protocols — and a deterministic distributed lower bound for k -edge connectivity.

12.0 The tool: communication complexity

The problem class is uniform across applications: there is a resource budget — streaming space, bits on the wire, rounds, per-node message

LEARNING OBJECTIVES

- Recognize when a question is really “must data move / must state be large?”, and name the players and the transcript.
- Prove the ladder rectangles $\rightarrow$ tiling $\rightarrow$ fooling sets $\rightarrow$ Yao, and state the canonical hard relations (GT, Index, DISJ).
- Transfer a communication bound into a space, round, message-length, or query lower bound by simulating the algorithm.
- Shape a reduction: embed the hard relation, argue correctness and transcript cost.

length, probes into a relation — and a claim that it cannot be small. To prove the claim, one defines a communication problem whose inputs are distributed exactly the way the algorithm’s input can be split, shows that an algorithm respecting the budget would yield a cheap protocol for it, and invokes a known lower bound. This section builds the ladder in that order: protocols and rectangles, the deterministic bounds, randomization and Yao’s minimax principle, the canonical hard relations, the restricted models that map onto systems settings, and the information-cost calculus [2, 10].

Notation

Alice holds X , Bob holds Y ; we freely view $X, Y \subseteq [n]$ as their characteristic bit vectors. A protocol Π produces a *transcript* $\Pi(X, Y)$; its cost is the worst-case total length in bits. $D(f)$: deterministic complexity; $R_\delta(f)$: public-coin randomized complexity with worst-case error δ ; $R_\delta^\rightarrow(f)$: the one-way variant where only Alice speaks; $D_\mu^\mu(f)$: *distributional* complexity under input distribution μ . All logarithms are base 2, costs in bits. For k parties (*number-in-hand*), the *blackboard* model lets every party see all messages; the *coordinator* model routes messages through a coordinator.

12.0.1 The deterministic rung: rectangles, tilings, fooling sets

Definition 12.1 (Two-party protocol, deterministic complexity). A (*deterministic*) *protocol* Π for $f: \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ is a binary tree: each internal node is owned by Alice or Bob, and the owner’s message is a function of the transcript so far and of that player’s input only; each leaf carries an output. The *cost* is the worst-case number of bits communicated, and $D(f)$ is the minimum cost of a protocol computing f exactly.

The entire deterministic theory falls out of one structural fact: what a transcript “knows” about the inputs is a rectangle.

Lemma 12.2 — Rectangle lemma. For every transcript t of a deterministic protocol, the set of input pairs producing t ,

$$R_t = \{(x, y) : \Pi(x, y) \text{ has transcript } t\},$$

is a *combinatorial rectangle*: $R_t = X_t \times Y_t$ with X_t a set of Alice inputs and Y_t a set of Bob inputs. ■

Proof. Induction on the transcript. The empty transcript gives both factors equal to the full input sets. Extend a transcript t by one message of Alice, $t' = t \cdot m$: then $(x, y) \in R_{t'}$ iff $x \in X_t$, $y \in Y_t$, and Alice’s

message function on (transcript t , input x) equals m — a condition on x alone, so $R_t = X_t \times Y_t$. The step for a message of Bob is symmetric. ■

Theorem 12.3 — Tiling bound. Let $t(f)$ be the minimum number of monochromatic rectangles — sets $X_i \times Y_i$ on which f is constant — that partition the whole input space. Then

$$D(f) \geq \log_2 t(f). \quad \blacksquare$$

Proof. The leaves of a protocol tree of cost c partition the inputs into at most 2^c transcript classes; by Lemma 12.2 each class is a rectangle, and correctness makes each monochromatic. So the protocol exhibits a monochromatic partition of size $\leq 2^{D(f)}$, whence $t(f) \leq 2^{D(f)}$. ■

The tiling bound is hard to apply directly (minimizing over partitions is difficult), but it has a corollary that is the workhorse of deterministic lower bounds: exhibit many input pairs that pairwise “cross”.

Corollary 12.4 — Fooling-set bound. Call $S = \{(x_1, y_1), \dots, (x_s, y_s)\}$ a *fooling set* for f if $f(x_i, y_i)$ takes the same value c for all i , while for all $i \neq j$ at least one of $f(x_i, y_j)$, $f(x_j, y_i)$ differs from c . Then $D(f) \geq \log_2 s$. ■

Proof. Let $R = X_R \times Y_R$ be a monochromatic rectangle of value c containing both (x_i, y_i) and (x_j, y_j) . Then $x_i, x_j \in X_R$ and $y_i, y_j \in Y_R$, so R also contains (x_i, y_j) and (x_j, y_i) , contradicting the fooling property. Hence each rectangle of a partition covers at most one point of S , and Theorem 12.3 gives $t(f) \geq s$. ■

12.0.2 Randomized protocols and distributional complexity

Randomized protocols share a public random string r and must succeed with probability at least $1 - \delta$ on *every* input. The bridge to the deterministic theory is distributional complexity.

Definition 12.5 (Public-coin and distributional complexity). $R_\delta(f)$ is the minimum worst-case cost of a protocol with shared randomness such that $\Pr_r[\Pi(X, Y; r) = f(X, Y)] \geq 1 - \delta$ for every input pair (X, Y) . Its distributional counterpart is $D_\delta^\mu(f)$: the minimum cost of a *deterministic* protocol whose error probability is at most δ when $(X, Y) \sim \mu$.

Theorem 12.6 — Yao’s minimax principle.

$$R_\delta(f) = \max_{\mu} D_\delta^\mu(f). \quad \blacksquare$$

Proof sketch. ($\geq$) A public-coin protocol is a distribution over deterministic protocols Π_r ; if it has worst-case error δ , then for every μ its average error over r is at most δ , so some r has μ -error at most δ — a deterministic protocol of no larger cost. ($\leq$) The converse is a minimax

(LP duality) argument: if for every μ there is a good deterministic protocol, a mixed strategy over deterministic protocols achieves error δ against every input. ■

¹ Randomness reshapes the ladder unevenly. Public coins collapse *equality* to $O(1)$ bits (hash and compare) and shrink greater-than to polylogarithmic cost (hashed equality tests locate the first differing bit), but they leave disjointness and one-way indexing linear, as the next two theorems record. When choosing the hard relation for a reduction, check which relations remain hard *in the model your simulation produces* — one-way, few rounds, or interactive.

1

12.0.3 The canonical hard relations

Definition 12.7 (Canonical hard relations). For inputs viewed as sets or vectors of size m : *disjointness* $\text{DISJ}_m(X, Y) = 1$ iff $X \cap Y = \emptyset$; *indexing* $\text{Index}_m(x, j) = x_j$, where Alice holds the vector $x \in \{0, 1\}^m$ and Bob holds the position $j \in [m]$; *greater-than* $\text{GT}_N(x, y) = 1$ iff $x > y$ on inputs $x, y \in [N]$.

Greater-than is the tutorial case for the deterministic machinery: the diagonal is a fooling set.

Proposition 12.8 — **GT is deterministic-logarithmic.** On inputs from $[N]$, $D(\text{GT}_N) = \Theta(\log N)$. ■

Proof. Upper: write x, y as $\log N$ -bit strings; the players scan from the most significant bit, exchanging the current bit until the first position where they differ, which decides the order — at most $2 \log N$ bits. Lower: $S = \{(i, i) : i \in [N]\}$ is a fooling set: $\text{GT}(i, i) = 0$ for every i , while $i < j$ gives $\text{GT}(i, j) = 1 \neq 0$. Corollary 12.4 yields $D(\text{GT}_N) \geq \log N$. ■

Indexing is the canonical *one-way* hard relation, and the one that maps onto streaming: a one-pass algorithm's memory state is Alice's single message.

Theorem 12.9 — **Indexing is hard one-way.** For every constant $\delta < 1/4$, any public-coin one-way protocol for Index_m with worst-case error at most δ communicates at least $(1 - H(2\delta))m - 1$ bits, H the binary entropy function. In particular, constant-error one-way indexing costs $\Omega(m)$ bits, and deterministically Alice must send her whole vector: $D^\rightarrow(\text{Index}_m) = m$. ■

Proof sketch. Fix the public coins to their average behavior under a uniform input (Yao, Theorem 12.6); the protocol is now a deterministic message function $m(x)$ and a decoder g . Each message stands for one candidate answer string $z_m = (g(m, 1), \dots, g(m, m))$, so all inputs mapping to a message must sit inside a Hamming ball around z_m ; a volume bound on such balls forces 2^{cost} balls to cover $\{0, 1\}^m$. The full proof is in the Appendix (Section 12.5). ■

Disjointness is the queen of the hard relations: it stays linear under public randomization, and it is the source of nearly every streaming and distributed space lower bound in this book's corpus.

Theorem 12.10 — Disjointness is hard, randomized. $R_{1/3}(\text{DISJ}_m) = \Omega(m)$ [9, 14]. ■

Proof sketch. No proof is given here — this is a genuinely deep theorem. Razborov's route [14] fixes a hard input distribution (each element placed in X and Y independently with small, correlated probabilities) and shows by rectangle-counting that any deterministic protocol with $o(m)$ communication must err on constant μ -mass; Yao's principle (Theorem 12.6) lifts this to randomized protocols. The information-cost proof reuses the same distribution (Section 12.0.5). ■

For multiparty computations the number-in-hand disjointness bound is the standard anchor; the rate quoted by the corpus paper of Section 12.1 is the following.

Proposition 12.11 — Multiparty disjointness. In the number-in-hand blackboard model with k parties and universe $[n]$, any randomized protocol with error 0.49 deciding whether $\bigcap_{\ell=1}^k X^{(\ell)} = \emptyset$ communicates $\Omega(n/(k \log k))$ bits [5]. ■

12.0.4 One-way protocols, sketching, and the streaming transfer

Systems settings rarely produce arbitrary interactive protocols; they produce restricted shapes, each with a matching hard relation. *One-way* protocols (only Alice speaks) model a one-pass stream split at a cut: Alice is the prefix, Bob the suffix, the message is the memory state. *Simultaneous messages* (each party sends one sketch to a referee) model mergeable sketches and distributed sketching. *Asymmetric* settings — Alice holds a huge table, Bob a tiny query — model data structures and external-memory probes [12]; indexing is precisely such a problem. The transfer back to the systems world is a formality worth stating once and reusing forever.

Proposition 12.12 — Streaming–communication transfer. Suppose a randomized streaming algorithm $\mathcal{A}$ computes g on every stream using at most S bits of space and r passes. Then for every k , the k -party number-in-hand problem in which party ℓ receives the ℓ -th contiguous chunk of a stream and the answer is g of the concatenation admits a public-coin blackboard protocol of cost at most $k \cdot r \cdot S$ bits. Consequently any communication lower bound C for that problem implies $S \geq C/(kr)$. ■

Proof. The parties share $\mathcal{A}$'s public coins. In pass p , party ℓ initializes the memory state broadcast by party $\ell - 1$, runs $\mathcal{A}$ over its own chunk,

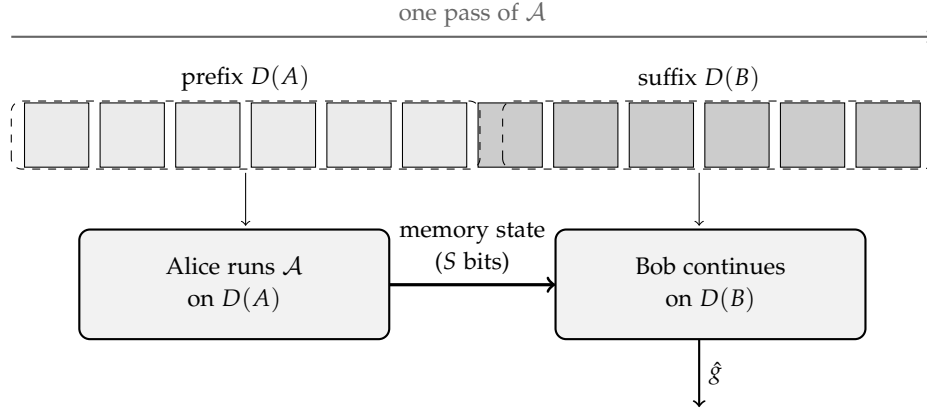


Figure 12.1: One pass of a space- S streaming algorithm over a concatenation $D(A) \cdot D(B)$ is a one-way protocol: Alice runs the algorithm on her prefix and forwards the final memory state; Bob continues and answers. With k parties and r passes the simulation costs $k \cdot r \cdot S$ bits (Proposition 12.12). Every space lower bound of this chapter flows through this picture.

and writes the resulting S -bit state to the blackboard; after k broadcasts the pass is simulated, and after r passes the last party holds $\mathcal{A}$'s output. Each of the r passes costs at most kS bits. ■

² The transfer is lossless up to the factor kr , so the lower-bound engineer's real freedom is in *where to cut*: choose k and the chunk boundaries so that each party's input is locally constructible from its share of the hard instance, and the objective g separates the YES and NO cases. In Section 12.1 the cut is between players' noisy substreams; in Section 12.3 it is Alice's set followed by Bob's set.

2

12.0.5 Information cost and simulation

The rectangle method saturates on direct-sum and multi-query questions ("k copies of the problem on one input"), which is exactly the shape of amortized streaming and multi-query database lower bounds. The modern calculus measures not bits sent but *information revealed*.

Definition 12.13 (External information cost). For a protocol Π and input distribution μ , the *external information cost* is the mutual information between the inputs and the transcript,

$$\text{IC}_\mu^{\text{ext}}(\Pi) = I_\mu(X, Y; \Pi(X, Y)),$$

and $\text{IC}_{\mu, \delta}^{\text{ext}}(f)$ is its infimum over protocols with worst-case error at most δ .

Proposition 12.14 — Information is bounded by communication. For every protocol Π and distribution μ ,

$$\text{IC}_\mu^{\text{ext}}(\Pi) \leq H(\Pi(X, Y)) \leq \text{cost}(\Pi),$$

$$\text{hence } R_\delta(f) \geq \text{IC}_{\mu, \delta}^{\text{ext}}(f) \text{ for every } \mu. \quad \blacksquare$$

Proof. The transcript is a function of the inputs and the public coins; since the coins are independent of (X, Y) , $I(X, Y; \Pi) \leq H(\Pi) \leq \text{worst-}$

case length. The second claim restricts the infimum defining $\text{IC}_{\mu,\delta}^{\text{ext}}(f)$ to protocols of cost $\leq R_\delta(f)$, a nonempty set. ■

Information cost earns its keep through two deep structural facts, both stated without proof.

Theorem 12.15 — Simulation and direct sum, informal. (i) (Compression [4].)

Low information cost converts back into low communication: f admits a public-coin protocol of cost $2^{O(\text{IC}_{\mu,\delta}^{\text{ext}}(f))}$, so information and communication complexities are exponentially close.

(ii) (Direct sum [3, 8].) Any protocol solving k independent copies of f under $\mu^{\otimes k}$ with constant error has information cost at least $k \cdot \text{IC}_\mu^{\text{ext}}(f)$ up to lower-order terms. ■

Proof idea. (i) Sample the transcript from its conditional distribution and abort when the sampled transcript is atypical; the rejection probability is controlled by the information cost. (ii) The k -copy information cost splits by the chain rule into k terms, each at least the single-copy information cost because the copies are independent and a protocol for all copies is, in particular, a protocol for each. ■

Combining (ii) with Proposition 12.14: any protocol for k copies communicates $\geq k \text{IC}(f)$ bits — how one proves tight direct-sum bounds for disjointness [3] and, through the transfer of Proposition 12.12, amortized space lower bounds for streams that must answer many queries (Section 12.4.1).

12.0.6 How to shape a communication reduction

The selection procedure is mechanical. *First, name the resource and the model:* bits of space (streaming), bits on the wire (coordinator/blackboard), rounds, per-node message length (sketching), or oracle queries (property testing) — the model determines which restricted protocol shape the simulation produces, and hence which hard relations remain hard. *Second, place the cut:* decide who holds what when the hard instance is split; the cut must respect locality (each party builds its share without seeing the others' shares) and separability (the objective is a gap function of the YES/NO answer). *Third, match the hard relation to the objective:* a frequency, cardinality, or join-output objective usually announces DISJ (an element present on both sides changes the count); a prefix-query/late-arrival structure announces Index; a threshold structure announces GT or a promise variant. *Fourth, budget the transcript:* each oracle query, message, or state forward costs $O(1)$ bits or $O(S)$ bits, and the simulation's total must be a clean multiple of

the budgeted resource. *Fifth, argue correctness with a gap:* verify that the approximation tolerance the algorithm is allowed cannot straddle the YES and NO values — otherwise the reduction silently fails, a check performed line-by-line in Sections 12.1 and 12.3.

When to Reach for Communication Complexity

Reach whenever the deliverable is an impossibility of the form “no algorithm/protocol/ sketch can do X with budget B ” — sublinear space in streaming, sublinear communication in distributed monitoring, sublinear query time in sampled access models, short messages in distributed sketching. Then run the recipe:

1. Identify the players and the transcript: who holds which part of a hypothetical hard input, and which artifact of the algorithm (memory state, message, sketch, query answer) plays the role of the communicated bits.
2. Pick the hard relation matching the model: DISJ for interactive and multiparty settings, Index for one-way and asymmetric ones, GT/promise variants for threshold structure (Section 12.0.3); deterministic settings use fooling sets (Corollary 12.4), randomized ones go through Yao (Theorem 12.6).
3. Embed the inputs: map each party’s share to a locally constructible sub-instance (sub-stream, relation, neighborhood), so that the objective differs by more than the approximation tolerance between the YES and NO cases.
4. Argue correctness and account for the transcript: the simulated algorithm’s answers are recoverable from the transcript bit-for-bit (distinct hard inputs force distinct, or too many, transcripts), so the budgeted resource times the blow-up factor still beats the known communication bound.
5. Transfer the bound: divide by the simulation overhead to get the space, round, message-length, or query lower bound (Propositions 12.12 and 12.11).

12.1 Noisy frequency moments: space and communication lower bounds

The exemplar is a PODS’26 paper on streaming statistics under noisy identity resolution [11]^{PODS}. Frequency moments, $F_p = \sum_j f_j^p$, are canonical streaming quantities behind cardinality, skew, and heavy-hitter analytics, and decades of sketching theory estimate them accu-

rately when every item is identified exactly. Real pipelines resolve identities imperfectly: after record matching, fuzzy deduplication, or noisy sensing, each arriving item is *ambiguous*, compatible with several ground-truth identities, and the frequency vector of the unknown ground truth is what analytics actually wants. The paper introduces a per-instance noise measure — the F_p -mismatch-ambiguity — and shows that F_p of the ground truth is still approximable, to a factor degraded by the noise, in one pass and $O(\varepsilon^{-2} m^{1-1/p})$ words over m items, and that this price is necessary: a reduction from multiparty disjointness proves that any constant-pass algorithm with the same guarantee needs $\Omega(\varepsilon^{-1/p} m^{1-1/p})$ bits of space, and any distributed protocol over k sites needs the same order of communication — in stark contrast with the polylogarithmic communication of the noiseless setting. This chapter dissects the lower-bound half; the estimator half is treated under the estimation-theory lens in Chapter 4.

Problem. A noisy dataset is a stream $\sigma = (\sigma_1, \dots, \sigma_m)$ together with a ground-truth assignment of positions to identities μ ; positions i, i' are *similar* ($\sigma_i \sim \sigma_{i'}$) when compatible with a common identity, and $G^\sigma = ([m], E^\sigma)$ is the resulting similarity graph. Writing B_i^σ for i 's clique of similar positions and B_i^μ for the positions sharing i 's true identity, the paper's per-instance noise parameter $\Lambda_p(\sigma, \mu)$ (notation as in Chapter 4; the paper writes η_p) measures how much p -clique mass the observed similarity graph and the true one disagree on: the symmetric-difference mass of ordered p -cliques between G^σ and G^μ is at most $O(\Lambda_p F_p(\mu))$ in both directions. An (α, β) -approximation of F_p outputs a value in $[(1 - \alpha)F_p, (1 + \alpha)F_p]$ with probability at least $1 - \beta$. The distributed variants are the *blackboard* model (k players, arbitrary rounds, all messages public) and the *coordinator* model (k sites with private channels to one coordinator); the stream is the concatenation of the players' substreams, and the communication budget counts all bits sent.

Guarantee.

Theorem 12.16 — Distributed lower bound, from the paper's blackboard-model theorem, notation cleaned. For any constants $p > 1$, $C > 0$, and $\varepsilon \in (0, 1/3)$: any $((\varepsilon + C\Lambda_p), 0.48)$ -approximation algorithm for computing F_p on any noisy dataset of up to m items distributed over $k \geq 10p(C + 1)$ players requires

$$\Omega(\varepsilon^{-1/p} m^{1-1/p})$$

bits of communication in the blackboard model. The same bound holds in the coordinator model regardless of the number of rounds, and any

³The $\beta = 0.48$ failure allowance is calibrated for a reduction from constant-error disjointness: the bound quoted below tolerates error 0.49, so a protocol built from the estimation algorithm must err strictly less often. The noise tolerance $\varepsilon + C\Lambda_p$ is what makes the theorem non-trivial: the hard instances carry genuine mismatch ambiguity of the same order as the allowed slack, so the algorithm cannot dodge by blaming noise.

$O(1)$ -pass algorithm of the same guarantee on a single stream of up to m items requires $\Omega(\varepsilon^{-1/p} m^{1-1/p})$ bits of space. ■

3

How communication complexity is applied. The reduction is from number-in-hand multiparty disjointness, using the anchor of Proposition 12.11 exactly as the paper quotes it. The embedding — the paper's input reduction, transcribed with cleaned notation — turns each player's set into a chunk of a noisy stream whose ground truth merges items across players.

Lemma 12.17 — Input reduction, from the paper's claim on the hard-instance structure, notation cleaned. Given k -party disjointness inputs $X^{(1)}, \dots, X^{(k)} \subseteq [n]$, construct a noisy dataset in which player ℓ 's substream contains, for every element $i \in X^{(\ell)}$, a group of t items $q_{i,1}^{(\ell)}, \dots, q_{i,t}^{(\ell)}$, with similarity pattern:

1. items created by the same player are pairwise dissimilar, even when created from the same element ($q_{i,j}^{(\ell)} \not\sim q_{i,j'}^{(\ell)}$ for $j \neq j'$);
2. if players $\ell \neq \ell'$ both hold element i , then every item of ℓ 's group for i is similar to every item of ℓ' 's group for i ;
3. if ℓ 's set has a zero at i , ℓ creates no items around i , and no further cross-player similarities exist;
4. items created from different elements are pairwise dissimilar.

Hence the ground-truth identity of element i collects all $t \cdot r_i$ items created from i , where $r_i = |\{\ell : i \in X^{(\ell)}\}|$, and $F_p(\mu) = \sum_i (t r_i)^p$. ■

Proof sketch. Each property holds by construction of the similarity relation: cross-player similarity is switched on exactly for shared elements (2), while intra-player items (1) and different-element items (4) are never similar; (3) records that nothing else is similar. Summing identity-class sizes per element gives $F_p(\mu)$. ■

The gap now does the work. If the disjointness instance is a NO instance (all sets pairwise disjoint), every $r_i \leq 1$ and $F_p(\mu) \leq n \cdot t^p$. If it is a YES instance (a common element i^* held by all k players), that element alone contributes $(tk)^p \geq t^p k^p$. Choosing the group size t and universe size n so that k^p exceeds n by more than the $(1 + \varepsilon + C\Lambda_p)^2$ slack factor — the construction's own ambiguity stays within the allowed $\Lambda_p = O(\varepsilon)$ (the coordinator-model hard input has ambiguity of order ε/k) — an $((\varepsilon + C\Lambda_p), 0.48)$ -approximation output separates the two cases with probability $0.52 > 0.51$, solving disjointness with error $0.48 < 0.49$. The players simulate the distributed algorithm on their own substreams; the

algorithm's total communication is the protocol's cost, which Proposition 12.11 forces to $\Omega(n/(k \log k))$; parameterizing the construction so the stream length is m evaluates the bound to $\Omega(\epsilon^{-1/p} m^{1-1/p})$. The streaming and coordinator statements of Theorem 12.16 follow by the transfer of Proposition 12.12: an r -pass space- S algorithm over the concatenated stream, cut into $k = O(1)$ players' chunks, yields a protocol of cost krS , so $S \geq \Omega(\epsilon^{-1/p} m^{1-1/p})/(kr)$.

What it buys. The lower bound certifies that the paper's one-pass upper bound is optimal in its dependence on m , and separates the noisy regime from the noiseless one, where F_p ($p > 2$) admits one-pass $O(\epsilon^{-2} m^{1-2/p} \log^2 m)$ -bit sketches and the distributed problem needs only polylogarithmic communication: under mismatch ambiguity the polylogarithmic free lunch provably disappears. For the practitioner it calibrates how much deduplication noise a pipeline can carry before frequency-based analytics must pay polynomial space.

12.2 Output-optimal sampling and counting for join-project

The exemplar is a PODS'26 paper on uniform sampling and approximate counting for join-project queries [7]^{PODS}. Worst-case optimal join algorithms evaluate conjunctive queries up to the AGM bound, but the join output can still be polynomially larger than the input N , and many analytics tasks would rather sample one uniform result, or count the results approximately, than enumerate them. The natural target is *output-sensitive* complexity: per-sample and counting time should degrade gracefully with the output size OUT , ideally $O(N/\sqrt{\text{OUT}})$ for the matrix query $\pi_{A,C}(R_1(A,B) \bowtie R_2(B,C))$ and analogous powers for star and chain queries. The paper delivers the first asymptotically optimal algorithms for matrix, star, and chain join-project queries — a hybrid heavy/light partition with intersection estimation, geometric search over OUT , and median amplification — and proves matching lower bounds, two families of them by communication reduction: two- and multi-party disjointness arguments in a query-access model, and a coupon-collector transfer showing that any faster sampler would enumerate join outputs faster than the known combinatorial listing bounds; a third, information-theoretic family shows sublinear algorithms for chain queries are outright impossible.

Problem. For a join-project query Q (e.g., the matrix query $Q_m = \pi_{A,C}(R_1(A,B) \bowtie R_2(B,C))$, or a k -leaf star query $Q_{s,k} = \pi_{A_1,\dots,A_k} \bowtie_i R_i(A_i, B)$), an instance I of total size N has output $\text{OUT} = |Q(I)|$. A W -uniform sampler returns each result with probability between $1/\text{OUT}$ and W/OUT ; an ϵ -approximate counter returns a value in $(1 \pm \epsilon) \text{OUT}$.

The algorithm first preprocesses (builds an index), then answers samples or counts; in the *property-testing query model* — the model of the lower bounds — the instance is accessed only through oracle queries: tuple tests, uniform tuple samples, degree, and neighbor queries, and the cost is the number of queries.

Guarantee.

Theorem 12.18 — Output-optimal matrix query, from the paper’s theorems, notation cleaned. For the matrix query $Q_m = \pi_{A,C}(R_1(A, B) \bowtie R_2(B, C))$:

- (i) there is an algorithm that builds an index in $O(N)$ time and returns a uniform sample from $Q_m(I)$ in $O(N/\sqrt{\text{OUT}})$ expected time, and an ε -approximation of OUT with probability $1 - \delta$ in $O(N \log(N/\delta) \log(1/\delta) / (\varepsilon^{2.5} \sqrt{\text{OUT}}))$ time;
- (ii) for any nonnegative integers N, OUT , any randomized algorithm that returns a constant-factor approximation of OUT with high probability requires $\Omega(N/\sqrt{\text{OUT}})$ time in the query model.

■

How communication complexity is applied. The workhorse is a reduction from two-party disjointness — the paper’s argument is transcribed here with cleaned notation, and its full proof is in the Appendix (Section 12.5).

Lemma 12.19 — Matrix-query counting bound, from the paper’s two-party lemma, notation cleaned. For Q_m , given any $N, \text{OUT}, n \in \mathbb{Z}^+$ (n the active domain size of attribute B), any randomized algorithm that takes an instance of input size N and output size OUT and outputs any multiplicative-factor approximation of OUT with high probability requires at least

$$\Omega\left(\min\{n, N/\sqrt{\text{OUT}}\}\right)$$

queries in the query model.

■

Proof sketch. Alice and Bob reduce a disjointness instance (S_A, S_B) on universe $[m]$, $|S_A| = |S_B| = m/4$, to a database: the active domain of B is $S_A \cup S_B$, while $R_1 = D(A) \times S_A$ and $R_2 = S_B \times D(C)$ for disjoint $D(A), D(C)$ of size $\sqrt{K}$. Then $N = \sqrt{K}(|S_A| + |S_B|) = \sqrt{K}m/2$, and $\text{OUT} = 0$ if $S_A \cap S_B = \emptyset$ but $\text{OUT} = K$ if they intersect — a single shared value b produces $|D(A)| \cdot |D(C)| = K$ distinct projected pairs. Any multiplicative approximation separates 0 from K ; every oracle query is answered by its owner with a small number of bits (a tuple test is one bit; a sample is drawn locally by the owner), so q

queries give a disjointness protocol of $o(q)$ communication, contradicting Theorem 12.10 unless $q = \Omega(m)$; with $K = \text{OUT}$ the arithmetic $N/\sqrt{\text{OUT}} = m/2$ evaluates the bound. ■

Two extensions round out the family. For the star query the reduction goes through k -party disjointness: player ℓ holds $S_\ell \subseteq [m]$ with $|S_\ell| = m/2k$, builds $R_\ell = D(A_\ell) \times S_\ell$ with $|D(A_\ell)| = K^{1/k}$ for disjoint $D(A_\ell)$; then $N = K^{1/k}m/2$, and $\text{OUT} = 0$ or $K = \prod_\ell |D(A_\ell)|$ according to disjointness, so Proposition 12.11 yields the $\Omega(N/\text{OUT}^{1/k})$ query bound for constant k , matching the paper's $O(N/\text{OUT}^{1/k})$ sampling and counting upper bounds. For uniform samplers the transfer is a *coupon-collector* argument: a sampler with preprocessing $O(N/\text{OUT}^{1/2-\gamma})$ and per-sample time $O(N/\text{OUT}^{1/2+\gamma})$ would first collect samples until enough collisions certify a constant-factor estimate of OUT , then draw $\Theta(\text{OUT} \log N)$ more samples to collect every result whp (the coupon collector), enumerating the full join in $O(N \text{OUT}^{1/2-\gamma} \log N)$ time and contradicting the $\Omega(N\sqrt{\text{OUT}})$ combinatorial listing bound for the matrix query (conditionally, via Boolean matrix multiplication hardness); the star-query variant, with the same γ -tradeoff around $N/\text{OUT}^{1/k}$, contradicts the $\Omega(N \text{OUT}^{1-1/k})$ listing bound the same way. Counting-from-sampling is the mirror reduction: Y independent invocations of a sampler with per-run success probability $1/W$, with $Y = \Omega(W/\varepsilon^2)$, give an ε -approximate count from the success fraction (a proposition the paper imports from the literature), which is why the counting and sampling lower bounds coincide.

On the upper-bound side, the algorithm meeting Theorem 12.18(i) is a hybrid partition at threshold $\sqrt{\text{OUT}}$: heavy values of B (degree at least $\sqrt{\text{OUT}}$) are detected by repeated uniform sampling with a union bound over values, and indexed exactly; the light part is sampled uniformly, and the fraction of light-part join tuples whose projection already appeared in the heavy part — the *intersection* — is estimated by Hoeffding's inequality from $|S| = \frac{1}{2\varepsilon^2} \ln \frac{2}{\delta}$ samples (the concentration rung of Chapter 2); since OUT is unknown a priori, the algorithm runs a geometric search over guesses $\lambda = N^2, N^2/2, \dots$, validating each guess with Markov (cheap accept) and Chebyshev (accuracy) medians over k_δ independent estimators — median amplification — while exact uniform sampling inside the light part uses rejection whose retry count follows a negative hypergeometric distribution, with expectation $r(P-K)/(K+1)$ for r successes among P balls with K marked, keeping the expected per-sample cost at the advertised bound.

What it buys. The reduction family certifies that $N/\sqrt{\text{OUT}}$ (and $N/\text{OUT}^{1/k}$ for stars) is the true complexity of sampled analytics over joins: sublinear per-sample time is possible, but not a penny cheaper,

and for chain queries not at all. The query-cost accounting doubles as a cost model for secondary storage — each oracle query is one page access — so the lower bounds also delimit any index that hopes to beat a constant number of scans.

12.3 Representation obliviousness and pseudodeterminism in streaming

The exemplar is a paper on pseudodeterministic streaming algorithms [6]. A randomized algorithm is *pseudodeterministic* (PD) when, on every input, it outputs a *canonical* value with high probability — different runs agree on the answer, which is what reproducibility-sensitive consumers (caching, auditing, downstream analytics) expect — and a streaming algorithm for a frequency moment is *representation oblivious* (RO) when its output distribution depends only on the frequency vector, not on the order in which the stream happens to present it. The paper asks what these two stability requirements cost for the most basic streaming quantity, the distinct count F_0 : ordinary randomized estimators use polylogarithmic space, and known algorithms (BJKST, AMS, and linear sketches generally) are representation oblivious, yet none of them is pseudodeterministic. The main result is an $\Omega(n)$ space lower bound for algorithms that are both — representation-oblivious pseudodeterministic (ROPD) — extended to a tight $\Omega(n/t(n))$ for $t(n)$ passes; the proof is a three-step reduction chain that contradicts $\text{CC}(\text{DISJ}) = \Omega(m)$, the chapter’s purest exhibit: no gadgets, just the definitions of the two stability properties doing the work.

Problem. For a stream D over universe $[n]$, let $F_0(D)$ be the number of distinct elements and $f^k(D)$ the vector of k -th powers of frequencies; an algorithm (ε, δ) -approximates F_k when $\Pr[|A(D) - F_k(D)| \leq \varepsilon F_k(D)] \geq 1 - \delta$ for every D (Definition 2.1 of the paper; the paper fixes $\delta = 1/3$).

Definition 12.20 (Pseudodeterminism and representation obliviousness, Definitions 2.2, 2.4 and 2.5 of the paper). A randomized algorithm A is *pseudodeterministic* if for every input D there is a canonical value $g(D)$ with $\Pr[A(D) = g(D)] \geq 2/3$. Two streams are R_k -equivalent if $f^k(D_1) = f^k(D_2)$ (for $k = 0$: the same set of distinct elements); A is *representation oblivious* if R_k -equivalent streams induce the same output distribution. A is *ROPD* (representation-oblivious pseudodeterministic) when both hold; then g is constant on R_k -equivalence classes.

The communication side needs one promised relation and its own obliviousness quotient (Definitions 2.10–2.13 of the paper): *Promise-Union_m* is the partial function of $A, B \subseteq [m]$ that outputs 1 when

$|A \cup B| \geq 3m/4$ and 0 when $|A \cup B| \leq m/4$; two instances are ρ_m -equivalent when $A_1 \cup B_1 = A_2 \cup B_2$; an ROPD protocol for Promise-Union $_m$ is a ρ_m -oblivious pseudodeterministic one.

Guarantee.

Theorem 12.21 — F_0 lower bound, Theorem 3.1 of the paper. Let $\varepsilon, \delta \leq 1/3$. Any ROPD streaming algorithm that (ε, δ) -approximates F_0 requires $\Omega(n)$ space, where n is the universe size. ■

Corollary 12.22 — Multipass tradeoff, Theorem 4.1 of the paper. Any ROPD $t(n)$ -pass streaming algorithm that (ε, δ) -approximates F_0 , $\varepsilon, \delta \leq 1/3$, requires $\Omega(n/t(n))$ space; this is tight, as a deterministic representation-oblivious $t(n)$ -pass algorithm computes F_0 exactly in $O(n/t(n))$ space (counting distinct elements in $t(n)$ blocks per pass, Observation 4.3 of the paper). ■

How communication complexity is applied. The proof chain has three links, each a black-box transformation of protocols; contradicting Theorem 12.10 at the end is the only external input.

Link 1: estimator to protocol (Lemma 3.2 of the paper). Suppose an ROPD algorithm A_n $(\frac{1}{3}, \frac{1}{3})$ -approximates F_0 over $[n]$ in space s . The protocol P_n for Promise-Union $_n$: Alice turns her set into the stream $D(A)$ (elements in lexicographic order), runs A_n , and communicates the final state — s bits; Bob continues on $D(B)$ and outputs 1 iff the estimate is at least $n/2$. Three claims (Claims 3.3–3.5 of the paper) verify the ROPD protocol conditions: *pseudodeterminism* — the canonical value is $h(A, B) = \mathbf{1}\{g(D(A) \cdot D(B)) \geq n/2\}$, where g is A_n 's canonical output, so Bob outputs h with probability $\geq 2/3$ on every input; *representation obliviousness* — ρ_n -equivalent instances concatenate to R_0 -equivalent streams (same set of distinct elements), so the output distributions coincide; *correctness* — on promised inputs, if $|A \cup B| \leq n/4$ then $g \leq \frac{4}{3} \cdot \frac{n}{4} = \frac{n}{3} < \frac{n}{2}$, while if $|A \cup B| \geq 3n/4$ then $g \geq \frac{2}{3} \cdot \frac{3n}{4} = \frac{n}{2}$, and the $(\frac{1}{3}, \frac{1}{3})$ -approximation guarantee keeps the estimate on the correct side of the threshold. Note the *canonical-value argument*: it is g , not the algorithm's random output, that the threshold reads, which is what makes the protocol PD rather than merely correct.

Link 2: protocol to disjointness (Lemma 3.7 of the paper).

Lemma 12.23 — Promise-Union lower bound, Lemma 3.7 of the paper. The communication complexity of any ROPD protocol for Promise-Union $_m$ is $\Omega(m)$. ■

Proof sketch. From a family of ROPD Promise-Union protocols with $o(m)$ communication, build a pseudodeterministic protocol for

DISJ_m with $o(m)$ communication. Because the canonical output of a ρ -oblivious protocol depends only on the union, the function $h(X) = g(X, \emptyset)$ determines everything; take an inclusion-minimal set Y with $h(Y) = 1$ (so $h(Z) = 0$ for every proper subset $Z \subsetneq Y$), which exists with $m < |Y| \leq 4m$. Complements inside Y then turn disjointness into a union question by De Morgan, $A \cap B = \emptyset \iff \overline{A} \cup \overline{B} = Y$, and the two cases are separated by h itself. The full proof, including why the players may agree on Y in advance, is in the Appendix (Section 12.5). ■

Link 3: contradiction. If an ROPD F_0 estimator used $o(n)$ space, Link 1 with $m = n$ would give ROPD Promise-Union $_n$ protocols of $o(n)$ communication, contradicting Lemma 12.23, since $\Omega(n) = \Omega(m)$. The multipass version (Lemma 4.2 of the paper) simulates $t(n)$ passes with $t(n)$ alternations of the state hand-off at cost $2s \cdot t(n)$, and plugging $s = o(n/t(n))$ contradicts the same lemma; the transfer is exactly Proposition 12.12 in the two-party case. The *output-distribution argument* is the thread running through both links: representation obliviousness is precisely the statement that output distributions are invariant under input re-arrangement, and both the streaming-to-protocol step and the Y -construction exploit that invariance to pin canonical values.

What it buys. The theorem locates the price of reproducibility: each stability requirement alone is affordable (BJKST and AMS are representation oblivious in polylog space, Theorems 5.1 and 5.4 of the paper, and ordinary PD estimators for F_0 are not excluded), but together they force linear space. The paper also identifies the exact communication question that would extend the bound to all pseudodeterministic estimators — a lower bound for New-Promise-Union (gap $0.6m$ vs $0.4m$) against protocols oblivious only under the identity relation — and shows a curious phase transition: against the coarser τ_m -equivalence (union *and* intersection fixed), a trivial protocol solves Promise-Union, so the choice of obliviousness quotient is the whole game. For systems designers the message is sharp: order-invariant, run-stable distinct-count reporting cannot be sketched.

12.4 Advanced usage and further reading

Three refinements extend the method past the two-party, constant-error, single-relation template: information cost for amortized and multi-query bounds, round elimination for round/pass tradeoffs, and asymmetric protocols for data-structure and probe lower bounds. This section sketches each in prose, one further corpus exhibit, then the reading list.

Deterministic distributed lower bounds for k -edge connectivity [15]. In the distributed sketching model, every node of a graph sends one message (a sketch of its neighborhood) to a referee who must decide a global property. The paper proves that every *deterministic* algorithm deciding k -edge connectivity on n -node graphs has a worst-case message length of $\Omega(k)$ bits, for every super-constant $k \leq \gamma\sqrt{n}$ (Theorem 1 of the paper). The hard graph is built by the probabilistic method: a family of $e^{\Theta(k)}$ neighborhoods (an almost-disjoint set family with pairwise intersections at most $\epsilon d/2$, instantiated at $d = 2k - 1$; Lemma 3 of the paper) is paired with a pigeonhole argument on nested partitions (Claim 1) showing that if every node sends $o(k)$ bits, then for at least a $3/4$ fraction of nodes there exist two neighborhoods that are *indistinguishable* yet *separated* — one forcing k -edge-connectivity, the other not (Lemma 4 and Corollary 1 of the paper). The communication anchor is a new relation, $\text{UniqueOverlap}_{m,s}$ (three parties with support- s vectors must decide whether their inputs share a common element), whose deterministic complexity in the *simultaneous* three-party model is $\Omega(m)$ (Theorem 2 of the paper): the three parties simulate the sketching algorithm's messages, and the indistinguishable pair supplies the fooling set for the tiling bound of Theorem 12.3 — deterministic complexity, so the rectangle machinery of Section 12.0.1 applies verbatim rather than Yao's principle.

12.4.1 Information cost and amortized lower bounds

The information-cost calculus of Section 12.0.5 becomes indispensable when the question is amortized: k queries over one input, k copies of a stream problem, or protocols where only the *total* information leaked matters. Theorem 12.15(ii) turns a single-copy information bound into a k -fold communication bound, and Bar-Yossef et al.'s information-statistics approach [3] executes this for disjointness under Razborov's distribution, yielding the tight direct-sum theorem and, via the streaming transfer, the space lower bounds for frequency moments and distances [1, 13]. The practical recipe: pick the input distribution under which the transcript's information about the hidden object is easiest to bound from below (usually the mixture that correlates YES and NO cases as in Section 12.1), prove $\text{IC} \geq c$ for one copy, conclude $\Omega(kc)$ total communication for k copies, and divide by the simulation overhead as in step 5 of the reachbox recipe.

12.4.2 Round elimination

Many systems bounds are stated in rounds (distributed passes, multi-pass streaming, cell-probe with amortized probes), and the matching communication technique deletes the first round of a protocol: if the

first message carries little information about the queries that arrive only at the end, then a protocol for the $(r - 1)$ -round problem exists at slightly worse parameters, and induction drives the complexity to a contradiction at zero rounds. The asymmetric setting — one party holds a large table, the other a short query — is the natural habitat: Miltersen, Nisan, Safra, and Wigderson used round elimination on data-structure problems to prove that certain query times force large space [12], and the same peeling argument gives $\Omega(1/\varepsilon^2)$ -type pass-space tradeoffs in streaming: eliminate the first pass by arguing its state cannot anticipate which of the ε -scale blocks the hard relation hides in. When your reduction yields a *few-round* protocol rather than a one-way one — as in the multipass simulation of Corollary 12.22 — round elimination is the tool that converts “cheap in r rounds” into “costly in space”.

12.4.3 *Asymmetric protocols and sketching*

The asymmetric model charges the large-side player (Alice, holding the database or stream prefix) and lets the small side (Bob, holding the query or suffix) speak for free; it matches any setting where a big immutable object is summarized once and queried many times — sketching, materialized samples, cell-probe structures. Indexing (Theorem 12.9) is the canonical hard relation here, and its one-way structure is exactly what a one-pass streaming cut produces (Figure 12.1). Simultaneous-message (sketching) protocols are the extreme point: all parties speak once, to a referee; hard relations there are harder to find, and the three-party UniqueOverlap relation of [15] is a good corpus example of tailoring the relation to the model rather than forcing DISJ into a shape it does not fit.

12.4.4 *Further reading*

- Kushilevitz and Nisan [10] — the comprehensive monograph; the deterministic theory (rectangles, tilings, rank), the canonical relations, and the multiparty models all live here.
- The originals: Kalyanasundaram and Schnitger [9] and Razborov [14] for disjointness; Babai, Frankl, and Simon [2] for the complexity classes; Chakrabarti, Khot, and Sun [5] for the multiparty rate used in Section 12.1.
- The streaming consequences: Alon, Matias, and Szegedy [1] where the frequency-moment program began, and Muthukrishnan’s survey [13] for the catalogue of space lower bounds by communication reduction.

- Information complexity: Bar-Yossef, Jayram, Kumar, and Sivakumar [3] for the information-statistics method; Jain, Radhakrishnan, and Sen [8] for message compression and direct sums; Braverman [4] for the interactive theory and the simulation theorem.
- Asymmetric communication and data structures: Miltersen, Nisan, Safra, and Wigderson [12], the round-elimination source.

Chapter Recap

- **What the tool is.** Communication complexity proves that data must move or state must be large: cut the input among parties, let the algorithm's memory and messages be the transcript, and invoke a lower bound for a hard relation — DISJ, Index, GT, or a tailored cousin.
- **Which relation when.** Interactive or multiparty two-sided settings: DISJ ($\Omega(m)$ randomized; $\Omega(m/(k \log k))$ for k parties). One-pass streams and asymmetric summarize-then-query settings: Index ($\Omega(m)$ even one-way). Deterministic settings and threshold structure: fooling sets and GT ($\Theta(\log N)$). Amortized/multi-copy questions: information cost with the direct-sum and simulation theorems.
- **Why the transfer works.** A pass of a space- S algorithm over k chunks is a kS -bit protocol round (Proposition 12.12); oracle queries, sketch messages, and state hand-offs are all transcripts at known bit cost, so communication bounds divide into space, round, message-length, or query bounds.
- **What the corpus showed.** Noisy frequency moments (Section 12.1): a multiparty-DISJ embedding with similarity-pattern gadgets proves $\Omega(\epsilon^{-1/p} m^{1-1/p})$ space and communication. Join-project sampling and counting (Section 12.2): two- and k -party DISJ reductions in the query model, plus a coupon-collector transfer from listing hardness, certify $N/\sqrt{\text{OUT}}$ optimality. Representation obliviousness (Section 12.3): a three-link chain — estimator to Promise-Union protocol to DISJ — forces $\Omega(n)$ space for ROPD F_0 .
- **Advanced usage.** Information cost for amortized bounds (Section 12.4.1), round elimination for pass/space tradeoffs, asymmetric and simultaneous-message protocols for data structures and distributed sketching (including the deterministic $\Omega(k)$ message bound of [15]). Deferred proofs are collected in the Appendix (Section 12.5).

12.5 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 12.9 (indexing, one-way)

Statement (recalled). For every constant $\delta < 1/4$, any public-coin one-way protocol for Index_m with worst-case error at most δ communicates at least $(1 - H(2\delta))m - 1$ bits, H the binary entropy. ■

Proof. Let Π be such a protocol of cost c , and feed it a uniformly random input (x, j) . Averaging over the public coins, some fixed random string r leaves the deterministic protocol Π_r with error at most δ under this distribution. In Π_r , Alice's message is a function $m(x) \in \{0, 1\}^c$ and Bob decodes $g(m, j) \in \{0, 1\}$. For a message m , let $z_m = (g(m, 1), \dots, g(m, m)) \in \{0, 1\}^m$ be the answer string it induces, and let $X_m = \{x : m(x) = m\}$. The distributional error is

$$\frac{1}{2^m} \sum_m \sum_{x \in X_m} \frac{d_H(x, z_m)}{m} \leq \delta,$$

where d_H is Hamming distance — the total mismatch mass is at most $\delta m 2^m$. Let B_m be the Hamming ball of radius $2\delta m$ around z_m ; since $\delta < 1/4$ the standard volume bound gives $|B_m| \leq \sum_{i \leq 2\delta m} \binom{m}{i} \leq 2^{H(2\delta)m}$ (the binomial entropy bound; cf. Chapter 11). Every $x \in X_m \setminus B_m$ contributes at least $2\delta m$ mismatches, so

$$2\delta m \cdot |\{x : x \notin B_{m(x)}\}| \leq \delta m 2^m \implies |\{x : x \in B_{m(x)}\}| \geq 2^{m-1}.$$

But $\{x : x \in B_{m(x)}\} \subseteq \bigcup_m B_m$, a union of at most 2^c balls, hence $2^{m-1} \leq 2^c \cdot 2^{H(2\delta)m}$ and therefore $c \geq (1 - H(2\delta))m - 1$. For the deterministic claim, set $\delta = 0$: correctness forces $d_H(x, z_{m(x)}) = 0$ for every x , so each message is used by at most one input, i.e. $2^c \geq 2^m$. ■

Proof of Lemma 12.19 (matrix-query bound)

Statement (recalled). For the matrix query

$$Q_m = \pi_{A,C}(R_1(A, B) \rtimes R_2(B, C)),$$

any randomized algorithm that outputs any multiplicative-factor approximation of OUT with high probability requires $\Omega(\min\{n, N/\sqrt{\text{OUT}}\})$ queries in the query model. ■

Proof. The proof (of the paper's two-party lemma, transcribed with cleaned notation) reduces disjointness on universe $[m]$. Alice and Bob receive $S_A, S_B \subseteq [m]$ with $|S_A| = |S_B| = m/4$ and construct a database over attributes A, B, C : the domain of B is $[m]$ with active domain

$S_A \cup S_B$ (of size n); $D(A)$ and $D(C)$ are disjoint value sets of size $\sqrt{K}$ each, for a parameter $K > 1$. Alice sets $R_1 = D(A) \times S_A$; Bob sets $R_2 = S_B \times D(C)$. The input size is $N = |R_1| + |R_2| = \sqrt{K}(|S_A| + |S_B|) = \sqrt{K}m/2$. For the output: if $S_A \cap S_B = \emptyset$, no b -value joins, so $\text{OUT} = 0$; if some $b \in S_A \cap S_B$, then every pair $(a, c) \in D(A) \times D(C)$ appears, so $\text{OUT} = K$. Any multiplicative-factor approximation separates 0 from K with high probability, deciding disjointness.

Now the transcript accounting. Each oracle query is answered by its owner with a small number of bits: a tuple test $(a, b) \in R_1$ is answered by Alice checking $b \in S_A$ (one bit); degree and neighbor queries on $a \in D(A)$ are answered from S_A (degree $|S_A|$; the neighbors are the elements of S_A); a uniform sample from R_1 Alice draws locally from $D(A) \times S_A$; Bob handles R_2 symmetrically; and b -side queries on R_1 use the public set $D(A)$. So q queries yield a two-party disjointness protocol of $o(q)$ communication (a constant number of bits per query). Since $n = |S_A \cup S_B| \geq m/4$ and, when $\text{OUT} = K$, $N/\sqrt{\text{OUT}} = m/2$, an algorithm making $o(\min\{n, N/\sqrt{\text{OUT}}\}) = o(m)$ queries would solve disjointness with $o(m)$ communication, contradicting Theorem 12.10. When $\text{OUT} = 0$ the bound reads $\Omega(n)$, which holds since the active domain of B must be read to certify emptiness (the paper's hard instances for this case set $n = \Theta(N)$). ■

Proof of Lemma 12.23 (Promise-Union)

Statement (recalled). The communication complexity of any ROPD protocol for Promise-Union_m is $\Omega(m)$. ■

Proof. The proof (of the paper's Lemma 3.7, transcribed with cleaned notation) contradicts Theorem 12.10. Suppose there is a family $P = (P_\ell)_{\ell \in \mathbb{N}}$ of ROPD protocols solving $\text{Promise-Union}_\ell$ with communication $o(\ell)$. Set $n = 4m$ and consider P_n . Since P_n is pseudodeterministic, it has a canonical output $g(A, B)$ with $\Pr[P_n(A, B) = g(A, B)] \geq 2/3$ for all inputs; since it is ρ_n -oblivious, g is constant on ρ_n -classes, i.e. $g(A, B)$ depends only on the union $A \cup B$. Define $h(X) = g(X, \emptyset)$, so that $g(A, B) = h(A \cup B)$ for all A, B . Because P_n solves Promise-Union_n , $h(X) = 1$ whenever $|X| \geq 3n/4$ and $h(X) = 0$ whenever $|X| \leq n/4$.

Since $h([n]) = 1$, there exists an inclusion-minimal $Y \subseteq [n]$ with $h(Y) = 1$ and $h(Z) = 0$ for every proper subset $Z \subsetneq Y$ — minimality over the full subset lattice, not just by cardinality. Any $|X| \leq n/4$ has $h(X) = 0$, so $|Y| = n' > n/4 = m$; trivially $n' \leq n = 4m$. The set Y , and a bijection identifying Y with $[n']$, depends only on the protocol P_n — public information the players fix before seeing their inputs. The disjointness protocol Q_m is: Alice and Bob receive $A, B \subseteq [m] \subseteq [n']$, form the complements $\bar{A} = [n'] \setminus A$ and $\bar{B} = [n'] \setminus B$, and run $P_{n'}$ on $(\bar{A}, \bar{B})$; Bob outputs its answer. By De Morgan, $A \cap B = \emptyset \iff$

$\overline{A} \cup \overline{B} = [n']$. In the disjoint case the union is all of $[n'] = Y$, whose canonical value is $h(Y) = 1$, so the output is 1 with probability $\geq 2/3$. In the intersecting case the union is a *proper* subset of $[n'] = Y$, whose canonical value is 0 by minimality, so the output is 0 with probability $\geq 2/3$. Thus Q_m is a pseudodeterministic protocol for DISJ_m of cost $o(n') = o(m)$ (as $n' \leq 4m$), contradicting $\text{CC}(\text{DISJ}_m) = \Omega(m)$. ■

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Online Decision-Making: Competitive Analysis and Regret

Online decision-making is the theory of *commitment under ignorance*. The input arrives — a workload, a stream, a sequence of oracle answers — and the system must act *now*: adapt a filter entry or keep paying for false positives, trust an ML prediction or verify it, repair a data point it will never revisit, pick a configuration for the next round. What is judged later is the committed sequence, against a comparator that saw the whole input in advance. Two classical yardsticks quantify the gap. *Competitive analysis* bounds the cost of every run by a multiple of the offline optimum: “within factor c of the best possible, on every input, no matter how adversarial.” *Regret* sums the per-round losses of a repeated decision and bounds the excess over the best fixed action in hindsight: “average loss converging to the best configuration’s, at rate $O(1/\sqrt{T})$.” For a systems paper these convert empirical tuning folklore into worst-case contracts, and choosing between the two frames is the craft this chapter teaches.

In the SIGMOD’26 corpus this tool family is sparse — two papers carry it via competitive analysis [8, 9] and two via online regret [10, 13], none an online-algorithms paper per se. The pattern recurs wherever a system must commit before the future reveals itself: adversarial arrival order or worst-case oracle errors, irrevocable choices, and a guarantee quoted against a hindsight optimum. All four corpus exhibits double as this chapter’s exemplars. This chapter presents the two frames and their ladder first; Section 13.5 adds learning-augmented algorithms, adaptive adversaries, and the primal-dual method as extension reading.

13.0 The tool: online decisions

The problem class is uniform across applications: an algorithm interacts with an input $\sigma = (\sigma_1, \sigma_2, \dots)$ *piece by piece*, and the action at piece t is fixed before σ_{t+1} is seen — the filter meets the next query before

LEARNING OBJECTIVES

- Recognize the online-decision shape: a system that must commit before the input ends.
- Prove the ladder: rent-or-buy at ratio 2, its randomized $e/(e-1)$ twin, Hedge and UCB at $\sqrt{T}$ -type regret.
- Match the frame to the structure: irrevocable state $\rightarrow$ competitive ratio; repeated game with feedback $\rightarrow$ regret.
- Dissect a paper’s argument: comparator, adversary class, feedback, bound.

knowing how often it recurs, the repair engine commits a cleaned point before the series ends, the search routine prunes on an oracle's say-so. The analysis compares the committed sequence against a *comparator with foresight*. Two comparators define the two frames of this chapter: the *offline optimum* — the best decision sequence over the whole input — and the *best fixed action* — the single decision best over all rounds. Everything else is ladder craft: which comparator, which adversary, which feedback.

Notation

$\text{ALG}(\sigma)$ is the cost of the online algorithm on input σ and $\text{OPT}(\sigma)$ the cost of the offline optimum; for randomized algorithms the bound is on $\mathbb{E}[\text{ALG}(\sigma)]$, expectation over the algorithm's coins. An *oblivious* adversary fixes σ before seeing any coin flip; an *adaptive* adversary may choose σ_{t+1} as a function of the algorithm's past behavior. In the regret frame, T is the number of rounds, N (or K) the number of actions, losses and rewards are normalized to $[0, 1]$, and R_T denotes regret.

13.0.1 The ladder: ratios, then regrets

The competitive frame descends from the amortized analysis of list update and paging [17], and its definitions are compact.

Definition 13.1 (Competitive ratio). An online algorithm A for a cost-minimization problem is c -competitive if there is an input-independent constant b with

$$\text{ALG}(\sigma) \leq c \cdot \text{OPT}(\sigma) + b \quad \text{for every input sequence } \sigma;$$

for randomized A the bound is on $\mathbb{E}[\text{ALG}(\sigma)]$, read against a specified adversary class.

The canonical exhibit is *ski rental* (rent-or-buy): skis rent for 1 per day and cost B to buy; each morning the skier learns only whether the season continues, and must decide rent-or-buy before skiing. Every “small repeated cost versus one large permanent cost” trade-off in a system is this problem in disguise.

Theorem 13.2 — Deterministic ski rental. Rent for $B - 1$ days, then buy. This algorithm is $(2 - \frac{1}{B})$ -competitive, and no deterministic algorithm achieves a smaller competitive ratio. ■

Proof. Let the season last t days. If $t < B$ the algorithm rents throughout, paying $t = \text{OPT}(t)$. If $t \geq B$ it pays $(B - 1) + B = 2B - 1$ while $\text{OPT}(t) = \min(t, B) = B$, a ratio of $2 - 1/B$. The matching lower bound is Theorem 13.3 below. ■

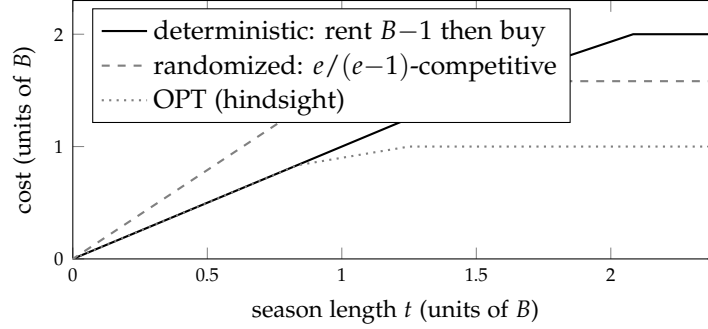


Figure 13.1: Ski rental geometry (prices in units of B): every algorithm tracks OPT while renting and flattens when it buys. The deterministic threshold at B flattens at $2B$; the randomized buy time flattens, in expectation, at $\frac{e}{e-1}B$ — the whole game is where to place the elbow.

Theorem 13.3 — Deterministic lower bound. Every deterministic ski-rental algorithm has an input on which its ratio is at least $2 - 1/B$. In particular no deterministic algorithm is c -competitive for any $c < 2$ and all B . ■

Proof sketch. A deterministic algorithm is a buy day b . The adversary ends the season exactly on day b : buying too early wastes the purchase, too late wastes the rent. The full two-case accounting is in the Appendix (Section 13.6). ■

Randomization helps once the adversary is fixed to be *oblivious*: it must commit to the season length before seeing the algorithm's coins, so a distribution over buy days cannot be targeted [4, 12].

Theorem 13.4 — Randomized ski rental, continuous relaxation. Let the season end at any real time $t \geq 0$, with rental rate 1 and purchase price B . Against an oblivious adversary, the algorithm that buys at a random time $x \in [0, B]$ drawn from the density $p(x) = e^{x/B} / (B(e-1))$ (renting until x or the season's end, whichever comes first) satisfies, for every t ,

$$\mathbb{E}[\text{ALG}(t)] \leq \frac{e}{e-1} \cdot \text{OPT}(t). \quad \blacksquare$$

The ratio $e/(e-1) \approx 1.582$ is also the optimal randomized competitive ratio for ski rental against oblivious adversaries [12].¹

Proof. The expected cost is $\mathbb{E}[\text{ALG}(t)] = \int_0^t p(x)(x+B)dx + t \Pr[x > t]$. Write $s = t/B$ and substitute $u = x/B$; using $\int e^u(u+1)du = e^u u$,

$$\frac{\mathbb{E}[\text{ALG}(t)]}{B} = \frac{s e^s}{e-1} + s \left(1 - \frac{e^s - 1}{e-1}\right) = \frac{s}{e-1} + s = \frac{e}{e-1} \cdot \frac{t}{B} \quad (t \leq B),$$

while for $t \geq B$ the same expression at $s = 1$ gives $\mathbb{E}[\text{ALG}(t)] = \frac{e}{e-1}B \leq \frac{e}{e-1} \text{OPT}(t)$, since $\text{OPT}(t) = B$ there. ■

¹ The discrete problem attains $e/(e-1) + O(1/B)$ with a geometric distribution over buy days. The pattern to remember: choose the coin distribution so the expected cost is a constant multiple of the optimum for every input — an equalization condition, not a tail bound.

The second frame changes the protocol: instead of one commitment against one adversarial input, the same decision repeats over T rounds with feedback after each round.

Definition 13.5 (External regret). At each round $t \in [T]$ the learner picks an action a_t from a set $\mathcal{A}$ of size N , then suffers loss $\ell_t(a_t)$ for a loss vector $\ell_t \in [0, 1]^{\mathcal{A}}$. The *external regret* is

$$R_T = \sum_{t=1}^T \ell_t(a_t) - \min_{a \in \mathcal{A}} \sum_{t=1}^T \ell_t(a),$$

the excess loss over the *best fixed action chosen once, in hindsight*. An algorithm is *no-regret* if $R_T = o(T)$: its average loss converges to the best fixed action's. In *full-information* feedback the whole ℓ_t is revealed after round t ; in *bandit* feedback only $\ell_t(a_t)$.

The naive candidate — follow the leader (FTL), play the action with the least past loss — is not no-regret: two actions with alternating losses make FTL wrong at every round, for $\Theta(T)$ regret. Every no-regret algorithm is FTL stabilized: by exponential damping (multiplicative weights, below), by randomization (follow-the-perturbed leader), or by a proximal term (online gradient descent) [6, 18]. The workhorse is:

Theorem 13.6 — Multiplicative weights / Hedge. Let $N \geq 2$, $T \geq 4 \ln N$, and let the loss vectors $\ell_1, \dots, \ell_T \in [0, 1]^N$ be arbitrary — possibly chosen adaptively, after seeing the learner's past actions. Exponential weights initializes $w_1(i) = 1$ for all i , updates $w_{t+1}(i) = w_t(i) (1 - \varepsilon)^{\ell_t(i)}$, and plays action i with probability $w_t(i) / \sum_j w_t(j)$. With $\varepsilon = \sqrt{\ln N / T}$,

$$\mathbb{E}[R_T] \leq 2\sqrt{T \ln N}. \quad \blacksquare$$

Proof idea. A potential argument on the total weight W_t : Jensen's inequality charges the expected loss of round t to the log-decline of W_t , and the weights telescope; the best action's weight lower-bounds W_{T+1} . (Full proof in Section 13.6; the scheme originates with [11, 14] and is surveyed in [1].) $\blacksquare$

When the environment is *stochastic* rather than adversarial — each action i is an arm with a fixed unknown reward distribution of mean $\mu_i \in [0, 1]$, and only the played arm is observed — the bandit rung improves on Hedge by concentrating plays on plausible-optimal arms only:

Theorem 13.7 — UCB1, Theorem 1 of 2, notation cleaned. Play each of the K arms once; thereafter play the arm maximizing the index $\bar{r}_i + \sqrt{2 \ln t / n_i}$ ($\bar{r}_i$: empirical mean, n_i : play count). With $\Delta_i = \max_j \mu_j - \mu_i$ the gap of

arm i , the expected regret satisfies

$$\mathbb{E}[R_T] \leq \sum_{i: \Delta_i > 0} \frac{8 \ln T}{\Delta_i} + \left(1 + \frac{\pi^2}{3}\right) \sum_{i: \Delta_i > 0} \Delta_i = O(\sqrt{KT \log T}). \blacksquare$$

Proof idea. Optimism: the index is an upper confidence bound, so by Lemma 13.8 a suboptimal arm is played only while its empirical mean still *could* be optimal; each such play shrinks its confidence width by $1/\sqrt{n_i}$, and summing the widths yields the $\ln T/\Delta_i$ terms. The confidence lemma is proved in Section 13.6; the completion is Theorem 1 of [2]. $\blacksquare$

The workhorse behind the index is a concentration statement — the same Hoeffding-plus-union-bound skeleton as the concentration chapter of this book, uniform over arms and play counts.

Lemma 13.8 — UCB confidence width. Fix a round $t \geq 1$; let $\bar{r}_i$ be arm i 's empirical mean after $n_i \geq 1$ plays. With probability at least $1 - 2K/t^3$, simultaneously for all arms,

$$|\bar{r}_i - \mu_i| \leq \sqrt{2 \ln t / n_i}. \blacksquare$$

The two regret rungs differ in what they assume: Hedge needs no stochasticity at all, while UCB1 needs independent rewards and pays with a gap-structured — often much smaller — bound. Adversarial bandit feedback is handled by $\exp3$ at $O(\sqrt{KT \log N})$ [3].

The newest rung changes what the algorithm may know. Modern systems carry predictors — learned cost models, ML oracles, LLM hints — and the question is how to use a prediction that may be wrong [15, 16].

Definition 13.9 (Consistency and robustness). An online algorithm A equipped with a prediction (advice) $\hat{\tau}$ is α -consistent if $\text{ALG} \leq \alpha \cdot \text{OPT} + b$ whenever the prediction is correct, and β -robust if $\text{ALG} \leq \beta \cdot \text{OPT} + b$ for every input and every prediction, however wrong.

Proposition 13.10 — Advice-following is fragile. For ski rental with a predicted season length $\hat{\tau}$, the advice-following algorithm — rent forever if $\hat{\tau} \leq B$, buy immediately if $\hat{\tau} > B$, which is what OPT would do if the prediction were true — is exactly 1-consistent and has robustness B (unbounded as a function of the instance). $\blacksquare$

Proof. If the prediction is correct, the algorithm plays OPT's own strategy, so its cost is OPT: consistency 1. For robustness, let the prediction be $\hat{\tau} = B + 1 > B$ while the true season lasts $t = 1$ day: the algorithm buys at once and pays B , while $\text{OPT}(1) = 1$, a ratio of B . $\blacksquare$

The learning-augmented program asks for both at once — near-1 consistency *and* bounded robustness, typically by hedging the advice with a classical schedule; the trade-offs are developed in Section 13.5.1.

13.0.2 How to choose the frame

Frame selection is mechanical once two questions are answered. *First, does the decision change persistent state?* If the choice is irrevocable — adapt a filter entry, buy the skis, rebuild an index, commit a repair, prune a search branch — a wrong commitment cannot be averaged away and the guarantee must hold on every input: use the competitive frame, comparator OPT, adversary explicit (workload order is rarely under the system’s control, so assume adversarial). If the same choice repeats over rounds, each ending with feedback, early mistakes amortize: use the regret frame, comparator the best fixed action. *Second, what does the feedback reveal?* Full-information feedback (every candidate’s score eventually observable) admits Hedge at $O(\sqrt{T \ln N})$; bandit feedback (only the chosen candidate’s score) forces UCB/exp3-style exploration at $O(\sqrt{KT \log T})$.

Hybrids are common and legitimate: a windowed streaming algorithm compared to the offline optimum with an *additive, decaying* per-step gap (Section 13.3) borrows the competitive comparator and the regret bookkeeping; a competitive sub-routine can sit inside a high-probability bound (Section 13.1); a stochastic tuner can be reduced to a bandit wholesale (Section 13.4). Name the frame in the paper’s own terms — corpus papers say “regret” for hindsight-optimal-sequence comparators too — but the comparator and the adversary class, not the word, determine which proof technique applies.

When to Reach for Online Analysis

Reach whenever a component must commit while the input is still arriving — adapt-or-not choices, windowed repair, oracle trust, repeated tuning — and the deliverable is a contract quoted against a hindsight optimum. Then:

1. Name the decision, what each arrival reveals, and which choices are irrevocable.
2. Name the comparator — offline optimum (ratio) or best fixed action (regret) — and the adversary class (oblivious or adaptive) if randomized.
3. Pick the rung: irrevocable one-shot state → deterministic competitive analysis (ratio 2 the classic barrier; randomization buys $e/(e-1)$ vs an oblivious adversary); repeated rounds with feed-

back $\rightarrow$ Hedge ($2\sqrt{T \ln N}$, adversarial, full information) or UCB ($O(\sqrt{KT \log T})$, stochastic, bandit); windowed streaming vs an offline DP $\rightarrow$ additive per-step regret with a decay argument.

4. Verify prerequisites: losses/rewards in $[0, 1]$; feedback model fixed; i.i.d. rewards for UCB-type bounds; adversary class stated for randomized ratios.
5. If a learned predictor is available, define consistency/robustness targets and hedge the advice rather than trusting it (Section 13.5.1).

13.1 Ski rental inside a distinct-count filter

The exemplar is a SIGMOD'26 paper on adaptive approximate data structures [8]. Approximate membership and distinct-count filters answer queries in constant space, but they return false positives, and the standard cure is verification: each suspect answer is checked against the database, at exactly the cost the filter was meant to avoid. Many such filters can also *adapt* — materialize a reverse map that eliminates a recurring false positive once and for all — but adaptation carries a one-time cost, and it is wasted if the false positive never recurs. The paper's observation is that this adapt-or-not decision is ski rental verbatim: verification is rent, the reverse map is the purchase, and the season length is how often the same false positive recurs. It derives the optimal deterministic adapt schedule for every cost ratio, proves it 2-competitive, exposes a golden-ratio threshold in the cheap-adaptation regime, and composes the schedule with a filter bound for an end-to-end high-probability guarantee.

Problem. A filter answers queries over a stream; a false positive costs C_{DB} — one database verification — each time it recurs. The filter may *adapt*: for a one-time cost C_{RM} it materializes a reverse map eliminating that false positive permanently. Write $\alpha = C_{RM}/C_{DB}$. The input is the sequence of false-positive recurrences, revealed one at a time and chosen adversarially; after each false positive the filter irrevocably decides adapt-or-not. The offline optimum knows the sequence and adapts at the best moment; the question is how much a blind schedule can lose against it.

Guarantee.

Theorem 13.11 — Adapt threshold, Theorem 1 of the paper, notation cleaned. If $\alpha \geq 1$ (adapting costs at least one verification), the optimal deterministic

strategy adapts immediately after the same false positive recurs $\lfloor \alpha \rfloor$ times, and it is 2-competitive — no deterministic schedule achieves a smaller ratio. ■

Theorem 13.12 — Cheap adaptation, Theorem 5 of the paper. If $\alpha < 1$, the optimal deterministic strategy adapts immediately when $\alpha < 0.618$ and on the second recurrence otherwise. ■

How competitive analysis is applied. The reduction is literal: rent = C_{DB} per recurrence, buy = C_{RM} , season = recurrence count of the worst false positive; Theorem 13.11 is the $\alpha \geq 1$ rung of Theorem 13.2, and its proof re-runs the two-case ski-rental accounting: for a season of $k \leq \lfloor \alpha \rfloor$ recurrences the schedule pays $k C_{DB}$ and is optimal; for $k > \lfloor \alpha \rfloor$ it pays $\lfloor \alpha \rfloor C_{DB} + (C_{DB} + C_{RM})$ against an offline optimum of $C_{DB} + C_{RM}$, a ratio of $1 + \lfloor \alpha \rfloor / (1 + \alpha) \leq 2$. In the $\alpha < 1$ regime two candidate schedules compete — adapt immediately (worst-case ratio $1 + \alpha$) and adapt on the second recurrence (ratio $(2 + \alpha) / (1 + \alpha)$) — and equating them gives $\alpha^2 + \alpha - 1 = 0$, i.e. the golden ratio $\alpha = (\sqrt{5} - 1) / 2 \approx 0.618$ of Theorem 13.12; adapting on the third recurrence or later is dominated (ratio $\geq (3 + \alpha) / (1 + \alpha)$). The composition step closes the loop with the data structure itself: by Lemma 6 of the paper, a strongly adaptive filter driven by the ski-rental schedule never returns more than $(1 + \lceil \alpha \rceil)(n\epsilon + O(\sqrt{n\epsilon \log n + \log n}))$ false positives with high probability — the filter’s whp false-positive budget, times the at most $1 + \lceil \alpha \rceil$ recurrences the schedule grants any one of them.

What it buys. The filter inherits a worst-case cost contract with no assumptions on the workload, and the adapt threshold is a closed form computed from two measured constants — verification latency and reverse-map build cost — rather than a tuned parameter. Because the composition multiplies, any filter whose false positives are whp-bounded plugs in and inherits the same end-to-end guarantee.

13.2 Paying for oracle predictions in pattern enumeration

The exemplar is a SIGMOD’26 paper on graph pattern matching with learned oracles [9]. Subgraph isomorphism — enumerate all matches of a pattern Q in a graph G — sits behind motif search, fraud detection, and knowledge-graph queries, and it is hopeless in the worst case. The paper accelerates a VF3-style enumerator with an ML model M that predicts, at each partial match, whether it can still extend to a full match, so that hopeless branches are pruned without exploration. The online content is not arrival order but *irrevocable trust*: once the enumerator skips a branch on the oracle’s word the matches beneath it

are lost, and the oracle is wrong some of the time — the adversary that must be survived is worst-case oracle error, not workload order. The paper’s answer is a guarantee package imported from online analysis wholesale: a recall floor under oracle errors, an impossibility barrier from enumeration complexity, and a competitive ratio for total work against the offline optimal enumeration.

Problem. Given G and Q , enumerate the match set $Q(G)$. The oracle M is invoked on partial matches ρ and predicts whether ρ extends to a full match; the enumerator prunes on “false”. M has error rate η and makes N_N false predictions in a run. Skipping is irrevocable: matches under a wrongly pruned branch never return. The paper measures accuracy by recall, packaged as its robustness notion:

Definition 13.13 (β -robustness, the paper’s notion, notation cleaned). An enumeration algorithm is β -robust if its recall is at least $1/\beta$: it returns at least a $1/\beta$ fraction of all full matches. ■

Guarantee.

Theorem 13.14 — Robustness barrier, Theorem 2 of the paper. No algorithm for SubIso is simultaneously output-polynomial and β -robust for any positive constant β , unless $\text{fewP} = \text{P}$. ■

Theorem 13.15 — Recall under oracle errors, Theorem 4 of the paper, notation cleaned. For constant $\varepsilon \in (0, 1)$, with probability at least $1 - \varepsilon$ the recall of the pruning enumerator VF3_M^N is at least

$$\frac{\text{Output}_T}{\text{Output}_T + N_N B'} \quad \blacksquare$$

where Output_T is the number of full matches returned, N_N the number of false predictions of M , and B an explicit quantity depending on $|G|$, the error rate η , and ε .

Theorem 13.16 — Competitive ratio, Corollary 7 of the paper, notation cleaned. With probability at least $1 - \varepsilon$, the total cost of the oracle-guided enumeration is at most $2|G| C f(Q, M) \cdot \text{OPT}$, where OPT is the offline optimal enumeration cost, C depends on $|G|$, η and ε , and $f(Q, M)$ is the cost of one oracle prediction. ■

How online decisions are applied. Three moves, each mapped to the recipe of Section 13.0.2. *Comparator:* the offline optimal enumeration cost; since the enumerator has precision 1, at least Output_T full matches exist, so $\text{Output}_T \leq \text{OPT}$, and a work bound of $|G|(\text{Output}_T + 1) C f(Q, M)$ becomes the competitive ratio of Theorem 13.16. *Adversary:* worst-case oracle errors — the guarantees hold for any prediction

sequence with the stated error count, an oblivious-style assumption on M 's mistakes rather than on input order, and the honest frame of this exemplar. *Workhorse*: the generalized Bonferroni inequality converts a per-false-prediction bound — each false prediction of M misses at most B matches with probability $1 - \varepsilon/N_N$ — into the global recall floor of Theorem 13.15 at probability $1 - \varepsilon$: union-bound craft, in enumeration clothing.

What it buys. ML-accelerated enumeration gets a contract, not an ablation: precision always; recall within an explicit factor that improves as the oracle does, so model quality pays off in guaranteed currency; a barrier (Theorem 13.14) showing that robustness and output-polynomial time cannot both be bought; and a work bound competitive with offline optimal whenever predictions are cheap.

13.3 Windowed time-series repair with decaying regret

The exemplar is a SIGMOD'26 paper on cleaning multivariate time series [10]. Real sensor streams violate both soft constraints — smoothness, forecast consistency — and hard constraints such as conservation laws, and repairs often must be made *as the stream arrives*: a fix committed now cannot be revisited when late evidence shows up. The paper, SHoTClean, first shows that offline repair under both constraint types is exactly a maximum-weight path in a DAG — nodes are anchor points, edges are interpolations, and weight is a strictly decreasing function of gap length — and then runs the same construction inside a sliding window of width W , committing each repair with only W steps of lookahead. Its guarantee is hybrid exactly as Section 13.0.2 anticipated: the comparator is the offline whole-stream optimum, and the bound is an additive per-step regret that decays *exponentially* in the window width.

Problem. A time series $X = (x_1, \dots, x_n)$ must be repaired to X' satisfying the hard constraints while staying close to X under $\Delta(X, X') = \sum_j \|x_j - x'_j\|_2$. Offline, the problem is solved on a DAG: a path $P = (i_0, \dots, i_m)$ of anchor points induces the repaired series X'_P by piecewise interpolation between consecutive anchors, edges carry weight $s_i \gamma(i_k - i_{k-1})$ with $s_i > 0$ and γ strictly decreasing (the paper uses $\gamma(t) = e^{-\beta t}$), and the maximum-weight path is the repair. Online, the algorithm at position i sees only the last W points and commits anchors irrevocably.

Guarantee.

Theorem 13.17 — Offline optimality, Theorem 2.1 of the paper. Let P^* be a maximum-weight path in the constructed DAG. Then the repaired series X'_{P^*} is exactly an optimal solution of $\min_{X'} \Delta(X, X')$, over repairs satisfying the hard constraints. ■

Theorem 13.18 — Online regret, Theorem 4.1 of the paper, notation cleaned. Let R_n be the total score gap between the offline algorithm and the online algorithm with window W . Then

$$R_n \leq n e^{-\beta(W+1)}, \quad \text{in particular} \quad \frac{R_n}{n} \leq e^{-\beta(W+1)}. \quad \blacksquare$$

How regret analysis is applied. The comparator is the offline maximum-weight path certified optimal by Theorem 13.17 — a competitive-style comparator — while the bookkeeping is per-step and additive, i.e. regret; the paper itself calls R_n the regret. The decomposition is the point. Fix an anchor choice of the offline optimum that connects x_{k^*} back to anchor x_{o_g} . If $k^* - o_g \leq W$ the online algorithm has the same point inside its window and incurs no regret on that step. If $k^* - o_g > W$ it must choose some $\hat{k} \in (o_g, i]$ within reach, and its one-step regret is $r_{o_g} = s_{o_g} [\gamma(k^* - o_g) - \gamma(\hat{k} - o_g)] \leq s_{o_g} \gamma(W + 1) \leq e^{-\beta(W+1)}$, using that γ is decreasing and $s \leq 1$. Summing over the $\leq n$ steps gives the display of Theorem 13.18. The structure doing the work is that scores decay geometrically, so beyond-window mistakes cost exponentially little — window width is a regret dial, not a correctness parameter.

What it buys. The operator tunes a single dial: W trades memory and detection latency against repair quality, and because average regret decays as $e^{-\beta(W+1)}$, a modest window already matches offline repair on average. The stream is processed once, with no reprocessing of committed repairs and no model of the data needed for the guarantee.

13.4 Discretization search as a bandit

The exemplar is a SIGMOD'26 paper on interpretable attribute discretization [13]. Turning numeric attributes into a few human-readable bins is a preprocessing step everywhere — dashboards, analytics, ML pipelines — but the “best” binning depends on a downstream utility such as visualization interestingness or causal analysis, and that utility is a black box: evaluable, expensive, with no exploitable functional form. The paper’s system, Quill, searches partition sets by first clustering candidate partitions of each attribute by their induced data distribution, then selecting cluster-per-attribute assignments with a Monte Carlo control agent whose reward is the blended semantic/utility objective. The online-decision content is the search itself: each candidate evaluation

is a pull of an arm with unknown reward distribution, and the paper proves a genuine bandit regret guarantee — via an exact reduction in the single-attribute case, a boundary it states honestly.

Problem. Given a dataset D with numeric attributes $\mathcal{A}_{\text{num}}$, a partition set B (one partition per attribute) is scored by $J(B, D) = \alpha M_{\text{sem}}(B) + (1 - \alpha) M_{\text{util}}(B, D)$, with M_{sem} grading interpretability and M_{util} the black-box utility; the Pareto frontier of $(M_{\text{sem}}, M_{\text{util}})$ defines optimal partition sets. Candidate partitions of each attribute are grouped into clusters $C_1, \dots, C_K$ (DBSCAN over Earth-Mover’s distances of bin distributions), and the search is posed as an MDP: states are partial attribute-to-cluster assignments, an action assigns a cluster to one attribute (a representative partition is sampled from it), transitions are deterministic, an episode is $d = |\mathcal{A}_{\text{num}}|$ steps, and the reward $J(B, D) \in [0, 1]$ arrives at episode end.

Guarantee.

Theorem 13.19 — Bandit reduction, Theorem 1 of the paper, notation cleaned. If (i) $d = 1$ (single attribute, one action per episode) and (ii) the Monte Carlo control agent selects and updates by the UCB index, then Algorithm 1 of the paper is a UCB-guided variant that reduces exactly to a stochastic multi-armed bandit with K arms (one per cluster) and reward $J \in [0, 1]$. Running it for T episodes inherits the UCB regret guarantees — gap-free $O(\sqrt{KT \log T})$ and gap-dependent $\sum_i O(\log T / \Delta_i)$, where Δ_i is how far cluster C_i ’s expected reward falls below the optimal expected reward. The equivalence does not extend to the multi-attribute case $d > 1$. ■

How online regret is applied. This is the reduction school: no new bandit algorithm is invented; the system is engineered so that Theorem 13.7 applies verbatim. Three choices carry the load: rewards normalized to $[0, 1]$ (the objective J is bounded); arm = cluster rather than raw partition, keeping K small enough for the $\sqrt{KT \log T}$ rate to bite; and stochastic rewards, which hold because a fixed cluster’s sampled partitions induce a fixed reward distribution observed i.i.d. across episodes. The proof is an induction on episodes showing selected arms, maintained statistics, and update rules coincide with UCB1’s. The $d > 1$ boundary is honest and instructive: with several attributes per episode the delayed joint reward couples the arms, the bandit equivalence fails, and the paper claims no regret bound there — the prerequisite check of the recipe in Section 13.0.2, done in public.

What it buys. A principled exploration budget: sublinear regret means the average quality of chosen partition sets converges to the best cluster's at rate $O(\sqrt{K \log T/T})$, so expensive utility evaluations are spent only where they pay off. The single-attribute contract covers exactly the common case of discretizing one column for one dashboard — and the paper is explicit about where it ends.

13.5 Advanced usage and further reading

Everything so far either compared one committed run against the offline optimum or averaged losses over rounds. Three extensions matter for systems work: predictions that may be wrong, adversaries that observe the algorithm, and online covering problems with LP structure.

13.5.1 Learning-augmented online algorithms

Definition 13.9 asked for α -consistency and β -robustness at once, and Proposition 13.10 showed naive advice-following fails the second clause. The repair is to *hedge* the advice: blend the advice-following schedule with a classical competitive schedule, so that with correct predictions the blend tracks the advice (asymptotic consistency near 1) while with adversarial predictions it degrades only to the classical ratio — a constant independent of the instance. The original construction for ski rental is due to [16], and the program was put on the map for caching by [15], yielding algorithms that are near-optimal when the predictor is good and never much worse than the classical bound when it is not. The craft is choosing the blend: it decides where on the consistency/robustness curve the system sits, and the ε of the blend is a dial in exactly the sense the δ -budgets of the concentration chapter are. For a systems paper the import is operational: any learned cost model in the loop — predicted season length, predicted re-access time, predicted selectivity — should be quoted as (α, β) advice, not trusted.

13.5.2 Adaptive adversaries

Section 13.0 fixed the adversary class only for the randomized ratio; the taxonomy deserves precision [4]. An *oblivious* adversary commits the whole input up front; an *adaptive-online* adversary chooses each piece as a function of the algorithm's behavior so far and must itself answer online; an *adaptive-offline* adversary observes everything and then answers with hindsight. The classical fact: against adaptive-offline adversaries randomization buys *nothing* — the optimal randomized competitive ratio equals the optimal deterministic one — so the $e/(e-1)$ of Theorem 13.4 is specific to the oblivious class, and a system whose clients can observe and react to its choices cannot buy that ratio. In the

regret frame, Hedge and exp3 remain no-regret even against adaptive adversaries [3, 6], which is one reason repeated-game guarantees travel better to contested environments than one-shot ratios.

13.5.3 *Primal-dual for online covering*

Ski rental is secretly a pair of linear programs: rent-or-buy per day is a fractional covering LP whose dual packs the season's days. The primal-dual method [7] turns this observation into an algorithm schema for online covering problems: maintain a feasible dual solution as a potential, raise dual variables whenever cost is incurred, and buy (constrain) whenever a dual constraint becomes tight. The schema yields monotone, deterministic algorithms with competitive ratios matching the integrality gap of the LP relaxation, and it powers many of the classical results: fractional ski rental at $e/(e-1)$, online set cover at $O(\log^2 n)$, paging and k -server relaxations, and online packing problems (matching, ad allocation) by duality symmetry. Database admissions — online index selection, admission control, resource allocation under arriving workloads — are covering-shaped more often than not, and the primal-dual template is usually the shortest route from “we have an LP” to “we have a competitive ratio.”

13.5.4 *Further reading*

- Borodin and El-Yaniv [5] — the comprehensive treatment of competitive analysis, adversary classes included.
- Bubeck and Cesa-Bianchi [6] — the survey of regret analysis for stochastic and adversarial bandits; UCB, exp3, and lower bounds in one place.
- The originals: Littlestone and Warmuth [14] and Freund and Schapire [11] for multiplicative weights; Auer, Cesa-Bianchi, Freund, and Schapire [2] for UCB and [3] for exp3; Zinkevich [18] for online convex programming.
- Learning-augmented algorithms: Lykouris and Vassilvitskii [15] and Purohit and Svitkina [16].
- Buchbinder and Naor [7] — the primal-dual method for online covering and packing, in full generality; Arora, Hazan, and Kale [1] for multiplicative weights as an algorithm-design meta-method.

Chapter Recap

- **What the tool is.** Online analysis handles components that must commit while the input is still arriving: either one irrevocable

run judged against the offline optimum (competitive ratio), or repeated decisions with feedback judged against the best fixed action in hindsight (regret).

- **Which frame when.** Irrevocable state, adversarial input $\rightarrow$ competitive ratio (deterministic 2 the classic ski-rental barrier; randomization buys $e/(e-1)$ against oblivious adversaries only). Repeated rounds with feedback $\rightarrow$ Hedge $2\sqrt{T \ln N}$ (adversarial, full information) or UCB $O(\sqrt{KT \log T})$ (stochastic, bandit). Windowed streaming $\rightarrow$ additive per-step regret against the offline optimum.
- **Why it works.** Deterministic ratios by two-case season accounting; randomized ratios by equalizing coin distributions; Hedge by a potential on the total weight; UCB by optimism plus confidence widths.
- **What the corpus showed.** Filter adaptation (Section 13.1): ski rental verbatim, threshold at the golden ratio, composed into a whp false-positive bound. Oracle-guided enumeration (Section 13.2): irrevocable pruning under worst-case oracle error; recall floors and a competitive ratio, plus a few $P = P$ barrier. Windowed repair (Section 13.3): regret $\leq n e^{-\beta(W+1)}$ against the offline DAG optimum. Discretization search (Section 13.4): an exact bandit reduction, honestly bounded to $d = 1$.
- **Advanced usage.** Learning-augmented (α, β) guarantees (Section 13.5.1) for predictors in the loop; adversary taxonomy (Section 13.5.2); the primal-dual schema for online covering (Section 13.5.3). Deferred proofs are in the Appendix (Section 13.6).

13.6 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 13.3 (ski-rental lower bound)

Statement (recalled). Every deterministic ski-rental algorithm has an input on which its ratio is at least $2 - 1/B$. ■

Proof. A deterministic algorithm is a buy day b (one that never buys loses unboundedly on long seasons, since its cost is t against $\text{OPT}(t) = B$ for $t \geq B$). The adversary ends the season on day b , so the algorithm pays $(b-1) + B$ while $\text{OPT}(b) = \min(b, B)$. If $b \leq B$, the ratio is $\frac{b-1+B}{b} = 1 + \frac{B-1}{b} \geq 1 + \frac{B-1}{B} = 2 - \frac{1}{B}$. If $b > B$, then $\text{OPT}(b) = B$

and the algorithm pays at least $B + B = 2B$, for a ratio of at least $2 \geq 2 - 1/B$. Hence no deterministic algorithm beats $2 - 1/B$, and in particular none is c -competitive for $c < 2$ and all B . ■

Proof of Theorem 13.6 (Hedge regret bound)

Statement (recalled). Exponential weights with $\varepsilon = \sqrt{\ln N/T} \leq 1/2$ on arbitrary adversarial losses in $[0, 1]^N$ satisfies $\mathbb{E}[R_T] \leq 2\sqrt{T \ln N}$. ■

Proof. Let $W_t = \sum_i w_t(i)$ be the total weight and $\bar{\ell}_t = \sum_i p_t(i) \ell_t(i)$ the expected loss of round t conditional on the past; since the adversary may be adaptive, $\mathbb{E}[\text{total loss}] = \mathbb{E}[\sum_t \bar{\ell}_t]$ by the tower property, so it suffices to bound $\sum_t \bar{\ell}_t$ pathwise. Because $x \mapsto (1 - \varepsilon)^x$ is convex, it lies below its chord on $[0, 1]$: $(1 - \varepsilon)^x \leq 1 - \varepsilon x$. Hence

$$W_{t+1} = W_t \sum_i p_t(i) (1 - \varepsilon)^{\ell_t(i)} \leq W_t (1 - \varepsilon \bar{\ell}_t),$$

and $\ln(1 - u) \leq -u$ gives $\varepsilon \bar{\ell}_t \leq \ln W_t - \ln W_{t+1}$. Summing over t and using $W_1 = N$:

$$\sum_{t=1}^T \bar{\ell}_t \leq \frac{\ln N - \ln W_{T+1}}{\varepsilon}.$$

For any fixed action i^* (chosen as the best in hindsight), $W_{T+1} \geq w_{T+1}(i^*) = (1 - \varepsilon)^{L^*}$ with $L^* = \sum_t \ell_t(i^*)$, so $-\ln W_{T+1} \leq L^* (-\ln(1 - \varepsilon)) \leq L^*(\varepsilon + \varepsilon^2)$ for $\varepsilon \leq \frac{1}{2}$. Therefore

$$\mathbb{E}[\text{total loss}] \leq \frac{\ln N}{\varepsilon} + (1 + \varepsilon) L^*,$$

$$\text{so } \mathbb{E}[R_T] \leq \frac{\ln N}{\varepsilon} + \varepsilon T \leq 2\sqrt{T \ln N},$$

using $L^* \leq T$ (losses are in $[0, 1]$) and $\varepsilon = \sqrt{\ln N/T}$. ■

Proof of Lemma 13.8 (UCB confidence width)

Statement (recalled). With probability at least $1 - 2K/t^3$, simultaneously for all K arms, $|\bar{r}_i - \mu_i| \leq \sqrt{2 \ln t / n_i}$. ■

Proof. Condition on the (random) play counts n_i ; for each arm i the n_i observed rewards are i.i.d. in $[0, 1]$ with mean μ_i , so additive Hoeffding gives, for any $u > 0$, $\Pr[|\bar{r}_i - \mu_i| \geq u] \leq 2 \exp(-2n_i u^2)$. Setting $u = \sqrt{2 \ln t / n_i}$ yields $2 \exp(-4 \ln t) = 2t^{-4}$ per arm, and a union bound over the K arms gives failure probability at most $2Kt^{-4} \leq 2K/t^3$ for $t \geq 1$; the conditioning is discharged by averaging over the play counts. ■

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